LAPLACIAN ALGEBRAS AND SUBCALIBRATIONS FOR MINIMAL HYPERCONES

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Dedicated to Anguo Cui

Abstract. We develop an algebraic mechanism based on Laplacian algebras for constructing minimal hypercones and determining their variational properties. Generator counts and degrees organize the constructions; homogeneous multipliers depending on the generators yield subcalibration fields, stability bounds and quantitative area-minimizing inequalities. Two-generator isoparametric algebras give polynomial subcalibrations for their minimizing hypercones. In ambient dimension at least three, strict gradient-square closure forces nonlinear odd degree to be cubic. We then study algebras of degrees $(2, 2, g)$ and quadratic compositions, placing OT–FKM splitting quartics and Riedler's pullbacks in a common quadratic-composition framework. It yields two further infinite quartic families with a sharp minimizing threshold, as well as harmonic sextics from complex Gram coordinates. For an irreducible harmonic minimal cubic on its essential space, we prove a sharp bound of five generators for its smallest Laplacian algebra. We determine the minimizing range of Clifford cubics and construct irreducible nonradial harmonic examples in ambient dimensions $8, 14, 26$; the last is strictly area minimizing. Finally, spectral families whose smallest algebras have arbitrarily many generators provide examples in every degree at least two: real parts of octonionic products, complex Pfaffians and complex determinants. Explicit potentials establish strict area minimality and stability bounds, including across nonisolated singularities. The algebraic identities and certificates are supported by reproducible exact symbolic computations.

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1. Introduction

Area-minimizing hypercones arise as tangent cones at singularities of area-minimizing hypersurfaces and play a central role in their regularity theory [39, 75, 79, 33]. Recent work of Chodosh and collaborators [18, 19] establishes generic regularity in ambient dimensions nine, ten, and eleven for Plateau solutions and homology minimizers. Their arguments rely on the analysis of minimizing cones and Jacobi fields. In particular, the treatment of cylindrical singularities in [19] uses the strict stability and strict area minimality of minimizing quadratic cone factors. We study three closely connected aspects of algebraic minimal hypercones: their variational properties, the structure of their associated Laplacian algebras, and their construction through polynomial realizations of algebraic closure relations. Laplacian algebras provide a common framework for these questions: their closure identities encode both the minimality equation for a defining polynomial and the differential expressions governing its subcalibrations.

The construction of algebraic minimal hypercones and their classification in low degree have a long history. In 1967, Hsiang [42] classified the quadratic case and initiated the study of cubic minimal hypercones. To formulate these problems algebraically, write $N = n + 1$, $n \geq 2$, $r = |z|$, and $R = r^2$, and set

$$\mathcal{L}(H) = |\nabla H|^2 \Delta H - \frac{1}{2} \left\langle \nabla |\nabla H|^2, \nabla H \right\rangle, \tag{1.1}$$

for $H \in C^2$. If F is a square-free homogeneous polynomial whose real irreducible factors have nonsingular real zeros, the regular part of $\{F = 0\}$ is minimal exactly when

$$\mathcal{L}(F) = \lambda_F F, \tag{1.2}$$

for a polynomial λ_F; see Proposition 3.14.

Without additional structural assumptions, both classification and explicit construction remain difficult. For example, the Cartan–Münzner identities describe isoparametric families whose minimal members yield algebraic hypercones [62, 63]. For radial minimal cubics, where λ_F is proportional to R, Peng–Xiao [66] and, independently, Tkachev [84, 83] developed a systematic classification theory.

Once a minimal hypercone is given, a further question is whether it minimizes area. The classical minimizing quadratic cones [6, 48, 72] provide a model for an algebraic approach to this question. Building on earlier calibration and subsolution arguments, De Philippis and Paolini [25] gave a short proof of the area minimality of Simons cones using subcalibrations. Suitable divergence inequalities on the two sides replace the divergence-free condition and give direct perimeter comparisons; see Remark 2.3.

Motivated by these quadratic constructions, we seek potentials $G = aF$, where a is a positive homogeneous multiplier on an appropriate conical domain. Its normalized gradient satisfies

$$X_G = \frac{\nabla G}{|\nabla G|}, \qquad \operatorname{div} X_G = \frac{\mathcal{L}(G)}{|\nabla G|^3}, \tag{1.3}$$

where $\nabla G \neq 0$. On the regular zero set in this domain, $\nabla G = a \nabla F$, so X_G agrees with the exterior normal to $\{F < 0\}$. The following definition includes the removable singular sets needed in our applications.

Definition 1.1 (Homogeneous subcalibration potential)**.** Let F be a nonzero homogeneous polynomial on $\mathbb{R}^N$. A continuous function G on $\mathbb{R}^N$ is a *homogeneous subcalibration potential* for F if there are a closed conical set Z with $\mathcal{H}^{N-1}(Z) = 0$ and a positive homogeneous function $a \in C^2(\mathbb{R}^N \setminus Z)$ such that $G = aF$ on $\mathbb{R}^N \setminus Z$, G is homogeneous of degree $\ell > 0$, and

$$\nabla G \neq 0, \qquad G\mathcal{L}(G) \geq 0 \quad \text{on } \mathbb{R}^N \setminus Z. \tag{1.4}$$

Since polynomial sublevel sets have locally finite perimeter, Corollary 2.8 gives perimeter comparison for $\{F < 0\}$ across Z.

For the Simons cone, with $x, y \in \mathbb{R}^m$, $m \geq 4$, the potential in [25, Section 2] is, up to a positive constant,

$$F = |x|^2 - |y|^2, \qquad G = |x|^4 - |y|^4 = (|x|^2 + |y|^2)F = RF.$$

For Lawson cones in $\mathbb{R}^k \times \mathbb{R}^h$, $k, h \geq 2$, put $U = |x|^2/(k-1)$, $V = |y|^2/(h-1)$, and $F = U - V$. In the dimension ranges of [24, Lemma 4.1, equation (4.7)], the quartic subcalibration potential factors as

$$G = \frac{U^2 - V^2}{4} = aF, \qquad a = \frac{U+V}{4}. \tag{1.5}$$

Phase-dependent alternatives are discussed in [52] and Remark 2.10. In the two polynomial examples above, both F and a lie in $\mathbb{R}[|x|^2, |y|^2] = \mathbb{R}[R, F]$. This algebra is closed under Δ and the gradient pairing $\Gamma(u, v) = \langle \nabla u, \nabla v \rangle$, so the subcalibration calculation stays in the same two quadratic variables.

More generally, a subalgebra $\mathcal{A} \subset \mathbb{R}[z_1, \ldots, z_N]$ is a *Laplacian algebra* if $R \in \mathcal{A}$ and $\Delta\mathcal{A} \subset \mathcal{A}$. It is graded and finitely generated [57, Corollary 22 and Lemma 24], and the product rule gives closure under Γ. For homogeneous generators $I = (I_1, \ldots, I_k)$, closure is recorded by polynomial identities

$$\Delta I_i = b_i(I), \qquad \Gamma(I_i, I_j) = c_{ij}(I),$$

which form the *closure table* [57, Proposition 37(a)]. For $F \in \mathcal{A}$ and a homogeneous multiplier $a = \alpha \circ I$, with α smooth on an open set $W \subset \mathbb{R}^k$, the chain rule expresses $\mathcal{L}(F)$, $\mathcal{L}(aF)$, and $|\nabla(aF)|^2$ in the generator variables on $I^{-1}(W)$. These closure tables may therefore guide the construction and recognition of minimal algebraic hypercones; see Section 7.

We use the structural theory of Mendes–Radeschi [57, 59, 58] to study cubic and quartic models and develop new polynomial realizations. We begin with algebras generated by the radius squared and one defining polynomial.

1.1. **Two generators and radial multipliers.**

In this two-generator case, the Mendes–Radeschi characterization [57, Proposition 37(c)] leads to an isoparametric family. We construct polynomial subcalibrations for all of its nonflat minimizing cones.

Theorem A (Polynomial subcalibrations for isoparametric hypercones)**.** *Let $C \subset \mathbb{R}^N$ be a nonflat area-minimizing isoparametric hypercone, and let F be its centered Cartan–Münzner defining polynomial, as in (6.5). There is an explicit homogeneous polynomial a_F, positive on $\mathbb{R}^N \setminus \{0\}$, such that $G = a_F F$ satisfies*

$$\deg G \leq 10, \qquad \nabla G \neq 0 \quad \text{on } \mathbb{R}^N \setminus \{0\},$$

and

$$\operatorname{sgn}(G) \operatorname{div} X_G > 0 \qquad \text{off } C.$$

The field X_G is a subcalibration of C, and its divergence proves strict area minimality.

Lawlor [47], Ferus–Karcher [31] and Wang [88] established the minimizing results used here; see Section 6.3 for their precise scope. Our explicit potentials have a dimension-independent degree bound and recover strict area minimality. The complete list is in Theorem 6.5. For harmonic defining polynomials, a radial criterion applies in arbitrary degree.

Theorem B (A radial criterion for harmonic polynomials)**.** *Let $F \not\equiv 0$ be a homogeneous polynomial of degree $e \geq 2$ on $\mathbb{R}^N$, $N \geq 3$, and suppose*

$$\Delta F = 0, \qquad S = |\nabla F|^2, \qquad \Gamma(F, S) = 2qF$$

for a polynomial q. If, for a constant $\kappa \geq 0$,

$$Rq \leq \kappa S \quad \text{on } \mathbb{R}^N, \qquad N \geq 2e + 1 + 2\sqrt{\kappa},$$

then $G = r^{(N-1-2e)/2}F$ generates a two-sided subcalibration. The cone $C = \{F = 0\}$, oriented as the boundary of $\{F < 0\}$, is strictly area minimizing, and

$$Q_C(u) \geq \left[\frac{(N-1-2e)^2}{4} - \kappa\right] \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C \setminus \{0\}).$$

The stability estimate follows from the multiplier–Jacobi identity and the homogeneous ground-state bound (Theorems 4.4 and 4.6). A finite κ exists exactly when ∇F vanishes only at the origin. Its optimal value is at least $e(e-1)$, with equality precisely for normalized harmonic Cartan–Münzner polynomials (Propositions 4.19 and 4.20).

Remark 1.2 (Realizability and regular algebraic cones). This regularity requirement connects Theorem B with Perdomo's question about non-isoparametric regular algebraic minimal hypersurfaces of spheres [67, Section 1 and Main theorem]; his nonexistence theorem concerns irreducible cubics in $\mathbb{S}^3$. We conjecture that, for $N \geq 3$, a nonconstant real homogeneous irreducible polynomial F whose zero link is nonempty and minimal, with $\nabla F \neq 0$ everywhere on that link, defines an isoparametric hypersurface. In the harmonic case, we expect F itself, up to scaling, to be a harmonic Cartan–Münzner polynomial. Proposition 4.22 proves this harmonic conclusion when $q = \lambda R^{e-2}$, an additional condition on the minimality weight, distinct from choosing the radial multiplier $a = r^\sigma$ in Theorem B.

1.2. **Gradient-square closure and quadratic compositions.** Adjoining $S = |\nabla F|^2$ closes the algebra in a distinguished class. For $N \geq 3$, odd degree at least three then forces $\deg F = 3$. If the critical set consists only of the origin, F is a Cartan cubic; the other cubic members are described in Proposition 7.5. In even degree, we record the closure identities and a short isoparametric consequence on the regular link, followed by a matrix quartic example.

For three-generator degrees $(2, 2, g)$, $g > 2$, subtracting a polynomial in the two block radii from the higher-degree generator makes it separately harmonic and bihomogeneous. The motivating OT–FKM splitting quartics, developed from [21], are quadratic compositions of Clifford coordinates, as are Riedler's quartic pullbacks. Section 8 develops their general orthogonal-block criterion. These compositions have generator degrees $(2, 4, 4)$, distinguished from strict gradient-square closure $(2, 4, 6)$.

Theorem C (Subcalibrations from orthogonal quartic blocks)**.** *Let E_0, E_1, E_2 be nonzero homogeneous quartic polynomials on $\mathbb{R}^N$, with $R = |x|^2$, satisfying*

$$E_0 + E_1 + E_2 = R^2, \qquad \Gamma(E_i, E_j) = 16RE_i\delta_{ij}, \qquad \Delta E_i = 8r_iR.$$

Assume $r_i > 0$, $r_1, r_2 \geq 2$, and either

$$r_1 + r_2 \geq 9, \qquad \text{or} \qquad r_1 = r_2 = 4 \quad \text{and} \quad r_0 \geq 5.$$

Set $F = (r_2 - 1)E_1 - (r_1 - 1)E_2$ and $\Sigma = \{E_1 = E_2 = 0\}$. Then there is a homogeneous polynomial potential $G = AF$ of degree at most ten, with $A > 0$ off Σ and $\operatorname{Crit} G = \Sigma$, such that

$$\operatorname{sgn}(G) \operatorname{div} X_G > 0 \qquad \text{where } F \neq 0.$$

The exceptional set Σ is removable for this field. The oriented boundary of $\{F < 0\}$ has support $\{F = 0\}$ and is strictly area minimizing and strictly stable.

Together with the instability results, the theorem gives the minimizing classification for the splitting family and recovers Riedler's Clifford–Simons range. The same table also treats the octonionic Hopf pairing and the balanced $(3, 7, 7)$ realization. Their weights, potentials and ranges are collected in Table 2; their different singular strata distinguish the geometric realizations. In particular, Theorem 8.27 constructs two further infinite families in $\mathbb{R}^{8k}$ with an unused block of weight two; they are area minimizing exactly for $k \geq 3$. Quadratic Jordan and Gram coordinates also give the degree-independent profile equation of Proposition 8.4 and the harmonic sextic family in Example 8.33.

1.3. **Cubic structure and a common subcalibration.** The historical and algebraic background is recalled in Section 9. For harmonic minimal cubics, we obtain a sharp bound on the smallest algebra. For $\mathcal{A} = \mathbb{R}[I_1, \dots, I_k]$, its *spherical quotient dimension* is $\dim I(\mathbb{S}^{N-1})$, where $I = (I_1, \dots, I_k)$.

Theorem D (Five generators for harmonic minimal cubics)**.** *Let F be a nonzero irreducible homogeneous harmonic cubic on its essential Euclidean space, with minimal regular zero set. The smallest Laplacian algebra $\mathcal{A}_R(F)$ has at most five homogeneous generators. For a radial cubic at most four suffice. For a nonradial cubic the algebra is polynomial on five algebraically independent generators of degrees $(2, 2, 3, 3, 4)$, with spherical quotient dimension four; its ambient dimension is 8, 14, or 26.*

Here radial means that $\mathcal{L}(F) = -qF$ has q proportional to R. The intrinsic closure recovers a Cartan polynomial and forces the dimension list $8, 14, 26$. Explicit Cartan deformations realize all three dimensions: the models in $\mathbb{R}^8$ and $\mathbb{R}^{14}$ are unstable, whereas the model in $\mathbb{R}^{26}$ is strictly area minimizing. For radial cubics, the following criterion is read from ΔS.

Theorem E (A cubic trace criterion)**.** *Let $F \not\equiv 0$ be a homogeneous harmonic cubic on $\mathbb{R}^N$, $N \geq 8$, satisfying $\Gamma(F, S) = 4RF$. If*

$$\Delta S \geq \max\left\{16, \frac{2(40 - N)}{3}\right\} R,$$

then there exists a continuous homogeneous potential $G = AF$ of degree six, with $A > 0$ on $\{S > 0\}$ and G smooth there, whose gradient is nonzero there and whose normalized gradient has the strict divergence sign of F off its zero set. The critical set $\{S = 0\}$ is removable. Hence $\{F < 0\}$ is locally perimeter minimizing, and its oriented boundary has support $\{F = 0\}$ and is strictly area minimizing.

The stronger form, Theorem 10.2, also covers the low-dimensional Clifford cases outside this sufficient trace bound. It yields the common degree-six potential $F\sqrt{RS}$ for rank-four trace cubics, complexified rank-three Jordan norms and the G_2 trilinear cubic. The nonradial model in $\mathbb{R}^{26}$ uses the same potential with its five-generator defect calculation.

For the Clifford family, let $(A_0, \dots, A_q)$ be a symmetric Clifford system on $\mathbb{R}^{2m}$, $q \geq 1$, and put

$$F(y, z) = \sum_{i=0}^{q} z_i \langle A_i y, y \rangle.$$

Theorem F (Clifford cubic classification)**.** *The oriented boundary of $\{F < 0\}$ is area minimizing exactly when $m \geq 4$, $q \geq 3$, and $(m, q) \neq (4, 3)$. In that range it is strictly area minimizing, and, after replacing F by $F/2$, its Laplacian data satisfy Theorem 10.2. For $(m, q) = (4, 3)$ the cone is unstable although every tangent cone at a nonzero point is area minimizing.*

The classification uses the common positivity criterion, singular-stratum obstructions and an explicit negative variation for the exceptional pair. The examples and their criteria are summarized in Table 1.

TABLE 1. Selected families and the criteria used for their minimizing members. The third column counts generators of the smallest Laplacian algebra $\mathcal{A}_R(F)$; ≤ 4 is an upper bound. Explicit potentials are constructed in the cited results.

Family	deg F	Number	Criterion	Minimizing range
Isoparametric	g	2	Two-generator classification; Theorem A	Full list in Theorem 6.5
Orthogonal quartic families	4	3	Three-weight criterion; Theorem C	Splitting, Hopf, pullback and balanced ranges in Table 2
Rank-four trace	3	3	Cubic trace bound; Theorem E	$N = 27$; Corollary 10.9
Complexified rank-three Jordan norms	3	4	Cubic trace bound; Theorem E	$N = 18, 30, 54$; Theorem 10.14
G_2 trilinear	3	4	Cubic trace bound; Theorem E	$N = 21$; Theorem 10.15
Clifford cubics	3	≤ 4	Full cubic criterion; Theorem 10.2	$m \geq 4$, $q \geq 3$, $(m, q) \neq (4, 3)$
Nonradial Cartan deformation	3	5	Five-generator positivity; Theorem 11.5	$N = 26$
Octonionic products; complex Pfaffians	$d \geq 3$	$d + 1$	Spectral threshold; Theorem H	Every $d \geq 3$; Theorem H
Complex determinants	$m \geq 3$	$m + 1$	Minor-norm inequalities; Section 14	Every $m \geq 3$; Theorem I

1.4. **Pullbacks and arbitrary-degree spectral families.** Riedler's pullback construction and question [69, Theorem 5.2.5 and Question 5.3.3(2)] motivate the transfer theorem in Theorem 5.4. A homogeneous target subcalibration H with $H\Delta H \geq -M|\nabla H|^2$ transfers through a quadratic Clifford map P to $R^a H(P)$ in sufficiently large admissible dimensions, under the singular-set hypotheses of that theorem. The corresponding smallest algebra has an exact description.

Theorem G (The smallest algebra under Clifford pullback)**.** *Let $A_1, \dots, A_m$ be symmetric endomorphisms of $\mathbb{R}^N$, $m \geq 2$, with $A_iA_j + A_jA_i = 2\delta_{ij}\,\mathrm{Id}$, and set*

$$P(x) = (\langle A_1x, x\rangle, \dots, \langle A_mx, x\rangle), \qquad R = |x|^2, \quad q(y) = |y|^2, \quad Q = q(P).$$

For every nonzero essential homogeneous polynomial f of degree $s \geq 2$ on $\mathbb{R}^m$,

$$\mathcal{A}_R(f \circ P) = \mathbb{R}[R, P^*\mathcal{A}_q(f)].$$

If $Q < R^2$ somewhere, this algebra is isomorphic to the polynomial extension $\mathcal{A}_q(f)[t]$, with target degrees doubled and $\deg t = 2$. In this case, its minimal number of homogeneous generators is exactly one more than that of $\mathcal{A}_q(f)$.

Iteration doubles the defining degree at each step and, under the nondegeneracy hypothesis, adds exactly one minimal generator per step (Corollary 5.7). Cubic applications appear in Corollary 11.6 and Theorem 11.7.

For arbitrary degree, nonnegative spectral variables give a second uniform construction. Write e_k for elementary symmetric polynomials, with $e_0 = 1$.

Theorem H (A spectral criterion and its octonionic and Pfaffian applications)**.** *Let $F, E_1, \dots, E_d$ be real homogeneous polynomials on $\mathbb{R}^N$ of degrees $d, 2, \dots, 2d$, where $d \geq 2$, $F \not\equiv 0$, and $E_1 = R$. Assume that at every point $E_k = e_k(r_1, \dots, r_d)$ for some nonnegative r_i,*

and that

$$\Delta F = 0, \qquad S := \Gamma(F,F) = E_{d-1},$$
$$\Gamma(F,E_k) = 2(d-k+1)E_{k-1}F,$$
$$\Gamma(E_i,E_j) = 4\sum_{\alpha=1}^{d} r_\alpha\, \partial_\alpha e_i(r)\partial_\alpha e_j(r).$$

If

$$\Delta S \geq \left(24 - \frac{16}{d}\right) E_{d-2} \qquad \text{on } \{S>0\},$$

there exists a homogeneous potential $G = AF$ of degree $2d-1$, with $A>0$ and $\nabla G \neq 0$ there, satisfying $G\mathcal{L}(G) > 0$ wherever $F \neq 0$. The hypotheses force $N \geq 4d$. The critical set $\{S=0\}$ is removable, and the normalized gradient of G gives a two-sided subcalibration. The phase and boundary current are locally minimizing. The cone is strictly area minimizing, and its regular part satisfies

$$Q_C(u) \geq \frac{(N-4d+1)^2}{4}\int_C |x|^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}).$$

In particular, for every $d \geq 2$, each of the following harmonic degree-d cones is the support of a strictly area-minimizing integral boundary and has strictly stable regular part:

(i) $\{\mathrm{Re}(x_1 \cdots x_d) = 0\} \subset \mathbb{O}^d$, *for every fixed parenthesization;*
(ii) $\{\mathrm{Re}\,\mathrm{Pf}\, Z = 0\}$ *on the complex skew-symmetric $2d \times 2d$ matrices, with Euclidean norm $\sum_{i<j}|z_{ij}|^2$.*

Their critical sets have real codimension sixteen and twelve, respectively. The Hardy lower bounds are $(4d+1)^2/4$ and $(4d^2-6d+1)^2/4$, respectively.

The proof uses Newton inequalities and monotonicity in reciprocal spectral variables. Octonionic products and Pfaffians realize the same gradient-pairing table with Laplacian coefficients 32 and 24. For their earlier harmonic-morphism and regular-minimality results, see [69, Section 4.5.2] and [41, Corollary 5.2 and Remark 5.5].

1.5. **Complex determinants.** Real parts of complex determinants have this gradient table with the smaller Laplacian coefficient 16. The preceding multiplier fails in orders at least three. A product of proper minor norms supplies a different certificate in every order, including across nonisolated singularities. Here E_j is the sum of squared absolute values of all $j \times j$ minors and $F_m = \mathrm{Re}\det_{\mathbb{C}} Z$.

Theorem I (Real parts of complex determinants)**.** *For every $m \geq 1$, the boundary $T_m = \partial[\![\{F_m < 0\}]\!]$ is locally area minimizing in $M_m(\mathbb{C}) \cong \mathbb{R}^{2m^2}$. For $m \geq 2$,*

$$G_m = F_m\left(\prod_{j=1}^{m-1} E_j\right)^{1/2}$$

gives a subcalibration smooth off the rank-at-most-$m-2$ locus, which has real codimension eight. For $m \geq 2$, the cone is strictly area minimizing, and

$$Q_{C_m}(u) \geq \frac{(m^2-m-1)^2}{4}\int_{C_m} r^{-2}u^2, \qquad u \in C_c^\infty((C_m)_{\mathrm{reg}} \setminus \{0\}). \tag{1.6}$$

Regular minimality was proved in [41, Corollary 5.2 and Example 5.4]. For $m \geq 3$, the smallest algebra has exactly $m+1$ independent generators of maximal degree $2m$. The proof of global positivity is uniform in m; for $m=3$ its potential is $F\sqrt{RS}$, linking the spectral construction to the cubic one.

Sections 2 to 4 develop singular-set comparison, finite algebraic calculus and the multiplier–Jacobi identity; Section 5 develops pullback transfer. Structural restrictions in Sections 6, 7

and 9 support the quartic criteria and cubic minimizing classification in Sections 8, 10 and 11. Sections 12 to 14 extend the method to arbitrary degree and unbounded generator number.

Remark 1.3 (Smooth minimizing hypersurfaces). These new area-minimizing algebraic hypercones also give smooth minimizing hypersurfaces. Wang's extension [91, Theorem 1.1] of the Hardt–Simon theorem [39] provides, on each side of such a cone, a properly embedded smooth area-minimizing hypersurface having that cone as its tangent cone at infinity. This applies also to the examples with nonisolated singularities.

2. Geometric Comparison and Subcalibrations across Singular Sets

All ambient metrics in this section are Euclidean. For background on minimal surfaces, functions of bounded variation, and minimizing boundaries, see [37].

2.1. **Stationarity and stability on the regular part.** A smooth submanifold is *minimal* when its mean curvature vector vanishes. For singular objects we use integral currents, with the boundary convention $\partial T(\eta) = T(d\eta)$, mass measure $\|T\|$, and restriction $T \llcorner A$ [29]. The associated varifold is *stationary* when

$$\delta|S|(Y) = \int \operatorname{div}_{T_xS} Y \, d\|S\|(x) = 0, \qquad Y \in C^1_c(\Omega \setminus \operatorname{spt} \partial S; \mathbb{R}^N), \tag{2.1}$$

where T_xS is the approximate tangent plane and its divergence is the trace of the ambient derivative [73, Section 16]. We write S_{reg} for the regular part and $\operatorname{Sing} S$ for the remaining support, away from the boundary.

For a smooth minimal M, let A be its second fundamental form and $\nabla^\perp$ its normal connection. The second variation in a compactly supported normal direction V is

$$Q^\perp_M(V) = \int_M \left(|\nabla^\perp V|^2 - \sum_{i,j=1}^{n} \langle A(e_i, e_j), V\rangle^2 \right), \tag{2.2}$$

where $e_1, \dots, e_n$ is a local orthonormal tangent frame. With the Laplacian convention $\int u\Delta u = -\int |\nabla u|^2$, the normal Jacobi operator is

$$J^\perp_M V = \Delta^\perp_M V + \sum_{i,j} \langle A(e_i, e_j), V\rangle A(e_i, e_j), \qquad Q^\perp_M(V) = -\int_M \langle V, J^\perp_M V\rangle. \tag{2.3}$$

Stability means $Q^\perp_M(V) \geq 0$ for every such V. For a stationary singular current the tests are compactly supported smooth normal fields on S_{reg}, away from its boundary; see [77] and [73, Lemma 9.7 and Remark 9.8]. Each test has compact support away from the singular set, but no common positive distance is prescribed for the whole test class. Extension across it uses intrinsic energy cutoffs, whereas divergence comparison below uses ambient L^1 cutoffs. For cones, strict stability means the scale-invariant gap defined below.

2.2. **Area minimization and the strict notions for cones.** An integral current S is *locally mass minimizing* if

$$\mathbf{M}_U(S) \leq \mathbf{M}_U(S + \partial R)$$

whenever $U \Subset \Omega$ and R is an integral current compactly supported in U of one higher dimension. Here $\mathbf{M}_U(S) = \|S\|(U)$. For multiplicity-one submanifolds we also call this *area minimization*. Such minimizers are stationary and stable, by first and second variation.

A measurable set $E \subset \Omega$ has *locally finite perimeter* if $\mathbf{1}_E \in BV_{\mathrm{loc}}(\Omega)$: its distributional derivative is a vector-valued Radon measure of locally finite mass. For $U \Subset \Omega$,

$$\operatorname{Per}(E; U) = |D\mathbf{1}_E|(U), \qquad D\mathbf{1}_E = -\nu_E \, \mathcal{H}^{N-1} \llcorner \partial^* E, \tag{2.4}$$

where $\partial^* E$ is the reduced boundary and ν_E its measure-theoretic exterior unit normal; see [55].

The current $[\![E]\!]$ denotes integration over E with the standard orientation, and $\mathbf{M}_U(\partial[\![E]\!]) = \operatorname{Per}(E; U)$. Its boundary has multiplicity one on $\partial^* E$, with support $\overline{\partial^* E} \cap \Omega$. For polynomial

hypersurfaces we take $C = \partial[\![\{F < 0\}]\!]$. Its support equals the full zero set when both signs occur near every zero, in particular for nonzero harmonic F; see Remark 2.7.

A set E is *locally perimeter minimizing* if $\mathrm{Per}(E;U) \le \mathrm{Per}(H;U)$ for every locally finite-perimeter set H with $H \triangle E \Subset U \Subset \Omega$. Local perimeter minimization of E is equivalent to local mass minimization of $\partial[\![E]\!]$. This is a direct consequence of the codimension-one decomposition theorem [73, Theorem 27.6 and Corollary 27.8]; see also [22, Proposition 2.15].

For the cone statements let C be a stationary multiplicity-one integral cone of real dimension $n \ge 2$, with $\partial C = 0$. Put $r = |x|$ and $C_1 = C \llcorner B_1$. We also use C for its support. By slicing and homogeneity, $\partial C_1 = \langle C, r, 1\rangle$ is the oriented spherical slice on ∂B_1 [73, Lemma 28.5(2)].

Definition 2.1 (Minimality and stability, with their strict forms)**.** The cone is *area minimizing* if C_1 has least mass among integral n-currents with boundary ∂C_1. It is *strictly area minimizing* in the vertex-avoidance sense if it is area minimizing and there is $\Theta > 0$ such that

$$\mathbf{M}(T) - \mathbf{M}(C_1) \ge \Theta \varepsilon^n \quad \text{whenever} \quad \partial T = \partial C_1, \quad \operatorname{spt} T \cap B_\varepsilon = \varnothing, \quad 0 < \varepsilon < 1. \tag{2.5}$$

The competitors are integral currents of finite mass and arbitrary integer multiplicity in the ambient Euclidean space. For $C = \partial[\![E]\!]$ with E conical, area minimization is equivalent to local perimeter minimization.

For a two-sided hypersurface cone and $V = u\nu_C$, (2.2) becomes

$$Q_C(u) = \int_{C_{\mathrm{reg}}} \big(|\nabla_C u|^2 - |A_C|^2 u^2\big), \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}). \tag{2.6}$$

The cone is *stable* if $Q_C(u) \ge 0$ for every such u, and *strictly stable* if there is $\delta > 0$ such that

$$Q_C(u) \ge \delta \int_{C_{\mathrm{reg}}} r^{-2} u^2 \qquad \text{for every } u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}). \tag{2.7}$$

In arbitrary codimension the same definition uses $Q_C^\perp(V)$ and $|V|^2$ in place of $Q_C(u)$ and u^2.

The vertex-avoidance convention is that of Hardt–Simon [39, Section 3, (3.1)], stated for integral currents in [50, p. 210]. These strict notions enter constructions with prescribed conical singularities [10], metric deformations [90], and Simon's Liouville theorem [76, equations (1.4), (1.12) and Corollary 1.13].

Strict stability does not imply area minimization, as Proposition 6.3 and Theorem 6.4 illustrate; Proposition E.1 gives calibrated area-minimizing cones of real codimension two without a positive Hardy gap. For a singular link $\Lambda = C \cap \mathbb{S}^{N-1}$, angular tests are compactly supported in Λ_{reg}; the corresponding spectral formulation is recorded in (4.16).

2.3. **Calibrations and the Stokes comparison.** A *calibration* is a closed differential n-form whose comass, the supremum of its absolute value on orthonormal n-frames, is at most one. It calibrates an integral current S if $\omega(\tau_S) = 1$ at $\|S\|$-almost every point, where τ_S is the oriented unit tangent n-vector [40].

If ω calibrates S and $T - S = \partial R$ with $\operatorname{spt} R \Subset U$, Stokes' theorem gives

$$\mathbf{M}_U(T) - \mathbf{M}_U(S) \ge (T - S)(\omega) = \partial R(\omega) = R(d\omega) = 0. \tag{2.8}$$

The common contributions outside U cancel; a cutoff equal to one near $\operatorname{spt} R$ makes the evaluation of the difference compactly supported. Thus a calibrated current is mass minimizing.

2.4. **Hypersurface forms and vector-field subcalibrations.** In codimension one, the identification $\omega_X = \iota_X dV$ gives

$$\|\omega_X\|_* = |X|, \qquad d\omega_X = (\operatorname{div} X) dV, \qquad \omega_X|_{\partial E} = (X \cdot \nu_E)\, d\mu_{\partial E}. \tag{2.9}$$

Evaluation on an oriented orthonormal frame and the coordinate formula for divergence give these identities. A hypersurface calibration is equivalently a field with $|X| \le 1$, $\operatorname{div} X = 0$, and $X \cdot \nu_E = 1$ on the boundary.

Definition 2.2 (Two-sided subcalibration, smooth formulation)**.** Let $E \subset \Omega \subset \mathbb{R}^N$ have smooth boundary. A C^1 field X is a two-sided subcalibration for E if

$$|X| \le 1, \qquad X \cdot \nu_E = 1 \text{ on } \partial E, \qquad \operatorname{div} X \le 0 \text{ in } E, \quad \operatorname{div} X \ge 0 \text{ in } \Omega \setminus E. \tag{2.10}$$

The equivalent form conditions follow from (2.9).

These phase signs give the comparison used by De Philippis and Paolini [25]. Applying the Stokes argument above to a smooth competitor H with $H \triangle E \Subset U \Subset \Omega$ gives

$$\begin{aligned} \operatorname{Per}(H;U) - \operatorname{Per}(E;U) &\ge \int_U (\mathbf{1}_H - \mathbf{1}_E) \operatorname{div} X \, dx \\ &= \int_{H \triangle E} q_X \, dx \ge 0, \qquad q_X = (1 - 2\mathbf{1}_E) \operatorname{div} X. \end{aligned} \tag{2.11}$$

The boundary flux is saturated for E and bounded by the perimeter for H. Below we extend this comparison to singular and phase-dependent fields.

Remark 2.3 (Historical background on quadratic cones). Fleming's regularity theorem in $\mathbb{R}^3$ [35] and Almgren's theorem in $\mathbb{R}^4$ [2] exclude nonflat area-minimizing hypercones in those dimensions. Simons established stability of the balanced quadratic cones in $\mathbb{R}^{2m}$, $m \ge 4$, [77]; Bombieri–De Giorgi–Giusti proved their area minimality [6, Sections II–III]. The homogeneous barriers in [6, Section IV, p. 260] occur in their construction of entire minimal graphs, separately from the cone-minimization proof. Lawson reduced the equivariant Plateau problem to weighted orbit spaces and constructed exact calibrating one-forms through explicit homogeneous trial functions [48, Section 5, pp. 244–247]. In his 1973 Berkeley thesis, Simões completed the remaining quadratic cases: the cone over the minimal product $S^1 \times S^5$ is not area minimizing, whereas the cone over the minimal product $S^2 \times S^4$ is area minimizing [72]. Here the factors have the radii making their product minimal in the unit sphere. Miranda and collaborators developed new proofs through the nonparametric minimal surface equation. In particular, Massari–Miranda used graphical subsolutions and cylinder limits [56]; the quartic function $G = (|x|^4 - |y|^4)/4$ already appears in this approach [60, pp. 146–147]. Benarros–Miranda gave two one-sided homogeneous functions [4, Section 2, pp. 46–47]. After orienting both normalized gradient fields consistently, their construction gives potentials $G_\pm = a_\pm F$ with positive, phase-dependent multipliers. De Philippis–Paolini combined the earlier approaches in a short subcalibration proof, using divergence inequalities directly in the ambient space [25, pp. 234–235]. De Philippis–Maggi used the exact perimeter-deficit identity [24, Proposition 4.1, equation (4.5)] to obtain quantitative stability estimates for Lawson cones. Liu completed their six exceptional parameter pairs using different subcalibrations [52, Theorems 1–2 and Section 2.1], explicitly acknowledging the inspiration of Benarros–Miranda [52, p. 1002].

2.5. **Comparison across removable exceptional sets.** The Stokes comparison (2.11) extends across locally $\mathcal{H}^{N-1}$-null singular sets, also for different phase potentials.

Boundary flux away from an exceptional set. For a locally finite-perimeter set E, a relatively closed set Z, and $X \in C^1(\Omega \setminus Z; \mathbb{R}^N)$, the boundary condition is $X \cdot \nu_E = 1$ almost everywhere on $\partial^* E \setminus Z$. Extending ηX by zero for $\eta \in C_c^1(\Omega \setminus Z)$, Gauss–Green gives

$$\int_E \operatorname{div}(\eta X) \, dx = \int_{\partial^* E \setminus Z} \eta \, X \cdot \nu_E \, d\mathcal{H}^{N-1}.$$

To extend this calculation across Z, we use cutoffs below. For singular subcalibration fields in a Riemannian metric, see [53, Fact 2.3 and Section 7].

Cutoffs near the singular set.

Lemma 2.4 (Cutoffs near sets of zero Hausdorff measure)**.** *Let $Z \subset \Omega \subset \mathbb{R}^N$ be relatively closed and satisfy*

$$\mathcal{H}^{N-1}(Z \cap K) = 0 \quad \text{for every compact } K \Subset \Omega. \tag{2.12}$$

For every compact $K \Subset U \Subset \Omega$, every open set O with $Z \cap K \subset O \Subset U$, and every $\varepsilon > 0$, there is $\psi \in C_c^1(O)$ such that $0 \le \psi \le 1$, $\psi = 1$ on a neighborhood of $Z \cap K$, and

$$\int_U \big(\psi + |\nabla \psi|\big) < \varepsilon. \tag{2.13}$$

Proof. Cover the compact set $Z \cap K$ by finitely many balls $B_{r_i}(x_i)$ whose doubled balls lie in O, with $r_i < 1$ and $\sum_i r_i^{N-1} < \varepsilon/(2C_N)$. Choose smooth cutoffs ψ_i supported in the doubled balls, equal to one on the covering balls, with $0 \le \psi_i \le 1$ and $|\nabla \psi_i| \le C/r_i$. Then $\psi = 1 - \prod_i (1 - \psi_i)$ equals one near $Z \cap K$, and satisfies $\psi \le \sum_i \psi_i$ and $|\nabla \psi| \le \sum_i |\nabla \psi_i|$. Hence

$$\int_U \big(\psi + |\nabla \psi|\big) \le C_N \sum_i (r_i^N + r_i^{N-1}) < \varepsilon. \qquad \square$$

We extend the quantitative subcalibration identity of De Philippis–Maggi [24, Proposition 4.1, equation (4.5)], also used in [52, Section 2, equation (3)], to bounded fields that are C^1 outside a relatively closed, locally $\mathcal{H}^{N-1}$-null set.

Theorem 2.5 (Two-sided comparison)**.** *Let $E \subset \Omega$ have locally finite perimeter, and let $Z \subset \Omega$ be relatively closed with $\mathcal{H}^{N-1}(Z \cap K) = 0$ for every compact $K \Subset \Omega$. Suppose $X \in C^1(\Omega \setminus Z; \mathbb{R}^N)$ satisfies*

(i) $|X| \le 1$;
(ii) $X \cdot \nu_E = 1$ *for $\mathcal{H}^{N-1}$-almost every point of $\partial^* E \setminus Z$;*
(iii) $\operatorname{div} X \le 0$ *almost everywhere in $E \setminus Z$, and* $\operatorname{div} X \ge 0$ *almost everywhere in $(\Omega \setminus E) \setminus Z$.*

Define $q_X = (1 - 2\mathbf{1}_E) \operatorname{div} X \ge 0$ off Z and set it to zero on Z. Then $q_X \in L^1_{\mathrm{loc}}(\Omega)$. For every locally finite-perimeter competitor H with $H \triangle E \Subset U \Subset \Omega$,

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) = \int_{\partial^* H \cap U} (1 - X \cdot \nu_H)\, d\mathcal{H}^{N-1} + \int_{H \triangle E} q_X \, dx. \tag{2.14}$$

The values of X on Z do not affect the boundary integral. In particular,

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) \ge \int_{H \triangle E} q_X \, dx. \tag{2.15}$$

Thus E is locally perimeter minimizing and $\partial[\![E]\!]$ is locally mass minimizing. If $q_X > 0$ almost everywhere, equality in perimeter comparison forces $H = E$ up to a null set. When $|X| = 1$, the first integral in (2.14) is $\frac{1}{2} \int_{\partial^* H \cap U} |\nu_H - X|^2 \, d\mathcal{H}^{N-1}$.

Proof. Choose $K_0 \Subset K_1 \Subset U$ containing the support of $\mathbf{1}_H - \mathbf{1}_E$ in K_0, and put $\mu = |D\mathbf{1}_E| + |D\mathbf{1}_H|$. The null-set assumption and (2.4) give $\mu(Z \cap K_1) = 0$. Take open neighborhoods $O_j \Subset U$ of $Z \cap K_1$ with $\mu(O_j) \to 0$, and use Lemma 2.4 to choose $\psi_j \in C_c^1(O_j)$ with $\int_U (\psi_j + |\nabla \psi_j|) \to 0$. Pass to a subsequence so that $\psi_j \to 0$ almost everywhere, and set $\eta_j = 1 - \psi_j$. Fix $\chi \in C_c^1(U)$ equal to one near K_0, with support in K_1. The field $\chi \eta_j X$, extended by zero near Z, is an admissible C^1 field. The distributional derivative identity (2.4), applied to $\mathbf{1}_H - \mathbf{1}_E$, gives

$$\int_{H \triangle E} \eta_j q_X = \int_{\partial^* H} \chi \eta_j X \cdot \nu_H - \int_{\partial^* E} \chi \eta_j - \int_U (\mathbf{1}_H - \mathbf{1}_E) X \cdot \nabla \eta_j. \tag{2.16}$$

We have used $\chi = 1$ on the difference set and $(\mathbf{1}_H - \mathbf{1}_E) \operatorname{div} X = q_X$ there. The final integral is bounded in absolute value by $\int_U |\nabla \psi_j|$. The boundary integrals converge because their

cutoff errors are at most $\mu(O_j)$. Thus the right-hand side of (2.16) has a finite limit. Since $0 \le \eta_j q_X \to q_X$ almost everywhere, Fatou's lemma gives

$$\int_{H\triangle E} q_X \le \liminf_j \int_{H\triangle E} \eta_j q_X < \infty.$$

Now $0 \le \eta_j q_X \le q_X$, so dominated convergence gives equality in the limit of (2.16). Since $H = E$ outside K_0, the boundary fluxes cancel there. Consequently

$$\int_{H\triangle E} q_X = \int_{\partial^* H\cap U} X\cdot\nu_H - \mathrm{Per}(E;U),$$

which proves (2.14). For almost every ball $B \Subset \Omega$, both $H = E\setminus B$ and $H = E\cup B$ have locally finite perimeter. The integrability just proved for these two comparisons gives integrability of q_X on the two parts of B, hence local integrability in Ω. The current conclusion follows from the equivalence in Section 2.2; the other conclusions follow from nonnegativity. □

Remark 2.6 (Semialgebraic singular sets). A closed semialgebraic set of dimension at most $N-2$ satisfies (2.12): its finite smooth stratification [5, Sections 2.3 and 2.8] has strata of zero $\mathcal{H}^{N-1}$-measure.

Remark 2.7 (Finite perimeter of polynomial sublevel sets). Let F be a nonzero real polynomial of degree d, and put $E = \{F < 0\}$ and $Q_L = (-L, L)^N$. On each coordinate line, the restriction of F is either zero or a nonzero univariate polynomial of degree at most d. Thus the restriction of $\mathbf{1}_E$ has one-dimensional variation at most d. For $\varphi \in C_c^1(Q_L)$, Fubini and one-dimensional integration by parts give

$$\left|\int_{Q_L} \mathbf{1}_E\,\partial_i\varphi\,dx\right| \le d(2L)^{N-1}\|\varphi\|_\infty.$$

Each distributional derivative $D_i\mathbf{1}_E$ is therefore a finite measure on Q_L, and

$$\mathrm{Per}(E;Q_L) \le \sum_{i=1}^N |D_i\mathbf{1}_E|(Q_L) \le Nd(2L)^{N-1}.$$

In particular, polynomial sublevel sets have locally finite perimeter. This also applies to $\{G<0\}$ whenever G has the same sign almost everywhere as a polynomial.

We also recall that $\{F=0\}$ has Lebesgue measure zero. Write F as a polynomial in one coordinate. Outside the zero set of a nonzero coefficient polynomial, its fibers are finite. Induction on N and Fubini prove the assertion.

If both signs of F occur in every neighborhood of each zero, then

$$\partial E = \operatorname{spt}\partial[\![E]\!] = \{F=0\}.$$

Indeed, both phases have positive volume in every ball centered at a zero. The perimeter in such a ball cannot vanish, since otherwise $\mathbf{1}_E$ would be constant almost everywhere there. Away from the zero set, $\mathbf{1}_E$ is locally constant. For nonzero harmonic F, the strong maximum principle gives this sign condition.

Corollary 2.8 (Gradient subcalibration criterion). *Let $G \in C^2(\Omega\setminus Z)\cap C^0(\Omega)$, assume that $E = \{G<0\}$ has locally finite perimeter, and suppose that Z contains the critical set of G and the singular part of $\{G=0\}$. If Z is relatively closed, $\mathcal{H}^{N-1}(Z\cap K) = 0$ for every compact $K \Subset \Omega$, and*

$$G\mathcal{L}(G) \ge 0 \qquad \text{on } \Omega\setminus Z,$$

then E is locally perimeter minimizing.

Proof. Use $X = \nabla G/|\nabla G|$ and (1.3). On $G<0$ its divergence is nonpositive, and on $G>0$ it is nonnegative. Along the regular zero set it is the exterior unit normal. □

We call G a *sub-calibration potential* for $E = \{G < 0\}$ when its normalized gradient satisfies Definition 2.2, or Theorem 2.5 when an exceptional set is present. Definition 1.1 specifies the homogeneous potentials used for polynomial defining equations.
Applying the criterion to homogeneous potentials. In the algebraic applications, $G = aF$ has positive degree ℓ, and a is smooth and positive outside a closed conical exceptional set Z containing the singular zeros of F. Once G extends continuously with the signs of F, its negative phase has locally finite perimeter by Remark 2.7. Moreover, outside Z,

$$F \neq 0 \implies \langle x, \nabla G\rangle = \ell G \neq 0, \qquad F = 0 \implies \nabla G = a\nabla F \neq 0.$$

It remains to check the null-set condition on Z and the global sign $G\mathcal{L}(G) \geq 0$. The exceptional sets and their codimensions are collected in Appendix E.2.

One-sided comparison and two different fields. A set E is *inward minimizing* if perimeter comparison holds for competitors contained in E, and *outward minimizing* if it holds for competitors containing E. All changes are compactly supported. These are the subminimality and complementary subminimality of [25, Definition 1.1]. The paired comparison retains both divergence remainders.

Theorem 2.9 (One-sided and paired comparison)**.** *Let $E \subset \Omega$ have locally finite perimeter, and let $Z \subset \Omega$ be relatively closed with $\mathcal{H}^{N-1}(Z \cap K) = 0$ for every compact $K \Subset \Omega$. Suppose $X_-, X_+ \in C^1(\Omega \setminus Z; \mathbb{R}^N)$ satisfy*

$$|X_\pm| \leq 1, \qquad X_\pm \cdot \nu_E = 1 \quad \mathcal{H}^{N-1}\text{-a.e. on } \partial^* E \setminus Z.$$

Assume only the respective phase signs

$$\operatorname{div} X_- \leq 0 \text{ in } E \setminus Z, \qquad \operatorname{div} X_+ \geq 0 \text{ in } (\Omega \setminus E) \setminus Z.$$

Define, with zero values on Z,

$$q_- = -\mathbf{1}_E \operatorname{div} X_-, \qquad q_+ = \mathbf{1}_{\Omega\setminus E} \operatorname{div} X_+.$$

Then $q_-, q_+ \in L^1_{\mathrm{loc}}(\Omega)$, and the following conclusions hold for changes compactly supported in $U \Subset \Omega$.

(i) *The assumptions on X_- alone imply*

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) \geq \int_{E\setminus H} q_- \qquad (H \subset E).$$

The assumptions on X_+ alone imply the corresponding estimate

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) \geq \int_{H\setminus E} q_+ \qquad (H \supset E).$$

(ii) *Together they imply, for every locally finite-perimeter H,*

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) \geq \int_{E\setminus H} q_- + \int_{H\setminus E} q_+. \tag{2.17}$$

Thus E is locally perimeter minimizing, and its oriented boundary is locally mass minimizing among integral currents. If each weight is positive almost everywhere on its phase, equality implies $H = E$ up to a null set.

Proof. For $H \subset E$, the cutoff proof of Theorem 2.5 uses the divergence sign only on $E \setminus H$. With X_-, it gives

$$\operatorname{Per}(H; U) - \operatorname{Per}(E; U) = \int_{\partial^* H \cap U} (1 - X_- \cdot \nu_H) + \int_{E\setminus H} q_-.$$

The analogous identity with X_+ holds for $H \supset E$. Their nonnegative boundary terms prove (i); taking $H = E \setminus B$ and $H = E \cup B$ for almost every relatively compact ball proves local integrability of the respective weights.

For an unrestricted competitor, apply these inequalities to $H\cap E$ and $H\cup E$. The perimeter inequality for unions and intersections [55] gives

$$\mathrm{Per}(H\cap E;U)+\mathrm{Per}(H\cup E;U)\le \mathrm{Per}(H;U)+\mathrm{Per}(E;U).$$

Adding the two one-sided estimates proves (2.17). The current conclusion follows from the equivalence in Section 2.2, and positivity gives the equality assertion. □

Remark 2.10 (Fields defined on their respective sides). When E is open and $\partial E\setminus Z$ is smooth, it is enough to construct X_- on an open neighborhood of $\overline{E}\setminus Z$, and X_+ on an open neighborhood of $\overline{\Omega\setminus E}\setminus Z$, relative to $\Omega\setminus Z$. A smooth cutoff equal to one on the relevant closed phase extends each bounded field by zero to the unused side. This preserves its norm, boundary values, and the divergence on its own phase. In a conical setting the cutoff may be chosen on the unit sphere, so homogeneity is preserved. No matching of the two potentials is required; their normalized gradients both equal ν_E on the regular boundary.

Liu's subcalibrations for the six exceptional Lawson-cone parameter pairs [52, Sections 2.1 and 3] use powers of different block radii, and sometimes different homogeneous degrees, on the two sides. The positive multipliers $G_\pm = a_\pm F$ give the common normal ν_E along the regular cone, so Theorem 2.9 applies.

Phase-dependent multipliers also appear in Liu's application of Simon's analytic estimates to hypersurfaces with prescribed closed singular sets [53, Theorem 1 and Definition 1.3].

Strict minimality and distance-weighted comparison.

Theorem 2.11 (Strict minimality from two-sided divergence integrals)**.** *Let $E\subset\mathbb{R}^{n+1}$, $n\ge 2$, be conical, and suppose the hypotheses of Theorem 2.5 hold globally with a conical exceptional set Z and a zero-homogeneous field X. Suppose*

$$A_-:=\int_{B_1\cap E} q_X\,dx>0,\qquad A_+:=\int_{B_1\setminus E} q_X\,dx>0. \tag{2.18}$$

Then $C=\partial[\![E]\!]$ is strictly area minimizing in Definition 2.1, with

$$\Theta=\min\{A_-,A_+\}. \tag{2.19}$$

In particular the assertion permits nonisolated singularities.

Proof. Both integrals are finite by Theorem 2.5. Homogeneity gives $q_X(rx)=r^{-1}q_X(x)$, so their values over B_ε are $\varepsilon^n A_-$ and $\varepsilon^n A_+$. If a set competitor H has no perimeter in the connected ball B_ε, then $\mathbf{1}_H$ is constant there. Accordingly, $H\triangle E$ contains either $B_\varepsilon\cap E$ or $B_\varepsilon\setminus E$, up to a null set. Equation (2.15) proves the desired mass gap for set-boundary competitors.

For an arbitrary integral-current competitor, consider the nearest-point projection onto the closed unit ball,

$$\pi(x)=\frac{x}{\max\{1,|x|\}}.$$

This map is 1-Lipschitz, fixes ∂C_1, and maps $\mathbb{R}^{n+1}\setminus B_\varepsilon$ into itself. Although π is not proper, the integral representation of the finite-mass current T defines $\pi_\#T$, since the n-Jacobian of π is bounded by one. The pushforward mass estimate and the identity $\partial(\pi_\#T)=\pi_\#(\partial T)$ give

$$\mathbf{M}(\pi_\#T)\le\mathbf{M}(T),\qquad \partial(\pi_\#T)=\partial C_1;$$

see [73, Remark 26.21(1), Lemma 26.25, and Remark 27.2(3)]. For finite-mass currents with noncompact support, these statements follow from the compactly supported case by cutoffs: the additional boundary term is bounded by $O(L^{-1})\mathbf{M}(T)$ for a cutoff at radius L, and tends to zero. The same projection is used in [23, proof of Proposition 1.2]. We may therefore replace T by $\pi_\#T$ and assume $\operatorname{spt}T\subset\overline{B}_1$.

Since $\partial(T-C_1)=0$, coning $T-C_1$ to the origin gives an integral $(n+1)$-current R with $\partial R=T-C_1$ and support in $\overline{B}_1$ [73, equation (26.26) and Remark 27.2(2)–(3)]. A

top-dimensional integral current is integration against an integer-valued multiplicity. Since its boundary has finite mass, this multiplicity has bounded variation [73, Remark 26.28]. Thus

$$R = [\![k]\!], \qquad k \in BV_c(\mathbb{R}^{n+1};\mathbb{Z}), \qquad v := \mathbf{1}_E + k \in BV_{\mathrm{loc}}(\mathbb{R}^{n+1};\mathbb{Z}).$$

We have $\partial[\![v]\!] = T + C - C_1$. In B_ε, the right-hand side vanishes because T avoids this ball and $C = C_1$ there. The Constancy Theorem [73, Theorem 26.27], applied on the connected open set B_ε, therefore gives $v = m$ almost everywhere there for some $m \in \mathbb{Z}$.

Set $H = \{v \geq 1\}$. The integer-level decomposition [73, Theorem 27.6] gives, for every relatively compact open set $U \subset \mathbb{R}^{n+1}$,

$$|Dv|(U) = \sum_{j\in\mathbb{Z}} \mathrm{Per}(\{v \geq j\}; U) \geq \mathrm{Per}(H; U).$$

Thus H has locally finite perimeter. Moreover, $H = E$ outside $\overline{B}_1$, and $\mathbf{1}_H$ is constant in B_ε. Since $\|C\|(\partial B_1) = 0$, the mass measures of T and $C - C_1$ are mutually singular. Hence, for $\rho > 1$,

$$\mathrm{Per}(H; B_\rho) \leq |Dv|(B_\rho) = \mathbf{M}(T) + \mathrm{Per}(E; B_\rho) - \mathbf{M}(C_1).$$

Apply the set comparison in B_ρ and cancel the common exterior perimeter. This yields (2.5) with (2.19). □

Corollary 2.12 (Strict mass gap from paired fields)**.** *Suppose Theorem 2.9 holds globally for a conical set $E \subset \mathbb{R}^{n+1}$, $n \geq 2$, a conical exceptional set Z, and fields that are zero-homogeneous on their respective phases. If*

$$A_- := \int_{B_1\cap E} q_- > 0, \qquad A_+ := \int_{B_1\setminus E} q_+ > 0,$$

then every integral-current competitor in (2.5) satisfies

$$\mathbf{M}(T) - \mathbf{M}(C_1) \geq \min\{A_-, A_+\}\varepsilon^n.$$

Proof. The degree--1 weights are locally integrable by Theorem 2.9. A set competitor avoiding B_ε fills or empties it, so (2.17) gives the gap. The current-to-set reduction in the proof of Theorem 2.11 preserves ball avoidance and does not increase the mass excess. □

Corollary 2.13 (Polynomial potentials with nonzero divergence)**.** *Let $N \geq 3$. Suppose a homogeneous polynomial G satisfies the global hypotheses of Corollary 2.8. If both $\{G < 0\}$ and $\{G > 0\}$ are nonempty and $\mathcal{L}(G) \not\equiv 0$, then $\partial[\![\{G < 0\}]\!]$ is strictly area minimizing.*

Proof. The normalized gradient is zero-homogeneous. Its critical set is conical and satisfies (2.12). Off this set,

$$q_X = \frac{|\mathcal{L}(G)|}{|\nabla G|^3}.$$

A nonzero polynomial has a Lebesgue-null zero set by Remark 2.7. Thus $q_X > 0$ almost everywhere on both open phases. Equation (2.18) follows, and Theorem 2.11 applies. □

For nonpolynomial potentials, verify positive flux in both phases and apply Theorem 2.11, as for the radical potentials in Example 8.25 and Section 14.

For regular area-minimizing hypercones, Niu [65, Theorem 3] characterizes simultaneous strict stability and strict minimality by (2.21). We prove sufficiency across the singular sets of Theorem 2.5, using the quantitative subcalibration method of De Philippis–Maggi [24, Section 4] and Liu [52, Section 2.2].

Proposition 2.14 (Distance-weighted sufficient criterion)**.** *Under the geometric and comparison hypotheses of Theorem 2.11, suppose the cone is two-sided and minimal on its regular part and, for some $c > 0$,*

$$q_X(x) \geq c\,\frac{\mathrm{dist}(x, C)}{|x|^2} \quad \text{for almost every } x \notin C. \tag{2.20}$$

Assume each phase contains points at positive distance from C. Then the cone is strictly area minimizing and strictly stable, and

$$\mathrm{Per}(H;U)-\mathrm{Per}(E;U)\geq c\int_{H\triangle E}\frac{\mathrm{dist}(x,C)}{|x|^2}\,dx. \tag{2.21}$$

More precisely, $Q_C(u)\geq c\int_C r^{-2}u^2$ for the regular-part test class in Definition 2.1.

Proof. The weighted comparison follows from (2.15). The two phase integrals are positive, so Theorem 2.11 gives strict minimality. For a smooth compactly supported regular-part function u, perturb the boundary by the normal graph tu in a fixed regular tubular neighborhood. Minimality and tubular coordinates give

$$\mathrm{Per}(E_{tu};U)-\mathrm{Per}(E;U)=\tfrac{1}{2}t^2Q_C(u)+o(t^2),$$
$$\int_{E_{tu}\triangle E}\frac{\mathrm{dist}(x,C)}{|x|^2}\,dx=\tfrac{1}{2}t^2\int_C r^{-2}u^2+o(t^2).$$

Indeed, on each normal segment the leading integral is $r^{-2}\int_0^{|tu|}s\,ds$; its tubular Jacobian and the radial weight contribute only higher-order terms on this compact support. Divide (2.21) by $t^2/2$ and let $t\to 0$. □

2.6. **Algebraic tangent cones and local instability.** Let $C=\{F=0\}$ for a nonzero polynomial F, and let $p\in C$. Writing $F(p+z)=H_m(z)+\cdots+H_d(z)$ with $H_m\not\equiv 0$ homogeneous of degree m, we have

$$r^{-m}F(p+rz)\longrightarrow H_m(z)\quad\text{in } C^\infty_{\mathrm{loc}}\quad(r\downarrow 0).$$

The implicit function theorem gives smooth graphical convergence of $(C-p)/r$ near compact subsets of $\{H_m=0,\ \nabla H_m\neq 0\}$. If H_m changes sign near every zero, then $\{H_m=0\}$ is the unique local Hausdorff tangent: uniform convergence gives one inclusion, and sign change gives the other. This condition holds when F is harmonic or H_m is an indefinite quadratic form. Thus an indefinite Hessian at a critical point gives the quadratic tangent $\{z:\langle \mathrm{Hess}\,F(p)z,z\rangle=0\}$. For homogeneous F and $p\neq 0$, Euler's identity gives $D_pH_m=0$, so this leading zero set splits off the line $\mathbb{R}p$. The cubic Hessian criterion in Proposition 9.6 identifies these quadratic factors.

Lemma 2.15 (Instability under a regular tangent limit)**.** *Let C be an n-dimensional hypersurface, minimal on its two-sided regular part. Suppose that $C_j=(C-p)/r_j$, $r_j\downarrow 0$, converges locally graphically in C^2 on compact regular subsets of $\Sigma=Q\times\mathbb{R}^k$, where Q is a two-sided minimal cone. If Q is unstable on its regular part, then C admits a compactly supported regular-part variation with negative second variation in every neighborhood of p.*

Proof. Write $Q_\Sigma(\phi)=\int_\Sigma(|\nabla_\Sigma\phi|^2-|A_\Sigma|^2\phi^2)$. Choose $\phi\in C^\infty_c(Q_{\mathrm{reg}})$ with $Q_Q(\phi)<0$. For $k>0$, take a nonzero $\chi\in C^\infty_c(\mathbb{R}^k)$ and set $\chi_T(w)=\chi(w/T)$. The product metric gives

$$Q_{Q\times\mathbb{R}^k}(\phi\chi_T)=T^k\left(\int_{\mathbb{R}^k}\chi^2\right)Q_Q(\phi)+T^{k-2}\left(\int_{\mathbb{R}^k}|\nabla\chi|^2\right)\left(\int_Q\phi^2\right)<0$$

for sufficiently large T; for $k=0$ use ϕ itself. Fix the resulting test function ψ and a relatively compact regular neighborhood of its support. Transport ψ by the graphical parametrizations to ψ_j on C_j, extending by zero. The induced metrics and second fundamental forms converge, so $Q_{C_j}(\psi_j)\to Q_\Sigma(\psi)<0$. For $u_j(x)=\psi_j((x-p)/r_j)$,

$$Q_C(u_j)=r_j^{n-2}Q_{C_j}(\psi_j)<0.$$

Its support lies in the regular part and shrinks to p. □

A leading form $a(|u|^2-|v|^2)$ in orthogonal coordinates, $a\neq 0$, with $u,v\in\mathbb{R}^s$ and $s=2,3$, gives an unstable quadratic factor: $|A|^2=(2s-2)/\rho^2$, where $\rho^2=|u|^2+|v|^2$, and the sharp radial Hardy constant is $(2s-3)^2/4<2s-2$. For example, the minimal

cubic $F = \mathrm{Re}(z_1 z_2 z_3)$ on $\mathbb{C}^3$ has leading term $\mathrm{Re}(z_2 z_3)$ at $p = (1, 0, 0)$, giving the tangent $\mathbb{R}^2 \times \{(u, v) \in \mathbb{R}^2 \oplus \mathbb{R}^2 : |u| = |v|\}$. The lemma proves local instability at p; see Section 10.7 for the Clifford and Hurwitz cubic models.

3. Laplacian Algebras and Quotient Calculus

3.1. **Laplacian algebras: differential closure and finite generation.** Following Mendes–Radeschi [57, Section 5], we develop closure and finite generation to reduce differential calculations to finite tables and identify the smallest algebra containing a defining polynomial.

Let $V = \mathbb{R}^N$ have its Euclidean inner product, let $\mathbb{R}[V]$ be its ring of real polynomial functions, and put $R(x) = |x|^2$. For smooth functions define

$$\Gamma(f, g) = \langle \nabla f, \nabla g \rangle.$$

All polynomial algebras contain the constants; their generators need not be algebraically independent.

Definition 3.1 (Mendes–Radeschi)**.** A subalgebra $\mathcal{A} \subset \mathbb{R}[V]$ is a *Laplacian algebra* if $R \in \mathcal{A}$ and $\Delta f \in \mathcal{A}$ for every $f \in \mathcal{A}$.

This definition is orthogonally invariant and requires no group action.

Lemma 3.2 (Gradient closure and homogeneity)**.** *A Laplacian algebra is closed under Γ and contains every homogeneous component of each of its elements. In particular, $\mathcal{A} = \bigoplus_{d \geq 0} \mathcal{A}_d$, where $\mathcal{A}_d = \mathcal{A} \cap \mathbb{R}[V]_d$ and $\mathbb{R}[V]_d$ denotes the homogeneous polynomials of degree d.*

Proof. The product rule gives

$$2\Gamma(f, g) = \Delta(fg) - f\Delta g - g\Delta f. \tag{3.1}$$

Thus $\Gamma(f, g) \in \mathcal{A}$ for $f, g \in \mathcal{A}$. The Euler operator $E = x \cdot \nabla$ consequently preserves $\mathcal{A}$, since $Ef = \Gamma(R, f)/2$. If $f = \sum_{j=0}^{D} f_j$ is its homogeneous decomposition, then $Ef_j = jf_j$. For each $0 \leq d \leq D$, choose the interpolation polynomial p_d satisfying $p_d(j) = \delta_{dj}$ for $j = 0, \ldots, D$. Then $f_d = p_d(E)f \in \mathcal{A}$. □

Example 3.3 (Orthogonal invariants)**.** If a group $H \subset O(V)$ acts orthogonally, its invariant algebra $\mathbb{R}[V]^H$ is Laplacian. Indeed, R is invariant and $\Delta(f \circ h) = (\Delta f) \circ h$ for $h \in H$. For compact H, normalized Haar averaging gives

$$\Pi_H g(x) = \int_H g(hx)\, dh, \qquad \Pi_H(fg) = f\Pi_H(g) \quad (f \in \mathbb{R}[V]^H).$$

The projection below extends this module property to every Laplacian algebra.

Example 3.4 (Two orthogonal blocks)**.** Let $V = \mathbb{R}^p \oplus \mathbb{R}^q$, where $p, q \geq 1$, and write $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$, $u = |x|^2$, $v = |y|^2$. Then $R = u + v$, and

$$\Delta u = 2p, \quad \Delta v = 2q, \qquad \Gamma(u, u) = 4u, \quad \Gamma(u, v) = 0, \quad \Gamma(v, v) = 4v.$$

For a polynomial $h(u, v)$, direct differentiation therefore gives $\Delta h = 4uh_{uu} + 4vh_{vv} + 2ph_u + 2qh_v$. Thus $\mathbb{R}[u, v]$ is Laplacian, with image $[0, \infty)^2$.

3.1.1. *The Reynolds projection.* We recall the properties needed for finite generation, following [57, Lemma 21, Corollary 22, and Theorem 23].

For polynomials f, g and an integer $j \geq 0$, set

$$B_j(f, g) = \sum_{i_1, \ldots, i_j = 1}^{N} (\partial_{i_1} \cdots \partial_{i_j} f)(\partial_{i_1} \cdots \partial_{i_j} g), \qquad B_0(f, g) = fg.$$

The product rule and commutation of derivatives give

$$2B_{j+1}(f, g) = \Delta B_j(f, g) - B_j(\Delta f, g) - B_j(f, \Delta g). \tag{3.2}$$

Induction shows that $B_j(f,g) \in \mathcal{A}$ whenever $f, g \in \mathcal{A}$. If f is homogeneous of degree d, write $f = \sum_{|\alpha|=d} f_\alpha x^\alpha$ and define the constant-coefficient operator $f(\partial) = \sum_{|\alpha|=d} f_\alpha \partial^\alpha$. Counting ordered derivatives gives

$$f(\partial)g = \frac{1}{d!} B_d(f,g). \tag{3.3}$$

In particular, $f(\partial)$ takes $\mathcal{A}$ into $\mathcal{A}$.

On the finite-dimensional space $\mathbb{R}[V]_d$, use the Fischer inner product

$$(f,g)_d = f(\partial)g = \sum_{|\alpha|=d} \alpha! f_\alpha g_\alpha.$$

For degrees $a, b, a+b$, the coefficient formula also gives

$$(fg,h)_{a+b} = (g, f(\partial)h)_b. \tag{3.4}$$

Proposition 3.5 (Reynolds projection; Mendes–Radeschi)**.** *Let $\Pi_d : \mathbb{R}[V]_d \to \mathcal{A}_d$ be orthogonal projection for the Fischer inner product and let $\Pi = \bigoplus_d \Pi_d$. Then $\Pi|_{\mathcal{A}} = \mathrm{Id}$, Π preserves degree, and*

$$\Pi(fg) = f\Pi(g) \qquad (f \in \mathcal{A},\ g \in \mathbb{R}[V]). \tag{3.5}$$

Proof. For homogeneous $f \in \mathcal{A}_a$ and $g \in \mathbb{R}[V]_b$, write $g = \Pi_b g + g^\perp$. For $h \in \mathcal{A}_{a+b}$, (3.3)–(3.4) give

$$(fg^\perp, h)_{a+b} = (g^\perp, f(\partial)h)_b = 0,$$

since $f(\partial)h \in \mathcal{A}_b$. Thus $\Pi(fg^\perp) = 0$ and $\Pi(f\Pi_b g) = f\Pi_b g$; sum over degrees. □

Proposition 3.6 (Finite generation and polynomial quotients)**.** *Every Laplacian algebra $\mathcal{A}$ has finitely many homogeneous generators. Moreover,*

$$\mathcal{A} = \mathrm{Frac}(\mathcal{A}) \cap \mathbb{R}[V]. \tag{3.6}$$

Equivalently, if $f, g \in \mathcal{A}$, $g \neq 0$, and f/g is a polynomial on V, then $f/g \in \mathcal{A}$.

Proof [57, Lemma 24(a)–(b)]. Put $\mathcal{A}_+ = \bigoplus_{d>0} \mathcal{A}_d$. Hilbert's basis theorem [45, Chapter I, Section 2, Proposition 2.3 and Corollary 2.4, p. 11] gives homogeneous $I_1, \dots, I_k \in \mathcal{A}_+$ generating the ideal $(\mathcal{A}_+)\mathbb{R}[V]$. For homogeneous $f \in \mathcal{A}_d$, $d > 0$, write $f = \sum_i a_i I_i$ with $\deg a_i = d - \deg I_i < d$. The Reynolds property gives

$$f = \sum_i \Pi(a_i) I_i.$$

Degree induction proves algebraic generation. If $h = f/g$ is a polynomial, applying Π to $f = gh$ gives $f = g\Pi(h)$, hence $h = \Pi(h) \in \mathcal{A}$. □

3.1.2. *Finite closure tables and the smallest algebra.*

Proposition 3.7 (Checking closure on generators)**.** *Let $I_1, \dots, I_k$ be homogeneous polynomials and suppose that $R \in \mathcal{A} = \mathbb{R}[I_1, \dots, I_k]$. Then $\mathcal{A}$ is Laplacian if and only if*

$$\Delta I_i \in \mathcal{A}, \qquad \Gamma(I_i, I_j) \in \mathcal{A} \quad \textit{for every } i, j.$$

Proof. Necessity follows from Lemma 3.2. For sufficiency, the chain rule for a polynomial p gives

$$\Delta(p \circ I) = \sum_i p_i(I) \Delta I_i + \sum_{i,j} p_{ij}(I) \Gamma(I_i, I_j) \in \mathcal{A}.$$

□

For a prescribed homogeneous polynomial F, define

$$\mathcal{A}_R(F) = \bigcap \{\mathcal{A} \subset \mathbb{R}[V] : \mathcal{A} \text{ is Laplacian and } F \in \mathcal{A}\}. \tag{3.7}$$

The collection contains $\mathbb{R}[V]$; its Laplacian intersection $\mathcal{A}_R(F)$ is therefore the smallest Laplacian algebra containing R, F. Finite generation is uniform in fixed dimension and degree.

Proposition 3.8 (Uniform finite generation)**.** *For every $N, d \geq 1$, there is an integer $B = B(N, d)$ such that every homogeneous polynomial $F \in \mathbb{R}[x_1, \dots, x_N]$ of degree d has*

$$\mathcal{A}_R(F) = \mathbb{R}[I_1, \dots, I_k], \qquad 1 \leq \deg I_i \leq B, \qquad k \leq \binom{N+B}{N} - 1,$$

with homogeneous I_i. In particular, these bounds apply uniformly to irreducible degree-d equations of minimal cones in $\mathbb{R}^N$, and to the subclass of harmonic equations.

Proof. We follow [57, proof of Lemma 24(a)], using all coefficients at once. Put $T = \mathbb{R}[t_\alpha : |\alpha| = d]$, $\mathscr{F} = \sum_{|\alpha|=d} t_\alpha x^\alpha$, and let $\mathscr{B} \subset T[x_1, \dots, x_N]$ be the smallest T-algebra containing $R, \mathscr{F}$ and closed under Δ_x. It is graded by x-degree: its elements are sums of expressions obtained from these homogeneous inputs by multiplication and Δ_x. Hilbert's basis theorem [45, Chapter I, Section 2, Proposition 2.3 and Corollary 2.4, p. 11] therefore gives positive-degree, x-homogeneous elements $b_1, \dots, b_s \in \mathscr{B}$ generating the ideal $J = (\mathscr{B}_+)T[x]$.

For every real coefficient vector c, evaluation $t = c$ commutes with multiplication and Δ_x. Consequently $\mathscr{B}(c) = \mathcal{A}_R(F_c) =: \mathcal{A}_c$: every evaluated expression lies in $\mathcal{A}_c$, and the evaluated algebra is itself Laplacian and contains R, F_c. Evaluation also preserves x-degree, so the image ideal is $J(c) = (\mathcal{A}_c)_+\mathbb{R}[x] = (b_1(c), \dots, b_s(c))$. For homogeneous $f \in \mathcal{A}_c$ of positive degree, write $f = \sum_i h_i b_i(c)$ with homogeneous coefficients of degree $\deg f - \deg_x b_i < \deg f$, omitting zero terms. Apply the graded Reynolds projection of this real Laplacian algebra, Proposition 3.5, or [57, Theorem 23]:

$$f = \sum_i \Pi_c(h_i) b_i(c).$$

Induction on degree shows that the nonzero $b_i(c)$ generate $\mathcal{A}_c$. The bound $B = \max_i \deg_x b_i$ therefore depends only on N, d. Finally, bases of the homogeneous pieces of $\mathcal{A}_c$ in degrees $1, \dots, B$ give at most $\sum_{j=1}^{B} \binom{N+j-1}{j} = \binom{N+B}{N} - 1$ generators. □

The uniform bound $B(N, d)$ is nonexplicit.

Generator counts use the essential Euclidean space

$$V_{\mathrm{ess}}(F) = V_{\mathrm{cyl}}(F)^\perp, \qquad V_{\mathrm{cyl}}(F) = \{v : D_v F \equiv 0\},$$

unless a cylindrical extension is specified; R and $\mathcal{A}_R(F)$ refer to this space.

Harmonic minimal equations already require generators of increasing degree: for $F = \operatorname{Re}\det Z$ on $M_d(\mathbb{C})$, $d \geq 3$, the smallest algebra needs $d+1$ generators and its least possible maximal homogeneous generator degree is $2d$. Thus $B(2d^2, d) \geq 2d$. The proof is given with the determinant calculations in Example 14.2.

Since $S = |\nabla F|^2 \in \mathcal{A}_R(F)$, the generator test proves $\mathcal{A}_R(F) = \mathbb{R}[R, F, S]$ whenever this algebra is Laplacian. Otherwise quantities such as ΔS or $\Gamma(S, S)$ may force additional generators.

3.2. **The quotient geometry of a Laplacian algebra.** Choose homogeneous generators $I_1, \dots, I_k$ for $\mathcal{A}$, including $I_1 = R$, and write $I = (I_1, \dots, I_k)$. Its fibers $I^{-1}(z)$ lie on spheres and are the common levels of all elements of $\mathcal{A}$.

The images

$$K = I(V), \qquad K_{\mathbb{S}} = I(\mathbb{S}^{N-1})$$

are semialgebraic by the Tarski–Seidenberg theorem [5, Section 2.2]. The induced map identifies the topological quotient of the sphere with $K_{\mathbb{S}}$, since it is a continuous bijection from

a compact space to a Hausdorff space. Changing generators gives mutually inverse polynomial maps between the images. Hence $d_{\mathbb{S}}(\mathcal{A}) := \dim K_{\mathbb{S}}$, the *spherical quotient dimension*, is intrinsic; it need not equal $k-1$.

By Mendes–Radeschi [57, Theorem A and Theorem 25], [58, Theorem A and Corollary B], these fibers define a *manifold submetry*: the quotient projection maps each closed ball onto the ball of the same radius about its image, and its fibers are smooth embedded submanifolds, possibly disconnected and of different dimensions. Moreover, $\mathcal{A}$ contains every polynomial constant on its fibers. The quotient metric, given by distance between fibers, generally differs from the Euclidean metric inherited by the coordinate image $K \subset \mathbb{R}^k$.

3.2.1. *The metric and the induced differential operator.* Write a closure table as $\Delta I_i = b_i(I)$ and $\Gamma(I_i, I_j) = g_{ij}(I)$. At every $x \in V$,

$$\big(g_{ij}(I(x))\big) = dI_x\, dI_x^T. \tag{3.8}$$

Thus $\mathbf{g} \geq 0$ and $\operatorname{rank}\mathbf{g} = \operatorname{rank} dI$. On a regular quotient chart, an independent subset of the I_i has inverse metric given by its Gram submatrix: its gradients span the orthogonal complement of the fiber. This is the Riemannian-submersion description of [57, Proposition 26].

The induced Laplacian has second-order coefficients g_{ij} and first-order coefficients b_i, as in Theorem 3.10. It need not be the quotient Laplace–Beltrami operator: the b_i also record fiber-volume variation.

Example 3.9 (The interval quotient and its weight)**.** Return to Example 3.4, with $N = p+q$ and $p, q \geq 2$. On the unit sphere, $u + v = 1$, so $t = u \in [0,1]$ is the single independent invariant. Euler's identity and the polar-coordinate expression for Δ give

$$|\nabla_{\mathbb{S}} t|^2 = 4t(1-t), \qquad \Delta_{\mathbb{S}} t = 2p - 2Nt.$$

The quotient metric is therefore

$$ds^2 = \frac{dt^2}{4t(1-t)}.$$

The substitution $t = \cos^2\theta$ turns it into $d\theta^2$, with $0 \leq \theta \leq \pi/2$. For a function $h(t)$, however,

$$\Delta_{\mathbb{S}} h(t) = 4t(1-t)h''(t) + (2p - 2Nt)h'(t).$$

Equivalently this operator is

$$\frac{1}{w(t)}\frac{d}{dt}\left(4t(1-t)w(t)\frac{dh}{dt}\right), \qquad w(t) = t^{p/2-1}(1-t)^{q/2-1},$$

where $w(t)\,dt$ is proportional to the pushforward of spherical volume. The rank drops at the endpoints $x = 0$ and $y = 0$.

For the chosen generators, the finite closure table has the form

$$\Delta I_i = b_i(I), \qquad \Gamma(I_i, I_j) = g_{ij}(I). \tag{3.9}$$

For smooth functions of $z = (z_1, \dots, z_k)$ on a neighborhood of the image, write $p_i = \partial p/\partial z_i$ and $p_{ij} = \partial^2 p/\partial z_i \partial z_j$, and define

$$\widehat{\Delta} p = \sum_i b_i p_i + \sum_{i,j} g_{ij} p_{ij}, \qquad \widehat{\Gamma}(p,q) = \sum_{i,j} g_{ij} p_i q_j.$$

Theorem 3.10 (Exact finite-dimensional reduction)**.** *If* (3.9) *holds, then for* $H = p \circ I$ *and* $K = q \circ I$*,*

$$\Delta H = (\widehat{\Delta} p) \circ I, \qquad \Gamma(H, K) = \widehat{\Gamma}(p,q) \circ I.$$

Writing $Q_p = \widehat{\Gamma}(p,p)$*, one has*

$$\boxed{\mathcal{L}(p \circ I) = \left(Q_p \widehat{\Delta} p - \frac{1}{2}\widehat{\Gamma}(p, Q_p)\right) \circ I.} \tag{3.10}$$

Thus the curvature inequality is an exact differential inequality on the semialgebraic image $I(\mathbb{R}^N)$.

Proof. The first two identities follow from the chain rule and (3.9). In particular, $|\nabla H|^2 = Q_p \circ I$, so

$$\operatorname{Hess} H(\nabla H, \nabla H) = \tfrac{1}{2}\Gamma(H, |\nabla H|^2) = \tfrac{1}{2}\widehat{\Gamma}(p, Q_p) \circ I.$$

Together these formulas give (3.10). □

The sign is tested on the actual image $K = I(\mathbb{R}^N)$. These identities hold also at rank-drop points; for $G = p \circ I$, X_G is defined wherever $Q_p \circ I > 0$. The comparison theorem imposes no bound on quotient dimension.

For polarization of these contractions, see Appendix E.1 and Section 9. Section 4 reduces the divergence sign to an inequality on K.

3.3. **Area minimization and irreducible defining equations.** The full zero set $C = \{F = 0\}$ may differ from the support of $\partial[\![\{F < 0\}]\!]$. For a square-free equation, area minimization permits only one irreducible factor with a nonsingular real zero, hence only one contributing a smooth hypersurface patch. We use this effective equation and write $C_{\mathrm{reg}} = \{F = 0,\ \nabla F \neq 0\}$.

Recall that a degree-d homogeneous harmonic polynomial restricts to a *spherical harmonic* $f = F|_{\mathbb{S}^{N-1}}$, with $F(r\theta) = r^d f(\theta)$ [3, Chapter 5]. The link of its zero cone is the nodal set $\{f = 0\}$ on the sphere; Figure 1 illustrates this restriction.

Lemma 3.11 (Square-free equations and their critical sets)**.** *Let F be a nonzero homogeneous polynomial of positive degree on $\mathbb{R}^N$, $N \geq 2$. If F is square-free, then $Z = \{\nabla F = 0\}$ has dimension at most $N-2$, and hence $\mathcal{H}^{N-1}(Z \cap K) = 0$ for every compact K. If F is harmonic, it has no nonconstant nonnegative polynomial factor; in particular, it is square-free and each of its real irreducible factors has a nonsingular real zero.*

For the harmonic square-freeness and critical-set statements, see [38, Proposition 2.2 and Theorem 2.1]. The following argument also identifies the possible real factors.

Proof. If the critical set had a component of codimension one, its irreducible defining polynomial P would divide every partial derivative of F. Euler's identity then gives $P \mid F$. Writing $F = PH$, square-freeness gives $P \nmid H$, so $P \mid \partial_i F$ forces $P \mid \partial_i P$ for every i, contradicting the degree of a nonzero derivative of P. The Hausdorff-measure assertion follows from the semialgebraic stratification used in Section 2.5.

For the harmonic assertion, factors of a homogeneous polynomial are homogeneous: the lowest and highest degrees in a product add, so their equality forces equality for each factor. Suppose $F = PQ$, with $P \geq 0$ nonconstant. Then $\deg Q < \deg F$, and spherical harmonic orthogonality gives

$$0 = \int_{\mathbb{S}^{N-1}} FQ = \int_{\mathbb{S}^{N-1}} PQ^2 > 0,$$

a contradiction; the orthogonality permits arbitrary lower-degree Q, not necessarily harmonic [3, Proposition 5.9, pp. 79–80]. Strict positivity follows because PQ^2 is a nonzero nonnegative homogeneous polynomial, positive on a nonempty open part of the sphere. Taking $P = H^2$ excludes repeated factors. Every real irreducible factor must also change sign, since otherwise it or its negative would be a nonnegative factor. Such a factor has a nonsingular real zero by [5, Theorem 4.5.1]. □

Proposition 3.12 (Irreducible effective equations for minimizing cones)**.** *Let $F \in \mathbb{R}[x_1, \ldots, x_N]$, $N \geq 2$, be nonconstant, homogeneous and square-free. Suppose $E = \{F < 0\}$ is locally perimeter minimizing in $\mathbb{R}^N$ and $\partial[\![E]\!] \neq 0$. Then exactly one real irreducible factor*



FIGURE 1. A degree-six spherical harmonic $f = \Phi|_{\mathbb{S}^2}$, where $\Phi = \mathrm{Re}((x+iy)^6 + \frac{1}{2}(y+iz)^6 + \frac{1}{3}(z+ix)^6)$. Left: values on the unit sphere, with the nodal set in black. Right: the radial amplitude plot $\theta \mapsto c^{-1}|f(\theta)|\theta$, with a positive display normalization c. Red and blue indicate positive and negative values. The link of the zero cone is the nodal set in the left panel. Visualization inspired by Matthew de Courcy-Ireland's illustration in Logunov's lecture slides [54, slide 4].

of F has a nonsingular real zero. After absorbing a nonzero constant and signs into this factor, one can write

$$F = PQ, \qquad P \text{ real irreducible}, \qquad Q \geq 0,$$
$$\{F < 0\} = \{P < 0\} \quad \text{almost everywhere}.$$

In particular, if every real irreducible factor has a nonsingular real zero, then F is irreducible. Under the same minimizing hypotheses, harmonic F is therefore irreducible.

Proof. Let $V = |\partial[\![E]\!]|$ and $M = \operatorname{reg} V$. Its nonempty support is a closed cone, hence connected. Regularity of minimizing boundaries makes M real analytic and gives $\mathcal{H}^{N-2}(\operatorname{sing} V) = 0$ [55, Part III, introductory theorem, pp. 275–277]. Thus M is connected: for $N = 2$ this follows from smoothness, and for $N \geq 3$ from [92, Section 5, proof of Theorem 1.1; Theorem 3.2]. The singular set here belongs to the boundary, not to its polynomial representation.

We use the following elementary divisibility fact. If a polynomial H vanishes on a nonempty smooth open patch of the zero set of a real irreducible polynomial P_j, then $P_j \mid H$. Indeed, express the patch as a graph $t = \varphi(x')$. If $P_j \nmid H$, Gauss' lemma [44, Chapter VIII, Section 5] makes P_j, H coprime over $\mathbb{R}(x')$, so their resultant in t is a nonzero polynomial in x'. The common root $\varphi(x')$ makes it vanish on an open set, a contradiction.

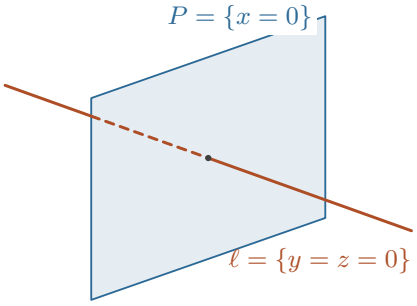
Write $F = c\prod_j P_j$ with distinct irreducible factors. For each factor with a nonsingular real zero, the divisibility fact and coprimality give a nonempty smooth patch $U_j \subset \{P_j = 0\}$ avoiding all other factors. On U_j, $\nabla F \neq 0$ and F changes sign, so $U_j \subset M$. If two such factors P_1, P_2 existed, the analytic function $P_1|_M$, zero on U_1, would vanish on connected M, contradicting $P_1 \neq 0$ on U_2.

For an irreducible real polynomial, having a nonsingular real zero is equivalent to changing sign [5, Theorem 4.5.1]. There must be one such factor, since otherwise F is of fixed sign and $\partial[\![E]\!] = 0$. Choose the signs of the remaining factors to make their product $Q \geq 0$. Its zero set is Lebesgue null, proving equality of the phases almost everywhere. The harmonic assertion follows from Lemma 3.11. □

Example 3.13 (A redundant line and two intersecting hyperplanes)**.** For $F = x(y^2 + z^2)$ in $\mathbb{R}^3$, the zero set is the plane $P = \{x = 0\}$ together with the line $\ell = \{y = z = 0\}$. The

negative phase agrees almost everywhere with $\{x < 0\}$; its boundary current is the plane calibrated by $dy \wedge dz$, with support P. The extra line contributes no two-dimensional mass. For $F = xy$, instead, $\Delta F = 0$ and $\mathcal{L}(F) = -2F$, so the regular zero set is minimal, but both factors contribute hypersurface branches. Their boundary is not area minimizing by Proposition 3.12. For the elementary crossing-lines comparison and its product with a line, see [61, Figures 10.1.3 and 10.7.1, Lemma 10.6]. See also the regularity obstruction in [55, Theorem 28.1] and [30, Section 1 and Theorem 1].

(a) A plane and a transverse line

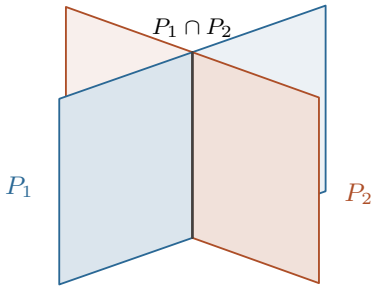


$F = x(y^2 + z^2)$

Area-minimizing current: $[\![P]\!]$

The added line has zero two-dimensional mass.

(b) Two intersecting hyperplanes



$F = xy$

Not area minimizing

Both planes contribute two-dimensional mass.

FIGURE 2. Left: $\{x(y^2 + z^2) = 0\} = P \cup \ell$, but the minimizing boundary current has support P. Right: both intersecting hyperplanes in $\{xy = 0\}$ carry area; their union is stationary as a varifold but the associated boundary current is not area minimizing. Both pictures are illustrated in $\mathbb{R}^3$.

For structural results, unless stated otherwise, we choose a nonconstant real homogeneous irreducible defining polynomial with $C_{\mathrm{reg}} \neq \varnothing$. Removing a nonnegative factor preserves the boundary current, but may change the full zero set, the degree and $\mathcal{A}_R(F)$; all degree and generator statements refer to the chosen effective equation.

The classical algebraic minimality criterion now takes the following form; see [64, Section 6.2.1, equation (6.2.2)].

Proposition 3.14 (Minimality and polynomial divisibility)**.** *Let F be real irreducible and homogeneous of degree $e \geq 1$, with $C_{\mathrm{reg}} \neq \varnothing$. Then C_{reg} is minimal if and only if*

$$\mathcal{L}(F) = F\lambda_F, \tag{3.11}$$

where λ_F is homogeneous of degree $2e-4$, possibly zero, for $e \geq 2$, and $\lambda_F = 0$ for $e = 1$. The same equivalence holds for a square-free homogeneous F when every real irreducible factor has a nonsingular real zero.

Proof. By (1.3), minimality means $\mathcal{L}(F) = 0$ on C_{reg}. The patch divisibility argument in Proposition 3.12 gives $F \mid \mathcal{L}(F)$. For the square-free extension, take for each factor a nonsingular patch avoiding the others, apply the same argument, and use pairwise coprimality. Since $\deg \mathcal{L}(F) = 3e - 4$ when it is nonzero, the degree follows. For $e = 1$, all second derivatives vanish. The converse follows from the same curvature formula. □

The factor condition cannot be omitted in the square-free extension: for $t = y^2 + z^2$ and $F = x(2x^2 + t)$, the regular zero set is planar, but $\mathcal{L}(F) = 8(9x^2 - t)F + 4xt^2$ is not divisible by F. Its effective equation is simply x.

Under (3.11), differential closure and polynomial division [57, Lemma 24(b)], or Proposition 3.6, give $\lambda_F \in \mathcal{A}_R(F)$. In the harmonic case this becomes

$$\Delta F = 0, \qquad S = |\nabla F|^2, \qquad \Gamma(F, S) = 2qF, \qquad q = -\lambda_F, \tag{3.12}$$

with $S, q \in \mathcal{A}_R(F)$. These identities simplify the closure calculation, while Proposition 3.8 supplies the uniform generator bound.

4. Multipliers, Jacobi Identities, and Global Positivity

The multiplier expansion relates global divergence to its Jacobi trace; Section 3 reduces the sign test to the invariant image.

4.1. The global multiplier defect.

Proposition 4.1 (Odd-power invariance and regular defining potentials)**.** *Let $\Omega \subset \mathbb{R}^N \setminus \{0\}$ be open and conical, and let F be smooth and homogeneous with $\nabla F \neq 0$ on $C \cap \Omega = \{F = 0\} \cap \Omega$.*

(i) *Let $a > 0$ be smooth and homogeneous on Ω, and let $m \geq 1$ be an odd integer. Set*

$$\widetilde{F} = F^m, \qquad G = aF, \qquad \widetilde{G} = a^m \widetilde{F} = G^m.$$

Then

$$\nabla \widetilde{G} = m G^{m-1} \nabla G, \qquad \mathcal{L}(\widetilde{G}) = m^3 G^{3m-3} \mathcal{L}(G). \tag{4.1}$$

Thus $X_{\widetilde{G}} = X_G$ wherever $F \neq 0$ and $\nabla G \neq 0$, and X_G gives its smooth extension across $C \cap \Omega$. Conversely, every positive smooth homogeneous multiplier b for $\widetilde{F}$ is obtained by setting $a = b^{1/m}$. Hence the two defining equations admit the same multiplier subcalibration fields, with the same regular boundary extensions and the same remaining exceptional sets.

(ii) *If a smooth homogeneous function G has the same zero set and signs as F on Ω, and $\nabla G \neq 0$ on $C \cap \Omega$, then $G = bF$ for a positive smooth homogeneous function b on Ω.*

Proof. For (i), the chain rule gives $\mathcal{L}(\psi(G)) = (\psi'(G))^3 \mathcal{L}(G)$. Take $\psi(t) = t^m$. Its derivative is positive off $t = 0$, which proves the field identity. On the regular cone, $\nabla G = a \nabla F \neq 0$, so X_G provides the exterior normal extension. Off the cone, the two gradients vanish at exactly the same points. The inverse correspondence follows from $bF^m = (b^{1/m} F)^m$; the positive root is smooth and homogeneous.

For (ii), use local coordinates (y, t) with $t = F$. Since $G(y, 0) = 0$,

$$G(y, t) = t \int_0^1 \partial_t G(y, st)\, ds.$$

The integral extends $b = G/F$ smoothly, with $b|_C = |\nabla G|/|\nabla F| > 0$ by the common phase signs and regularity. These extensions agree on overlaps; the quotient also gives $\deg b = \deg G - \deg F$. □

Part (i) preserves the divergence and its phase integrals. For $m > 1$, $\nabla \widetilde{G} = 0$ on the cone; the Jacobi–Hardy formulas use its regular field extension X_G, the degree $\ell = \deg \widetilde{G}/m$, and $w = |\nabla G|^{-1}|_C$.

The displayed algebra $\mathbb{R}[R, F^m]$ need not itself be Laplacian closed, even when $\mathbb{R}[R, F]$ is: for $F = x_1$, $\Delta(F^3) = 6F \notin \mathbb{R}[R, F^3]$. The radial and gradient powers used below are discussed in Appendix A.

Let $\Omega \subset \mathbb{R}^N$ be open and $F, a \in C^\infty(\Omega)$, with $a > 0$. Suppose $\nabla F \neq 0$ on $C = \{F = 0\} \cap \Omega$ and C is minimal. Put $G = aF$. The product rule gives $\mathcal{L}(G)|_C = a^3 \mathcal{L}(F)|_C = 0$. Smooth division by F, as in Proposition 4.1(ii), therefore defines

$$\lambda_F = \frac{\mathcal{L}(F)}{F}, \qquad \mathcal{K}_F(a) = \frac{\mathcal{L}(aF)}{F}.$$

Then $G\mathcal{L}(G) = aF^2 \mathcal{K}_F(a)$. Polynomial divisibility gives the same λ_F as (3.11). Write

$$P = \nabla F, \quad Q = \nabla a, \qquad S = |P|^2, \quad T = \langle P, Q \rangle, \quad V = |Q|^2.$$

Theorem 4.2 (Exact multiplier expansion)**.** *Under the preceding smooth hypotheses, the multiplier defect equals*

$$\begin{aligned}\mathcal{K}_F(a) = a^3\lambda_F &\\ &+ a^2\big(S\Delta a + 2T\Delta F - 2\operatorname{Hess} F(P,Q) - \operatorname{Hess} a(P,P)\big)\\ &+ 2a(T^2 - SV)\\ &+ Fa\big(2T\Delta a + V\Delta F - \operatorname{Hess} F(Q,Q) - 2\operatorname{Hess} a(P,Q)\big)\\ &+ F^2\mathcal{L}(a). \end{aligned} \tag{4.2}$$

If, in addition, F, a are polynomials in a Laplacian algebra $\mathcal{A}$ and $F \mid \mathcal{L}(F)$ in $\mathbb{R}[x_1, \dots, x_N]$, then every term in (4.2) belongs to $\mathcal{A}$. Proposition 3.14 gives sufficient geometric conditions for this polynomial divisibility.

Proof. Expand $\nabla(aF) = aP + FQ$, $\Delta(aF) = a\Delta F + F\Delta a + 2T$, and

$$\operatorname{Hess}(aF) = a\operatorname{Hess} F + F\operatorname{Hess} a + dF \otimes da + da \otimes dF.$$

The constant term in $\mathcal{L}(aF)$ is $a^3\mathcal{L}(F) = a^3F\lambda_F$; collecting the remaining powers of F gives the formula. The identity

$$2\operatorname{Hess} u(\nabla v, \nabla w) = \Gamma(v, \Gamma(u,w)) + \Gamma(w, \Gamma(u,v)) - \Gamma(u, \Gamma(v,w))$$

expresses the Hessian terms through gradient inner products.

By closure, $\mathcal{L}(F) = S\Delta F - \Gamma(F,S)/2 \in \mathcal{A}$. Polynomial division, Proposition 3.6, then gives $\lambda_F \in \mathcal{A}$, proving the algebra assertion. □

Lemma 4.3 (The defect at a critical zero)**.** *Let F be a nonzero polynomial satisfying $\mathcal{L}(F) = F\lambda_F$, with polynomial λ_F, and let a be a C^2 function near a point p with $F(p) = 0$ and $\nabla F(p) = 0$. The right-hand side of (4.2) defines a continuous extension of $\mathcal{L}(aF)/F$, and*

$$\mathcal{K}_F(a)(p) = a(p)^3\lambda_F(p). \tag{4.3}$$

If $\lambda_F(p) < 0$, no such multiplier with $a(p) > 0$ can satisfy $(aF)\mathcal{L}(aF) \geq 0$ throughout a punctured neighborhood of p. In particular, a nonnegative C^2 multiplier satisfying this sign condition must vanish at p.

Proof. The expansion is a product-rule identity wherever $F \neq 0$, and every term on its right-hand side is continuous. At p, one has $F = 0$, $P = 0$, $S = 0$, and $T = 0$, so only $a^3\lambda_F$ remains. If $a(p) > 0$ and $\lambda_F(p) < 0$, continuity gives $a > 0$ and $\mathcal{K}_F(a) < 0$ near p. A nonzero polynomial has a zero set with empty interior, so there are nearby points with $F \neq 0$, where

$$(aF)\mathcal{L}(aF) = aF^2\mathcal{K}_F(a) < 0.$$

This contradicts the required sign. □

Logarithmic differentiation of a positive multiplier. On the open domain where $a > 0$, including the regular cone, $h = \log a$ is smooth and satisfies

$$\nabla h = \frac{\nabla a}{a}, \qquad \Delta h = \frac{\Delta a}{a} - \frac{|\nabla a|^2}{a^2}, \qquad \Gamma(h,q) = \frac{\Gamma(a,q)}{a}. \tag{4.4}$$

If a has degree d, then $h(tx) = h(x) + d\log t$ for $t > 0$.

4.2. **The Jacobi boundary condition.** Let $J_{C_{\mathrm{reg}}} = \Delta_{C_{\mathrm{reg}}} + |A_C|^2$ be the Jacobi operator associated with the second variation; see [77] and [73, Remark 9.8]. With scalar mean curvature defined by $H = \operatorname{div}_{C_{\mathrm{reg}}} \nu$, a variation with normal speed w has linearization $-J_C w$.

Theorem 4.4 (Exact Jacobi trace)**.** *For the smooth F, a of Theorem 4.2, along $C_{\mathrm{reg}} \cap \Omega$,*

$$\boxed{\mathcal{K}_F(a)|_{C_{\mathrm{reg}}} = -a^4|\nabla F|^3 J_{C_{\mathrm{reg}}}\left(\frac{1}{a|\nabla F|}\right).} \tag{4.5}$$

Consequently, $\mathcal{K}_F(a) \geq 0$ near this hypersurface requires the positive Jacobi supersolution

$$w = (a|\nabla F|)^{-1} > 0, \qquad J_C w \leq 0. \tag{4.6}$$

It is a Jacobi field exactly when the boundary defect vanishes.

Proof. The flow of $\nabla G/|\nabla G|^2$ parametrizes the level sets $M_t = \{G = t\}$, with normal speed $w = |\nabla G|^{-1} = (a|\nabla F|)^{-1}$ at $t = 0$. The mean-curvature variation at the minimal zero level is $\frac{d}{dt} H_{M_t}|_0 = -J_C w$. On the other hand, (1.3) gives, along this flow,

$$H_{M_t} = \frac{F\mathcal{K}_F(a)}{|\nabla G|^3}, \qquad \left.\frac{dF}{dt}\right|_0 = \left.\frac{\langle \nabla F, \nabla G\rangle}{|\nabla G|^2}\right|_C = \frac{1}{a}.$$

Since $F = 0$ on C, differentiation of the other factors contributes zero at $t = 0$. Using $|\nabla G|^3 = a^3|\nabla F|^3$ there yields

$$-J_C w = \left.\frac{d}{dt} H_{M_t}\right|_0 = \left.\frac{\mathcal{K}_F(a)}{a^4|\nabla F|^3}\right|_C,$$

which is (4.5). □

Proposition 4.5 (Harmonic power multipliers on the boundary)**.** *Let $F \not\equiv 0$ be a homogeneous polynomial of degree $e \geq 2$ satisfying (3.12). Put $R = r^2$, $K = \Delta S = 2|\operatorname{Hess} F|^2$, and $T = |\nabla S|^2$. For real σ, γ, define $G = r^\sigma S^\gamma F$ on $\{S > 0\}$. Its boundary speed is $w = r^{-\sigma} S^{-\gamma - 1/2}$. On C_{reg},*

$$-\frac{J_C w}{w} = \frac{\sigma\big(N - 1 - 2e - \sigma - 4\gamma(e-1)\big)}{R} + \frac{\gamma K - (2\gamma+1)q}{S} - \frac{\gamma(2\gamma+3)}{2}\frac{T}{S^2}. \tag{4.7}$$

In particular, a two-sided subcalibration generated by G requires the right-hand side to be nonnegative on C_{reg}.

Proof. Set $w_0 = S^{-1/2}$. Applying Theorem 4.4 with $a = 1$ and $\mathcal{L}(F) = -qF$ gives $J_C w_0/w_0 = q/S$. On C_{reg}, equation (3.12) implies $\nabla S \perp \nabla F$ and $\operatorname{Hess} F(\nu_C, \nu_C) = 0$. Since $\operatorname{Hess} F\, \nu_C = \nabla S/(2\sqrt{S})$, decomposition into normal and tangential components gives

$$|A_C|^2 = \frac{K}{2S} - \frac{T}{2S^2}, \qquad |\nabla_C \log w_0|^2 = \frac{T}{4S^2}.$$

For $p = 2\gamma + 1$, the chain rule reads

$$\frac{J_C(w_0^p)}{w_0^p} = p\frac{J_C w_0}{w_0} + (1-p)|A_C|^2 + p(p-1)|\nabla_C \log w_0|^2.$$

Finally, C has dimension $N - 1$, so $\Delta_C r^{-\sigma}/r^{-\sigma} = \sigma(\sigma - N + 3)/R$. The homogeneity of w_0^p, of degree $-p(e-1)$, gives the cross term $2\sigma p(e-1)/R$. Substitution proves the formula. □

Theorem 4.6 (Positive-solution identity and homogeneous Hardy gap)**.** *Let $C^n \subset \mathbb{R}^{n+1}$, $n \geq 2$, be a cone whose regular part is two-sided and minimal. Let G be a smooth homogeneous defining function of degree ℓ on a conical neighborhood of $C_{\mathrm{reg}} \setminus \{0\}$, with $\nabla G \neq 0$ there. Set*

$$w = |\nabla G|^{-1}\big|_C > 0, \qquad V_G = -\frac{J_C w}{w} = r^{-2}\beta_G(\theta). \tag{4.8}$$

For every $u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\})$,

$$Q_C(u) = \int_C w^2 \left|\nabla_C\left(\frac{u}{w}\right)\right|^2 + \int_C V_G u^2. \tag{4.9}$$

If $\beta_G \geq \beta_0$ on the regular link for a finite constant β_0, then

$$Q_C(u) \geq \left[\left(\ell - \frac{n}{2}\right)^2 + \beta_0\right] \int_C r^{-2}u^2. \tag{4.10}$$

For $G = aF$ in Theorem 4.4,

$$V_G = \left.\frac{\mathcal{K}_F(a)}{a^3|\nabla F|^2}\right|_C = \partial_{\nu_C} \operatorname{div}\left(\frac{\nabla G}{|\nabla G|}\right)\Big|_C. \tag{4.11}$$

Consequently a nonnegative Jacobi boundary defect implies stability; if also $\ell \neq n/2$, it implies strict stability with constant at least $(\ell - n/2)^2$.

Proof. Substituting $u = wv$ and integrating by parts gives (4.9). Homogeneity gives $w = r^{1-\ell}\phi(\theta)$, $\phi > 0$. Writing $\Lambda_{\mathrm{reg}} = C_{\mathrm{reg}} \cap \mathbb{S}^n$, the cone volume element $r^{n-1}\,dr\,d\mu_\Lambda$ gives

$$\int_C w^2 |\nabla_C v|^2 \geq \int_{\Lambda_{\mathrm{reg}}} \phi^2 \int_0^\infty r^{n+1-2\ell} |\partial_r v|^2 \, dr\, d\mu_\Lambda.$$

For every real p and $f \in C_c^\infty(0,\infty)$, completion of the square proves

$$\int_0^\infty r^p (f')^2 \, dr \geq \frac{(p-1)^2}{4} \int_0^\infty r^{p-2} f^2 \, dr.$$

Apply this with $p = n+1-2\ell$ for each angular point. The resulting term is $(\ell - n/2)^2 \int_C r^{-2}u^2$. Adding $\int_C V_G u^2 \geq \beta_0 \int_C r^{-2}u^2$ proves (4.10). The first equality in (4.11) follows by dividing (4.5) by $a^4|\nabla F|^3 w$. For the second, differentiate $\operatorname{div} X_G = F\mathcal{K}_F(a)/|\nabla G|^3$ in the unit normal direction on $F = 0$, where $\partial_\nu F = |\nabla F|$. □

Remark 4.7 (Vanishing trace and the Hardy gap). If $J_C w = 0$, formula (4.9) still contains the Hardy gap $(\ell - n/2)^2$. This lower bound vanishes only at the central degree $\ell = n/2$. Failure of strict stability requires a sequence with Hardy quotient tending to zero; see Proposition E.1.

Remark 4.8 (Subcalibrations as paired geometric barriers). For $G = aF$, $a > 0$, and $\mathcal{K}_F(a) \geq 0$, the regular levels $M_t = \{G = t\}$, oriented by X_G, satisfy

$$H_{M_t} = \frac{F\mathcal{K}_F(a)}{|\nabla G|^3}, \qquad tH_{M_t} \geq 0.$$

Thus G is a supersolution on $F < 0$ and a subsolution on $F > 0$ of the level-set equation $\operatorname{div} X_G = 0$. The level-set variation in the proof of Theorem 4.4 has normal graph height $s_t = tw + O(t^2)$, with $w = (a|\nabla F|)^{-1}$, and gives

$$H_{M_t} = -tJ_C w + O(t^2). \tag{4.12}$$

Thus $J_C w < 0$ gives strict local barriers on both sides.

The sign of $\mathcal{K}_F(a)$ away from the cone requires a separate calculation, as the stable non-minimizing examples in Section 6.3 illustrate.

The admissible degree. Let $C^{N-1} \subset \mathbb{R}^N$, $N \geq 3$, be a minimal cone, let $\Sigma = C \cap \mathbb{S}^{N-1}$, and suppose a homogeneous potential $G = aF$ has total degree ℓ. Then $w = (a|\nabla F|)^{-1}$ has degree $1 - \ell$. If the entire link Σ is smooth, it is compact and connected; see the maximum-principle argument of Hardt–Simon [39, p. 104]. Let $\lambda_1(\Sigma)$ be the first eigenvalue of $-\Delta_\Sigma - |A_\Sigma|^2$. Existence of a positive homogeneous Jacobi supersolution requires

$$\ell \in I_F := \left[\frac{N-1}{2} - \beta_F, \frac{N-1}{2} + \beta_F\right], \qquad \beta_F^2 = \frac{(N-3)^2}{4} + \lambda_1(\Sigma). \tag{4.13}$$

If the radicand is negative, no positive homogeneous Jacobi supersolution with positive angular part exists. More precisely, if $\phi_1 > 0$ is the first eigenfunction and $w = r^{1-\ell}\phi_1$, then

$$J_C w = r^{-1-\ell}\left[(1-\ell)(N-2-\ell) - \lambda_1(\Sigma)\right]\phi_1.$$

For any positive angular supersolution, integrating its inequality against ϕ_1 gives the same necessary degree condition. Using ϕ_1, interior degrees give $J_C w < 0$ and endpoints give $J_C w = 0$. A prescribed angular multiplier requires its own sign test.

For a smooth compact connected link, the zero-gap case is rigid. Write $n = N - 1$, $H_\Sigma = -\Delta_\Sigma - |A_\Sigma|^2$, and $\delta_C = (n-2)^2/4 + \lambda_1(\Sigma)$, the exact Hardy gap in (4.16). If $w = r^{1-\ell}\phi > 0$ and $-J_C w/w = r^{-2}\beta \geq 0$, then

$$\delta_C = 0 \quad \Longrightarrow \quad \ell = n/2, \qquad \beta \equiv 0, \qquad \phi \text{ is a positive first eigenfunction of } H_\Sigma. \tag{4.14}$$

Indeed, (4.13) forces $\ell = n/2$. Then $(H_\Sigma + (n-2)^2/4)\phi = \beta\phi$. Pairing with a positive first eigenfunction ψ gives $0 = \int_\Sigma \beta\phi\psi$, hence $\beta = 0$. Simplicity of the first eigenvalue now gives the last assertion. In particular, a nonnegative boundary defect that is positive at one point suffices for strict stability on such a link.

Remark 4.9 (Two admissible power multipliers). Let F be a homogeneous polynomial of degree $e \geq 2$ on $\mathbb{R}^{n+1}$, $n \geq 3$, whose zero set C is minimal, with $\nabla F \neq 0$ on $C \setminus \{0\}$ and nonempty smooth compact connected link Σ. Put $S = |\nabla F|^2$, and suppose that $s = S|_\Sigma$ is nonconstant. If two distinct real parameter pairs (α_i, b_i) give potentials

$$G_i = r^{\alpha_i} S^{b_i/2} F, \qquad \operatorname{sgn}(F) \operatorname{div} X_{G_i} \geq 0 \quad \text{where } F \neq 0$$

in a two-sided neighborhood of $C \setminus \{0\}$ on which both fields are defined, then C is strictly stable. Indeed, by Theorem 4.4, the positive boundary speeds $w_i = r^{-\alpha_i} S^{-(b_i+1)/2}$ are Jacobi supersolutions. Theorem 4.6 first gives stability. Their potential degrees are $\ell_i = e + \alpha_i + b_i(e-1)$. If the Hardy gap were zero, (4.14) would force $\ell_i = n/2$ and both angular parts $s^{-(b_i+1)/2}$ to be proportional to the same first eigenfunction. Thus $s^{-(b_1-b_2)/2}$ would be constant. Since s is nonconstant, $b_1 = b_2$, and equality of the degrees then gives $\alpha_1 = \alpha_2$, a contradiction.

Example: admissible degrees on the Simons cone. Let

$$F(x,y) = |x|^2 - |y|^2, \qquad (x,y) \in \mathbb{R}^4 \times \mathbb{R}^4, \qquad R = r^2.$$

Its zero link is $\Sigma = \mathbb{S}^3(1/\sqrt{2}) \times \mathbb{S}^3(1/\sqrt{2})$, with $|A_\Sigma|^2 = 6$. The constant function attains $\lambda_1(\Sigma) = -6$, so (4.13) gives

$$\beta_F^2 = \frac{25}{4} - 6 = \frac{1}{4}, \qquad I_F = [3,4].$$

The quadratic F itself lies outside this interval. Choose instead $G = RF = |x|^4 - |y|^4$, of degree $\ell = 4$, as in [25, Section 2]; see Remark 2.3 for its historical antecedents. Direct calculation gives

$$\begin{aligned} |\nabla G|^2 &= 4R(R^2 + 3F^2), & \Delta G &= 24F, \\ \mathcal{L}(G) &= 192RF^3, & \mathcal{K}_F(R) &= 192RF^2, \\ G\mathcal{L}(G) &= 192R^2F^4 \geq 0. \end{aligned}$$

This is the strictly minimizing $g = 2$, $m_1 = m_2 = 3$ case of Section 6.3.

On the other hand,

$$w = |\nabla G|^{-1}\big|_C = \frac{1}{2r^3}, \qquad J_C w = 0, \qquad \mathcal{K}_F(R)\big|_C = 0.$$

Nevertheless Theorem 4.6 gives

$$Q_C(u) \geq \left(4 - \frac{7}{2}\right)^2 \int_C r^{-2}u^2 = \frac{1}{4}\int_C r^{-2}u^2,$$

and (4.16) shows this constant is optimal. Thus an endpoint degree and zero Jacobi trace coexist with both strict area minimality and strict stability. For $G_\varepsilon = r^{-\varepsilon}G$, $0 < \varepsilon < 1$, the boundary coefficient is $\beta_{G_\varepsilon} = \varepsilon - \varepsilon^2 > 0$. For all sufficiently small $\varepsilon > 0$, G_ε retains the subcalibration sign and its normalized gradient satisfies the linear distance bound; see Corollary 6.10.

The spectral obstruction for a singular link. For a singular link, we use the angular Rayleigh infimum, without assuming connectedness of Σ_{reg} or existence of a first eigenfunction.[1] Define

$$\lambda_{\mathrm{F}}(\Sigma) = \inf_{0\neq\phi\in C_c^\infty(\Sigma_{\mathrm{reg}})} \frac{\int_{\Sigma_{\mathrm{reg}}}(|\nabla\phi|^2 - |A_\Sigma|^2\phi^2)}{\int_{\Sigma_{\mathrm{reg}}}\phi^2}. \tag{4.15}$$

This infimum may equal $-\infty$; when finite, it is the bottom of the Friedrichs extension and need not be attained. Zhu's spherical stability convention [93, Section 2.1.5, equation (2.8)] includes the Ricci term:

$$\lambda_1^Z(\Sigma) = \lambda_{\mathrm{F}}(\Sigma) - (N-2).$$

The exact cone stability quotient is

$$\begin{aligned}\inf_{0\neq u\in C_c^\infty(C_{\mathrm{reg}}\setminus\{0\})} \frac{Q_C(u)}{\int_C r^{-2}u^2} &= \frac{(N-3)^2}{4} + \lambda_{\mathrm{F}}(\Sigma)\\ &= \frac{(N-1)^2}{4} + \lambda_1^Z(\Sigma).\end{aligned} \tag{4.16}$$

By radial–angular separation [77, 9], the sharp radial Hardy inequality and the angular Rayleigh bound give the lower bound. For a product test $u(r,\theta) = \eta(r)\varphi(\theta)$, the cone quotient is the sum of the radial Hardy quotient and the angular Rayleigh quotient. Independent minimizing sequences give equality, including $\lambda_{\mathrm{F}}(\Sigma) = -\infty$. If the right-hand side is negative, no fields satisfying the comparison criterion can exist.

Remark 4.10 (What the boundary trace determines). Equality $a_1|_{C_{\mathrm{reg}}} = a_2|_{C_{\mathrm{reg}}}$ is equivalent to local smooth divisibility of $a_1 - a_2$ by F. By (4.5), it implies

$$\mathcal{K}_F(a_1)|_{C_{\mathrm{reg}}} = \mathcal{K}_F(a_2)|_{C_{\mathrm{reg}}}. \tag{4.17}$$

Thus the boundary defect depends only on $a|_C$.

For calibrated complex cones with zero Hardy gap, see the Jacobi-field construction in Proposition E.1.

4.3. **A multiplier criterion for harmonic defining functions.** The following criterion permits multipliers that degenerate at the singular set. Its spectral application appears in Section 12.

Let $F \not\equiv 0$ be a homogeneous polynomial of positive degree on $\mathbb{R}^N$, $N \geq 2$, satisfying (3.12), and set $U = \{S > 0\}$. Let h be smooth on U, put $a = e^h$, and suppose that a smooth function τ on U satisfies

$$\Gamma(F,h) = \tau F. \tag{4.18}$$

Thus ∇h is tangent to the regular zero set. Introduce

$$H_h = \Gamma(h,h), \qquad Q_h = H_h + 2\tau, \qquad P_h = Q_h\Delta h - \tfrac{1}{2}\Gamma(h,Q_h), \tag{4.19}$$

and the boundary coefficient

$$D_h = S(\Delta h - \tau - H_h) - q - \tfrac{1}{2}\Gamma(h,S). \tag{4.20}$$

These functions are defined on U without division by F.

Lemma 4.11 (Harmonic multiplier identity)**.** *With the preceding notation, $G = aF$ satisfies*

$$|\nabla G|^2 = a^2(S + F^2Q_h), \tag{4.21}$$

$$\mathcal{L}(G) = a^3F\left(D_h + F^2P_h - \tfrac{F}{2}\Gamma(F,Q_h)\right). \tag{4.22}$$

Along $C_{\mathrm{reg}} = \{F = 0, S > 0\}$, one has

$$\frac{D_h}{S} = -\frac{J_C w}{w}, \qquad w = (a\sqrt{S})^{-1}. \tag{4.23}$$

[1] If Σ is the connected support of a stationary integral $(N-2)$-varifold in $\mathbb{S}^{N-1}$, with $\mathcal{H}^{N-3}(\operatorname{Sing}\Sigma) = 0$, regular-part connectedness follows from [93, Lemma 2.2].

If in addition $\Gamma(F,Q_h)=\kappa F$ *for a smooth function* κ*, then* (4.22) *becomes*

$$\mathcal{L}(G)=a^3F(D_h+F^2D_{1,h}),\qquad D_{1,h}=Q_h\Delta h-\tfrac{1}{2}\kappa-\tfrac{1}{2}\Gamma(h,Q_h). \tag{4.24}$$

Proof. The product and chain rules give

$$\nabla G=a(\nabla F+F\nabla h),\qquad \Delta G=aF(\Delta h+Q_h).$$

The first formula gives (4.21). Set $V=S+F^2Q_h$. Differentiating $|\nabla G|^2=a^2V$ yields

$$a^{-3}\Gamma(G,|\nabla G|^2)=2FV(\tau+H_h)+\Gamma(F,V)+F\Gamma(h,V).$$

Here

$$\Gamma(F,V)=2qF+2FSQ_h+F^2\Gamma(F,Q_h),$$
$$\Gamma(h,V)=\Gamma(h,S)+2\tau F^2Q_h+F^2\Gamma(h,Q_h).$$

Substitute these identities in $\mathcal{L}(G)=|\nabla G|^2\Delta G-\frac{1}{2}\Gamma(G,|\nabla G|^2)$, and use $Q_h=H_h+2\tau$. The terms proportional to F and F^3 give (4.22). The additional divisibility assumption gives (4.24).

On the regular zero set, $\mathcal{K}_F(a)=a^3D_h$, so (4.11) gives (4.23). □

Theorem 4.12 (A monotonicity criterion for harmonic multipliers)**.** *In the setting of Lemma 4.11, suppose on* U *that*

$$Q_h\geq 0,\qquad D_h\geq 0,\qquad \Delta h\geq 0,\qquad F\Gamma(F,Q_h)\leq 0,\qquad \Gamma(h,Q_h)\leq 0. \tag{4.25}$$

Then $\nabla(aF)\neq 0$ *on* U*, and* $F\mathcal{L}(aF)\geq 0$ *there. With* $Z=\mathbb{R}^N\setminus U=\{\nabla F=0\}$*, the field*

$$X=\frac{\nabla(aF)}{|\nabla(aF)|}\quad \text{on } U$$

is a two-sided subcalibration of $E=\{F<0\}$ *across* Z*. Consequently* E *is locally perimeter minimizing and* $\partial[\![E]\!]$ *is locally mass minimizing. No regularity of the spherical link is assumed.*

Proof. By (4.21), $|\nabla(aF)|^2\geq a^2S>0$. In the bracket in (4.22), each of D_h, $F^2Q_h\Delta h$, $-F^2\Gamma(h,Q_h)/2$, and $-F\Gamma(F,Q_h)/2$ is nonnegative. Thus $F\mathcal{L}(aF)\geq 0$, and (1.3) gives the required divergence signs. At every regular zero, $\nabla(aF)=a\nabla F$, so X is the exterior unit normal to E.

By Remark 2.7 and Lemma 3.11, E has locally finite perimeter and its algebraic critical set $Z\subset\{F=0\}$ has locally zero $\mathcal{H}^{N-1}$-measure. Apply Theorem 2.5 to X; no extension of a across Z is needed. □

Remark 4.13 (Alternative coefficient tests). If $\Gamma(F,Q_h)=\kappa F$, the monotonicity conditions may be replaced by $Q_h\geq 0$, $D_h\geq 0$, and $D_{1,h}\geq 0$. Without this divisibility assumption, the conditions

$$Q_h\geq 0,\quad D_h\geq 0,\quad P_h\geq 0,\quad \Gamma(F,Q_h)^2\leq 16D_hP_h$$

also suffice: the last inequality gives $\frac{1}{2}|F\Gamma(F,Q_h)|\leq D_h+F^2P_h$. All inequalities must hold throughout U.

For the natural choice $a=\sqrt{S}$, the quantities in the criterion use only q,S and their derivatives. Writing $K=\Delta S$ and $T=\Gamma(S,S)$, one obtains

$$\begin{aligned} h&=\tfrac{1}{2}\log S, & \tau&=q/S, & Q_h&=\frac{T}{4S^2}+\frac{2q}{S},\\ \Delta h&=\frac{K}{2S}-\frac{T}{2S^2}, & & & D_h&=\frac{K}{2}-2q-\frac{T}{S}. \end{aligned} \tag{4.26}$$

Under the spectral identities below, the two monotonicity signs hold, and the remaining condition reduces to a lower bound for ΔS.

Remark 4.14 (The nonharmonic correction). Suppose instead that $\mathcal{L}(F) = -qF$, without assuming $\Delta F = 0$, and retain $U = \{S > 0\}$, (4.18), and the auxiliary functions above. The term $2S\Delta F$ in $\Gamma(F, S)$ changes only the defect:

$$\mathcal{L}(aF) = a^3 F\left(D_h + FQ_h\Delta F + F^2 P_h - \tfrac{F}{2}\Gamma(F, Q_h)\right).$$

The gradient identity is unchanged, so the square-completion test of Remark 4.13 becomes

$$Q_h \geq 0, \qquad D_h \geq 0, \qquad P_h \geq 0, \qquad \big(\Gamma(F, Q_h) - 2Q_h\Delta F\big)^2 \leq 16 D_h P_h.$$

It gives comparison when $Z = \{S = 0\}$ is locally $\mathcal{H}^{N-1}$-null, in particular for square-free homogeneous F by Lemma 3.11. The natural choice $h = \frac{1}{2}\log S$ must still satisfy the compatibility assumption: now $\Gamma(F, h) = \Delta F + qF/S$, which need not be divisible smoothly by F across its regular zero set.

4.4. **Radial multipliers in arbitrary degree.** Let F be a nonzero homogeneous polynomial of degree $e \geq 2$, put $S = |\nabla F|^2$, and suppose $\mathcal{L}(F) = -qF$ for a polynomial q. The identity in Proposition A.2 gives the following radial specialization. For $\sigma > 0$, set

$$A_\sigma = \sigma(N - 1 - 2e - \sigma), \qquad c_\sigma = \sigma(\sigma + 2e),$$
$$B_\sigma = c_\sigma\big(e + \sigma(N - 1)\big). \tag{4.27}$$

Then, on $r > 0$,

$$|\nabla(r^\sigma F)|^2 = r^{2\sigma}\left(S + c_\sigma \frac{F^2}{R}\right), \tag{4.28}$$

$$\mathcal{L}(r^\sigma F) = r^{3\sigma-2} F\,\mathscr{H}_\sigma, \tag{4.29}$$

$$\mathscr{H}_\sigma = A_\sigma S - Rq + c_\sigma F\Delta F + B_\sigma \frac{F^2}{R}. \tag{4.30}$$

The exact sign condition is $\mathscr{H}_\sigma \geq 0$ on $r > 0$. A sufficient test follows.

Proposition 4.15 (A radial criterion without harmonicity)**.** *Let F be a homogeneous polynomial of degree $e \geq 2$ on $\mathbb{R}^N$, $N \geq 3$, taking both signs, and suppose $\mathcal{L}(F) = -qF$ for a polynomial q. Assume that $Z = \{\nabla F = 0\}$ has dimension at most $N - 2$, as holds whenever F is square-free by Lemma 3.11. If constants $\kappa, \eta \geq 0$ and $\sigma > 0$ satisfy*

$$Rq \leq \kappa S, \qquad F\Delta F \geq -\eta S \quad \text{on } \mathbb{R}^N, \tag{4.31}$$

and

$$\sigma\big[N - 1 - 2e(1 + \eta) - (1 + \eta)\sigma\big] \geq \kappa, \tag{4.32}$$

then $G = r^\sigma F$ generates a two-sided subcalibration and $\partial[\![\{F < 0\}]\!]$ is strictly area minimizing. Moreover, on $C_{\mathrm{reg}} = \{F = 0, S > 0\}$,

$$Q_C(u) \geq \left[\frac{(N - 1 - 2e)^2}{4} - \kappa\right]\int_C r^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}}). \tag{4.33}$$

Proof. The assumptions give

$$\mathscr{H}_\sigma \geq (A_\sigma - c_\sigma\eta - \kappa)S + B_\sigma\frac{F^2}{R} \geq 0.$$

Since $\sigma, c_\sigma, B_\sigma > 0$, the bracket is positive off the cone. Euler's identity and (4.28) give $Z \subset \{F = 0\}$ and $\nabla G \neq 0$ off Z. The continuous potential has the signs of F, and the algebraic set Z is locally $\mathcal{H}^{N-1}$-null. Thus Corollary 2.8 and Theorem 2.11 give minimization and its strict form.

The Jacobi trace (4.5) and (4.29) give, on C_{reg},

$$-\frac{J_C w}{w} = \frac{A_\sigma}{R} - \frac{q}{S} \geq \frac{A_\sigma - \kappa}{R}, \qquad w = r^{-\sigma}S^{-1/2}. \tag{4.34}$$

Apply Theorem 4.6 with $\ell = e + \sigma$ and use

$$\left(e + \sigma - \frac{N-1}{2}\right)^2 + A_\sigma = \frac{(N-1-2e)^2}{4}$$

to obtain (4.33). □

Writing $b_\eta = N - 1 - 2e(1+\eta)$, the left side of (4.32) is maximized, when $b_\eta > 0$, at $\sigma = b_\eta/[2(1+\eta)]$. This yields the explicit sufficient dimension condition

$$b_\eta > 0, \qquad N \geq 1 + 2e(1+\eta) + 2\sqrt{(1+\eta)\kappa}. \tag{4.35}$$

Separating the terms in (4.30) can lose sharpness:

Example 4.16 (Nonsharpness of the simplified criterion)**.** Let $F = 7u - v$, where $u = |x|^2$, $v = |y|^2$, and $(x, y) \in \mathbb{R}^2 \times \mathbb{R}^8$. This is a classical Lawson cone; see [52] for quantitative subcalibrations in this family. Here

$$\Delta F = 12, \qquad S = 196u + 4v, \qquad \mathcal{L}(F) = -56F.$$

The least nonnegative constants in (4.31) are $\kappa = 14$ and $\eta = 3$, both attained at $u = 0 < v$. Consequently (4.32) would require $\sigma(-7-4\sigma) \geq 14$, which fails for every $\sigma > 0$. Nevertheless, the radial polynomial potential $G = RF$ satisfies the exact identity

$$\mathcal{L}(G) = 64RF\big[(v - 11u)^2 + 96u^2\big]. \tag{4.36}$$

Since $S > 0$ away from the origin, this gives a two-sided subcalibration with strict signed divergence off the cone. Thus (4.32) need not hold for any positive exponent, even with the least admissible comparison constants.

Finite comparison constants also need not exclude nonzero critical points without harmonicity. For this explicitly reducible comparison, outside the introductory convention, put $u = |x|^2$, $v = |y|^2$, with $(x, y, z) \in \mathbb{R}^4 \times \mathbb{R}^4 \times \mathbb{R}^2$, the square-free but reducible polynomial $F = u^2 - v^2$ satisfies

$$S = 16(u^3 + v^3), \qquad \Delta F = 24(u - v), \qquad \mathcal{L}(F) = 192F(u - v)^2.$$

Hence $\kappa = \eta = 0$ are admissible in (4.31), and $\sigma = 1/2$ satisfies (4.32). Its zero set is the Simons cone in $\mathbb{R}^8$ times $\mathbb{R}^2$, with critical set $\{x = y = 0\}$. The quartic defining equation includes the positive factor $u + v$ away from that set.

If F belongs to a Laplacian algebra $\mathcal{A}$ containing R, then $S, \Delta F \in \mathcal{A}$, and the division property of Proposition 3.6 gives $q \in \mathcal{A}$. Thus $R\mathscr{H}_\sigma \in \mathcal{A}$, and its sign on $R > 0$ can be tested on the real quotient using the closure table.

Corollary 4.17 (A harmonic radial criterion in arbitrary degree)**.** *Under the hypotheses of Proposition 4.5, suppose that for a constant $\kappa \geq 0$,*

$$Rq \leq \kappa S \quad \text{on } \mathbb{R}^N, \qquad N \geq 2e + 1 + 2\sqrt{\kappa}. \tag{4.37}$$

Then, with $\sigma = (N - 1 - 2e)/2$, the potential $G = r^\sigma F$ generates a two-sided subcalibration. The phase $\{F < 0\}$ is locally perimeter minimizing, and its oriented boundary current is locally mass minimizing with support $C = \{F = 0\}$. The cone is strictly area minimizing, and

$$Q_C(u) \geq \left[\frac{(N-1-2e)^2}{4} - \kappa\right] \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C \setminus \{0\}). \tag{4.38}$$

Proof. Write $b = N - 1 - 2e$. By Proposition 4.19 below, $S > 0$ off the origin and $\kappa \geq e(e-1) > 0$. The dimension assumption gives $b > 0$, and $\sigma = b/2$ satisfies $\sigma(b - \sigma) = b^2/4 \geq \kappa$. Harmonicity gives both nonempty phases and $F\Delta F = 0$. Thus Proposition 4.15 applies with $\eta = 0$ and $Z = \{0\}$, proving the subcalibration, strict minimality, and Hardy bound. The support of the boundary current is all of $\{F = 0\}$ by Remark 2.7. □

Any $\sigma \geq 0$ with $\sigma(N-1-2e-\sigma) \geq \kappa$ works; even integers give polynomial potentials, as in the Simons example. A local radial subcalibration already implies strict stability for regular harmonic cones:

Corollary 4.18 (Strict stability from a harmonic radial potential)**.** *Let F be a nonzero real homogeneous harmonic polynomial of degree $e \geq 2$ on $\mathbb{R}^{n+1}$, $n \geq 3$. Suppose $C = \{F = 0\}$ is minimal, $\nabla F \neq 0$ on $C \setminus \{0\}$, and its link is nonempty, smooth, compact, and connected. If for some $\alpha \in \mathbb{R}$ the potential $G = r^\alpha F$ satisfies*

$$\operatorname{sgn}(F) \operatorname{div} X_G \geq 0 \qquad \text{where } F \neq 0$$

in a two-sided neighborhood of $C \setminus \{0\}$ on which X_G is defined, then C is strictly stable.

Proof. The hypotheses already imply that F is real irreducible. Indeed, Lemma 3.11 shows that F is square-free and each real irreducible factor has a nonempty zero set on the unit sphere. These closed sets are pairwise disjoint, since a common zero of two factors would give $\nabla F = 0$. Their union is the connected link, so there can be only one factor. Set $R = r^2$ and $S = |\nabla F|^2$. By Proposition 3.14, $\mathcal{L}(F) = -qF$ for a homogeneous polynomial q of degree $2e-4$, possibly zero. The sign hypothesis and Theorem 4.4 make $w = r^{-\alpha} S^{-1/2}$ a positive Jacobi supersolution. Suppose that the Hardy gap is zero. Equation (4.14) gives $e + \alpha = n/2$ and $J_C w = 0$. With $c = \alpha(\alpha+2e)(e+\alpha n)$, the radial identity of Proposition A.2 and $\Delta F = 0$ give

$$\mathcal{L}(G) = r^{3\alpha-4} F B, \qquad B = R(\alpha^2 S - Rq) + cF^2. \tag{4.39}$$

Thus $B \geq 0$ near the cone, and $J_C w = 0$ together with Theorem 4.4 gives $B|_{C\setminus\{0\}} = 0$. The divisibility argument in Proposition 3.14 yields $B = FB_1$. At a regular zero, F changes sign, so $B \geq 0$ on both sides forces B_1 to vanish there as well. Applying the same argument again gives $F^2 \mid B$. Since F has nonzero real zeros, it is coprime to R. Equation (4.39) therefore implies $F^2 \mid \alpha^2 S - Rq$. Its degree is at most $2e-2 < 2e$, so

$$Rq = \alpha^2 S \qquad \text{identically.} \tag{4.40}$$

Choose an extremum p of F on the unit sphere with $F(p) \neq 0$. Euler's identities give $\nabla F(p) = eF(p)p$ and $\operatorname{Hess} F(p)p = (e-1)\nabla F(p)$. Using harmonicity and $\mathcal{L}(F) = -qF$ gives $q(p) = e(e-1)S(p)$, with $S(p) > 0$. Consequently (4.40) forces $\alpha^2 = e(e-1)$. Together with $\alpha = n/2 - e$, this becomes $n^2 = 4e(n-1)$, which is impossible for integers $n \geq 3$, since it would imply $n - 1 \mid 1$. □

Proposition 4.19 (Global comparison and the critical set)**.** *Let $F \not\equiv 0$ be a homogeneous polynomial of degree $e \geq 2$ on $\mathbb{R}^N$, $N \geq 3$, satisfying $\Gamma(F, S) = 2qF$, where $S = |\nabla F|^2$ and q is a polynomial. Then*

$$\exists\, \kappa < \infty : \; Rq \leq \kappa S \quad \Longleftrightarrow \quad \{\nabla F = 0\} = \{0\}. \tag{4.41}$$

In this case the least comparison constant is

$$\kappa_F = \max_{\mathbb{S}^{N-1}} \frac{q}{S} \geq e(e-1). \tag{4.42}$$

Consequently the hypotheses of Corollary 4.17 require $N \geq 4e$.

Proof. If $S > 0$ off the origin, compactness of the unit sphere and homogeneity give the finite maximum in (4.42). Conversely, suppose $c \neq 0$ and $\nabla F(c) = 0$. Euler's identity gives $F(c) = 0$. Write

$$F(c+y) = P_m(y) + O(|y|^{m+1}), \qquad m \geq 2,$$

where $P_m \not\equiv 0$ is the first nonzero homogeneous Taylor term. Choose a unit vector v maximizing $|P_m|$, and put $p = P_m(v) \neq 0$. Then $\nabla P_m(v) = mpv$ and $\operatorname{Hess} P_m(v)(v, v) = m(m-1)p$. Along $c + tv$,

$$S = m^2 p^2 t^{2m-2}(1 + O(t)),$$

$$q = \frac{\operatorname{Hess} F(\nabla F, \nabla F)}{F} = m^3(m-1)p^2 t^{2m-4}(1+O(t)).$$

It follows that

$$\frac{Rq}{S}(c+tv) = \frac{|c|^2 m(m-1)}{t^2} + O(t^{-1}) \longrightarrow +\infty. \tag{4.43}$$

This rules out a finite comparison constant, including when $q(c) = 0$.

For the lower bound, take a unit vector maximizing $|F|$. The same calculation, now with degree e at the origin, gives $q/S = e(e-1)$ at that vector. Finally, $4e - 1 < 2e + 1 + 2\sqrt{e(e-1)} < 4e$, so the integer dimension in the harmonic radial criterion is at least $4e$. □

Proposition 4.20 (Rigidity at the sharp comparison constant)**.** *Under the algebraic hypotheses of Proposition 4.19, let* $M = \max_{\mathbb{S}^{N-1}} |F|$*. Then*

$$Rq \le e(e-1)S \quad \Longleftrightarrow \quad S = e^2 M^2 R^{e-1}. \tag{4.44}$$

If also $\Delta F = 0$ *and* $N \ge 3$*, the equality case is precisely a harmonic Cartan–Münzner polynomial after normalization by* M*.*

Proof. Put $f = F|_{\mathbb{S}^{N-1}}$ and $s = S|_{\mathbb{S}^{N-1}}$. The restriction of F to any great circle is a real trigonometric polynomial of degree at most e. Applying the classical Bernstein–Szegő inequality [8, Section 5.1, Theorem 5.1.3] to a great circle in the direction of $\nabla_{\mathbb{S}} f$ gives

$$s = |\nabla_{\mathbb{S}} f|^2 + e^2 f^2 \le e^2 M^2. \tag{4.45}$$

On the other hand, homogeneity and $\Gamma(F, S) = 2qF$ give

$$\Gamma_{\mathbb{S}}(f^2, s) = 4f^2\big(q - e(e-1)s\big) \le 0.$$

Thus s is nonincreasing along the gradient ascent flow of f^2. The critical set of f^2 is semialgebraic with finitely many connected components, and f^2 is constant on each component. Choose $a < M^2$ above every other critical value. Every forward trajectory starting in the nonempty open set $\{f^2 > a\}$ has its accumulation points in the critical set at value M^2: the value increases along the trajectory, and every accumulation point of a gradient flow on this compact sphere is critical. At those points $s = e^2 M^2$. Monotonicity of s and (4.45) therefore force $s = e^2 M^2$ throughout $\{f^2 > a\}$. Homogeneity and the polynomial identity principle give the asserted formula for S on all of $\mathbb{R}^N$.

Conversely, differentiating that formula yields $q = e^3(e-1)M^2 R^{e-2}$, proving the comparison with equality. When F is harmonic, F/M satisfies both Cartan–Münzner equations with equal multiplicities. □

Lemma 4.21 (Degree of a nonnegative polynomial first integral)**.** *Let* $F \not\equiv 0$ *be a homogeneous polynomial of degree* $e \ge 1$ *on* $\mathbb{R}^N$*,* $N \ge 2$*, with* $\nabla F \ne 0$ *on* $\{F = 0\} \setminus \{0\}$*. If* $K \not\equiv 0$ *is a homogeneous polynomial of degree* $\ell > 0$*,* $K \ge 0$*, and* $\Gamma(F, K) = 0$*, then* $\ell \ge 2e$*.*

Proof. Choose $p \in \mathbb{S}^{N-1}$ with $K(p) = B = \max_{\mathbb{S}^{N-1}} K > 0$. The constrained maximum and Euler's identity give $\nabla K(p) = \ell B p$, hence

$$0 = \Gamma(F, K)(p) = e\ell B F(p).$$

Thus $F(p) = 0$ and $w = \nabla F(p) \ne 0$, with $w \perp p$. Differentiating Euler's identity gives $\operatorname{Hess} F(p)p = (e-1)w$. Differentiating $\Gamma(F, K) = 0$ at p then gives

$$\operatorname{Hess} K(p)w = -\operatorname{Hess} F(p)\nabla K(p) = -\ell B(e-1)w.$$

For $v = w/|w|$, the function $T(t) = K(p\cos t + v \sin t)$ is a real trigonometric polynomial of degree at most ℓ, with $0 \le T \le B$, and

$$T''(0) = \operatorname{Hess} K(p)(v, v) - \nabla K(p) \cdot p = -e\ell B.$$

The trigonometric Bernstein inequality, applied twice to $T - B/2$ [8, Section 5.1, Theorem 5.1.4], yields

$$e\ell B \le \|T''\|_\infty \le \ell^2 \|T - B/2\|_\infty \le \frac{\ell^2 B}{2}.$$

Consequently $\ell \ge 2e$. □

Proposition 4.22 (Rigidity for a radial minimality weight)**.** *Let $F \not\equiv 0$ be a homogeneous polynomial of degree $e \ge 2$ on $\mathbb{R}^N$, $N \ge 3$, with $\mathrm{Crit}(F) = \{0\}$. Put $S = |\nabla F|^2$ and $M_F = \max_{\mathbb{S}^{N-1}} |F|$. If*

$$\Gamma(F, S) = 2\lambda R^{e-2} F$$

for a constant λ, then

$$\lambda = e^3(e-1)M_F^2, \qquad S = e^2 M_F^2 R^{e-1} = \frac{\lambda}{e(e-1)} R^{e-1}.$$

If also $\Delta F = 0$, then F/M_F is a harmonic Cartan–Münzner polynomial. In particular,

$$\mathcal{A}_R(F) = \mathbb{R}[R, F], \qquad e \in \{2, 3, 4, 6\}, \qquad \kappa_F = e(e-1).$$

Proof. After replacing F by $-F$ if necessary, choose $p \in \mathbb{S}^{N-1}$ with $F(p) = M_F$. Then $\nabla F(p) = eM_F p$ and $S(p) = e^2 M_F^2$. Since S has degree $2e-2$,

$$2\lambda M_F = \Gamma(F, S)(p) = eM_F(2e-2)S(p) = 2e^3(e-1)M_F^3.$$

This determines λ. The spherical Bernstein–Szegő bound (4.45), extended by homogeneity, shows that $K := e^2 M_F^2 R^{e-1} - S \ge 0$. Using $\Gamma(F, R) = 2eF$, we obtain

$$\Gamma(F, K) = 2\big[e^3(e-1)M_F^2 - \lambda\big] R^{e-2} F = 0.$$

If $K \not\equiv 0$, its degree $2e-2$ contradicts Lemma 4.21. Hence $K = 0$.

If $\Delta F = 0$, the normalized equations $|\nabla(F/M_F)|^2 = e^2 R^{e-1}$ and $\Delta(F/M_F) = 0$ are the harmonic Cartan–Münzner equations, as in Proposition 4.20. Together with $\Gamma(R, F) = 2eF$ and $\Delta R = 2N$, they show that $\mathcal{A}_R(F) = \mathbb{R}[R, F]$. The nonempty zero link is regular by hypothesis and minimal since $\mathcal{L}(F) = -\lambda R^{e-2} F$; hence Theorem 6.1 gives the degree restriction. Finally, $Rq/S = e(e-1)$ off the origin. □

Example 4.23 (Harmonic isoparametric polynomials)**.** For a normalized harmonic Cartan–Münzner polynomial of degree $e \ge 2$ on $\mathbb{R}^N$, $N \ge 3$,

$$S = e^2 R^{e-1}, \qquad q = e^3(e-1)R^{e-2}, \qquad \kappa_F = e(e-1).$$

The dimension condition of Corollary 4.17 is therefore equivalent to $N \ge 4e$. In this range the choice $\sigma = 2\lfloor e/2 \rfloor \in \{e-1, e\}$ satisfies $\sigma(N - 1 - 2e - \sigma) \ge e(e-1)$, so

$$G = R^{\lfloor e/2 \rfloor} F$$

is a polynomial subcalibration potential. The realizable equal-multiplicity parameters satisfying this threshold are listed in Section 6.3.3, together with the Clifford-algebra explanation for the quartic restriction. By Proposition 4.22, they exhaust Corollary 4.17 when q is radial: harmonic quadrics in even dimensions $N \ge 8$, and Cartan cubics in dimensions 14 and 26.

Proposition 4.24 (Positive multipliers at a harmonic critical zero)**.** *Let $F \not\equiv 0$ be a harmonic polynomial, and suppose $F(c) = 0 = \nabla F(c)$. If a is C^2 near c and $a(c) > 0$, then $G = aF$ cannot satisfy $G\mathcal{L}(G) \ge 0$ throughout any neighborhood of c.*

Proof. Use the first nonzero Taylor term P_m, the vector v, and $p = P_m(v) \ne 0$ as in the proof of Proposition 4.19. Harmonicity gives $\Delta(aF) = 2\Gamma(a, F) + F\Delta a = O(t^{m-1})$ along $c + tv$. The leading Hessian term therefore yields

$$F(c+tv)\mathcal{L}(aF)(c+tv) = -a(c)^3 m^3 (m-1) p^4 t^{4m-4} + o(t^{4m-4}) < 0$$

for all sufficiently small positive t. Here $a > 0$ and $\nabla(aF) \ne 0$, so the required sign fails. □

Remark 4.25 (Quartic and quintic scope). The criterion of Corollary 4.17 applies in every degree but requires a regular link. In degrees four and five its conditions read, respectively, $N \geq 16$, $\kappa_F \leq (N-9)^2/4$, and $N \geq 20$, $\kappa_F \leq (N-11)^2/4$. Degree five is absent from the Cartan–Münzner list, so Proposition 4.20 forces $\kappa_F > 20$ in that case. Moreover, Proposition 4.22 forces q to be nonradial in any degree-five example satisfying the criterion. The determinant cones in Theorem I instead have nonzero critical points and use nonradial minor-norm multipliers. Likewise, adding cylindrical directions cannot produce new examples of the radial criterion, by Proposition 4.24.

4.5. **Homogeneous potentials on a spherical quotient.** Let $I = (I_1, \dots, I_j)$, where each invariant I_i is homogeneous of degree d_i, and write

$$z_i = \frac{I_i}{r^{d_i}}, \qquad K_{\mathbb{S}} = z(\mathbb{S}^{N-1}).$$

For an ambient closure containing R, Euler's identity $\nabla_{\mathbb{S}} I_i = \nabla I_i - d_i I_i \theta$ and the polar Laplacian give the spherical closure data

$$g^{\mathbb{S}}_{ij} = \Gamma(I_i, I_j)|_{R=1} - d_i d_j z_i z_j, \qquad b^{\mathbb{S}}_i = \Delta I_i|_{R=1} - d_i(d_i + N - 2) z_i. \tag{4.46}$$

More generally, assume the spherical data close as functions of z. Let $\widehat{\Gamma}_{\mathbb{S}}$, $\widehat{\Delta}_{\mathbb{S}}$ be their operators from Theorem 3.10, with signs tested on the actual image $K_{\mathbb{S}}$.

Proposition 4.26 (The spherical equation for a homogeneous potential)**.** *Let φ be smooth on a neighborhood of the relevant part of $K_{\mathbb{S}}$ in the ambient invariant-coordinate space. For $G = r^\ell \varphi(z)$, define*

$$W_\ell[\varphi] = \ell^2 \varphi^2 + \widehat{\Gamma}_{\mathbb{S}}(\varphi, \varphi), \tag{4.47}$$

$$\mathfrak{N}_\ell[\varphi] = W_\ell[\varphi]\big(\widehat{\Delta}_{\mathbb{S}}\varphi + (N-1)\ell\varphi\big) - \tfrac{1}{2}\widehat{\Gamma}_{\mathbb{S}}(\varphi, W_\ell[\varphi]). \tag{4.48}$$

Then

$$|\nabla G|^2 = r^{2\ell-2} W_\ell[\varphi], \qquad \mathcal{L}(G) = r^{3\ell-4}\mathfrak{N}_\ell[\varphi]. \tag{4.49}$$

Where $W_\ell[\varphi] > 0$, one has $r \operatorname{div} X_G = \mathfrak{N}_\ell[\varphi]/W_\ell[\varphi]^{3/2}$. These identities hold at singular quotient values whenever their pullbacks are smooth and $W_\ell[\varphi] > 0$.

Proof. Polar differentiation gives

$$\nabla G = r^{\ell-1}(\ell\varphi\, \partial_r + \nabla_{\mathbb{S}}\varphi), \qquad \Delta G = r^{\ell-2}[\widehat{\Delta}_{\mathbb{S}}\varphi + \ell(\ell + N - 2)\varphi].$$

The radial and angular terms give

$$\tfrac{1}{2}\Gamma(G, |\nabla G|^2) = r^{3\ell-4}\left[\ell(\ell-1)\varphi W_\ell[\varphi] + \tfrac{1}{2}\widehat{\Gamma}_{\mathbb{S}}(\varphi, W_\ell[\varphi])\right].$$

Subtracting this expression from $|\nabla G|^2 \Delta G$ proves the formula. □

For a nonzero homogeneous polynomial F of degree $e \geq 1$, put $R = r^2$ and $S = |\nabla F|^2$. When these quantities give a closed quotient as above, the angular variables are

$$f = \frac{F}{R^{e/2}}, \qquad s = \frac{S}{R^{e-1}}, \qquad K_{\mathbb{S}} = \{(f(x), s(x)) : r = 1\}. \tag{4.50}$$

If this image has dimension two, a homogeneous invariant potential with a positive multiplier has the form

$$G = r^\ell f\, a(f, s), \qquad a > 0.$$

The power family (A.1) corresponds to $\varphi(f, s) = f s^\gamma$ and $\ell = e + 2\alpha + 2\gamma(e-1)$. It is a restricted family of angular multipliers.

Remark 4.27 (Phase-dependent quotient potentials). Let $N \geq 3$. Suppose $E = \{F < 0\}$ has smooth boundary off a closed conical set Z containing the origin and satisfying the null-set hypothesis of Theorem 2.9. Use the quotient (4.50). Choose degrees ℓ_-, ℓ_+ and put $h_\pm = f a_\pm(f, s)$, where the positive multipliers $a_\pm$ are smooth on invariant-coordinate neighborhoods of the respective closed-phase images outside Z. Assume there that $W_{\ell_\pm}[h_\pm] > 0$, and assume

$$\mathfrak{N}_{\ell_-}[h_-] \leq 0 \quad (f < 0), \qquad \mathfrak{N}_{\ell_+}[h_+] \geq 0 \quad (f > 0).$$

Then $G_\pm = r^{\ell_\pm} h_\pm$ give (2.17) with weights on the respective phases

$$q_\pm = \frac{1}{r} \frac{|\mathfrak{N}_{\ell_\pm}[h_\pm]|}{W_{\ell_\pm}[h_\pm]^{3/2}},$$

extended by zero outside their phases. The boundary is area minimizing, and strictly area minimizing if both unit-ball phase integrals are positive. Indeed, on the regular boundary, $\nabla G_\pm = r^{\ell_\pm - e} a_\pm \nabla F$, so both normalized fields equal ν_E. The spherical equation gives their phase signs and weights. Apply Remark 2.10, Theorem 2.9, and Corollary 2.12. The two multipliers need not agree on $f = 0$.

A useful initial family is $G = R^\alpha S^\gamma F$, with $S = |\nabla F|^2$. Its full multiplier identities are collected in Appendix A, especially Proposition A.3.

5. Harmonic Pullbacks and Subcalibration Transfer

Riedler proves regular-locus minimality under umbilic harmonic morphisms and area minimization for Clifford–Simons quartics [69, Theorems 5.1.4 and 5.2.5], and asks about general minimal cones [69, Question 5.3.3(2), p. 88]. Theorems 5.4 and 5.5 control the dilation term with a source-radius factor and determine the smallest algebra, yielding Corollary 5.7. Quadratic and cubic applications appear in Sections 8 and 11.

5.1. **The pullback identity and the dilation term.** For a degree-d homogeneous polynomial harmonic morphism $\phi : \mathbb{R}^N \to \mathbb{R}^m$, $d \geq 1$, the Fuglede–Ishihara characterization, as recalled in [69, Theorem 2.1.2], gives, on its regular set, a dilation $\lambda > 0$ with

$$\Delta \phi_i = 0, \qquad \Gamma(\phi_i, \phi_j) = \lambda^2 \delta_{ij}.$$

In particular, $\|d\phi\|^2_{\mathrm{HS}} = m\lambda^2$. Subscripts x and y distinguish source and target operators.

Proposition 5.1 (Pullback criterion)**.** *Suppose that, on the regular set,*

$$d\phi\big(\nabla_x \lambda^2\big) = \eta(x)\phi(x). \tag{5.1}$$

Equivalently, $d\phi(\nabla_x \|d\phi\|^2_{\mathrm{HS}}) = m\eta\phi$; this records the factor m in the differential-norm formulation of [69, Definition 5.1.1(1)]. *Let H be a nonzero homogeneous polynomial of degree $\ell > 0$ on $\mathbb{R}^m$, put $G = H \circ \phi$, and write $W_H = |\nabla_y H|^2$. Then*

$$|\nabla_x G|^2 = \lambda^2 (W_H \circ \phi), \qquad \Delta_x G = \lambda^2 (\Delta_y H) \circ \phi, \tag{5.2}$$

$$\mathcal{L}_x(G) = \lambda^4 \mathcal{L}_y(H) \circ \phi - \frac{\ell\eta}{2} (H W_H) \circ \phi. \tag{5.3}$$

Suppose that a constant $\kappa \in \mathbb{R}$ satisfies

$$\frac{|\phi(x)|^2 \eta(x)}{2\lambda(x)^4} \leq \kappa \tag{5.4}$$

at regular points with $\phi(x) \neq 0$, and that

$$|y|^2 H(y) \mathcal{L}_y(H)(y) \geq \kappa \ell H(y)^2 W_H(y) \qquad (y \neq 0). \tag{5.5}$$

Then $G\mathcal{L}_x(G) \geq 0$ at regular points with $\phi(x) \neq 0$.

If $G \not\equiv 0$ and a closed algebraic set $Z \subset \mathbb{R}^N$ of dimension at most $N - 2$ contains the critical set of G, the singular zeros of G, and the points excluded from this calculation, then $\{G < 0\}$ is locally perimeter minimizing and $\partial[\![\{G < 0\}]\!]$ is locally mass minimizing.

Proof. The chain rule gives (5.2). Moreover,

$$\tfrac{1}{2}\Gamma_x(G,|\nabla_x G|^2) = \frac{W_H\circ\phi}{2}\left\langle d\phi(\nabla_x\lambda^2),(\nabla_y H)\circ\phi\right\rangle + \lambda^4\big(\mathrm{Hess}_y\, H(\nabla_y H,\nabla_y H)\big)\circ\phi.$$

Euler's identity $\langle y,\nabla_y H\rangle = \ell H$ and (5.1) prove (5.3). At a regular point with $y=\phi(x)\neq 0$, the two assumed inequalities give

$$G\mathcal{L}_x(G) \geq \ell H(y)^2 W_H(y)\left(\frac{\kappa\lambda^4}{|y|^2}-\frac{\eta}{2}\right)\geq 0.$$

The global conclusion follows from Remark 2.7 and Corollary 2.8, since the algebraic dimension bound makes Z locally $\mathcal{H}^{N-1}$-null. □

Both terms in (5.3) vanish on a regular minimal level cone $H=0$; away from the cone, the dilation term can change the sign. For example, if H is linear and $\eta>0$, then $\mathcal{L}_y(H)=0$ but $G\mathcal{L}_x(G) = -\eta H^2 W_H/2 < 0$ wherever $H\neq 0$.

Remark 5.2 (Weighted interpretation). Condition (5.5) gives the phase inequality for the normalized gradient with density $|y|^{-\kappa}$. Indeed, for $X_H = \nabla H/|\nabla H|$,

$$|y|^{\kappa}\,\mathrm{div}_y\big(|y|^{-\kappa}X_H\big) = \frac{\mathcal{L}_y(H)}{W_H^{3/2}} - \frac{\kappa\ell H}{|y|^2 W_H^{1/2}}.$$

It is the density of the Euler–Dierkes–Huisken functional $\mathcal{E}_\alpha(\Sigma)=\int_\Sigma |y|^\alpha\, d\mathcal{H}^{m-1}$ with $\alpha=-\kappa$. Minimal cones are stationary on their regular parts for area functionals with smooth positive radial densities. Subcalibration methods also apply to proving minimality for these functionals; see, for example, Dierkes [26]. For the Clifford maps below, $\alpha=-1/2$; the dilation bound converts the weighted inequality into ordinary-area comparison.

For the subclass $\lambda^2 = c|x|^{2d-2}$, with $c>0$, Euler's identity gives

$$\eta = 2d(d-1)c|x|^{2d-4}, \qquad d^2|\phi(x)|^2 = |d\phi_x(x)|^2 \leq \lambda^2|x|^2.$$

Hence (5.4) holds with $\kappa=(d-1)/d$.

5.2. **Area-minimizing Clifford pullbacks.** Let $A_1,\dots,A_m$ be a symmetric Clifford system on $\mathbb{R}^N$, where $m\geq 2$:

$$A_iA_j + A_jA_i = 2\delta_{ij}\,\mathrm{Id}\,.$$

Set $P_i(x)=\langle A_ix,x\rangle$, $P=(P_1,\dots,P_m)$, $R=|x|^2$, and $Q=|P|^2$. The Clifford relations imply

$$\Delta P_i = 0, \qquad \Gamma(P_i,P_j)=4R\delta_{ij}, \qquad \Gamma(R,P_i)=4P_i. \tag{5.6}$$

For example, a second Clifford matrix interchanges the two eigenspaces of A_i, so $\mathrm{tr}\,A_i=0$. The classification of quadratic harmonic morphisms by Ou and Wood, including its Clifford-system and direct-sum description, is recalled in [69, Theorem 4.5.3 and Remark 4.5.5]. The maps above are normalized umbilic maps: all positive eigenvalues of their component matrices equal one; see [69, Remark 5.1.12].

Lemma 5.3 (Laplacian algebra under Clifford pullback)**.** *If $\mathcal{A}\subset\mathbb{R}[y_1,\dots,y_m]$ is a graded Laplacian algebra, then $\mathbb{R}[R,P^*\mathcal{A}]$ is a Laplacian algebra on $\mathbb{R}^N$. Moreover, P is onto, $Q\leq R^2$, and $P^{-1}(0)\setminus\{0\}$, when nonempty, is a smooth submanifold of codimension m. For this map, Proposition 5.1 applies with $\lambda^2=4R$, $\eta=16$, and $\kappa=1/2$.*

Proof. For target polynomials u,v, the chain rule gives

$$\Delta_x(u\circ P) = 4R(\Delta_y u)\circ P, \qquad \Gamma_x(u\circ P, v\circ P) = 4R\Gamma_y(u,v)\circ P.$$

If u has degree j, then $\Gamma_x(R,u\circ P)=4j(u\circ P)$. Together with $\Delta R = 2N$ and $\Gamma(R,R)=4R$, these identities and the product rule prove the asserted closure.

For $x\neq 0$, the vectors A_ix are orthogonal and have length $|x|$. Projection of x onto their span gives $Q/R\leq R$. To prove surjectivity, take $u\in\mathbb{S}^{m-1}$ and put $A(u)=\sum u_iA_i$. This

symmetric involution has trace zero, since every A_i does. It therefore has a unit eigenvector x with eigenvalue one. For $v \perp u$, the matrices $A(v)$ and $A(u)$ anticommute, so $\langle A(v)x, x\rangle = 0$. Hence $P(x) = u$, and scaling proves that P is onto. Finally, (5.6) shows that dP has rank m at every nonzero point. The submersion theorem proves the statement about $P^{-1}(0)$. Since $dP(\nabla(4R)) = 16P$, the last assertion follows from $Q/(2R^2) \le 1/2$. □

In particular, if a target homogeneous polynomial f has degree at least two and $\nabla f(y) \neq 0$ for $y \neq 0$, then

$$|\nabla(f \circ P)|^2 = 4R(|\nabla f|^2 \circ P).$$

Thus its critical set is $Z = P^{-1}(0)$, and its zero set is regular off Z. The set Z is algebraic and has dimension at most $N - m$, apart from the possible isolated origin. Consequently it has zero $\mathcal{H}^{N-1}$-measure. Surjectivity preserves nonemptiness of both phases. These observations apply to the quadratic and Cartan targets below.

For a nonflat target hypersurface cone with both phases nonempty and the isolated critical point just assumed, this is also the geometric singular set:

$$\operatorname{Sing}\{f \circ P = 0\} = P^{-1}(0).$$

At every nonzero point of $P^{-1}(0)$, submersion coordinates identify the zero set with a product of the target cone and a smooth factor. The vertex is singular as well: a smooth cone would be a hyperplane, which cannot be the zero set of an even homogeneous polynomial taking both signs.

A source-radius factor compensates for the dilation term in (5.3) in sufficiently large admissible dimensions.

Theorem 5.4 (Area minimization under Clifford pullback)**.** *Let f be a nonzero homogeneous polynomial on $\mathbb{R}^m$, and let $Z \subset \{f = 0\}$ be a closed conical real algebraic set containing the origin and having codimension at least two. Suppose that $H \in C^3(\mathbb{R}^m \setminus Z) \cap C^0(\mathbb{R}^m)$ is homogeneous of degree $\ell > 0$, with*

$$\operatorname{sgn} H = \operatorname{sgn} f, \qquad |\nabla_y H| > 0 \quad \text{on } \mathbb{R}^m \setminus Z.$$

Put $W_H = |\nabla_y H|^2$, and suppose that, off Z,

$$H\mathcal{L}_y(H) \ge 0, \qquad H\Delta_y H \ge -MW_H$$

for some finite $M \ge 0$. Let $P : \mathbb{R}^N \to \mathbb{R}^m$ be the Clifford map above. If $a > 0$ and

$$N \ge 1 + 4\ell(1 + M) + 2a(1 + M) + \frac{\ell}{a}, \tag{5.7}$$

then

$$G = R^a H(P), \qquad E = \{f(P) < 0\},$$

defines a two-sided subcalibration: off $P^{-1}(Z)$, $\nabla_x G \neq 0$ and $G\mathcal{L}_x(G) \ge 0$, with strict inequality in the two open phases. Consequently E is locally perimeter minimizing and $\partial[\![E]\!]$ is locally mass minimizing. If $N \ge 3$ and both phases are nonempty, this current is strictly area minimizing. If both signs of f occur in every neighborhood of each target zero, then

$$\operatorname{spt} \partial[\![E]\!] = \{f(P) = 0\}.$$

In particular, this last condition holds when f is harmonic.

Proof. Set $h = H(P)$ and $b = a(a + 2\ell)$. The chain rule and Euler's identity give, on $\mathbb{R}^N \setminus P^{-1}(Z)$,

$$|\nabla_x G|^2 = 4R^{2a+1} W_H(P) + 4bR^{2a-1}h^2, \tag{5.8}$$

$$\begin{aligned}\mathcal{L}_x(G) &= 16R^{3a+2}(\mathcal{L}_y H)(P) + 8\big[a(N-1) - \ell - 2b\big]R^{3a}hW_H(P) \\ &\quad + 16bR^{3a}h^2(\Delta_y H)(P) + 8b\big[a(N-1) + \ell\big]R^{3a-2}h^3.\end{aligned} \tag{5.9}$$

For the second formula, use the radial identity (A.6) with the source function h, of degree 2ℓ, and substitute

$$|\nabla_x h|^2 = 4RW_H(P), \qquad \Delta_x h = 4R(\Delta_y H)(P),$$
$$\mathcal{L}_x(h) = 16R^2(\mathcal{L}_y H)(P) - 8\ell h W_H(P).$$

These identities follow from (5.6).

Multiplication of (5.9) by h yields

$$h\mathcal{L}_x(G) \geq 8\big[a(N-1) - \ell - 2b(1+M)\big]R^{3a}h^2 W_H(P) + 8b\big[a(N-1)+\ell\big]R^{3a-2}h^4.$$

The first coefficient is nonnegative by (5.7), and the last term is positive when $h \neq 0$. Since $R^a > 0$, this proves the claimed phase inequality. Formula (5.8) gives $\nabla_x G \neq 0$ off $Z' = P^{-1}(Z)$.

The extension of G is continuous with the signs of $f(P)$; at the origin, use $H(P(x)) = O(|x|^{2\ell})$. Surjectivity gives $f(P) \not\equiv 0$, hence local finite perimeter. Since P is a submersion off the origin, stratifying Z shows that $Z' = P^{-1}(Z)$ has codimension at least two. Thus Corollary 2.8 applies. For $N \geq 3$ and two nonempty phases, the strict divergence signs give strict minimality by Theorem 2.11.

The additional phase assumption transfers by submersion away from the origin and by surjectivity and homogeneity at the origin. The support conclusion follows from Remark 2.7. □

For harmonic nonconstant f, (5.6) and surjectivity make $F = f \circ P$ nonzero and harmonic. Area minimization then implies real irreducibility by Proposition 3.12. Nonharmonic targets still require the stated phase hypothesis for full support.

The ratio bound is preserved under iteration:

$$\Delta_x G = 4R^{a+1}(\Delta_y H)(P) + 2a(N+2a+4\ell-2)R^{a-1}h,$$

and (5.8) gives

$$G\Delta_x G + M|\nabla_x G|^2 = 4R^{2a+1}\big(H\Delta_y H + MW_H\big)(P) + \big[2a(N+2a+4\ell-2) + 4Mb\big]R^{2a-1}h^2 \geq 0. \tag{5.10}$$

Thus the same M is available for iteration; see Corollary 5.7. In particular, $a = 1$ preserves polynomial potentials and gives the sufficient bound $N \geq 5\ell + 3 + (4\ell+2)M$.

The harmonic radial criterion supplies further target potentials. Let f be homogeneous of degree e on $\mathbb{R}^m$ and satisfy the hypotheses of Corollary 4.17. Set $\sigma = (m-1-2e)/2 > 0$ and $H = |y|^\sigma f$. Harmonicity gives

$$H\Delta_y H = \sigma(\sigma + 2e + m - 2)|y|^{2\sigma-2}f^2 \geq 0 \qquad (y \neq 0).$$

Together with the corollary's phase sign and nonvanishing gradient, this verifies all target hypotheses with $Z = \{0\}$, $\ell = e + \sigma$, and $M = 0$. Thus Theorem 5.4 applies whenever the source-dimension bound (5.7) holds.

5.3. **The smallest algebra and iteration.** Write $q(y) = |y|^2$ on the target.

Theorem 5.5 (The smallest algebra under Clifford pullback)**.** *Let $P : \mathbb{R}^N \to \mathbb{R}^m$ be the Clifford map above and let f be a nonzero essential homogeneous polynomial of degree $s \geq 2$ on $\mathbb{R}^m$, so $V_{\mathrm{cyl}}(f) = \{0\}$. Then*

$$\mathcal{A}_R(f \circ P) = \mathbb{R}[R, P^*\mathcal{A}_q(f)]. \tag{5.11}$$

If $Q < R^2$ somewhere, this is isomorphic to the polynomial extension $\mathcal{A}_q(f)[t]$, with target degrees doubled and $\deg t = 2$. In particular, its minimal number of homogeneous generators is one more than that of $\mathcal{A}_q(f)$.

Proof. Set $\mathcal{A} = \mathcal{A}_q(f)$ and $\mathcal{B} = \mathcal{A}_R(f \circ P)$. By Lemma 5.3, $\mathbb{R}[R, P^*\mathcal{A}]$ is Laplacian and contains $f \circ P$, so $\mathcal{B} \subseteq \mathbb{R}[R, P^*\mathcal{A}]$. For the reverse inclusion put $\mathcal{C} = \{u \in \mathbb{R}[y_1, \dots, y_m] : u \circ P \in \mathcal{B}\}$. This is a unital algebra containing f. If $u \in \mathcal{C}$, the chain rule gives $4R(\Delta u) \circ P = \Delta(u \circ P) \in \mathcal{B}$. The quotient is a polynomial, so Proposition 3.6 gives $(\Delta u) \circ P \in \mathcal{B}$. Thus $\mathcal{C}$ is closed under Δ and, by the product rule, Γ.

It remains to recover q. The recursion (3.2) uses only multiplication and Δ, so $B_{s-1}(f, f) \in \mathcal{C}$. This quadratic form is $\langle My, y\rangle$ for a positive semidefinite M whose kernel is exactly the cylindrical directions of f: by polarization, $B_{s-1}(f, f)(v) = 0$ if and only if $\partial_v f \equiv 0$. Essentiality of f therefore makes M positive definite. Since

$$\Gamma(\langle My, y\rangle, \langle M^j y, y\rangle) = 4\langle M^{j+1} y, y\rangle,$$

all positive powers of M give quadratic forms in $\mathcal{C}$. Choose a polynomial p with $p(0) = 0$ and $p(\lambda) = 1$ at every eigenvalue of M. Then $p(M) = \mathrm{Id}$ gives $q \in \mathcal{C}$. Consequently $\mathcal{A} \subseteq \mathcal{C}$, proving (5.11).

At a point where $Q < R^2$, the Gram matrix of $\nabla R, \nabla P_1, \dots, \nabla P_m$ has determinant $4^{m+1} R^{m-1}(R^2 - Q) > 0$. Thus (R, P) has an open set in its image, and R is algebraically independent of the pulled-back polynomial ring. This proves the asserted isomorphism. For a finitely generated connected graded algebra, the minimal homogeneous generator number is $\dim(\mathcal{A}_+/\mathcal{A}_+^2)$, where $\mathcal{A}_+$ is its positive-degree ideal. Adjoining t adds exactly one independent class. □

When $Q = R^2$ identically, the relation $q(P) = R^2$ must be retained; the conclusion about one additional generator does not apply.

Choose minimal homogeneous generators $\mathcal{A}_q(f) = \mathbb{R}[u_1, \dots, u_r]$, with $\deg u_i = d_i$, and write

$$\Delta u_i = b_i(u), \qquad \Gamma(u_i, u_j) = c_{ij}(u), \qquad q = q_*(u).$$

The lifted variables $U_i = u_i(P)$ have the differential table

$$\begin{aligned} \Delta R &= 2N, \qquad \Gamma(R, R) = 4R, \qquad \Delta U_i = 4R\, b_i(U), \\ \Gamma(R, U_i) &= 4d_i U_i, \qquad \Gamma(U_i, U_j) = 4R\, c_{ij}(U). \end{aligned} \tag{5.12}$$

Thus the target table determines the calculation for any potential $G = R^a \phi(U)$; only ΔR depends on the source dimension. For a target potential factoring through $\mathcal{A}_q(f)$ as $H = \phi(u)$, this applies when ϕ is C^3 on an open neighborhood of $u(\mathbb{R}^m \setminus Z)$.

Proposition 5.6 (Target equivalence and dimension monotonicity)**.** *Fix a Clifford map $P_0 : \mathbb{R}^{N_0} \to \mathbb{R}^m$ and, for $k \geq 2$, set*

$$P_k(x_1, \dots, x_k) = \sum_{\alpha=1}^{k} P_0(x_\alpha), \qquad N_k = kN_0, \qquad R = \sum_{\alpha=1}^{k} |x_\alpha|^2.$$

Assume the target hypotheses on f, Z, H, M of Theorem 5.4, except for $H\mathcal{L}(H) \geq 0$. For fixed $a > 0$, put $G_k = R^a H(P_k)$. Then

$$H\mathcal{L}(H) \geq 0 \text{ off } Z \quad \Longleftrightarrow \quad G_k \mathcal{L}(G_k) \geq 0 \text{ off } P_k^{-1}(Z) \text{ for some } k \geq 2.$$

Once the source inequality holds for one such k, it holds for every larger k. In particular, the target inequality is equivalent to the source inequalities in all sufficiently large members of this family.

Proof. The image of (R, P_k) is exactly $\{(z, y) \in \mathbb{R} \times \mathbb{R}^m : z \geq |y|\}$. Indeed, $|P_k| \leq R$. Conversely, choose a unit vector e with $y = |y|e$ (arbitrary if $y = 0$), and write $y = y_1 + y_2$, where $y_1 = (z + |y|)e/2$ and $y_2 = -(z - |y|)e/2$. The eigenvector construction in Lemma 5.3 realizes $P_0(x_i) = y_i$ with $|x_i|^2 = |y_i|$. Two blocks therefore suffice.

For $y \notin Z$ and $z \geq |y|$, put $b = a(a + 2\ell)$ and

$$\Psi(z, y) = z^a H(y), \qquad \mathcal{W}(z, y) = 4z^{2a+1} W_H(y) + 4b z^{2a-1} H(y)^2.$$

For $N = N_k$, let $\mathcal{D}_N(z,y)$ be the value of $G_k\mathcal{L}(G_k)$ at any point with $(R,P_k) = (z,y)$. Formula (5.9) shows that this value is well-defined, and (5.8) gives $|\nabla G_k|^2 = \mathcal{W}(z,y) > 0$ there. For two block-sum dimensions $N' = N_{k'} \geq N = N_k$, the same formula gives the finite difference

$$\mathcal{D}_{N'}(z,y) - \mathcal{D}_N(z,y) = \frac{2a(N'-N)}{z}\Psi(z,y)^2\mathcal{W}(z,y) \geq 0. \tag{5.13}$$

The common image therefore proves monotonicity of the source inequalities.

The target inequality implies the source inequality for large k by Theorem 5.4. Conversely, fix $y \notin Z$. The image calculation permits $(R,P_k) = (z,y)$ for every $z \geq |y|$. By (5.9),

$$\lim_{z\to\infty} \frac{\mathcal{D}_{N_k}(z,y)}{16z^{4a+2}} = H(y)\mathcal{L}(H)(y).$$

If the source inequality holds, each expression on the left is nonnegative. This proves necessity. □

Composition preserves umbilic maps [69, Proposition 5.1.8(1)]. Induction from Theorems 5.4 and 5.5 gives the following conclusions.

Corollary 5.7 (Iteration)**.** *Let f be a nonzero essential homogeneous polynomial of degree $s \geq 2$ on the target $\mathbb{R}^m$, and suppose f, Z, H, M satisfy Theorem 5.4, with $H = \phi(u)$ as above. Successive Clifford block sums with at least two blocks and sufficiently large source dimensions yield minimizing phase boundaries. After j steps the defining degree is $2^j s$ and the smallest algebra is isomorphic to $\mathcal{A}_q(f)[t_1,\dots,t_j]$, with minimal generator degrees*

$$2^j d_1, \dots, 2^j d_r, \qquad 2, 4, \dots, 2^j.$$

Thus its minimal homogeneous generator number is exactly $r + j$. The same ratio constant M can be used at every step. If both signs of f occur near every zero, the minimizing supports are the complete successive zero sets.

Proof. The ratio estimate (5.10) preserves M. The hypotheses on regularity, signs, the exceptional set, and the nonvanishing of the gradient persist by the proof of Theorem 5.4; the new potential remains a function of the new generators. If $\partial_v(g \circ P) \equiv 0$ for a nonzero homogeneous g of degree $d > 0$, then $v = 0$: otherwise $(g \circ P)(v + x) = (g \circ P)(x)$ gives vanishing order $2d$ at v, whereas the submersion property of P makes this equal to the vanishing order of g at $P(v)$, which is at most d. The common-image calculation above gives $Q < R^2$ somewhere at each step, so the algebra and degree assertions follow inductively from Theorem 5.5. At each stage a symmetric Clifford system exists on $\Lambda^*\mathbb{R}^m$, with generators given by exterior multiplication plus contraction; direct sums make its source dimension arbitrarily large. Thus each required dimension bound can be met. The support assertion follows at every stage from Theorem 5.4. □

6. Two-Generator Algebras and Isoparametric Hypercones

When the radius and one defining polynomial generate a Laplacian algebra, isoparametric geometry identifies the minimal level and reduces the subcalibration inequality to one angular variable.

6.1. **Geometric background: parallel hypersurfaces and cones.** A connected complete embedded hypersurface of the unit sphere is *isoparametric* if its principal curvatures are constant. Münzner's theorem [62, 63] gives $g \in \{1,2,3,4,6\}$, with alternating multiplicities m_1, m_2 (equal for odd g). A degree-g Cartan–Münzner polynomial describes its parallel family, with regular spherical levels in $(-1,1)$ and two focal submanifolds at the endpoints. See Cecil–Ryan [13, Sections 3.2 and 3.4–3.6] for the geometry and Chi [17, Section 4] for the classification. The cone over a spherical hypersurface has curvatures zero radially and r^{-1} times the spherical curvatures otherwise; it is minimal exactly when its link is. We identify the unique minimal regular level below.

Lawlor's full curvature criterion is necessary and sufficient for area minimization of minimal isoparametric hypercones [47, Section 4.3]. Their nonflat strictly minimizing members also provide geometric models for non-affine entire solutions of the minimal surface equation, as shown by Simon [74, Theorem 6].

6.2. **The two-generator characterization.** For $\mathcal{A}_R(F) = \mathbb{R}[R, F]$, fixing the radius leaves one spherical invariant. The Mendes–Radeschi description gives:

Theorem 6.1 (One-dimensional quotients and the minimal level)**.** *Let $N \geq 3$, and let F be a nonconstant homogeneous polynomial on $\mathbb{R}^N$ with $\mathcal{A}_R(F) = \mathbb{R}[R, F]$. Suppose*

$$\Sigma = \{F = 0\} \cap \mathbb{S}^{N-1} \neq \varnothing, \qquad \nabla F \neq 0 \quad \text{on } \Sigma,$$

and suppose Σ is minimal in $\mathbb{S}^{N-1}$. Then $\{F = 0\}$ is the cone over the minimal regular member of a Cartan–Münzner isoparametric family. In particular, $\deg F \in \{1, 2, 3, 4, 6\}$.

Proof. Put $e = \deg F$. The case $e = 1$ is a hyperplane, so assume $e \geq 2$. By Mendes–Radeschi [57, Proposition 37(c) and its proof], there are constants $a \neq 0$ and b, with $b = 0$ for odd e, such that $\Psi = aF + br^e$ satisfies the normalized Cartan–Münzner equations. Their proof gives a Cartan–Münzner polynomial Φ_g, of degree $g \in \{1, 2, 3, 4, 6\}$, such that either $\Psi = \Phi_g$, or

$$\Psi = \epsilon(2\Phi_g^2 - r^{2g}), \qquad \epsilon \in \{-1, 1\}. \tag{6.1}$$

On the unit sphere, $\Sigma = \{\Psi = b\}$. Its regularity and $|\nabla_{\mathbb{S}^{N-1}}\Psi|^2 = e^2(1 - \Psi^2)$ imply $|b| < 1$.

Write $f_g = \Phi_g|_{\mathbb{S}^{N-1}}$ and $\Delta\Phi_g = c_g r^{g-2}$. The spherical Cartan–Münzner equations give the mean curvature of its regular levels:

$$\operatorname{div}_{\mathbb{S}^{N-1}} \frac{\nabla_{\mathbb{S}^{N-1}} f_g}{|\nabla_{\mathbb{S}^{N-1}} f_g|} = \frac{c_g - g(N-2)f_g}{g\sqrt{1 - f_g^2}}. \tag{6.2}$$

Since $N \geq 3$, only one regular level can be minimal. In the folded case (6.1),

$$\Sigma = \{f_g = t\} \cup \{f_g = -t\}, \qquad t = \sqrt{(1 + \epsilon b)/2} \in (0, 1).$$

Both levels are nonempty and regular, so they cannot both be minimal. Thus $\Psi = \Phi_g$, and Σ is its minimal regular level $f_g = c_g/[g(N-2)]$. The degree restriction follows. □

6.3. **One-variable reduction and classical classification.**

6.3.1. *Cartan–Münzner reduction.* Let Φ be a Cartan–Münzner polynomial of degree $g \in \{1, 2, 3, 4, 6\}$ on $\mathbb{R}^N$. For $g \geq 2$, write the two multiplicities as m_1, m_2, reverse the sign of Φ if necessary so that $1 \leq m_1 \leq m_2$, and put

$$s = m_1 + m_2, \qquad t = \frac{m_2 - m_1}{s}, \qquad d = N - 2 = \frac{g}{2}s. \tag{6.3}$$

Here d is the dimension of the minimal link. The Euclidean Cartan–Münzner equations are

$$|\nabla\Phi|^2 = g^2 r^{2g-2}, \qquad \Delta\Phi = \frac{g^2}{2}(m_2 - m_1)r^{g-2}. \tag{6.4}$$

Let

$$f = \frac{\Phi}{r^g}, \qquad z = f - t, \qquad F = r^g z = \Phi - tr^g. \tag{6.5}$$

The cone $C = \{F = 0\}$ is the cone over the unique minimal regular member of the isoparametric family. It is algebraic: for even g, r^g is polynomial, while for odd g the multiplicities are equal and $t = 0$. For $g \geq 2$, the cone is nonflat and smooth away from the origin, so $\operatorname{Sing} C = \{0\}$.

The coordinate z has a direct interpretation in the parallel isoparametric family. On its regular part put

$$\theta = \frac{1}{g}\arccos f \in (0, \pi/g), \qquad f = \cos(g\theta).$$

Here θ is spherical normal distance from the focal submanifold $M_+ = \{f = 1\}$ [17, Section 4, pp. 14–16]. The Cartan–Münzner equations give

$$|\nabla_{\mathbb{S}}\theta| = 1, \qquad \Delta_{\mathbb{S}}\theta = d\,\frac{\cos(g\theta) - t}{\sin(g\theta)}.$$

For the normal $\nabla_{\mathbb{S}}\theta$, the last expression is the divergence, or unaveraged scalar mean curvature. Thus the minimal leaf is characterized by

$$\cos(g\theta_*) = t, \qquad \tan^2(g\theta_*/2) = \frac{m_1}{m_2}, \qquad z = \cos(g\theta) - \cos(g\theta_*).$$

In particular z is a centered isoparametric coordinate, not a signed distance. For the local signed distance $q = \theta - \theta_*$,

$$z = -g\sin(g\theta_*)q + O(q^2).$$

We retain z because it preserves the polynomial form of the subcalibration calculation.

On $\mathbb{S}^{N-1}$, equations (6.4) give the one-variable diffusion closure

$$A(z) := |\nabla_{\mathbb{S}} z|^2 = g^2\big(1 - (z+t)^2\big), \tag{6.6}$$

$$B(z) := \Delta_{\mathbb{S}} z = -g^2 t - g(g+d)z. \tag{6.7}$$

The realizable interval is

$$I_t = [-1 - t, 1 - t]. \tag{6.8}$$

Proposition 6.2 (One-variable subcalibration equation)**.** *Let $\ell > 0$, let h be smooth on a neighborhood of I_t, and set*

$$G = r^\ell h(z), \qquad h(0) = 0, \qquad h'(0) > 0.$$

Define

$$E(z) = \ell^2 h(z)^2 + A(z)h'(z)^2 \tag{6.9}$$

and

$$\mathfrak{N}_\ell[h] = \big(Ah'' + Bh' + (d+1)\ell h\big)E - \frac{1}{2}Ah'E'. \tag{6.10}$$

Then $|\nabla G|^2 = r^{2\ell-2}E$ and $\mathcal{L}(G) = r^{3\ell-4}\mathfrak{N}_\ell[h]$, so the sign criterion becomes

$$G\mathcal{L}(G) = r^{4\ell-4}h(z)\mathfrak{N}_\ell[h](z) \geq 0. \tag{6.11}$$

If $E > 0$ throughout I_t and

$$\frac{h(z)}{z} > 0 \quad (z \neq 0), \qquad z\mathfrak{N}_\ell[h](z) \geq 0 \quad (z \in I_t), \tag{6.12}$$

then G generates a two-sided subcalibration and C is area minimizing. Its divergence is

$$r\,\mathrm{div}\, X_G = \frac{\mathfrak{N}_\ell[h]}{E^{3/2}}. \tag{6.13}$$

Proof. In Proposition 4.26, take $\varphi = h$. The spherical gradient pairing is $\Gamma_{\mathbb{S}}(u,v) = Au'v'$, the spherical Laplacian is $Au'' + Bu'$, and $N - 1 = d + 1$. Substitution gives the derivative identities and (6.11). For the angular profile h of G, the multiplier a_G is

$$G = a_G F, \qquad a_G = r^{\ell-g}\frac{h(z)}{z}, \qquad \frac{h(z)}{z}\bigg|_{z=0} = h'(0).$$

The quotient is smooth and positive, while $E > 0$ excludes critical points off the vertex. Since G is continuous at the vertex and has the signs of the polynomial F, Remark 2.7 and Corollary 2.8 give area minimality. Formula (1.3) gives (6.13). □

6.3.2. *Stability and the classical classification.* The squared norm of the second fundamental form of a minimal isoparametric hypersurface is constant:

$$|A_\Sigma|^2 = d(g-1). \tag{6.14}$$

The constant function attains the first eigenvalue $-d(g-1)$, so (4.16) gives the Hardy constant

$$D_{g,d} = \frac{(d-1)^2}{4} - d(g-1). \tag{6.15}$$

Wang's stability calculation uses the discriminant $(N-1)^2 - 4g(N-2) = 4D_{g,d}$ [88, Lemma 3.4, p. 216, and proof of Theorem B, p. 238]. The following is its sharp Hardy form.

Proposition 6.3 (Classical stability in Hardy form)**.** *For $g > 1$, stability and strict stability of the minimal isoparametric hypercone are equivalent, and hold precisely when*

$$d \geq 4g-2, \qquad \textit{equivalently} \qquad N \geq 4g. \tag{6.16}$$

The optimal constant in (2.7) is $D_{g,d}$ from (6.15). In the stable range,

$$D_{g,d} = \frac{1}{4} + \frac{d(d-4g+2)}{4} \geq \frac{1}{4}. \tag{6.17}$$

If (6.16) fails, the cone is unstable and hence not area minimizing.

Proof. The roots of $4D_{g,d}$, as a quadratic in d, are $2g-1 \pm 2\sqrt{g(g-1)}$. The lower root lies in $(0,1)$, and the upper root lies strictly between $4g-3$ and $4g-2$. Thus the integer link dimensions in question give exactly (6.16). Equation (6.17) gives the gap; negative second variation excludes minimization. □

For equal multiplicities, Corollary 4.17 with $e = g$, $\kappa = g(g-1)$ gives this same sharp constant.

Theorem 6.4 (Wang [88, Theorem A, p. 238])**.** *For $g \geq 2$, a minimal isoparametric hypercone is area minimizing if and only if*

$$N \geq 4g, \qquad (g, m_1, m_2) \notin \{(2,1,5), (4,1,6)\}, \tag{6.18}$$

up to interchanging the multiplicities. The $g = 1$ member is a hyperplane and is area minimizing.

The nonflat minimizing members are strictly minimizing by Lawlor's criterion [47, Section 4.3]; see the argument of Tang–Zhang [81, Section 2]. The two exceptions are strictly stable with optimal gap $1/4$, but not minimizing [88, Theorem B, p. 238].

6.3.3. *The harmonic part of the isoparametric classification.* For harmonic isoparametric polynomials, the dimension threshold in Example 4.23 selects the following parameters from the classical list. The Cartan–Münzner equations (6.4) and the dimension relation (6.3) give

$$\Delta F = 0 \iff m_1 = m_2 = m, \qquad N = gm + 2.$$

For $g = 4$, the possible equal multiplicities follow from the homogeneous list of Takagi–Takahashi [80], Stolz's multiplicity theorem [78], and the FKM construction [32, Theorem 4.1]. The sextic multiplicity restriction is due to Abresch [1]; see also [13, Chapter 3]. The first two columns give the multiplicity data; the last applies the radial criterion:

g	possible N	cases satisfying $N \geq 4g$
2	$2m+2,\ m \geq 1$	$N \geq 8$
3	$5, 8, 14, 26$	$14, 26$
4	$6, 10$	none
6	$8, 14$	none

The FKM restriction follows directly from the Clifford relations. A symmetric Clifford system $P_0, \ldots, P_a$ on $\mathbb{R}^{2l}$ has FKM multiplicities $(a, l-a-1)$, so equality requires $l = 2a+1$. If $a \geq 2$, the operator P_1P_2 preserves $E_+ = \ker(P_0 - I)$ and squares to $-I$. Since $\dim E_+ = l$,

this forces l to be even, a contradiction. Thus the only equal FKM pair is $(1,1)$, realized by $a=1, l=3$; the homogeneous list contributes the additional quartic pair $(2,2)$. The quartic and sextic rows lie below the stability threshold in Proposition 6.3.

6.4. **Explicit polynomial subcalibrations.** The one-variable calculation recovers every minimizing case, including unequal multiplicities and the required exceptional multipliers.

Theorem 6.5 (Explicit polynomial subcalibrations)**.** *Orient F as in (6.5), put $R=r^2$, and assume the isoparametric multiplicities exist. Every positive case in Theorem 6.4 admits the normalized-gradient subcalibration generated by the polynomial potential in the following table.*

g	*multiplicities*	*potential* G
1	*hyperplane*	F
2	$s\ge 7$, *or* $(m_1,m_2)=(3,3)$	RF
2	$(m_1,m_2)=(2,4)$	$\big(R+\frac{3}{10}F\big)F$
3	$m_1=m_2=4$ *or* 8	RF
4	$s\ge 9$, *or* $s=8$ *and* $m_1\ge 2$	R^2F
4	$(m_1,m_2)=(3,4)$	$\big(R^2+\frac{7}{54}F\big)F$
4	$(m_1,m_2)=(2,5)$	$\big(R^2+\frac{7}{15}F\big)F$
4	$(m_1,m_2)=(1,7)$	$R\big(R^2+\frac{1}{2}F\big)F$

In every row the total degree of G is at most ten, independently of N. Both sextic multiplicity pairs lie below the stability threshold, as recorded in Section 6.3.3. For every nonflat row, the field generated by the displayed polynomial potential proves strict area minimality, with the mass-gap constant in (2.19).

6.4.1. *Potentials obtained by a radial factor.* We first take $h(z)=z$. Formula (6.10) factors completely.

Proposition 6.6 (Radial polynomial defect)**.** *For $h=z$,*

$$\mathfrak{N}_\ell[z]=zQ_\ell(z), \tag{6.19}$$

where

$$Q_\ell(z)=g^2(1-t^2)\mu_\ell+g^2t(\ell^2-2\nu_\ell)z+(\ell^2-g^2)\nu_\ell z^2, \tag{6.20}$$

$$\mu_\ell=(d+1)\ell-\ell^2-gd,\qquad \nu_\ell=(d+1)\ell-gd. \tag{6.21}$$

Moreover,

$$\mu_\ell=D_{g,d}-\left(\ell-\frac{d+1}{2}\right)^2. \tag{6.22}$$

Thus the constant term is nonnegative precisely on the admissible degree interval obtained from the Jacobi equation.

Proof. Insert $h=z$, $h'=1$, $h''=0$, and $E=\ell^2z^2+A$ into (6.10); collect powers of z. Identity (6.22) follows from (6.15). □

The cases $g=2$ and $g=4$. For $g=2$, $d=s$, $\ell=4$, one has

$$Q_4=8P_2,\qquad P_2=(s-6)(1-t^2)-2(s-2)tz+3(s+2)z^2. \tag{6.23}$$

The vertex of this convex quadratic is

$$z_*=\frac{(s-2)t}{3(s+2)},$$

and at the right endpoint $z_+=2m_1/s$,

$$P_2(z_+)=\frac{16m_1((s+2)m_1-s)}{s^2}>0. \tag{6.24}$$

If $z_* \le z_+$, equivalently

$$m_1 \ge m_{1,*} := \frac{s(s-2)}{8(s+1)},$$

the minimum is nonnegative precisely when

$$K_2 := 3(s+2)(s-6)(1-t^2) - (s-2)^2 t^2 \ge 0. \tag{6.25}$$

For fixed s, this quantity increases with m_1. At the entry value,

$$K_2(m_{1,*}) = \frac{3(s-2)(s+2)(s^2-8s-12)}{4(s+1)^2} \ge 0 \quad (s \ge 10). \tag{6.26}$$

For $s = 7, 8, 9$, direct evaluation at $m_1 = 1$ gives, respectively,

$$\frac{23}{49}, \qquad 6, \qquad \frac{767}{81}.$$

If $z_* > z_+$, the minimum on I_t is the positive endpoint value (6.24). This proves $Q_4 \ge 0$ for every $s \ge 7$. For $s = 6$, the same formula handles $(m_1, m_2) = (3, 3)$, while $(2, 4)$ requires the correction in the next subsection.

For $g = 4$, $d = 2s$, $\ell = 8$,

$$Q_8 = 128 P_4, \qquad P_4 = (s-7)(1-t^2) - 2(s-3)tz + 3(s+1)z^2. \tag{6.27}$$

Now

$$z_* = \frac{(s-3)t}{3(s+1)}, \qquad P_4(z_+) = \frac{16 m_1((s+1)m_1 - s)}{s^2} > 0.$$

The vertex enters the interval at $m_{1,*} = (s-3)/8$. When it does, the minimum is controlled by

$$K_4 = 3(s+1)(s-7)(1-t^2) - (s-3)^2 t^2, \tag{6.28}$$

and

$$K_4(m_{1,*}) = \frac{3(s-3)(s+1)(s^2-10s-3)}{4s^2} \ge 0 \quad (s \ge 11). \tag{6.29}$$

For $s = 9, 10$, the values at $m_1 = 1$ are $52/27$ and $107/25$. For $s = 8$ and $m_1 \ge 2$, $K_4 \ge 27 - 52(1/2)^2 = 14$. This proves the $g = 4$ entries with a radial factor in Theorem 6.5. The case $g = 3$. Here $m_1 = m_2 = m$, $t = 0$, $d = 3m$, and we choose $\ell = 5$. Formula (6.20) becomes

$$Q_5(z) = 9(6m - 20) + 16(6m + 5)z^2. \tag{6.30}$$

It is positive for $m = 4, 8$, the two stable Cartan cases by Proposition 6.3 and the classification in [64, Section 3.1.1]. Since $r^5 z = RF$, this gives the stated polynomial potential.

6.4.2. *Borderline polynomial corrections.* At the endpoint, cancellation of the radial defect's leading odd term fixes the correction coefficient.

At the critical dimension $N = 4g$, one has $d = 4g - 2$. The natural endpoint degree is $\ell = 2g$, for which $\mu_\ell = 0$. Consider the first transverse correction

$$h = z(1 + \lambda z). \tag{6.31}$$

Proposition 6.7 (Forced critical correction)**.** *For $t > 0$, a two-sided sign near $z = 0$ forces*

$$\lambda = \lambda_* = \frac{2gt}{(2g+1)(1-t^2)}. \tag{6.32}$$

For this correction to preserve the sign of z on all of I_t, it is necessary and sufficient that

$$t < \frac{2g+1}{4g+1}. \tag{6.33}$$

Proof. At the critical degree, $d = 4g-2$ and $\ell = 2g$. Substitution in (6.10) gives

$$z\mathfrak{N}_{2g}[z(1+\lambda z)] = 2g^3\big[(2g+1)(1-t^2)\lambda - 2gt\big]z^3 + O(z^4).$$

The lower-order terms vanish. Nonnegativity for both signs of z therefore forces the cubic coefficient to vanish, which gives (6.32). Since $\lambda_* > 0$, the only possible sign loss for $h/z = 1+\lambda_* z$ is at $z = -1-t$. The inequality $1-\lambda_*(1+t) > 0$ is exactly (6.33). □

For $g = 2, s = 6$, condition (6.33) accepts $(3,3)$ and $(2,4)$, but rejects $(1,5)$. For $g = 4, s = 7$, it accepts $(3,4)$ and $(2,5)$, but rejects $(1,6)$. The quadratic correction $h = z+\lambda z^2$ therefore fails precisely for the two stable non-minimizing members of the classical list.

Substituting the three admissible unequal multiplicities in (6.32) gives the rational coefficients below. Their global signs are then checked on the exact interval I_t.

Proposition 6.8 (Three quadratic corrections)**.** *The following identities hold on the indicated realizable intervals.*

$$g = 2, \quad (m_1,m_2) = (2,4), \quad t = \tfrac{1}{3}, \quad I_t = [-\tfrac{4}{3}, \tfrac{2}{3}],$$
$$h = z(1+\tfrac{3}{10}z), \quad \ell = 4,$$
$$z\mathfrak{N}_4[h] = \frac{648}{125}z^4(11z+30); \tag{6.34}$$
$$g = 4, \quad (m_1,m_2) = (3,4), \quad t = \tfrac{1}{7}, \quad I_t = [-\tfrac{8}{7}, \tfrac{6}{7}],$$
$$h = z(1+\tfrac{7}{54}z), \quad \ell = 8,$$
$$z\mathfrak{N}_8[h] = -\frac{64}{19683}z^4(2548z^2 - 132237z - 910926); \tag{6.35}$$
$$g = 4, \quad (m_1,m_2) = (2,5), \quad t = \tfrac{3}{7}, \quad I_t = [-\tfrac{10}{7}, \tfrac{4}{7}],$$
$$h = z(1+\tfrac{7}{15}z), \quad \ell = 8,$$
$$z\mathfrak{N}_8[h] = -\frac{128}{1125}z^4(588z^2 - 8015z - 16550). \tag{6.36}$$

In every case $h/z > 0$ and $z\mathfrak{N}_\ell[h] \geq 0$ on I_t.

Proof. The three lower bounds for h/z are $3/5$, $23/27$, and $1/3$. In (6.34), $11z+30 > 0$ on I_t. The quadratics in parentheses in (6.35) and (6.36) are convex, and their endpoint values are both negative; hence they are negative on the entire interval. The identities follow by substitution into (6.10). □

Proposition 6.9 (The $(g, m_1, m_2) = (4,1,7)$ correction)**.** *For $t = 3/4$, $I_t = [-7/4, 1/4]$, $\ell = 10$, and $h = z(1+z/2)$,*

$$z\mathfrak{N}_{10}[h] = z^2P(z), \tag{6.37}$$

where

$$P(z) = 189z^5 + 1778z^4 + 5492z^3 + 4848z^2 - 763z + 42. \tag{6.38}$$

The polynomial P is positive on $[-7/4, 1/4]$.

Proof. Substitution into (6.10) gives (6.37). The rational Bernstein certificate in Example C.1 proves $P > 0$ on the whole interval. Also $h/z \geq 1/8$, and the potential is $r^{10}h = R(R^2 + F/2)F$. □

Proof of Theorem 6.5. The rows with a radial factor follow from Proposition 6.6 and the calculations in the preceding subsection. The remaining rows follow from Propositions 6.8 and 6.9. In each case $E > 0$ on I_t: if $z \neq 0$, then $h \neq 0$ and $E \geq \ell^2 h^2 > 0$; at $z = 0$, $E(0) = A(0)h'(0)^2 > 0$. Apply Proposition 6.2. For every nonflat row the displayed numerator is nonzero, and both phases are nonempty. The normalized gradient is smooth away from the vertex. Hence Corollary 2.13 proves strict minimality directly from the displayed polynomial potential. □

For every nonflat row except the quadratic case in ambient dimension $N = 9$, the displayed degree satisfies $\ell \neq (N-1)/2$. Thus Theorem 4.6 already gives strict stability from the subcalibration sign. In the exceptional case $\ell = 4 = (N-1)/2$, Proposition 6.3 gives the optimal Hardy constant 2. The following boundary calculations distinguish the degree gap from positivity of the Jacobi trace.

6.4.3. *Degenerate traces and quantitative strictness.*

Corollary 6.10 (Four polynomial potentials with vanishing Jacobi trace)**.** *For the four polynomial potentials in Theorem 6.5 corresponding to*

$$(g, m_1, m_2) = (2,3,3),\ (2,2,4),\ (4,3,4),\ (4,2,5), \tag{6.39}$$

one has $n = d+1 = 4g-1$, $\ell = 2g$, and

$$\beta_G \equiv 0, \qquad J_C(|\nabla G|^{-1}) = 0, \qquad \inf_{u \neq 0} \frac{Q_C(u)}{\int_C r^{-2}u^2} = \frac{1}{4}. \tag{6.40}$$

Their signed divergence satisfies, uniformly near the regular cone,

$$q_{X_G}(x) \asymp \frac{\operatorname{dist}(x,C)^3}{r^4}. \tag{6.41}$$

In particular the fields generated by these polynomial potentials do not satisfy a positive linear distance bound (2.20), although they prove strict minimality and strict stability. For every sufficiently small $\varepsilon > 0$, the perturbed potential $G_\varepsilon = r^{-\varepsilon}G$ has boundary coefficient

$$\beta_{G_\varepsilon} = \varepsilon - \varepsilon^2 > 0 \tag{6.42}$$

and satisfies the distance-weighted comparison of Proposition 2.14.

Proof. Along the minimal link the gradient length $|\nabla G|$ is constant, since $h'(0) = 1$ in each case and $A(0) = g^2(1-t^2)$. Here $\mu_\ell = 0$ in (6.22), so the boundary coefficient vanishes. Alternatively this follows from the absence of a linear term in $\mathfrak{N}_\ell[h]$ in the displayed factorizations. The optimal gap $1/4$ is (6.17) at $d = 4g-2$. For $(2,3,3)$, equations (6.23) and (6.19) give $\mathfrak{N}_4[z] = 192z^3$. For the other three potentials, (6.34)–(6.36) give $\mathfrak{N}_\ell[h] = z^3p(z)$ with $p > 0$ near zero and $E(0) > 0$. The spherical function z is a regular defining function of the compact minimal link. Thus $|z| \asymp \operatorname{dist}(x,C)/r$, and (6.13) proves (6.41). The same factorizations give strict signed divergence everywhere off the cone. All hypotheses of Proposition A.4 hold, yielding (6.42) and the claimed distance bound. □

For comparison, Niu [65, Lemma 5.3] obtains quantitative subcalibrations from the Hardt–Simon foliation under strict stability and strict minimality. Here the explicit potentials suffice, with a radial correction in the four cases above.

Corollary 6.11 (Distance-weighted comparison for the nonflat minimizing cases)**.** *Every nonflat cone in Theorem 6.5 admits a homogeneous potential, smooth and semialgebraic on $\mathbb{R}^N \setminus \{0\}$, whose normalized gradient satisfies (2.20). Except for (6.39), the polynomial potential in the table itself has this property. In the four remaining cases, a sufficiently small positive rational radial perturbation has it.*

Proof. The estimates (6.26) and (6.29) are strict in their indicated ranges, as are their low-dimensional checks and endpoint values. Thus $Q_\ell > 0$ on the entire realizable interval in all these nondegenerate rows; (6.30) is also strictly positive. The remaining nondegenerate correction has $P > 0$ by Proposition 6.9. In these cases $\mathfrak{N}_\ell[h] = zp(z)$ with $p > 0$ on the compact interval and $E > 0$. Near the link $|z|$ is comparable to spherical distance, and away from it compactness supplies a positive lower bound. Equation (6.13) and homogeneity prove (2.20). Apply Corollary 6.10 to the other four cases, choosing a sufficiently small positive rational ε. The factor $r^{-\varepsilon} = (|x|^2)^{-\varepsilon/2}$, and hence G_ε, is semialgebraic on $\mathbb{R}^N \setminus \{0\}$; see Proposition A.4. □

6.4.4. *The two classical exceptions.* Both exceptional cones possess positive Jacobi fields. Indeed the radial Jacobi equation is

$$\gamma(\gamma+d-1)+d(g-1)=0, \qquad \gamma_{\pm}=-\frac{d-1}{2}\pm\sqrt{D_{g,d}}. \tag{6.43}$$

For $(2,1,5)$ they are r^{-2}, r^{-3}; for $(4,1,6)$, they are r^{-6}, r^{-7}. These cases are stable but not minimizing, completing Theorem 6.4.

Corollary 6.12 (Lin's one-sided strict minimality [50, p. 210])**.** *Let $x\in\mathbb{R}^2$, $y\in\mathbb{R}^6$, and set*

$$E=\{|y|^2<5|x|^2\}, \qquad C=\partial[\![E]\!], \qquad C_1=C\llcorner B_1.$$

The cone over $S^1(\sqrt{1/6})\times S^5(\sqrt{5/6})$ is one-sided strictly area minimizing in $\overline{E}$. More precisely, it minimizes against finite-perimeter changes $H\subset E$ of compact support. There is $\Theta>0$ such that every finite-mass integral current T with

$$\partial T=\partial C_1, \qquad \operatorname{spt}T\subset\overline{E}, \qquad \operatorname{spt}T\cap B_\varepsilon=\varnothing$$

satisfies $\mathbf{M}(T)-\mathbf{M}(C_1)\geq\Theta\varepsilon^7$ for $0<\varepsilon<1$.

Proof. Put $u=|x|^2$, $v=|y|^2$, $R=u+v$, and $F=v-5u$. Take the radial potential $G=RF$. For functions of u,v,

$$\Delta P=4uP_{uu}+4vP_{vv}+4P_u+12P_v, \qquad \Gamma(P,Q)=4uP_uQ_u+4vP_vQ_v.$$

Hence

$$\begin{aligned}\Delta G&=16(v-8u),\\ |\nabla G|^2&=16(25u^3+24u^2v+v^3),\\ \mathcal{L}(G)&=64R(v-5u)^2(v-17u).\end{aligned}$$

The last identity follows by inserting the first two in

$$\mathcal{L}(G)=|\nabla G|^2\Delta G-\tfrac{1}{2}\Gamma(G,|\nabla G|^2),$$

the same radial calculation as Proposition 6.6. Thus $X=\nabla G/|\nabla G|$ is smooth off the origin, equals ν_E on the regular cone, and satisfies $\operatorname{div}X<0$ throughout E. Since $\mathcal{H}^{N-1}(\{0\})=0$, the one-sided part of Theorem 2.9 gives

$$\operatorname{Per}(H;U)-\operatorname{Per}(E;U)\geq\int_{E\setminus H}-\operatorname{div}X \qquad (H\subset E).$$

Set $\Theta=\int_{E\cap B_1}-\operatorname{div}X>0$. This integral is finite, and homogeneity makes its value over $E\cap B_\varepsilon$ equal to $\Theta\varepsilon^7$. To pass to currents, use radial projection and an integer-valued BV filling as in the proof of Theorem 2.11. The resulting function v has zero distributional gradient in the connected open set $\mathbb{R}^8\setminus\overline{E}$ and vanishes there outside a compact set, so it vanishes throughout that set. It is also constant in B_ε. Since it vanishes on the nonempty open set $B_\varepsilon\setminus\overline{E}$, this constant is zero. Hence $H=\{v\geq1\}$ satisfies $H\subset E$ and $H\cap B_\varepsilon=\varnothing$, up to null sets. Thus $E\setminus H$ contains $E\cap B_\varepsilon$, and the displayed one-sided comparison proves the claimed mass gap. □

7. Gradient-Square Closure and Cone Geometry

After the two-generator case, a natural candidate for the next closure is

$$R=|x|^2, \qquad S=|\nabla F|^2, \qquad \mathbb{R}[R,F,S].$$

The generator degrees are $(2,3,4)$ for cubics and $(2,4,6)$ for quartics. In odd degree, closure forces degree three and identifies the possible cubic types. In even degree, we record the closure identities, a regular-link equation and a rectangular matrix example. Here closure always means that this prescribed polynomial algebra is Laplacian.

7.1. **Odd-degree rigidity and cubic closure.** The identity

$$\mathcal{L}(F) = S\Delta F - \tfrac{1}{2}\Gamma(F,S)$$

relates the closure to minimality. In odd degree, the critical set determines which of the following two rigidity arguments applies.

Theorem 7.1 (Odd-degree rigidity)**.** *Let $N \geq 3$, and let $F \not\equiv 0$ be a homogeneous polynomial of odd degree $e \geq 3$ on $\mathbb{R}^N$. Suppose $\mathbb{R}[R,F,S]$ is a Laplacian algebra and $\nabla F(p) = 0$ at some $p \neq 0$. Then $e = 3$.*

Appendix B.1 proves the theorem by degree counting and Hessian comparisons at a nonzero critical point and a spherical maximum, forcing $(e-2)(e-3) = 0$.

With trivial critical set, the gradient square is radial:

Theorem 7.2 (Odd closure with trivial critical set)**.** *Let $N \geq 3$, and let $F \not\equiv 0$ be homogeneous of odd degree $e \geq 3$. If $\mathbb{R}[R,F,S]$ is Laplacian and $\{\nabla F = 0\} = \{0\}$, then*

$$S = cR^{e-1} \qquad (c > 0).$$

Consequently $e = 3$, and, up to an orthogonal change of variables and a nonzero constant factor, F is a Cartan cubic in one of the dimensions $N = 5, 8, 14, 26$.

A spherical gradient-flow first integral and incompatible Hessian traces force S to be radial. Odd closure also gives harmonicity; the polynomial eikonal classification [85, Theorem 1] and Cartan's cubic classification [11] then apply. See Appendix B.2. Thus both cases force degree three; the higher-degree pullbacks in Section 5 instead use $Q = |P|^2$ as a generator.

The restriction $N \geq 3$ is essential. On $\mathbb{R}^2$, $F(x,y) = \mathrm{Re}(x+iy)^e$ is harmonic and satisfies $S = e^2R^{e-1}$ for every positive integer e.

Lemma 7.3 (Three-generator cubic closure)**.** *Let F be a nonzero homogeneous cubic on $\mathbb{R}^N$. Suppose $\mathbb{R}[R,F,S]$ is a Laplacian algebra, where $S = |\nabla F|^2$. There are constants $\alpha, \beta, \gamma, \delta, \varepsilon$ such that*

$$\begin{aligned} \Delta F = 0, \qquad & \Delta S = \alpha R, \\ \Gamma(F,S) = \beta RF, \quad & \Gamma(S,S) = \gamma R^3 + \delta F^2 + \varepsilon RS. \end{aligned} \tag{7.1}$$

Moreover, $\alpha > 0$ and $\mathcal{L}(F) = -\beta RF/2$, so the regular zero set is minimal.

Proof. The generators have degrees $2, 3, 4$. Their homogeneous monomials of degrees $1, 2, 5, 6$ are, respectively, none, R, RF, and R^3, F^2, RS. Laplacian closure and the product rule (3.1) give (7.1). Bochner's identity [20, Proposition 1.47, equation (1.236), p. 61] gives $\alpha R = 2|\operatorname{Hess} F|^2$; the Hessian of a nonzero cubic is not identically zero, so $\alpha > 0$. The expression for $\mathcal{L}(F)$ follows from harmonicity. □

At a positive maximum $a = F(x)$ on the unit sphere, Euler's identity gives $\nabla F(x) = 3ax$ and $\operatorname{Hess} F(x)x = 6ax$, hence $\beta = 108a^2 > 0$. This gives a harmonic radial eigencubic. Its normalization and classification follow; variational properties are treated in Section 10.3.

7.1.1. *Normalized closure and the critical set.*

Proposition 7.4 (Normalized closure and the critical set)**.** *Assume the hypotheses of Lemma 7.3 and that $\nabla F(p) = 0$ for some $p \neq 0$. After multiplying F by a positive constant, the closure becomes*

$$\begin{aligned} \Delta F = 0, \qquad & \Delta S = \alpha R, \qquad \alpha = \frac{2(N+3)}{3}, \\ \Gamma(F,S) = 4RF, \quad & \Gamma(S,S) = 4RS + 12F^2. \end{aligned} \tag{7.2}$$

There is an integer $k \geq 1$ such that

$$N = 6k - 3, \qquad \alpha = 4k. \tag{7.3}$$

At every nonzero critical point, $\operatorname{Hess} F$ *has* k *positive and* k *negative eigenvalues, all of absolute value* $|p|$*, and* $N-2k$ *zero eigenvalues. The punctured critical set* $Z^* = \{x \neq 0 : \nabla F(x) = 0\}$ *is a real-analytic submanifold of dimension* $N-2k = 4k-3$*, with* $T_pZ^* = \ker \operatorname{Hess} F(p)$*. In orthonormal coordinates* $z = (u, v, w)$*, where* $u, v \in \mathbb{R}^k$ *are the positive and negative Hessian directions and* $w \in \mathbb{R}^{N-2k}$*,*

$$F(p+z) = \frac{|p|}{2}\big(|u|^2 - |v|^2\big) + O(|z|^3).$$

Proof. At the given nonzero critical point, Bochner's identity makes the trace-free Hessian nonzero, while the differentiated Euler identity places p in its kernel. Hence $N \geq 3$. The normalization (B.2), established in the proof of Theorem 7.1 in Appendix B.1, applies with $e = 3$. Together with the moment identity (B.9), it gives (7.2).

Let $p \neq 0$ be any critical point and put $H = \operatorname{Hess} F(p)$. The cubic Taylor expansion is

$$F(p+y) = \tfrac{1}{2}\langle Hy, y\rangle + F(y), \qquad S(p+y) = \langle H^2 y, y\rangle + O(|y|^3).$$

Comparing quadratic terms in $\Gamma(F, S) = 4RF$ gives $2\langle H^3 y, y\rangle = 2|p|^2\langle Hy, y\rangle$, and hence $H^3 = |p|^2 H$. Bochner's identity implies $H \neq 0$, and harmonicity gives $\operatorname{tr} H = 0$. Thus the nonzero eigenvalues are $\pm|p|$, with equal multiplicities, say k. Bochner's identity and (7.2) imply $2k|p|^2 = \operatorname{tr}(H^2) = \frac{1}{2}\alpha|p|^2$. Thus $\alpha = 4k$ and $N = 6k-3$, proving (7.3).

The implicit function argument in Appendix B.5 proves the asserted description of Z^*. The Hessian eigenvalues give the displayed Taylor expansion; Section 2.6 then identifies the quadratic tangent and the regular graphical convergence. □

7.1.2. *The cubic classification within this closure.*

Proposition 7.5 (Cubic types under gradient-square closure)**.** *Under the hypotheses of Proposition 7.4, the possible parameters are*

$$k \in \{1, 2, 3, 5\}, \qquad N \in \{3, 9, 15, 27\}. \tag{7.4}$$

These four cases are realized, respectively, by xyz *on* $\mathbb{R}^3$ *and*

$$F(A) = \tfrac{1}{3}\operatorname{Re}\operatorname{tr}(A^3) \quad \text{on } \operatorname{Herm}_4(\mathbb{F})_0, \qquad \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}.$$

In particular, for an irreducible defining polynomial only the dimensions $9, 15, 27$ *occur.*

Proof. We apply the existing radial-eigencubic classification, with its normalization made explicit. By Proposition 7.4,

$$\mathcal{L}(F) = -2RF, \qquad \operatorname{tr}\big((\operatorname{Hess} F)^2\big) = 2kR.$$

The Hessian-trace identity in [84, Lemma 5.1(i)] gives the cubic trace coefficient $-(N+6)\lambda - 6\beta_2$. Substituting $\lambda = -2$, $\beta_2 = 2k$, and $N = 6k-3$ gives $\operatorname{tr}\big((\operatorname{Hess} F)^3\big) = 6F$. The type parameters (n_1, n_2) are the characteristic dimensions of [64, Definition 6.7.1, Proposition 6.11.2, and Corollary 6.11.1]. The dimension and cubic Hessian-trace coefficient give

$$n_1 = 1 + \frac{6}{3(-2)} = 0, \qquad n_2 = \frac{N+1}{2} = 3k-1.$$

Under the additional quadratic Hessian-trace identity, a radial eigencubic is either exceptional or one of the four Hurwitz eigencubics of Clifford type [64, Corollary 6.11.2 and Section 6.11.5]. For the exceptional branch, the $n_1 = 0$ classification in [84, Proposition 6.1] gives exactly the three possible types $(n_1, n_2) = (0, 5),\ (0, 8),\ (0, 14)$. Among the four Clifford-type eigencubics, $n_1 = 0$ occurs only for the real product cubic in dimension three; the four algebras are listed as *mutants* in [64, Section 6.11.5]. This proves (7.4). The three exceptional cases are realized by the rank-four trace models listed in [64, Section 6.11.6(iii)]; their closure is also calculated directly in Corollary 10.9. The dimension-three model is reducible, so it is excluded by irreducibility. □

The general classification of radial eigencubics remains incomplete [34, Section 2.2]. Within this subclass, Theorem 10.8 selects only the dimension-27 quaternionic trace cone. General cubic algebras begin in Section 9.

7.2. **Even-degree closure and a regular-link identity.** Even-degree gradient-square closure need not imply harmonicity. We record its identities and a regular-link consequence, leaving further geometry and variational properties aside.

Proposition 7.6 (The even-degree closure and its signs)**.** *Let F be a real irreducible homogeneous polynomial of degree $d = 2m \geq 4$ on $\mathbb{R}^N$, with a nonempty minimal regular zero set. Suppose $\mathbb{R}[R, F, S]$ is Laplacian, where $R = |x|^2$ and $S = |\nabla F|^2$. There are constants $c, \alpha, \beta, \lambda, \mu$ such that*

$$\begin{aligned} \Delta F &= cR^{m-1}, & \Delta S &= \alpha R^{d-2} + \beta R^{m-2} F, \\ \mathcal{L}(F) &= (\lambda R^{d-2} + \mu R^{m-2} F)F, \\ \Gamma(F, S) &= 2cR^{m-1}S - 2\lambda R^{d-2}F - 2\mu R^{m-2}F^2. \end{aligned} \tag{7.5}$$

Moreover, $\alpha > 0$ and $\lambda < 0$.

Proof. The degrees of R, F, S are $2, d, 2d-2$. The homogeneous pieces of degrees $d-2$ and $2d-4$ are therefore spanned by R^{m-1} and by $R^{d-2}, R^{m-2}F$, respectively. Minimality and Proposition 3.14 make $\mathcal{L}(F)/F$ a polynomial of degree $2d-4$; division inside a Laplacian algebra, Proposition 3.6, places it in the same homogeneous piece. The identity $\mathcal{L}(F) = S\Delta F - \Gamma(F, S)/2$ proves the mixed entry.

Bochner's identity and $\Gamma(R, F) = 2dF$ give

$$|\operatorname{Hess} F|^2 = \frac{\alpha}{2} R^{d-2} + \left(\frac{\beta}{2} - cd(d-2)\right) R^{m-2} F. \tag{7.6}$$

At a nonzero regular zero this shows $\alpha \geq 0$. The polynomial F changes sign across that zero. If $\alpha = 0$, nonnegativity of the left side would force $\beta/2 - cd(d-2) = 0$, and hence $\operatorname{Hess} F = 0$, contrary to $d \geq 4$. Thus $\alpha > 0$.

Let $a > 0$ and $b_0 < 0$ be the maximum and minimum of F on the unit sphere. At either extremum, with value t, Euler's identity gives

$$\nabla F = dtx, \qquad S = d^2t^2, \qquad \nabla S = 2(d-1)Sx.$$

Substitution in the mixed entry, followed by division by $2t$, gives

$$d^3(d-1)t^2 + (\mu - cd^2)t + \lambda = 0.$$

The two distinct values a, b_0 are its roots. Consequently $\lambda = d^3(d-1)ab_0 < 0$. □

Remark 7.7 (Harmonic normalization). With $a = c/[d(N+d-2)]$, the polynomial $H = F - aR^m$ is harmonic, and

$$|\nabla H|^2 = S - 2ad^2R^{m-1}F + a^2d^2R^{d-1}.$$

Thus $\mathbb{R}[R, H, |\nabla H|^2] = \mathbb{R}[R, F, S]$. This change preserves the algebra, but generally changes the zero set and the critical set; it does not reduce the geometric classification to the harmonic case.

Proposition 7.8 (An isoparametric function on the regular link)**.** *Assume Proposition 7.6 and that F has a nonzero critical point. Put $\ell = d-1$ and $h = (\alpha - 2c^2)/(-\lambda)$. On $M = \{F = 0,\ S > 0\} \cap \mathbb{S}^{N-1}$, the function $u = -2\ell^2 S|_M/\lambda$ satisfies $0 < u \leq 1$ and*

$$|\nabla_M u|^2 = 4\ell^2 u(1-u), \qquad \Delta_M u = 2\ell^2(h-1) - 2\ell(N+2\ell-3)u. \tag{7.7}$$

Thus u is isoparametric in the intrinsic sense that its gradient square and Laplacian depend only on its value.

Proof. Let $p \neq 0$ be critical, $R_0 = R(p)$ and $H = \operatorname{Hess} F(p)$. Euler's identity gives $F(p) = 0$. Comparing quadratic terms at p in (7.5) and using (7.6) gives

$$H(2H^2 - 2cR_0^{m-1}H + \lambda R_0^{d-2}\operatorname{Id}) = 0, \quad \operatorname{tr} H = cR_0^{m-1}, \quad \operatorname{tr}(H^2) = \tfrac{\alpha}{2}R_0^{d-2}. \tag{7.8}$$

The normalized nonzero eigenvalues are among $\eta_\pm = (c \pm \sqrt{c^2 - 2\lambda})/2$, with opposite signs. Both occur: $H \neq 0$ since $\alpha > 0$, and if only η_+ occurred with multiplicity $a \geq 1$, its trace would give $(a-1)\eta_+ = \eta_-$, a contradiction. The other case is identical.

Closure and degree counting give

$$\begin{aligned}\Gamma(S,S) = {} & D_0R^{2d-3} + D_1R^{3m-3}F + D_2R^{d-3}F^2 + D_3R^{m-3}F^3 \\ & + D_4R^{d-2}S + D_5R^{m-2}FS,\end{aligned}$$

with the D_3 term omitted for $d = 4$. At p its constant term gives $D_0 = 0$, and its quadratic terms give $4H^4 = D_1R_0^{3m-3}H/2 + D_4R_0^{d-2}H^2$. For each of the two roots, this reads $4\eta^3 = D_1/2 + D_4\eta$. Using $\eta^2 = c\eta - \lambda/2$ yields

$$D_0 = 0, \qquad D_1 = -4c\lambda, \qquad D_4 = 4c^2 - 2\lambda. \tag{7.9}$$

On $C_{\mathrm{reg}} = \{F = 0, S > 0\}$ take $\nu = \nabla F/\sqrt{S}$. The mixed entry in (7.5) and (7.9) imply

$$\nu(S) = 2cR^{m-1}\sqrt{S}, \qquad |\operatorname{Hess} F\,\nu|^2 = (c^2 - \lambda/2)R^{d-2}.$$

Subtracting $\nu(S)^2$ from $\Gamma(S,S)$ gives the tangential gradient square. Differentiating $\Gamma(F,S)$ in direction ν, and using $\nu(R) = 0$ on C, gives $\operatorname{Hess} S(\nu,\nu) = (2c^2 - \lambda)R^{d-2}$. Since C_{reg} is minimal, $\Delta_C S = \Delta S - \operatorname{Hess} S(\nu,\nu)$, and hence

$$|\nabla_C S|^2 = -2\lambda R^{d-2}S, \qquad \Delta_C S = -(h-1)\lambda R^{d-2}. \tag{7.10}$$

Finally S has degree 2ℓ. Writing $s = S|_M$ and separating radial derivatives in the cone metric gives

$$|\nabla_M s|^2 = -2\lambda s - 4\ell^2 s^2, \qquad \Delta_M s = -(h-1)\lambda - 2\ell(N + 2\ell - 3)s.$$

Normalization proves (7.7); its first identity and $s > 0$ imply $0 < u \leq 1$. □

7.3. **Quartic closure and a rectangular matrix family.** The following example realizes generator degrees $(2,4,6)$ by rectangular matrices, with minimality selecting a single row dimension.

Proposition 7.9 (The rectangular matrix family)**.** *Let $p \geq 3$, give $M_{p,3}(\mathbb{R})$ its Frobenius inner product, and put*

$$B = X^{\mathsf{T}}X, \qquad P_j = \operatorname{tr}(B^j), \qquad R = P_1, \qquad F = 2P_2 - R^2, \qquad S = |\nabla F|^2. \tag{7.11}$$

Then $\mathcal{A}_R(F) = \mathbb{R}[R,F,S]$, with three algebraically independent generators of degrees $(2,4,6)$. Its complete closure table is

$$\Delta R = 6p, \qquad \Delta F = 4(6-p)R, \qquad \Delta S = 32(p+6)R^2 + 448F, \tag{7.12}$$

$$\begin{aligned}&\Gamma(R,R) = 4R, \quad \Gamma(R,F) = 8F, \quad \Gamma(R,S) = 12S,\\ &\Gamma(F,F) = S,\\ &\Gamma(F,S) = 8RS + 128(R^2+F)F,\\ &\Gamma(S,S) = 192(R^2+F)S + 1024R^3F + 2048RF^2.\end{aligned} \tag{7.13}$$

The regular zero set is minimal if and only if $p = 5$. In that case $N = 15$, F is nonharmonic, and

$$\Delta F = 4R, \qquad \mathcal{L}(F) = -64(R^2+F)F. \tag{7.14}$$

Proof. Differentiation of $B = X^{\mathsf{T}}X$ gives

$$\nabla P_j = 2jXB^{j-1}, \qquad \Gamma(P_i, P_j) = 4ijP_{i+j-1}. \tag{7.15}$$

For the required Laplacians, the product rule gives

$$\Delta P_1 = 6p, \qquad \Delta P_2 = 4(p+4)P_1, \qquad \Delta P_3 = 6\big((p+5)P_2 + P_1^2\big).$$

These formulas, derived in Appendix C.4.1, together with the product rule give

$$\nabla F = 4X(2B - R\,\mathrm{Id}), \qquad S = 64P_3 - 64RP_2 + 16R^3 = 64P_3 - 32RF - 16R^3, \tag{7.16}$$

and (7.12).

The characteristic equation of the 3×3 matrix B reduces P_4, P_5 to R, F, S. The resulting formulas in Appendix C.4.1 give the last two gradient entries of (7.13); Euler's identity supplies its first row. Thus Proposition 3.7 proves closure; conversely every Laplacian algebra containing R, F contains $S = \Gamma(F, F)$, proving the equality with $\mathcal{A}_R(F)$. On diagonal rectangular matrices with three distinct positive squared entries t_1, t_2, t_3, the Jacobian of (P_1, P_2, P_3) is a nonzero Vandermonde determinant times six. The triangular change in (7.16) is invertible, so R, F, S are algebraically independent.

Finally the displayed table gives

$$\mathcal{L}(F) = 4(5-p)RS - 64(R^2 + F)F.$$

At a matrix with $B = \mathrm{diag}(4, 1, 1)$, one has $F = 0$, $R = 6$, and $S = 768 > 0$. Thus the regular zero set can be minimal only for $p = 5$; for this value the curvature identity (1.3) proves minimality. The Laplacian in (7.12) proves the last assertion. For comparison, the member with $p = 6$ is harmonic and has the same gradient pairings, but its regular zero set is not minimal. □

7.4. **The boundary of gradient-square closure.** The quadratic compositions of the next section have a different three-generator structure:

Proposition 7.10 (An independent quartic generator)**.** *Let R, T, F be algebraically independent homogeneous polynomials of degrees $2, 4, 4$, with $R = |x|^2$, and put $S = |\nabla F|^2$. Suppose, for constants a, b, h with $b > 0$,*

$$S = 4R(aF + bT), \qquad \Delta S = 8(N+8)(aF + bT) + 8(a^2 + bh)R^2.$$

Then $\mathbb{R}[R, F, S]$ is not Laplacian, although

$$\mathbb{R}[R, F, S][R^{-1}] = \mathbb{R}[R, T, F][R^{-1}].$$

Proof. The degree-four part of $\mathbb{R}[R, F, S]$, whose displayed generators have degrees $2, 4, 6$, is contained in $\mathrm{span}\{R^2, F\}$. Algebraic independence places T outside this span. Its coefficient in ΔS is $8(N+8)b \neq 0$, so $\Delta S \notin \mathbb{R}[R, F, S]$. After adjoining R^{-1}, however, $T = b^{-1}(S/(4R) - aF)$, which proves both inclusions in the displayed equality. □

The orthogonal quartic blocks in Section 8 satisfy these identities by (8.25), with $\mathcal{A}_R(F) = \mathbb{R}[R, T, F]$. Thus the $(2, 4, 6)$ and $(2, 4, 4)$ closures differ as polynomial algebras despite agreeing after localization.

8. Quadratic Compositions and Quartic Hypercones

Starting with OT–FKM splitting quartics, we develop the two-block normal form, quadratic Jordan compositions, and a common theorem for orthogonal quartic data. Its realization table guides the quartic applications. A Gram sextic extends the construction, while higher-bidegree octonionic pullbacks exhibit transverse instability.

8.1. **OT–FKM splitting and the two-block normal form.** Allowing the two radii in the balanced OT–FKM sections of [21] to vary independently gives singular hypercones with a $(2,2,4)$ Laplacian algebra.

Let $x, y \in \mathbb{R}^\ell$. Choose skew-symmetric matrices $A_1, \dots, A_{m-1}$ satisfying

$$A_i A_j + A_j A_i = -2\delta_{ij}\,\mathrm{Id}\,.$$

Assume

$$2 \le m \le \ell - 2. \tag{8.1}$$

Only parameters for which this Clifford representation exists are included. In particular, A_1 is a complex structure, so ℓ is even. Set

$$u = |x|^2, \quad v = |y|^2, \quad R = u + v, \quad T = uv, \qquad Q = \langle x, y\rangle^2 + \sum_{i=1}^{m-1} \langle A_i x, y\rangle^2, \tag{8.2}$$

$$c = \frac{m-1}{\ell - 2}, \quad a = 1 - 2c, \quad b = c(1-c), \quad F = Q - cT. \tag{8.3}$$

Here $0 < c < 1$ and $b > 0$; write $C = \{F = 0\}$.

Remark 8.1 (Relation to the original OT–FKM cone). In these coordinates, the Cartan–Münzner OT–FKM polynomial [32] is

$$\Phi = R^2 - 2(u-v)^2 - 8Q = -R^2 + 8(1-c)T - 8F.$$

On $u = v$, the equation $F = 0$ is equivalent to $\Phi = (1-2c)R^2$. Without the balance condition it is instead $\Phi = -R^2 + 8(1-c)T$, whose normalized right-hand side varies with T/R^2. The codimension-two cone studied in [21] is exactly the balanced section $C \cap \{u = v\}$, whose only singular point is the vertex; the full hypercone has the two singular axes described below.

8.1.1. *Three generators of degrees* $(2,2,g)$. Quadratic Jordan closure [57, Proposition 37(b)] and Euler-operator bigrading [59, Proposition 19(a)] give a two-block, separately harmonic normal form for Laplacian algebras generated by two quadratics and one higher-degree polynomial. This includes the OT–FKM splitting quartics; higher-degree examples appear at the section's end.

Proposition 8.2. *Let $\mathcal{A} = \mathbb{R}[R, D, F]$ be a graded Laplacian algebra with independent quadratic generators R, D and a homogeneous generator F of degree $e > 2$, with $F \notin \mathbb{R}[R, D]$. After an orthogonal change of coordinates and a linear change of R, D, its quadratic part is generated by $u = |x|^2$, $v = |y|^2$, where $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$. There is a radial polynomial $H(u,v)$ such that $P = F - H$ is bihomogeneous of some bidegree (a,b), $a + b = e$, and $\Delta_x P = \Delta_y P = 0$.*

Suppose that $p, q \ge 2$ and both bidegrees are positive. Then closure is equivalent to identities

$$|\nabla_x P|^2 = a^2\big(K u^{a-1} v^b + L u^{a/2-1} v^{b/2} P\big), \tag{8.4}$$

$$|\nabla_y P|^2 = b^2\big(K u^{a} v^{b-1} + L u^{a/2} v^{b/2-1} P\big), \tag{8.5}$$

for constants $K > 0$ and $L \in \mathbb{R}$. The terms involving L are omitted unless a, b are both even. Necessarily

$$\frac{p-2}{a} = \frac{q-2}{b}. \tag{8.6}$$

In particular, P restricts to an isoparametric function on $\mathbb{S}^{p-1} \times \mathbb{S}^{q-1}$.

Proof. Write $D(z) = \langle Bz, z\rangle$, with B symmetric. The product rule gives $\Gamma(D, D) \in \mathcal{A}_2 = \operatorname{span}\{R, D\}$, so B^2 is a linear combination of B and the identity. Since D is independent of R, B has exactly two eigenvalues. Its orthogonal eigenspaces give u, v.

Put $\mathcal{B} = \mathbb{R}[u, v]$. The operator $x \cdot \nabla_x = \Gamma(u, \cdot)/2$ preserves $\mathcal{A}$, and $\mathcal{A}_e = \mathcal{B}_e + \mathbb{R}F$. Decomposing F by bidegree therefore shows that all its nonradial terms have the same

x-degree a. Subtract the other radial terms to obtain a bihomogeneous generator P_0 of bidegree (a,b). This algebra is now bigraded. The distinct bidegree components of ΔP_0 belong to $\mathcal{B}$, since they have degree less than e.

They vanish unless a,b are even. In the even mixed case write $\Delta_x P_0 = \lambda_x u^{a/2-1}v^{b/2}$ and $\Delta_y P_0 = \lambda_y u^{a/2}v^{b/2-1}$. Commuting these partial Laplacians gives $\lambda_x b(b+q-2) = \lambda_y a(a+p-2)$. Subtracting the appropriate multiple of $u^{a/2}v^{b/2}$ kills both. If one bidegree is zero, the subtraction only uses the active block. This proves the separately harmonic normal form.

The two bidegree components of $\Gamma(P,P)$ also lie in the algebra. Their degree is $2e-2$, so their expressions contain at most the first power of P. Matching bidegrees gives (8.4)–(8.5) initially with possibly different constant and linear coefficients. On the unit product spheres, let $f = P|_{\mathbb{S}^{p-1}\times\mathbb{S}^{q-1}}$. The separate tangential gradient squares are quadratics in f, with leading coefficients $-a^2$ and $-b^2$. Its mean is zero, by separate harmonicity. Its maximum $M>0$ and minimum $m<0$ are roots of both quadratics, since both partial gradients vanish at these points. Thus the two quadratics are respectively $a^2(K+Lf-f^2)$ and $b^2(K+Lf-f^2)$, with $K=-mM>0$ and $L=m+M$. This proves the common coefficients. Conversely, the two identities and the separate Euler and Laplace identities give every generator entry of the closure table, so Proposition 3.7 proves sufficiency.

If μ_2 is the average of f^2, integration by parts on each sphere gives

$$a^2K = a(2a+p-2)\mu_2, \qquad b^2K = b(2b+q-2)\mu_2.$$

This implies (8.6). Finally,

$$|\nabla f|^2 = (a^2+b^2)(K+Lf-f^2), \qquad \Delta f = -[a(a+p-2)+b(b+q-2)]f,$$

which are the isoparametric equations on the product. □

The block dimensions p,q need not agree; if $p=q>2$, (8.6) forces $a=b$. The change $F \mapsto F - H(u,v)$ preserves the algebra but generally changes its zero set, so the minimal level must still be determined within each explicit family. If one bidegree is zero, the active block reduces to [57, Proposition 37(c)], with an independent Euclidean factor adjoined.

8.2. **Quadratic Jordan algebras and the composition equation.** Quadratic-generated Laplacian algebras organize compositions of arbitrary even degree. For a real symmetric matrix A, write $q_A(x) = \langle Ax, x\rangle$. Then

$$\Delta q_A = 2\operatorname{tr} A, \qquad \Gamma(q_A, q_B) = 4q_{A\circ B}, \qquad A\circ B = \tfrac{1}{2}(AB+BA). \tag{8.7}$$

Thus a quadratic-generated algebra $\mathcal{A}$, containing $R=|x|^2$, is Laplacian exactly when its quadratic part $J=\mathcal{A}_2$ is a unital Jordan algebra under $f\circ g = \Gamma(f,g)/4$ [57, Proposition 37(b)]. The matrix realization matters: its multiplicities enter Δq_A. The Jordan classification gives Clifford and real, complex, or quaternionic Gram models; see [59, Theorems B–C, Section 3 and Appendix A].

The next proposition is the low-dimensional consequence of this classification; it concerns algebras generated entirely in degree two.

Proposition 8.3 (Three and four quadratic generators)**.** *Let $\mathcal{A}$ be generated by homogeneous quadratics, contain R, and be Laplacian. If $k=\dim \mathcal{A}_2$ is three or four, the following list is complete, up to orthogonal source coordinates and a linear change of quadratic generators.*

k	J	*Quadratic coordinates*
3	$\mathbb{R}^3$	$\lvert x_1\rvert^2, \lvert x_2\rvert^2, \lvert x_3\rvert^2$ *on three nonzero orthogonal blocks.*
3	$\operatorname{Sym}_2(\mathbb{R})$	$\lvert x\rvert^2, \lvert y\rvert^2, \langle x,y\rangle$, *with* $x,y\in\mathbb{R}^\ell$.
4	$\mathbb{R}^4$	*Four nonzero orthogonal block radii.*
4	$\mathbb{R}\oplus\operatorname{Sym}_2(\mathbb{R})$	*An independent block radius and the preceding real Gram coordinates.*
4	$\operatorname{Herm}_2(\mathbb{C})$	$\lvert z\rvert^2, \lvert w\rvert^2, \Re\langle z,w\rangle, \Im\langle z,w\rangle$, *with* $z,w\in\mathbb{C}^\ell$.

Here k is the number of minimal quadratic algebra generators; algebraic relations may occur among them.

Proof. The correspondence (8.7) embeds J as a unital Jordan subalgebra of the symmetric matrices. In the formally real Jordan classification recalled in [59, Appendix A], the simple factors of real dimension at most four are $\mathbb{R}$, $\mathrm{Sym}_2(\mathbb{R})$, and $\mathrm{Herm}_2(\mathbb{C})$, of dimensions $1, 3, 4$. The spin factors of dimensions three and four are the latter two matrix algebras. Taking direct sums gives precisely the displayed list. A primitive one-dimensional factor acts by an orthogonal projection; these projections split the source. The representation argument in [59, Section 3 and Appendix A] gives repeated standard real or complex modules for the remaining factors, hence their Gram coordinates. Finally, a degree-two element of a quadratic-generated algebra is a linear combination of its quadratic generators, proving the assertion about their minimal number. □

For the real two-column model use $R = |x|^2 + |y|^2$, $p = |x|^2 - |y|^2$, and $q = 2\langle x, y\rangle$. Its image is $\{R \geq 0 : p^2+q^2 \leq R^2\}$ for $\ell \geq 2$; for $\ell = 1$ only the boundary $R^2 = p^2+q^2$ is attained. The complex two-column model similarly has coordinates R, p, q_1, q_2, with $q_1 + iq_2 = 2\langle z, w\rangle$ and $p = |z|^2 - |w|^2$. Its image is $\{R \geq 0 : p^2 + q_1^2 + q_2^2 \leq R^2\}$ for $\ell \geq 2$; for $\ell = 1$ only its boundary is attained. These statements follow by realizing positive semidefinite 2×2 Gram matrices of the allowed rank.

An arbitrary quadratic span need not be closed: on $\mathbb{R}^3$, $\mathrm{span}\{R, x_1^2, x_1x_2\}$ omits $\Gamma(x_1x_2, x_1x_2) = x_1^2 + x_2^2$. Adjoining x_2^2 gives $\mathbb{R} \oplus \mathrm{Sym}_2(\mathbb{R})$. Every finite list has a unital matrix Jordan closure of dimension at most $N(N+1)/2$; restricting its size is essential, since all even-degree homogeneous polynomials are polynomials in the full collection x_ix_j.

Choose a basis $z = (z_1, \dots, z_k)$ of J and put

$$B_{ij}(z) = \tfrac{1}{4}\Gamma(z_i, z_j), \qquad b_i = \tfrac{1}{4}\Delta z_i,$$
$$A_P = \sum_{i,j} B_{ij}P_iP_j, \qquad \mathscr{D}P = \sum_{i,j} B_{ij}P_{ij} + \sum_i b_iP_i. \tag{8.8}$$

The entries of B are linear and the b_i are constant. The quotient calculus in Theorem 3.10 gives

$$|\nabla P(z)|^2 = 4A_P(z), \qquad \Delta P(z) = 4(\mathscr{D}P)(z), \qquad \mathcal{L}(P(z)) = 8\mathscr{T}(P)(z), \tag{8.9}$$

where

$$\mathscr{T}(P) = 2A_P\mathscr{D}P - \sum_{i,j} B_{ij}P_i\partial_jA_P. \tag{8.10}$$

These identities are evaluated on the actual generator image, also when the generators satisfy relations.

Proposition 8.4 (Composition criterion)**.** *Let P be a real homogeneous polynomial of degree d and set $F = P(z)$. Its regular zero set is minimal if and only if $\mathscr{T}(P) = 0$ on the part of the generator image where $P = 0$ and $A_P > 0$. In particular, $P \mid \mathscr{T}(P)$ is sufficient. It is also necessary if P is irreducible and has a regular zero in an open subset of the image where B is positive definite. In that case the quotient has degree $2d - 2$, while F has degree $2d$.*

Proof. The first assertion follows from (8.9) and the level-set mean-curvature identity. Positive definiteness makes the generator map a submersion and gives $A_P > 0$ at regular profile zeros. Minimality therefore makes $\mathscr{T}(P)$ vanish on a relatively open smooth real patch of $P = 0$. The patch-divisibility argument in Proposition 3.12 gives $P \mid \mathscr{T}(P)$. Its degree is $3d - 2$, which proves the last assertion. □

For orthogonal radii $u_i = |x_i|^2$, with $n_i = \dim V_i$, the data are

$$B = \mathrm{diag}(u_i), \qquad \mathscr{D}P = \sum_i u_iP_{ii} + \tfrac{1}{2}\sum_i n_iP_i. \tag{8.11}$$

Thus the criterion specializes to

$$P \;\Big|\; \Big(\sum_i u_i P_i^2\Big)\Big(2\sum_i u_i P_{ii} + \sum_i n_i P_i\Big) - \sum_i u_i P_i \partial_i \Big(\sum_j u_j P_j^2\Big), \tag{8.12}$$

under the irreducibility and interior-zero hypotheses above.

8.2.1. *Clifford profiles with a radial variable.* For a symmetric Clifford system on $\mathbb{R}^N$ with $s \geq 2$, let $p_i(x) = \langle P_i x, x\rangle$ for $1 \leq i \leq s$. The algebra $\mathbb{R}[R, p_1, \dots, p_s]$ has

$$B = \begin{pmatrix} R & p^T \\ p & R\,\mathrm{Id}_s \end{pmatrix}, \qquad \mathscr{D}P = R(P_{RR} + \Delta_p P) + 2\sum_i p_i P_{Rp_i} + \tfrac{N}{2} P_R. \tag{8.13}$$

This follows from (5.6); the real and complex two-column models have $(s, N) = (2, 2\ell)$ and $(3, 4\ell)$.

If a homogeneous profile $f(p)$ is independent of R, the identity (5.3) reads

$$\mathcal{L}_x(f(p)) = 16R^2(\mathcal{L}_p f)(p) - 8df(p)|\nabla_p f|^2(p), \qquad d = \deg f. \tag{8.14}$$

Thus a degree-d minimal target cone yields a degree-$2d$ minimal source equation, and a harmonic target gives a harmonic source. This is the Clifford specialization of Riedler's cone-pullback theorem [69, Theorem 5.1.4]; its subcalibration transfer is Theorem 5.4. In particular, minimal cubic targets supply sextic equations. Effective factors and the variational conclusions are determined by the corresponding results in Section 5.

Allowing the radial coordinate also includes the OT–FKM splitting in (8.3). For a homogeneous profile $P(t, y)$ of degree d, define

$$K_P = |\nabla_y P|^2 - P_t^2, \qquad \Box P = \Delta_y P - P_{tt}, \qquad \Gamma_L(P, K_P) = \nabla_y P \cdot \nabla_y K_P - P_t (K_P)_t.$$

On $F = P(R, p) = 0$, the same profile table gives

$$\begin{aligned} |\nabla F|^2 &= 4RK_P, \\ \mathcal{L}(F) &= 16R^2\left(K_P \Box P - \tfrac{1}{2}\Gamma_L(P, K_P)\right) + 8(N-2)RK_P P_t. \end{aligned} \tag{8.15}$$

For clarity, Euler's identity first gives, on the whole source,

$$|\nabla F|^2 = 4RK_P + 8dPP_t, \qquad \Delta F = 4R\Box P + (2N + 8d - 8)P_t.$$

Differentiate the first expression before setting $P = 0$; $\Gamma(F, R) = 4dF$ and the homogeneity of K_P, of degree $2d - 2$, then give (8.15). For $P = |y|^2 - ct^2$, $0 < c < 1$, this equation selects $c = 2(s-1)/(N-2)$, the familiar minimal isoparametric level. The splitting quartic instead has profile $\frac{1}{4}(\sum_{i=1}^m p_i^2 + cp_0^2 - cR^2)$, while Riedler's quartic is independent of R. Both therefore belong to the same composition calculus.

Remark 8.5 (An elementary radial profile). In the real two-column model, the radial variable gives quartic minimal levels: for $\ell > 2$ set

$$W = p^2 + q^2, \qquad c = \frac{1}{\ell - 1}, \qquad F = W - cR^2, \qquad \Phi = R^2 - 2W.$$

Equation (8.15) selects this c; $|\nabla\Phi|^2 = 16R^3$ and $\Delta\Phi = 8(\ell - 3)R$ identify $F = 0$ as its minimal isoparametric level. Its smallest algebra is $\mathbb{R}[R, F]$, since $W = F + cR^2$. This minimal level is covered by Section 6.3.

8.2.2. *Real and complex Gram profiles.* Let $X = [x_1 \;\cdots\; x_r] \in \mathbb{R}^{p\times r}$ and set $h_{ij} = \langle x_i, x_j\rangle$. The independent entries of $H = X^T X$ span $\mathrm{Sym}_r(\mathbb{R})$, with $R = \operatorname{tr} H$, and

$$\begin{aligned} \Delta h_{ij} &= 2p\delta_{ij}, \\ \Gamma(h_{ij}, h_{kl}) &= \delta_{ik}h_{jl} + \delta_{il}h_{jk} + \delta_{jk}h_{il} + \delta_{jl}h_{ik}. \end{aligned} \tag{8.16}$$

Indeed, $\nabla_{x_a} h_{ij} = \delta_{ai} x_j + \delta_{aj} x_i$; taking inner products and traces proves the table, including the diagonal entries. For $Z \in \mathbb{C}^{p\times r}$, use the real Euclidean metric and write $h_{ij} = \sum_a \overline{z_{ai}} z_{aj}$, so $H = Z^* Z$. The complex-bilinear extension of Γ satisfies

$$\Delta h_{ij} = 4p\delta_{ij}, \qquad \Gamma(h_{ij}, h_{kl}) = 2(\delta_{jk} h_{il} + \delta_{il} h_{kj}). \tag{8.17}$$

To verify it, use $\Delta = 4\sum_{a,b} \partial_{z_{ab}} \partial_{\bar{z}_{ab}}$ and $\Gamma(f,g) = 2\sum_{a,b} (f_{z_{ab}} g_{\bar{z}_{ab}} + f_{\bar{z}_{ab}} g_{z_{ab}})$. Taking real and imaginary parts gives the real table on $\mathrm{Herm}_r(\mathbb{C})$. These are the standard Gram realizations in [59, Section 3 and Section 7].

Their generator images consist of positive semidefinite Gram matrices of rank at most p. For $p \geq r$ the positive definite part is open and the Gram map is a submersion there: for a prescribed symmetric or Hermitian increment A, the variation $\dot{X} = \frac{1}{2} X H^{-1} A$, or $\dot{Z} = \frac{1}{2} Z H^{-1} A$, has $\dot{H} = A$. Consequently Proposition 8.4 applies directly to profiles with a regular zero in this positive definite part. For $p < r$ the same differential identities apply on the rank-constrained image. The constructions below select minimal profiles from these differential tables.

8.2.3. *Orthogonal quartic blocks and their weights.* The OT–FKM and Riedler quartics are quadratic profiles in Clifford coordinates and the radius. The spin-factor realization of Section 8.2 gives their common quartic table. The underlying Hurwitz matrix construction and its reduction to Clifford representations are described in [71, Chapter 0, pp. 2–3; Chapter 1, Lemma 1.5 and Proposition 1.7].

Let $\mathcal{C} = (P_0, \dots, P_m)$ be a symmetric Clifford system on $\mathbb{R}^{2\ell}$, with $m \geq 1$, and put

$$R = |z|^2, \qquad p_i(z) = \langle P_i z, z\rangle, \qquad p = (p_0, \dots, p_m).$$

Their differential table is (5.6), and Lemma 5.3 gives $|p|^2 \leq R^2$.

In the product $f \circ g = \Gamma(f,g)/4$, the quadratic space $J = \mathrm{span}\{R, p_0, \dots, p_m\}$ has multiplication

$$R \circ R = R, \qquad R \circ p_i = p_i, \qquad p_i \circ p_j = \delta_{ij} R. \tag{8.18}$$

Thus J is a spin-factor Jordan algebra. The Clifford relations give the following table for its quartic block norms.

Lemma 8.6 (Orthogonal quartic blocks)**.** *Partition $\{0, \dots, m\}$ into nonempty sets $I_1, \dots, I_s$, and set*

$$E_j = \sum_{i \in I_j} p_i^2 \quad (1 \leq j \leq s), \qquad E_* = R^2 - \sum_{i=0}^{m} p_i^2.$$

Then all these polynomials are nonnegative and

$$E_* + \sum_{j=1}^{s} E_j = R^2, \qquad \Gamma(E_i, E_j) = 16 R E_i \delta_{ij}, \tag{8.19}$$
$$\Delta E_j = 8|I_j| R, \qquad \Delta E_* = 8(\ell - m) R.$$

Here the pairing identity includes the index $*$.

Proof. For disjoint blocks the pairing in (5.6) gives $\Gamma(E_i, E_j) = 0$; for one block it gives $\Gamma(E_j, E_j) = 16 R E_j$ and $\Delta E_j = 8|I_j| R$. Also $\Gamma(R^2, E_j) = 16 R E_j$, so E_* is orthogonal to every E_j. Expanding its gradient square gives $\Gamma(E_*, E_*) = 16 R E_*$. Finally, $\Delta R^2 = 8(\ell + 1) R$ proves the asserted Laplacian of E_*. □

Remark 8.7 (Geometric interpretation). For $\ell > m + 1$, the Clifford map on the unit sphere has connected fibers and hemisphere quotient metric. If all $|I_j| \geq 2$, pulling back the target $\prod_j \mathrm{SO}(|I_j|)$-orbits gives the common quartic-block levels: this is Radeschi's composed foliation [68, Theorems A–B and Section 3]. The one-leaf target gives the OT–FKM foliation. In the splitting model the target $\mathrm{SO}(m)$-action fixes p_0; its leaves retain $u - v$, while the quartic subalgebra retains only $(u - v)^2$. The balanced $(3, 7, 7)$ model in Section 8.5.2 uses $\mathcal{C}_{9,1}$ and

target $\mathrm{SO}(3) \times \mathrm{SO}(7)$-orbits, with profile $E_+ = E_-$. A profile equation selects a union of leaves; the differential table requires no connectedness assumption.

For three quartic blocks, the weights in the table are the target block dimensions and the remaining weight $\ell - m$. The following criterion identifies every minimal linear combination with a regular zero meeting the positive part of this table.

Proposition 8.8 (Minimal levels of orthogonal quartic generators)**.** *Suppose that nonzero homogeneous quartic polynomials E_0, E_1, E_2 satisfy*

$$E_0 + E_1 + E_2 = R^2, \qquad \Gamma(E_i, E_j) = 16RE_i\delta_{ij}, \qquad \Delta E_i = 8r_iR \tag{8.20}$$

for constants $r_i > 0$. They generate a Laplacian algebra together with R, of generator degrees $(2, 4, 4)$ after eliminating one E_i. Let $F = \sum_i \alpha_i E_i$ have a regular zero point with all $E_i > 0$, and put $\sigma = \sum_i r_i\alpha_i$. Then $F = 0$ is minimal if and only if its nonzero coefficients have exactly two values $\alpha > 0 > \beta$, satisfying

$$\alpha(r_+ - 1) + \beta(r_- - 1) = 0, \qquad r_+ = \sum_{\alpha_i=\alpha} r_i, \quad r_- = \sum_{\alpha_i=\beta} r_i. \tag{8.21}$$

If $F = 0$ is minimal and none of the coefficients vanishes, $\mathbb{R}[R, F]$ is already Laplacian.

Proof. The norm identities give $E_i \geq 0$. Where they are positive, their gradient matrix is positive definite, so their image has interior. Set $A_j = \sum_i \alpha_i^j E_i$. The closure table gives

$$\Gamma(F, F) = 16RA_2, \qquad \mathcal{L}(F) = 128R^2(\sigma A_2 - A_3) - 64FA_2.$$

Thus minimality on the regular zero plane is equivalent to $\sigma A_2 - A_3 = \kappa F$ for a constant κ. The image contains an open piece of that plane, so comparison of its linear coefficients is valid. For each nonzero α_i this reads $\alpha_i(\sigma - \alpha_i) = \kappa$. There are at most two such values, and a zero point with positive E_i requires both signs. Their sum is σ, which is precisely (8.21). These conditions also make $\mathcal{L}(F)$ divisible by F globally and prove sufficiency. In the minimal case, if all coefficients are nonzero, then $A_2 = (\alpha + \beta)F - \alpha\beta R^2$; the displayed norm and Laplacian identities prove closure of $\mathbb{R}[R, F]$. □

Remark 8.9 (More quartic blocks). For any number of nonzero orthogonal quartic blocks summing to R^2, the same coefficient comparison permits only two nonzero coefficient values, provided the regular zero meets the positive-block region. Group equal positive, negative, and zero coefficients and add their weights. This reduces the minimal-level equation to (8.21), with two generators when no block is unused. Different representations can still yield different cones.

Remark 8.10 (Compatible Cartan–Münzner equations). Setting $\Phi_i = R^2 - 2E_i$ in (8.20) gives

$$\sum_i \Phi_i = R^2, \qquad |\nabla\Phi_i|^2 = 16R^3, \qquad \Delta\Phi_i = 4(N + 2 - 4r_i)R.$$

For distinct i, j, k it also gives $\Gamma(\Phi_i, \Phi_j) = -16R\Phi_k$. Conversely, the sum and the individual gradient identities imply these cross pairings; together with the displayed Laplacians they recover (8.20). These are compatible Cartan–Münzner-type equations, allowing degenerate quartic representations: for example, $E_i = (|x|^2 - |y|^2)^2$ with equal-dimensional blocks has $r_i = 1$.

For the quartic algebra $\mathcal{A} = \mathbb{R}[R, E_0, E_1, E_2]$ itself, $\mathcal{A}_2 = \mathbb{R}R$. In the Clifford construction above, it lies in a larger quadratic algebra. The criterion below determines its realizable weights.

Lemma 8.11 (Weights and image of the quartic blocks)**.** *For the nonzero blocks in (8.20), the weights are positive integers and $N = 2(r_0 + r_1 + r_2 - 1)$. Their spherical image is exactly*

$$(E_0, E_1, E_2)(\mathbb{S}^{N-1}) = \{(u_0, u_1, u_2) \in [0, 1]^3 : u_0 + u_1 + u_2 = 1\}. \tag{8.22}$$

Proof. Summing the Laplacian identities in (8.20) gives $N = 2(r_0 + r_1 + r_2 - 1)$. The norm identities imply $E_i \geq 0$. On the unit sphere, $u_i = E_i|_{\mathbb{S}^{N-1}}$ satisfies

$$|\nabla_{\mathbb{S}} u_i|^2 = 16u_i(1 - u_i), \qquad \Delta_{\mathbb{S}} u_i = 8r_i - 4(N + 2)u_i.$$

A constant u_i would be zero or one, contradicting the assumption that all three blocks are nonzero. Its minimum and maximum are therefore zero and one. Integration of the Laplacian identity gives $r_i > 0$. At a nonzero zero of E_i, write $H_i = \operatorname{Hess} E_i$. Comparing quadratic terms in $|\nabla E_i|^2 = 16RE_i$ gives

$$H_i^2 = 8RH_i, \qquad \operatorname{tr} H_i = 8r_i R.$$

Thus $H_i = 8RP_i$ for an orthogonal projection P_i of rank r_i; in particular, each weight is an integer.

Nonnegativity and the sum of the blocks place the spherical image in the displayed simplex. Its intersection with the simplex interior is nonempty, since $E_0E_1E_2$ is a nonzero polynomial. There the spherical gradient matrix $16(\operatorname{diag}(u) - uu^{\mathsf{T}})$ has rank two, so the intersection is relatively open by the submersion theorem. It is also relatively closed, since the spherical image is compact. Connectedness of the simplex interior, followed by taking its closure, proves (8.22). □

The Clifford-module restriction gives an exact numerical range for (8.20). We use the conventional function

$$(\delta(1), \ldots, \delta(8)) = (1, 2, 4, 4, 8, 8, 8, 8), \qquad \delta(k + 8) = 16\delta(k).$$

Here $\delta(k)$ is the real irreducible module dimension for the negative Clifford algebra on $k - 1$ generators. A symmetric Clifford system with $p \geq 2$ matrices on $\mathbb{R}^{2\ell}$ exists exactly when $\delta(p - 1) \mid \ell$ [32, Section 3.5]. See also [71, Chapter 0, pp. 3–4] for the equivalent Hurwitz–Radon restriction.

Proposition 8.12 (Realizable quartic weights)**.** *Let r_0, r_1, r_2 be positive integers, write their rearrangement as $a \leq b \leq c$, and set $\ell = a + b + c - 1$. Nonzero quartics satisfying* (8.20) *on $\mathbb{R}^{2\ell}$ exist if and only if $\delta(a + b - 1) \mid \ell$. Whenever this condition holds, a realization is obtained from one Clifford system by two blocks of squared quadratic coordinates and their complement in R^2.*

Construction and parameter restrictions.

Assume the divisibility condition. Take $a + b$ symmetric Clifford matrices, with quadratic coordinates $p_1, \ldots, p_{a+b}$. The three polynomials

$$\sum_{\alpha=1}^{a} p_\alpha^2, \qquad \sum_{\alpha=a+1}^{a+b} p_\alpha^2, \qquad R^2 - \sum_{\alpha=1}^{a+b} p_\alpha^2$$

have weights (a, b, c) by Lemma 8.6. Their positive Laplacian coefficients ensure that none is zero.

For any realization, summing the Laplacians forces $N = 2(r_0 + r_1 + r_2 - 1)$. Positivity and integrality of the weights follow from Lemma 8.11. The converse uses the completed classification with four principal curvatures and compatibility of the three blocks. See Appendix D and its weight-one, Clifford-joining, and exceptional-triple lemmas Lemmas D.1 to D.3. □

This is a classification of realizable weights, not of simultaneous orthogonal equivalence classes of triples. For example, $(3, 3, 5)$ is excluded since $\delta(5) = 8 \nmid 10$, although the individual Cartan–Münzner multiplicity pairs $(2, 7), (2, 7), (4, 5)$ all occur. The criterion is symmetric in the weights; the subcalibration range in Theorem 8.15 instead depends on which two blocks are chosen for the defining polynomial.

Corollary 8.13 (The weights with active sum nine)**.** *Let r_0, r_1, r_2 be positive integers with $r_1, r_2 \geq 2$ and $r_1 + r_2 = 9$. A quartic realization exists if and only if r_0 is a positive multiple of eight. Its ambient dimension is $N = 2r_0 + 16$.*

Proof. Up to exchanging the active weights, their pairs are $(2,7)$, $(3,6)$, and $(4,5)$. For $1 \leq r_0 \leq 7$, none of the corresponding module dimensions divides $\ell = r_0 + 8$; the seven checks are listed in Table 3. If $r_0 \geq 8$, the two smallest weights are the active pair. Hence $a + b - 1 = 8$ and the same criterion is $8 = \delta(8) \mid r_0 + 8$, equivalently $8 \mid r_0$. □

8.3. **Subcalibrations from orthogonal quartic data.** The orthogonal-quartic table determines the minimal level, singular set, and subcalibration of every realization in the following sufficient minimizing range. Exact thresholds for particular families also use the instability arguments below.

Throughout this subsection, let E_0, E_1, E_2 be nonzero homogeneous quartic polynomials on $\mathbb{R}^N$ satisfying (8.20), with $R = |x|^2$. Suppose $r_1, r_2 \geq 2$, and put

$$h = r_1 + r_2, \qquad \rho = r_0 + r_1 + r_2,$$

$$c = \frac{r_1 - 1}{h - 2}, \qquad a = 1 - 2c, \qquad b = c(1-c) > 0, \tag{8.23}$$

$$T = \frac{E_1 + E_2}{4}, \qquad F = \frac{(1-c)E_1 - cE_2}{4}. \tag{8.24}$$

By Lemma 8.11, $N = 2\rho - 2$. The choice of c is the minimal-level condition (8.21). Direct substitution gives

$$(\Gamma(I_i, I_j))_{I=(R,T,F)} = \begin{pmatrix} 4R & 8T & 8F \\ 8T & 4RT & 4RF \\ 8F & 4RF & 4R(aF + bT) \end{pmatrix}, \tag{8.25}$$

$$(\Delta R, \Delta T, \Delta F) = (2N, 2hR, 2aR).$$

Thus the gradient table is independent of r_0, while the ambient dimension records all three weights. By Proposition 3.7, $\mathbb{R}[R,T,F]$ is a Laplacian algebra. It is exactly $\mathcal{A}_R(F)$: writing $S = |\nabla F|^2$, one obtains

$$S = 4R(aF + bT), \qquad \Delta S = 8(N+8)(aF + bT) + 8(a^2 + bh)R^2.$$

Since $b > 0$, the second identity recovers T in $\mathcal{A}_R(F)$. The generators are algebraically independent. Indeed, the nonzero polynomial $E_0E_1E_2$ is positive somewhere; there the gradients of the three blocks are independent, and the change to (R,T,F) is locally invertible. Fixing $R = 1$ therefore leaves spherical quotient dimension two.

Lemma 8.14 (Geometry determined by the quartic data)**.** *At each fixed $R \geq 0$, the full image of (T,F) is described by*

$$0 \leq T \leq R^2/4, \qquad -cT \leq F \leq (1-c)T. \tag{8.26}$$

The critical set of F and the geometric singular set of $C = \{F = 0\}$ are both $Z = \{T = 0\}$, and $\dim Z \leq N - h$. The hypersurface C is minimal outside Z, both phases approach every point of C, and $\partial[\![\{F < 0\}]\!]$ has support C.

Proof. The simplex image in Lemma 8.11 and homogeneity prove (8.26). Moreover,

$$|\nabla F|^2 = R\big((1-c)^2E_1 + c^2E_2\big),$$

so the critical set of F is exactly Z. The reduced table gives

$$\mathcal{L}(F) = -8F\{b(R^2 + 2T) + 2aF\}, \tag{8.27}$$

which proves regular minimality.

Nonzero points of Z exist because E_0 attains one on the unit sphere. Fix such a point p and write $R_0 = R(p)$ and $\operatorname{Hess} E_i(p) = 8R_0P_i$ for $i = 1, 2$, using the projections obtained

in the proof of Lemma 8.11. The quadratic terms of $\Gamma(E_1, E_2) = 0$ give $P_1P_2 + P_2P_1 = 0$. Taking the trace shows $P_1P_2 = 0$. Hence

$$\operatorname{Hess} T(p) = 2R_0(P_1 + P_2), \qquad \operatorname{Hess} F(p) = 2R_0\big((1-c)P_1 - cP_2\big).$$

The first Hessian has rank h. Applying the implicit function theorem to h independent components of ∇T places Z locally in a smooth submanifold of codimension h. Consequently $\dim Z \le N - h$ and $\mathcal{H}^{N-1}(Z) = 0$. The second Hessian has positive and negative indices r_1, r_2, respectively, so both phases occur arbitrarily near every such p. If the zero set were a smooth hypersurface at p, its leading quadratic term would vanish on its tangent hyperplane and thus have rank at most two. Its actual rank is $h \ge 4$, a contradiction. Thus these points are geometrically singular.

Homogeneity gives the same two-phase conclusion at the origin, and regularity gives it on $C \setminus Z$. Polynomial finite perimeter and the local BV Poincaré inequality then identify the support of the boundary current with C. The vertex is singular as well: a cone smooth at its vertex would be a hyperplane, whereas C has nonzero singular points. □

Theorem 8.15 (Polynomial subcalibrations from orthogonal quartics)**.** *For the data* (8.20) *and* (8.23)–(8.24)*, suppose that either*

$$h \ge 9, \qquad \text{or} \qquad r_1 = r_2 = 4 \quad \text{and} \quad r_0 \ge 5.$$

There exists a homogeneous polynomial $G = AF$ of degree at most ten, with $A > 0$ on $T > 0$, whose critical set is $Z = \{T = 0\}$ and whose normalized gradient satisfies

$$\operatorname{sgn}(G)\ \operatorname{div} X_G > 0 \qquad \text{off } C = \{F = 0\}.$$

The current $\partial[\![\{F < 0\}]\!]$ has full support C and is strictly area minimizing. Its regular part is strictly stable, with the explicit lower bound

$$Q_C(\varphi) \ge \left[\frac{(N-17)^2}{4} + 8h - 72\right] \int_C r^{-2}\varphi^2, \qquad \varphi \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}), \quad r = \sqrt{R}. \tag{8.28}$$

8.3.1. *Selecting the multiplier.* At $p \in Z \setminus \{0\}$, (8.27) gives $\lambda_F(p) = -8bR(p)^2 < 0$. Hence Lemma 4.3 forces every admissible smooth multiplier to vanish on Z. For $A \in \mathbb{R}[R, T]$, scaling a nonzero point of Z gives $A(R, 0) = 0$ for all $R > 0$, so T divides A as a polynomial in R, T. The lowest two nonzero homogeneous choices are therefore T and RT, up to a positive constant.

These two candidates suffice in the stated ranges:

$$G = \begin{cases} TF, & h \ge 11, \text{ or } h = 10 \text{ and } r_1, r_2 \ge 3, \\ RTF, & h = 9, \text{ or } \{r_1, r_2\} = \{2, 8\}, \\ RTF, & r_1 = r_2 = 4,\ r_0 \ge 5. \end{cases} \tag{8.29}$$

With $s = T/R^2$ and $z = F/T$, these are $R^4 s^2 z$ and $R^5 s^2 z$: the radial factor changes the defect but preserves vanishing on Z. We use Theorems 3.10 and 4.2 and, for $h = 9$, the realizability restriction Corollary 8.13.

8.3.2. *The degree-eight potential and its range.* For $G_8 = TF$, the reduced table gives

$$|\nabla G_8|^2 = 4RT(bT^2 + aTF + 3F^2), \qquad \Delta G_8 = R\{2aT + 2(h+4)F\}, \tag{8.30}$$

$$\begin{aligned} \mathcal{L}(G_8) = 8TF\big\{&bT^2[(h-8)R^2 - 4T] \\ &+ aTF[(h-4)R^2 - 4T] + 3F^2[hR^2 - 4T]\big\}. \end{aligned} \tag{8.31}$$

Introduce the dimensionless variables

$$s = T/R^2 \in (0, 1/4], \qquad z = F/T \in [-c, 1-c].$$

The braces in (8.31), divided by R^2T^2, are

$$p_8(z, s) = b(h - 8 - 4s) + a(h - 4 - 4s)z + 3(h - 4s)z^2. \tag{8.32}$$

Since

$$b + az = (1-c)^2(z+c) + c^2(1-c-z) > 0,$$

we have $\partial_s p_8 = -4(b + az + 3z^2) < 0$. It is enough to prove

$$q(z) := p_8(z, 1/4) = b(h-9) + a(h-5)z + 3(h-1)z^2 > 0. \tag{8.33}$$

First take one of the degree-eight cases in (8.29), so $h \geq 10$. Interchanging E_1, E_2 preserves the problem, so we may assume $1/(h-2) \leq c \leq 1/2$ and $a \geq 0$. For $z \geq 0$, (8.33) is positive. On $[-c, 0]$ its vertex is $z_* = -a(h-5)/(6(h-1))$, which lies in this interval exactly when

$$c \geq c_* := \frac{h-5}{8(h-2)}.$$

If $c < c_*$, the minimum is attained at $-c$, where $q(-c) = 4c((h-1)c - 1) > 0$. If $c \geq c_*$, the minimum is positive exactly when

$$D_8(c) := 12(h-1)(h-9)c(1-c) - (h-5)^2(1-2c)^2 > 0. \tag{8.34}$$

This expression increases on $0 < c \leq 1/2$. For $h \geq 13$,

$$D_8(c_*) = \frac{3(h-1)(h-5)(h^2-14h+21)}{4(h-2)^2} > 0.$$

For $h = 12, 11$, the bound $c \geq 1/(h-2) > c_*$ gives, respectively,

$$D_8(1/10) = 107/25 > 0, \qquad D_8(1/9) = 52/27 > 0.$$

For $h = 10$ with $r_1, r_2 \geq 3$, one has $c \geq 1/4 > c_*$ and $D_8(1/4) = 14 > 0$. This proves positivity in every degree-eight case of (8.29).

The same candidate does not give a uniform result for $h = 10$. For $c = 1/8$, hence $a = 3/4$ and $b = 7/64$, one has

$$p_8(-1/16, 1/4) = -5/256 < 0. \tag{8.35}$$

By Lemma 8.14, this point is attained in every realization with $(r_1, r_2) = (2, 8)$. Thus TF has the wrong divergence sign at points of the complement. For $h = 9$, one has $q(z) = 4az + 24z^2$ with $a \neq 0$, so q is negative for small z of the opposite sign. For $r_1 = r_2 = 4$, one instead has $q(z) = -b + 21z^2$, negative near zero. The exact image assertion again realizes these negative values with $F \neq 0$. Hence, within the theorem's range, the degree-eight choices in (8.29) are precisely the cases in which TF has the required strict sign.

8.3.3. *The degree-ten potential in the remaining ranges.* For $G_{10} = RTF$, the same table gives

$$\mathcal{L}(G_{10}) = 8R^5T^3F\, p_{10}(z, s), \tag{8.36}$$

where

$$\begin{aligned} p_{10}(z,s) &= b[h - 8 + (2h-23)s] + a[h - 4 + (2h-14)s]z \\ &\quad + [3h + (15h-33)s + (18h+27)s^2]z^2 \\ &\quad + 2(r_0 - 1)s\big[b + az + 3(1+3s)z^2\big]. \end{aligned} \tag{8.37}$$

Because $r_0 \geq 1$ and $b + az > 0$ on the image interval, the last line is nonnegative. Thus increasing the unused block weight r_0 can only improve this certificate.

For $h = 9$, Corollary 8.13 gives $r_0 \geq 8$. At $r_0 = 8$, the full polynomial is

$$p_{10}(z, s) = b(1 + 9s) + a(5 + 18s)z + (27 + 144s + 315s^2)z^2. \tag{8.38}$$

Since $c \in [1/7, 6/7]$, one has $a^2 \leq 25/49$ and $4b \geq 24/49$. The negative discriminant is therefore bounded below by

$$\begin{aligned} &4b(1+9s)(27 + 144s + 315s^2) - a^2(5 + 18s)^2 \\ &\qquad \geq \frac{23 + 4788s + 30564s^2 + 68040s^3}{49} > 0. \end{aligned} \tag{8.39}$$

The leading coefficient is positive, so (8.38) is positive for all real z and $0 < s \le 1/4$. Increasing r_0 preserves positivity on the image interval.

When $h = 10$, the first three terms are

$$b(2-3s) + a(6+6s)z + (30+117s+207s^2)z^2.$$

Here $c \in [1/8, 7/8]$, so $a^2 \le 9/16$ and $b = (1-a^2)/4$. The negative of the discriminant is bounded below by

$$\begin{aligned} &4b(2-3s)(30+117s+207s^2) - a^2(6+6s)^2 \\ &\qquad \ge 6 + \frac{45}{2}s + \frac{117}{16}s^2 - \frac{4347}{16}s^3 \ge \frac{1797}{1024} > 0. \end{aligned}$$

The final inequality follows from $0 < s \le 1/4$ by discarding the positive linear and quadratic terms. The leading coefficient is positive, so this polynomial is positive for all real z and all allowed s.

Finally, for $r_1 = r_2 = 4$ one has $h = 8$, $a = 0$, and $b = 1/4$. The full polynomial becomes

$$p_{10}(z,s) = \frac{2r_0 - 9}{4}s + \big[24 + (81+6r_0)s + (153+18r_0)s^2\big]z^2. \tag{8.40}$$

This is positive for every $s > 0$ if $r_0 \ge 5$. Together these calculations prove the required sign in every case of (8.29).

8.3.4. *Global minimality and the Hardy bound.*

Completion of the proof of Theorem 8.15.

Both multipliers are positive on $T > 0$. Euler's identity excludes critical points where $F \neq 0$, and $|\nabla F|^2 = 4bRT$ excludes them on $F = 0$, $T > 0$. On Z, both potentials and their gradients vanish. Thus they have critical set Z and the same phases as F. The full-support and dimension conclusions of Lemma 8.14 verify the hypotheses of Corollary 2.8 and Theorem 2.5. Their comparison, followed by Corollary 2.13, gives strict integral-current mass minimization and the gap $\Theta\varepsilon^{N-1}$.

For stability, (4.11) gives

$$\beta_{TF} = 2(h-8)\frac{R^2}{T} - 8, \qquad \beta_{RTF} = 2(h-8)\frac{R^2}{T} + 4\rho - 50.$$

Since $R^2/T \ge 4$ and $h \ge 8$, the homogeneous Hardy estimate Theorem 4.6, with cone dimension $N - 1 = 2\rho - 3$, gives the same lower bound for both potential degrees:

$$\left(8 - \frac{N-1}{2}\right)^2 + 8(h-8) - 8 = \frac{(N-17)^2}{4} + 8h - 72,$$

$$\left(10 - \frac{N-1}{2}\right)^2 + 8(h-8) + 4\rho - 50 = \frac{(N-17)^2}{4} + 8h - 72.$$

For $h \ge 10$ this coefficient is positive. For $h = 9$, Corollary 8.13 gives $N = 2r_0 + 16 \ge 32$, so the coefficient $(N-17)^2/4$ is positive as well. In the remaining case, $h = 8$ and $r_0 \ge 5$ imply $N \ge 24$, so it is at least $17/4$. This proves (8.28). □

For the splitting specialization $r_0 = 1$, $h = \ell$, one has $N = 2\ell$, and the Hardy coefficient becomes $\ell^2 - 9\ell + 1/4$. For $r_0 > 1$ and unequal active weights, Proposition 8.12 determines realizability, and the inequalities above give a sufficient minimizing range.

8.3.5. *The balanced specialization.*

Corollary 8.16 (The degree-eight potential for balanced quartics)**.** *Suppose that the three nonzero quartics in* (8.20) *have weights* (r_0, ν, ν)*, with* $\nu \ge 5$*. Set*

$$F = \frac{E_+ - E_-}{4}, \qquad W = \frac{E_+ + E_-}{4}, \qquad G = WF.$$

The conclusions of Theorem 8.15 hold for G, with $\Sigma = \{W = 0\}$. In this normalization, $\Delta F = 0$, $|F| \le W \le R^2/4$, and

$$\mathcal{A}_R(F) = \mathbb{R}[R, W, F] \tag{8.41}$$

is a polynomial algebra with generator degrees $(2, 4, 4)$. Moreover,

$$|\nabla G|^2 = 4RW(W^2 + 3F^2), \qquad \Delta G = 4(\nu + 2)RF, \tag{8.42}$$

$$\mathcal{L}(G) = 16FW\left\{W^2\left[(\nu - 4)R^2 - 2W\right] + 3F^2\left[\nu R^2 - 2W\right]\right\}. \tag{8.43}$$

Proof. Take $T = W$, $F_{\mathrm{gen}} = F/2$, $h = 2\nu \ge 10$, and $c = 1/2$ in Theorem 8.15. Its table becomes

$$\Delta(R, W, F) = (2N, 4\nu R, 0), \qquad (\Gamma(I_i, I_j))_{I=(R,W,F)} = \begin{pmatrix} 4R & 8W & 8F \\ 8W & 4RW & 4RF \\ 8F & 4RF & 4RW \end{pmatrix}. \tag{8.44}$$

Substituting this normalization in (8.30)–(8.31) gives (8.42)–(8.43). □

The following table organizes the applications of the common quartic theorem. The splitting, Riedler, and weight-two families have exact thresholds, obtained by combining this theorem with instability arguments. Each construction also identifies its singular set, including the real rank-one strata in the weight-two families.

TABLE 2. Euclidean quartic realizations of Theorem 8.15. Only parameters admitting the specified Clifford realization are included. The intrinsic Simons-cone result in Theorem 8.24 uses a separate differential table.

Family	(r_0, r_1, r_2)	Minimizing parameters used here
OT–FKM splitting	$(1, m, \ell - m)$	$\ell \ge 10$, $2 \le m \le \ell - 2$
Riedler pullback	$(N/2 - 2k + 1, k, k)$	$N \ge 32$, $k \ge 4$
Hopf pairing	$(1, 8, 8)$	Ambient dimension 32
Balanced Clifford example	$(3, 7, 7)$	Ambient dimension 32
Additional block example	$(2, 5, 10)$	Ambient dimension 32
Weight-two families	$(2, m, 4k - m - 1)$	$m = 2, 3$, $k \ge 3$
Unequal Lawson pullback	$(22, 5, 6)$	Ambient dimension 64

8.4. **The splitting, Hopf and Riedler families.** We apply the common theorem to the splitting construction, the Hopf pairing and Riedler's pullbacks. The splitting and Riedler families admit complete variational classifications. The balanced Hopf section also illustrates intrinsic comparison inside a Simons cone, and the degree-nine Riedler potential supplies an additional Hardy improvement.

8.4.1. *The OT–FKM splitting criterion.* For the data (8.2)–(8.3), the exact minimizing threshold is as follows.

Theorem 8.17. *The hypersurface $C = \{F = 0\} \subset \mathbb{R}^{2\ell}$ is minimal on its regular part, and*

$$\operatorname{Sing} C = (\mathbb{R}^\ell \times \{0\}) \cup (\{0\} \times \mathbb{R}^\ell).$$

The oriented boundary $\partial[\![\{F < 0\}]\!]$ is area minimizing if and only if $\ell \ge 10$. In all these minimizing cases it is strictly area minimizing in the sense of Definition 2.1. The polynomial potentials and strict-stability bound of Theorem 8.15 apply, with degrees ten for $\ell = 10$ and eight for $\ell \ge 12$. For the remaining admissible dimensions $\ell = 4, 6, 8$, the regular part of C is unstable.

Closure and singularities. For $x \neq 0$, the vectors $x, A_1x, \dots, A_{m-1}x$ are orthogonal and have length $|x|$. Consequently $0 \leq Q \leq T$. Direct differentiation gives, in the variables $I = (u, v, Q)$,

$$(\Gamma(I_i, I_j)) = \begin{pmatrix} 4u & 0 & 4Q \\ 0 & 4v & 4Q \\ 4Q & 4Q & 4RQ \end{pmatrix}, \qquad \Delta I = (2\ell, 2\ell, 2mR). \tag{8.45}$$

For example, $\nabla_y Q = 2\sum_{i=0}^{m-1} \langle A_i x, y\rangle A_i x$, with $A_0 = \mathrm{Id}$, so $|\nabla_y Q|^2 = 4uQ$; differentiating in x gives $4vQ$. By Proposition 3.7,

$$\mathcal{A} = \mathbb{R}[u, v, Q] = \mathbb{R}[R, u - v, F] \tag{8.46}$$

is a graded Laplacian algebra, with generator degrees $(2, 2, 4)$. The determinant of its gradient matrix is $64RQ(T - Q)$, which is positive at points with $u, v > 0$ and $0 < Q < T$. These points exist by choosing the components of y both in and perpendicular to the span above. Thus the three generators are algebraically independent; the spherical quotient dimension is two. The separately harmonic normal form of Proposition 8.2 is $P = Q - (m/\ell)uv$; the minimal polynomial $F = Q - cuv$ differs from it by a radial term in the two blocks.

With $p_0 = u - v$ and $p_{i+1} = 2\langle A_i x, y\rangle$ for $A_0 = \mathrm{Id}$, the matrices in Remark 8.18 form a symmetric Clifford system, and

$$F = \frac{1}{4}\left(\sum_{i=1}^{m} p_i^2 + cp_0^2 - cR^2\right).$$

Its blocks $E_0 = p_0^2$, $E_1 = 4Q$, and $E_2 = 4(T - Q)$ have weights $(1, m, \ell - m)$. The substitutions in (8.3) therefore give (8.25) with $h = \ell$ and $N = 2\ell$. The calculation after (8.25) recovers $\mathcal{A}_R(F) = \mathbb{R}[R, T, F]$, of degrees $(2, 4, 4)$. Independence follows since $(u, v, Q) \mapsto (u + v, uv, Q - cuv)$ has Jacobian $u - v$. Both algebras have spherical quotient dimension two, but (8.46) also retains the sign of $u - v$.

The geometry follows from Lemma 8.14. For $\ell \geq 10$, evenness of ℓ and Theorem 8.15 give the variational conclusions. The remaining dimensions are excluded by the following local model and angular calculation.

Fix $x_0 \neq 0$, and write $W_{x_0} = \mathrm{span}\{x_0, A_1x_0, \dots, A_{m-1}x_0\}$. The first nonzero term of F at $(x_0, 0)$ is

$$|x_0|^2\big((1 - c)|y_\parallel|^2 - c|y_\perp|^2\big), \quad y_\parallel \in W_{x_0}, \quad y_\perp \in W_{x_0}^\perp. \tag{8.47}$$

The tangent cone is the product of $\mathbb{R}^\ell$ with this nonlinear Lawson cone. Its transverse link is

$$\mathbb{S}^{m-1}(\sqrt{c}) \times \mathbb{S}^{\ell-m-1}(\sqrt{1 - c}) \subset \mathbb{S}^{\ell-1}.$$

Both multiplicities are positive by (8.1). Smooth local orthonormal frames for W_x and $W_x^\perp$ identify a neighborhood of the stratum with a smoothly varying family of these quadratic cones. The same description holds on the other coordinate subspace.

Remark 8.18 (Clifford coordinates of the OT–FKM splitting). Put $A_0 = \mathrm{Id}$. The symmetric Clifford system

$$P_0 = \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & -\mathrm{Id} \end{pmatrix}, \qquad P_{i+1} = \begin{pmatrix} 0 & A_i^T \\ A_i & 0 \end{pmatrix} \quad (0 \leq i \leq m - 1)$$

has coordinates $p_0 = u - v$ and $p_{i+1} = 2\langle A_i x, y\rangle$. The common geometric interpretation is given in Section 8.2.3.

8.4.2. *The unstable OT–FKM splitting dimensions.* For $\ell = 4, 6$, the transverse Lawson cone in (8.47) is unstable: its link has dimension $\ell - 2$ and constant squared curvature $\ell - 2$, giving the Hardy margin

$$\frac{(\ell-3)^2}{4} - (\ell-2) < 0. \tag{8.48}$$

Lemma 2.15 transfers a negative variation from this transverse cone to C_{reg}, using the local model (8.47).

For $\ell = 8$, the transverse Lawson cone can be area minimizing; instability follows instead from the angular calculation below.

Lemma 8.19. *On the regular part of the quartic cone,*

$$|A_C|^2 = (\ell-2)\frac{R}{T} + \frac{2(\ell-1)}{R}. \tag{8.49}$$

On $\Lambda = C \cap \mathbb{S}^{2\ell-1}$, put $t = T|_\Lambda$. Then

$$|A_\Lambda|^2 = \frac{\ell-2}{t} + 2(\ell-1), \qquad |\nabla_\Lambda t|^2 = 4t(1-4t), \tag{8.50}$$
$$\Delta_\Lambda t = 2(\ell-1) - 8\ell t.$$

Proof. Put $W = |\nabla F|^2$ and $\nu = \nabla F/\sqrt{W}$. The Euclidean Bochner identity gives $|\nabla^2 F|^2 = \Delta W/2 - \Gamma(F, \Delta F)$. On $F = 0$, (8.25) yields

$$\begin{aligned} W &= 4bRT, \\ |\nabla^2 F|^2 &= 4(a^2 + b\ell)R^2 + 8b(\ell+4)T, \\ |\nabla\sqrt{W}|^2 &= 4[(a^2+b)R^2 + 5bT], \\ \nabla^2 F(\nu,\nu) &= \Delta F = 2aR. \end{aligned}$$

Now use

$$|A_C|^2 = \frac{|\nabla^2 F|^2 - 2|\nabla\sqrt{W}|^2 + (\nabla^2 F(\nu,\nu))^2}{W}$$

to obtain (8.49). The normal Hessian of any polynomial V can also be computed from the closure table, using

$$\operatorname{Hess} V(\nu,\nu) = \frac{\Gamma(F,\Gamma(F,V)) - \frac{1}{2}\Gamma(V,W)}{W}.$$

On $F = 0$, this gives

$$\Gamma(T,F) = 0, \qquad \operatorname{Hess} T(\nu,\nu) = 2R - \frac{4T}{R}, \qquad \Delta_C T = 2(\ell-1)R + \frac{4T}{R}.$$

Subtracting the radial contribution of the degree-four function T on the $(2\ell-1)$-dimensional cone gives the last identity in (8.50). Subtracting the squared radial derivative from $\Gamma(T,T) = 4RT$ gives the gradient identity. □

For $\ell = 8$, the positive function $\phi = t^{-1}$ satisfies

$$(\Delta_\Lambda + |A_\Lambda|^2)\phi = 46\phi. \tag{8.51}$$

To apply this identity, we cut off near the singular set of the link. Its two singular strata have dimension seven and transverse dimension seven in Λ. The local quadratic-cone description after (8.47) implies $t \asymp \rho^2$ and transverse volume density comparable to $\rho^6\, d\rho$, where ρ is distance to the singular stratum. Thus $\phi \asymp \rho^{-2}$, its gradient is $O(\rho^{-3})$, and ϕ, its gradient and the potential term have the required finite integrals.

Choose cutoffs η_ε vanishing for $\rho < \varepsilon$ and equal to one for $\rho > 2\varepsilon$, with gradient bounded by $O(\varepsilon^{-1})$. Then

$$\int_\Lambda \phi^2 |\nabla\eta_\varepsilon|^2 = O(\varepsilon^{-4}\varepsilon^{-2}\varepsilon^7) = O(\varepsilon) \longrightarrow 0.$$

Multiplying (8.51) by $\eta_\varepsilon^2\phi$ and integrating by parts on the regular part proves that the angular Rayleigh quotients of $\eta_\varepsilon\phi$ tend to -46. The cone has dimension fifteen, so (4.16) gives compactly supported regular variations whose scale-invariant quotients tend to

$$\frac{(15-2)^2}{4} - 46 = -\frac{15}{4} < 0. \tag{8.52}$$

This proves instability. Together with (8.48) and the evenness of ℓ, it completes Theorem 8.17.

8.4.3. *The Hopf pairing and its balanced section.* The Hopf pairing realizes weights $(1,8,8)$ and has two linear singular strata. Its balanced section also gives an intrinsic comparison inside a Simons cone.

Let $h : \mathbb{R}^{16} \to \mathbb{R}^9$ be the quadratic extension of the standard octonionic Hopf map $\mathbb{S}^{15} \to \mathbb{S}^8$, normalized by

$$|h(x)|^2 = |x|^4, \qquad Dh(x)Dh(x)^{\mathsf{T}} = 4|x|^2\,\mathrm{Id}_9, \qquad \Delta h = 0. \tag{8.53}$$

For polarization of the spherical norm identity and the Hopf construction from norm-preserving bilinear multiplication, see [71, Chapter 15]. The coordinates are $h_i(x) = \langle A_i x, x\rangle$ for a nine-matrix Clifford system on $\mathbb{R}^{16}$; see [68, Theorem A and Section 1.2]. On $\mathbb{R}^{16} \oplus \mathbb{R}^{16}$, set

$$u = |x|^2, \quad v = |y|^2, \quad R = u + v, \quad V = u - v, \quad T = uv, \qquad F(x,y) = \langle h(x), h(y)\rangle. \tag{8.54}$$

In particular, $|F| \le T$, and F has bidegree $(2,2)$.

Proposition 8.20 (The pairing closure)**.** *The algebra $\mathcal{A} = \mathbb{R}[u,v,F] = \mathbb{R}[R,V,F]$ is a graded Laplacian algebra of spherical quotient dimension two. Its closure table, in the ordered generators (u,v,F), is*

$$(\Delta u, \Delta v, \Delta F) = (32, 32, 0), \qquad \big(\Gamma(I_i, I_j)\big) = \begin{pmatrix} 4u & 0 & 4F \\ 0 & 4v & 4F \\ 4F & 4F & 4uv(u+v) \end{pmatrix}. \tag{8.55}$$

The quartic zero set $C = \{F = 0\}$ is minimal on its regular part, and its geometric singular set is

$$Z = (\mathbb{R}^{16} \times \{0\}) \cup (\{0\} \times \mathbb{R}^{16}). \tag{8.56}$$

Proof. Equation (8.53) gives

$$|\nabla_x F|^2 = 4uv^2, \qquad |\nabla_y F|^2 = 4u^2 v, \qquad \Delta F = 0.$$

Euler's identity in each block gives the remaining entries of (8.55), and Proposition 3.7 proves Laplacian closure. Surjectivity of the Hopf map gives the full image

$$\{(u,v,f) : u,v \ge 0,\ |f| \le uv\}.$$

Its interior proves algebraic independence and spherical quotient dimension two. Since uv varies at fixed R and $F = 0$, $\Gamma(F,F) = 4Ruv \notin \mathbb{R}[R,F]$.

Since $\Gamma(F,R) = 8F$ and $\Gamma(F,T) = 4RF$, we obtain

$$\mathcal{L}(F) = -\tfrac{1}{2}\Gamma(F, 4RT) = -8(R^2 + 2T)F.$$

This proves regular minimality and identifies the critical set as Z. At $(0, y_0)$, $y_0 \neq 0$, the transverse quadratic term is $\langle A(h(y_0))x, x\rangle$, where $A(a) = \sum_i a_i A_i$. Its eigenvalues are $\pm|y_0|^2$, each with multiplicity eight. Hence the tangent cone is the product of a quadratic cone of signature $(8,8)$ with $\mathbb{R}^{16}$, and the point is geometrically singular. The other axis is identical; the vertex is singular as well. □

Remark 8.21 (A second realization of the pairing table). The Clifford construction in Section 8.2.3 also realizes this table. Put $Q = u^2 + v^2 - 2F$. For a nine-matrix Clifford system $P_0, \dots, P_8$ on $\mathbb{R}^{32}$, the generators

$$\widehat{R} = |z|^2, \qquad \widehat{V} = \langle P_0 z, z\rangle, \qquad \widehat{Q} = \sum_{i=0}^{8} \langle P_i z, z\rangle^2$$

have the same gradient and Laplacian identities as (R, V, Q). Both images are $\{(R, V, Q) : R \geq 0,\ V^2 \leq Q \leq R^2\}$. The inverse relations $u = (R + V)/2$, $v = (R - V)/2$, and $F = (R^2 + V^2 - 2Q)/4$ therefore identify their graded differential algebras.

Theorem 8.22 (A polynomial subcalibration of the pairing cone)**.** *The potential $G = TF = |x|^2|y|^2F$ has degree eight and subcalibrates $C = \{F = 0\}$. Its oriented boundary current $\partial[\![\{F < 0\}]\!]$ is strictly area minimizing in $\mathbb{R}^{32}$. Moreover,*

$$Q_C(\varphi) \geq \frac{449}{4} \int_C r^{-2}\varphi^2, \qquad \varphi \in C_c^\infty(C_{\mathrm{reg}}), \quad r = \sqrt{R}. \tag{8.57}$$

Proof. The three blocks

$$E_0 = (u - v)^2, \qquad E_1 = 2(T + F), \qquad E_2 = 2(T - F)$$

have weights $(1, 8, 8)$. The centered generator $F/2$ gives $h = 16$, $r_0 = 1$, and $c = 1/2$ in Theorem 8.15. Its balanced specialization Corollary 8.16 gives $G = TF$ and

$$\Delta G = 40RF, \qquad |\nabla G|^2 = 4RT(T^2 + 3F^2), \tag{8.58}$$

$$\mathcal{L}(G) = 32TF\big[T^2(2R^2 - T) + 3F^2(4R^2 - T)\big]. \tag{8.59}$$

The Hardy constant in (8.28) is $449/4$ at $(N, h) = (32, 16)$. □

Remark 8.23 (Relation with the OT–FKM and Riedler constructions). The matrices $B_i = \mathrm{diag}(A_i, -A_i)$ form a Clifford system on $\mathbb{R}^{32}$. Its OT–FKM polynomial [68, Section 1.3] is

$$\Psi = R^2 - 2\sum_{i=1}^{9} \langle B_i(x, y), (x, y)\rangle^2 = 4F - V^2.$$

It has multiplicities $(8, 7)$. Applying the centering formula (6.5) to $-\Psi$, with ordered multiplicities $(7, 8)$, shows that its minimal regular hypercone is $\{\Psi + R^2/15 = 0\}$. Nonvertex singularities distinguish C from every cone over a regular isoparametric hypersurface. They also distinguish it from Riedler's family: its singular dimension is $N - 2k$, so congruence would require $N = 32$, $2k = 16$, but sixteen Clifford matrices require ambient dimension at least 256 [69, Remark 5.2.6(1) and Table 4.1]. Although $F = \langle \cdot, \cdot \rangle \circ (h, h)$, the product map $\Phi(x, y) = (h(x), h(y))$ satisfies

$$D\Phi D\Phi^{\mathsf{T}} = 4\,\mathrm{diag}(u\,\mathrm{Id}_9, v\,\mathrm{Id}_9).$$

It is not horizontally conformal on $u \neq v$, so the pullback criterion in Proposition 5.1 does not apply to this map on $\mathbb{R}^{32}$.

The balanced section inside a Simons cone. For the pairing polynomial above, the ambient space for comparison is now itself a minimal cone. Let

$$\mathcal{B} = \{V = 0\} = \{|x| = |y|\}, \qquad D = \{V = F = 0\}, \qquad E_{\mathcal{B}} = \mathcal{B} \cap \{F < 0\}.$$

The Simons cone $\mathcal{B}$ is a singular space of dimension 31, smooth away from the origin. We use its induced metric and the orientation determined by ∇V. Intrinsic perimeter is variation in the smooth part, extended across the vertex by the cutoff below. Write it as $\mathrm{Per}_{\mathcal{B}}$, and set $D_a = \partial[\![E_{\mathcal{B}}]\!] \llcorner B_a(0)$.

Theorem 8.24 (The pairing section of a Simons cone)**.** *The cone D has dimension 30, is smooth away from the origin, and is minimal in $\mathbb{R}^{32}$. The intrinsic potential $G_{\mathcal{B}} = RF$ subcalibrates it in $\mathcal{B}$. For every $a > 0$, the integral current D_a minimizes mass among integral 30-currents supported in $\mathcal{B} \cap \overline{B}_a(0)$ with boundary ∂D_a. Thus D_a solves the Plateau problem in the singular ambient space $\mathcal{B}$, with its Euclidean induced area.*

Proof. On $\mathcal{B} \setminus \{0\}$, the gradients of V and F are nonzero and orthogonal by (8.55). Hence $D \setminus \{0\}$ is smooth of codimension two. The unit normal to $\mathcal{B}$ is $n_{\mathcal{B}} = (x, -y)/\sqrt{R}$. The

bidegree of F gives $\operatorname{Hess} F((x,-y),(x,-y)) = -4F$. Since $\mathcal{B}$ is minimal, the induced operators satisfy

$$\begin{aligned} \Delta_{\mathcal{B}} R = 62, \qquad \Delta_{\mathcal{B}} F = 4F/R, \\ \Gamma_{\mathcal{B}}(R,R) = 4R, \qquad \Gamma_{\mathcal{B}}(R,F) = 8F, \qquad \Gamma_{\mathcal{B}}(F,F) = R^3. \end{aligned} \tag{8.60}$$

Here ∇F is tangent to $\mathcal{B}$, and $T = R^2/4$. The same chain rule as in Theorem 3.10 now yields

$$\begin{aligned} \Delta_{\mathcal{B}}(RF) = 82F, \qquad |\nabla_{\mathcal{B}}(RF)|^2 = R^5 + 20RF^2, \\ \mathcal{L}_{\mathcal{B}}(RF) = 16RF(2R^4 + 85F^2), \end{aligned} \tag{8.61}$$

where $\mathcal{L}_{\mathcal{B}}$ is (1.1) for the induced metric. The normalized gradient

$$X = \frac{\nabla_{\mathcal{B}}(RF)}{|\nabla_{\mathcal{B}}(RF)|}$$

is smooth on $\mathcal{B} \setminus \{0\}$, equals the exterior normal along $D \setminus \{0\}$, and its divergence has the strict sign of F off D.

We apply the proof of Theorem 2.5 in the smooth part of $\mathcal{B}$. Only the vertex requires a cutoff. For a radial cutoff η_ε, zero on $B_\varepsilon(0)$ and one outside $B_{2\varepsilon}(0)$,

$$\int_{\mathcal{B}\cap B_{2\varepsilon}(0)} |\nabla_{\mathcal{B}} \eta_\varepsilon| \, d\mathcal{H}^{31} = O(\varepsilon^{30}).$$

Hence extending a bounded characteristic function across the vertex adds no perimeter mass. The estimate $|\operatorname{div}_{\mathcal{B}} X| \le c/r$ from homogeneity makes the divergence error vanish as well. For every relatively compact open set $U \subset \mathcal{B}$ and every locally finite-perimeter set $H \subset \mathcal{B}$ with $H \triangle E_{\mathcal{B}} \Subset U$,

$$\operatorname{Per}_{\mathcal{B}}(H;U) - \operatorname{Per}_{\mathcal{B}}(E_{\mathcal{B}};U) \ge \int_{H\triangle E_{\mathcal{B}}} |\operatorname{div}_{\mathcal{B}} X| \, d\mathcal{H}^{31} \ge 0. \tag{8.62}$$

The set $E_{\mathcal{B}}$ has locally finite perimeter because its boundary is a cone with a smooth compact link.

For the asserted integral-current comparison, cone a competitor minus D_a to the origin inside $\mathcal{B} \cap \overline{B}_a(0)$. This gives an integral 31-dimensional filling. On the oriented smooth part of $\mathcal{B}$, write this filling as $[\![k]\!]_{\mathcal{B}}$, with integer-valued locally BV multiplicity, and set $v = \mathbf{1}_{E_{\mathcal{B}}} + k$ and $H = \{v \ge 1\}$. The integer-level decomposition used in Theorem 2.11 gives $|D\mathbf{1}_H| \le |Dv|$ on the smooth part. Writing T for the competitor, for $b > a$ we obtain

$$0 \le \operatorname{Per}_{\mathcal{B}}(H;B_b) - \operatorname{Per}_{\mathcal{B}}(E_{\mathcal{B}};B_b) \le \mathbf{M}(T) - \mathbf{M}(D_a).$$

The inequalities use (8.62) and cancellation of the common exterior mass; D has no mass on a sphere. The representation and integer-level decomposition hold in smooth charts. The vertex cutoff extends the characteristic function, and no positive-dimensional integral current is supported at the vertex. Thus the comparison includes competitors meeting it.

Finally, the divergence in (8.61) vanishes on D, so D is minimal in $\mathcal{B}$. For its unit normal ν within $\mathcal{B}$, the two block lengths satisfy $|\nu_x| = |\nu_y|$, by the partial gradient identities above and $u = v$. Hence $\mathrm{II}_{\mathcal{B}}(\nu,\nu) = 0$. Since $\operatorname{tr}_{T\mathcal{B}} \mathrm{II}_{\mathcal{B}} = 0$, its trace on TD is also zero. The second fundamental form composition formula proves Euclidean minimality of $D \setminus \{0\}$. The smooth compact link gives $\int_{D\cap B_{2\varepsilon}} |\nabla_D \eta_\varepsilon| \, d\mathcal{H}^{30} = O(\varepsilon^{29}) \to 0$, so the Euclidean first variation also vanishes across the vertex. □

The Euclidean area-minimizing cones of codimension two in [21, Theorem B] are also area-minimizing hypersurfaces within the Simons cones containing them. Indeed, let $K \subset \mathcal{S}$ be such a cone and put $K_a = K \llcorner B_a(0)$. Every integral current T supported in $\mathcal{S} \cap \overline{B}_a(0)$ with $\partial T = \partial K_a$ is also an admissible Euclidean competitor, so $\mathbf{M}(K_a) \le \mathbf{M}(T)$. The induced area on $\mathcal{S}$ is the Euclidean area.

8.4.4. *Riedler's Clifford–Simons quartics.* The construction and area-minimizing range $N \geq 32$, $k \geq 4$ are due to Riedler [69, Section 5.2 and Theorem 5.2.5]. The common quartic theorem gives polynomial potentials; the degree-nine potential below improves their Hardy bound. The sixteen-dimensional endpoint is a splitting cone.

Let $A_1, \dots, A_{2k}$ be a symmetric Clifford system on $\mathbb{R}^N$, and write $P_i(x) = \langle A_i x, x\rangle$ and $Q = \sum_{i=1}^{2k} P_i^2$. Define

$$F = \sum_{i=1}^{k} P_i^2 - \sum_{i=k+1}^{2k} P_i^2, \qquad C^4_{N,k} = \{x \in \mathbb{R}^N : F(x) = 0\}.$$

This particular Clifford realization requires $\delta(2k-1) \mid N/2$ [32, Section 3.5]. Here $F = f \circ P$, where $f(y) = \sum_{i\leq k} y_i^2 - \sum_{i>k} y_i^2$, so the composition is the pullback of a Simons cone. The unequal-target example Example 8.32 extends this connection with the transfer theorem of Section 5. At the comparison endpoint $k = 1$, $F = (P_1 - P_2)(P_1 + P_2)$ is reducible and harmonic; Proposition 3.12 excludes area minimization. For the minimizing parameters below, the same proposition gives irreducibility. The chain rule gives the identities recorded in [69, Lemma 5.2.11]:

$$\big(\Gamma(I_i, I_j)\big)_{I=(R,Q,F)} = \begin{pmatrix} 4R & 8Q & 8F \\ 8Q & 16RQ & 16RF \\ 8F & 16RF & 16RQ \end{pmatrix}, \qquad \Delta I = (2N, 16kR, 0). \tag{8.63}$$

The determinant is $1024R(R^2 - Q)(Q^2 - F^2)$, so the spherical quotient has dimension two wherever $0 < Q < R^2$ and $|F| < Q$. For a concrete example, take the quadratic octonionic Hopf map $\widetilde{P} = (P_1, \dots, P_9) : \mathbb{R}^{16} \to \mathbb{R}^9$, normalized by $|\widetilde{P}|^2 = R^2$, and retain its first eight components. Then $k = 4$ and $Q = R^2 - P_9^2$. Choose an image with $P_9 \neq 0$ and nonzero components in both four-dimensional blocks. This gives $0 < Q < R^2$ and $|F| < Q$. Taking two copies of this eight-matrix Clifford system gives $N = 32$, $k = 4$, in the minimizing range below; the same point, placed in one copy, retains the strict inequalities. When $Q \not\equiv R^2$, these generators correspond to (8.20) via

$$(E_0, E_1, E_2) = \left(R^2 - Q, \frac{Q+F}{2}, \frac{Q-F}{2}\right), \qquad (r_0, r_1, r_2) = \left(\frac{N}{2} - 2k + 1, k, k\right).$$

For $N \geq 32$, $k \geq 4$, normalize $T = Q/4$, $F_{\mathrm{gen}} = F/8$. Lemma 8.14 gives regular minimality, the singular set $\{Q = 0\}$, and full support. If $k = 4$, then $r_0 \geq 9$ and Theorem 8.15 gives $G = RQF$; if $k \geq 5$, Corollary 8.16 gives $G = QF$. Thus the boundary current is strictly area minimizing in this range, with the Hardy bound (8.28). Positive constant rescalings do not affect normalized gradients.

The defining polynomial itself satisfies

$$|\nabla F|^2 = 16RQ, \qquad \Delta F = 0, \qquad \mathcal{L}(F) = -64F(Q + 2R^2).$$

The last identity follows from $\mathcal{L}(F) = -\Gamma(F, \Gamma(F, F))/2$ and (8.63).

Example 8.25 (Riedler's subcalibration potential)**.** For $N \geq 32$ and $k \geq 4$, Riedler's thesis uses [69, Section 5.2.2, Lemmas 5.2.12–5.2.13]

$$G = RQ^{3/4}F = 16^{-3/4}R^{1/4}S^{3/4}F, \qquad S = |\nabla F|^2 = 16RQ. \tag{8.64}$$

This potential has degree nine. The parameters in Proposition 4.5 are $(e, \sigma, \gamma) = (4, 1/2, 3/4)$. Using (8.63), we obtain the boundary Jacobi coefficient

$$\beta_G = (2N - 41) + (12k - 47)\frac{R^2}{Q} > 0. \tag{8.65}$$

Since $Q \leq R^2$, Theorem 4.6 gives

$$Q_C(u) \geq \left[\left(9 - \frac{N-1}{2}\right)^2 + 2N + 12k - 88\right] \int_C r^{-2}u^2.$$

This lower bound exceeds the general quartic bound (8.28), with $h = 2k$, by

$$\left[\frac{(N-19)^2}{4} + 2N + 12k - 88\right] - \left[\frac{(N-17)^2}{4} + 16k - 72\right] = N + 2 - 4k = 2r_0 > 0.$$

Thus Riedler's degree-nine potential gives the stronger Hardy bound.

Corollary 8.26 (Instability of Riedler's sixteen-dimensional quartic)**.** *The cone $C^4_{16,4}$ is unstable on its regular part and is not area minimizing. More precisely, there are compactly supported variations on the regular part with scale-invariant quotients of the second variation tending to $-15/4$.*

Proof. Write F_R for the defining quartic of $C^4_{16,4}$ and let $A_1, \dots, A_8$ be its symmetric Clifford matrices. The volume matrix $A_9 = A_1 \cdots A_8$ satisfies

$$A_9^T = A_9, \qquad A_9^2 = I, \qquad A_9 A_i = -A_i A_9.$$

Its two eigenspaces have dimension eight, since A_1 exchanges them isometrically. Choosing compatible orthonormal bases gives coordinates $(x, y) \in \mathbb{R}^8 \oplus \mathbb{R}^8$ in which

$$A_{i+1} = \begin{pmatrix} 0 & B_i \\ B_i^T & 0 \end{pmatrix} \quad (0 \le i \le 7), \qquad B_0 = I.$$

The matrices $B_1, \dots, B_7$ are skew-symmetric and satisfy $B_i B_j + B_j B_i = -2\delta_{ij} I$. Thus the eight vectors $B_i y$ $(0 \le i \le 7)$ are pairwise orthogonal, all of length $|y|$, and

$$\sum_{i=0}^{7} \langle x, B_i y\rangle^2 = |x|^2|y|^2, \qquad Q = 4|x|^2|y|^2,$$

$$\frac{F_R}{8} = \sum_{i=0}^{3} \langle x, B_i y\rangle^2 - \frac{1}{2}|x|^2|y|^2 = F_{\text{split}}.$$

Consequently $C^4_{16,4}$ is orthogonally congruent to a splitting cone (8.3) with $(\ell, m) = (8, 4)$. Its Clifford representation is the balanced one: the symmetric involution $B_1 B_2 B_3$ anticommutes with B_4, so its two eigenspaces have dimension four.

In the original variables, $\Lambda = C^4_{16,4} \cap \mathbb{S}^{15}$ and $q = Q|_\Lambda = 4t$, so (8.50) reads

$$|A_\Lambda|^2 = \frac{24}{q} + 14, \qquad |\nabla_\Lambda q|^2 = 16q(1-q), \qquad \Delta_\Lambda q = 56 - 64q. \tag{8.66}$$

The $\ell = 8$ argument above transfers under this orthogonal identification, and (8.52) gives the asserted quotients and instability. □

This is the case left undecided in [69, Remark 5.2.6(3)]. Together with Theorem 5.2.5 and the exclusions in Remark 5.2.6(1)–(2) of that thesis, it gives

$$C^4_{N,k} \text{ is area minimizing} \quad \Longleftrightarrow \quad N \ge 32 \text{ and } k \ge 4$$

among the admissible Clifford–Simons quartics.

8.5. **Further quartic realizations.** The remaining realizations in Table 2 have different singular geometry. Their subcalibrations and Hardy bounds follow from the same quartic theorem.

8.5.1. *Two infinite families with real rank-one singular sets.* An unused block of weight two gives two explicit nonharmonic families in $\mathbb{R}^{8k}$. Their singular set is the real rank-one matrix cone, and their area-minimizing range has the sharp threshold $k \ge 3$.

Identify $\mathbb{R}^{4k}$ with $\mathbb{H}^k$ and let J_1, J_2, J_3 be componentwise right multiplication by the quaternionic units $\mathbf{i}, \mathbf{j}, \mathbf{k}$. For $x, y \in \mathbb{H}^k$, $k \ge 2$, put $R = |x|^2 + |y|^2$ and $W = |x|^2|y|^2 - \langle x, y\rangle^2$, and define

$$F_{k,2} = (4k-3)(\langle x\mathbf{i}, y\rangle^2 + \langle x\mathbf{j}, y\rangle^2) - W, \tag{8.67}$$

$$F_{k,3} = (4k-3)(\langle x\mathbf{i}, y\rangle^2 + \langle x\mathbf{j}, y\rangle^2 + \langle x\mathbf{k}, y\rangle^2) - 2W.$$

Here the displayed pairings are real Euclidean inner products. For $h = \sum_\alpha \bar{x}_\alpha y_\alpha$, one has $\langle x\mathbf{i}, y\rangle = \Im_{\mathbf{i}} h$ and similarly for the other units. Thus the second family is also the Gram composition

$$F_{k,3} = (4k-3)|h|^2 - (4k-5)(\Re h)^2 - 2|x|^2|y|^2. \tag{8.68}$$

Theorem 8.27 (Two families and their sharp threshold)**.** *For $k \geq 2$ and $m = 2, 3$, the polynomial $F_{k,m}$ defines an irreducible minimal hypercone $C_{k,m} \subset \mathbb{R}^{8k}$ with the algebra and singular geometry in Proposition 8.28. The cone is area minimizing if and only if $k \geq 3$. For $k \geq 3$, the degree-eight potential*

$$G_{k,m} = W F_{k,m} \tag{8.69}$$

subcalibrates the full boundary current, which is strictly area minimizing. The regular cone satisfies

$$Q_{C_{k,m}}(\varphi) \geq \left(16k^2 - 36k - \frac{31}{4}\right) \int_{C_{k,m}} r^{-2}\varphi^2. \tag{8.70}$$

At $k = 2$ both cones are unstable on their regular parts.

Both sequences arise from the following construction, which gives their algebra, irreducibility, and transverse singular geometry.

Let $J_1, \dots, J_m$ be skew-symmetric orthogonal endomorphisms of $\mathbb{R}^\ell$ satisfying

$$J_i J_j + J_j J_i = -2\delta_{ij}\,\mathrm{Id}, \qquad m \geq 2, \qquad \ell - m - 1 \geq 2.$$

For $x, y \in \mathbb{R}^\ell$, put

$$R = |x|^2 + |y|^2, \qquad W = |x|^2|y|^2 - \langle x, y\rangle^2, \qquad Q = \sum_{i=1}^m \langle J_i x, y\rangle^2, \tag{8.71}$$

$$\begin{gathered} a = \ell - m - 2, \qquad b = m - 1, \qquad d = a - b = \ell - 2m - 1, \\ F = (\ell - 3)Q - bW. \end{gathered} \tag{8.72}$$

The vectors $x, J_1 x, \dots, J_m x$ are mutually orthogonal and have length $|x|$, so $0 \leq Q \leq W \leq R^2/4$.

Proposition 8.28 (Construction and exact algebra)**.** *The polynomial F is real irreducible and nonharmonic, has nonempty regular zero set, and has essential space $\mathbb{R}^{2\ell}$. Its regular zero set is minimal, and*

$$|\nabla F|^2 = 4R(dF + abW), \qquad \mathcal{L}(F) = -8F\big(2dF + 2abW + abR^2\big). \tag{8.73}$$

Moreover,

$$\mathcal{A}_R(F) = \mathbb{R}[R, W, F]$$

is a polynomial algebra on three generators of degrees $(2, 4, 4)$. Its critical set is the geometric singular set of $C = \{F = 0\}$:

$$\operatorname{Sing} C = \{W = 0\} = \{(x, y) : \operatorname{rank}_{\mathbb{R}}[x\ y] \leq 1\}. \tag{8.74}$$

This set has dimension $\ell + 1$ and is smooth away from the origin. At each nonzero singular point the transverse tangent cone is the Lawson quadratic cone of signature $(m, \ell - m - 1)$.

Proof. The symmetric Clifford system associated with $\mathrm{Id}, J_1, \dots, J_m$ has quadratic coordinates $|x|^2 - |y|^2$, $2\langle x, y\rangle$, and $2\langle J_i x, y\rangle$; see [32, Section 3]. The blocks

$$E_0 = R^2 - 4W, \qquad E_1 = 4Q, \qquad E_2 = 4(W - Q) \tag{8.75}$$

satisfy (8.20) with weights $(2, m, \ell - m - 1)$ by Lemma 8.6. In the normalization of (8.24), $T = W$ and $F_{\mathrm{gen}} = F/(\ell - 3)$. Thus (8.25) gives

$$(\Delta R, \Delta W, \Delta F) = (4\ell, 2(\ell-1)R, 2dR),$$

$$(\Gamma(I_i, I_j))_{I=(R,W,F)} = \begin{pmatrix} 4R & 8W & 8F \\ 8W & 4RW & 4RF \\ 8F & 4RF & 4R(dF + abW) \end{pmatrix}. \tag{8.76}$$

Equation (8.73) follows from (8.27). The smallest-algebra and independence assertions follow from the argument after (8.25). In this case R is already obtained from F by one Laplacian: ℓ is even because $J_1^2 = -\mathrm{Id}$, whereas $2m+1$ is odd, so $d \neq 0$ and $R = \Delta F/(2d)$. In particular F is nonharmonic.

For the remaining assertions, fix $x \neq 0$ and decompose

$$\mathbb{R}^\ell = \mathbb{R}x \oplus U_x^+ \oplus U_x^-, \qquad U_x^+ = \mathrm{span}\{J_1 x, \dots, J_m x\}.$$

For $y = y_0 + y_+ + y_-$, the exact fibre formulas are

$$W = |x|^2(|y_+|^2 + |y_-|^2), \qquad F = |x|^2(a|y_+|^2 - b|y_-|^2). \tag{8.77}$$

They exhibit regular zeros and both phases. Writing $F = y^{\mathsf{T}} M(x) y$, the matrix $M(x)$ has rank $\ell - 1 \geq 4$ for every real $x \neq 0$, with kernel $\mathbb{R}x$. It has the same rank over $\mathbb{R}(x)$, so this quadratic in y cannot factor into linear forms over that field. Its coefficients, which are homogeneous quadratics in x, have no nonconstant common divisor. A linear common divisor would give $M(x) = 0$ at a nonzero point of a real hyperplane. A quadratic common divisor would give $M(x) = p(x)B$ for a constant matrix B, and $M(x)x = 0$ would force $B = 0$. Both contradict the rank assertion. Gauss's lemma therefore proves real irreducibility.

A constant annihilating direction (v_1, v_2) would give two vanishing derivatives of distinct bidegrees $(1,2)$ and $(2,1)$. The second implies $M(x)v_2 = 0$ for every $x \neq 0$, hence $v_2 = 0$; interchanging x, y gives $v_1 = 0$. Thus the ambient space is essential. Finally $dF + abW = a^2Q + b^2(W - Q)$, so (8.73) identifies the critical set with $W = 0$. Equality in real Cauchy–Schwarz gives (8.74). The charts $y = tx$ with $x \neq 0$, and their counterparts with $y \neq 0$, give its smoothness and dimension. Smooth local frames in (8.77) give the transverse quadratic cone. At (x, tx) an isometric normal coordinate is $w = \Pi_{x^\perp}(\delta y - t\delta x)/\sqrt{1+t^2}$; the nonzero Hessian eigenvalues are $2aR$ and $-2bR$, with multiplicities m and $\ell - m - 1$. Their sum is at least four, so the zero set cannot be smooth there. This also proves that these critical points are geometric singularities, as in Lemma 8.14. □

Proof of Theorem 8.27. The weights are $(2, m, 4k-m-1)$, and their active sum is $h = 4k-1$. For $k \geq 3$, this is at least eleven. The normalization $T = W$, $F_{\mathrm{gen}} = F_{k,m}/(4k-3)$ therefore gives (8.69) from (8.29), with $W > 0$ off the rank-one singular set. The strict-minimality and full-support conclusions follow from Theorem 8.15. In this normalization,

$$\begin{aligned} \mathcal{L}(WF) = 8WF\big\{&[3(\ell-1)R^2 - 12W]F^2 \\ &+ d[(\ell-5)R^2 - 4W]WF + ab[(\ell-9)R^2 - 4W]W^2\big\}, \end{aligned} \tag{8.78}$$

which is (8.31) after the same constant rescaling. The Hardy coefficient in (8.28), with $N = 8k$ and $h = 4k-1$, is exactly

$$\frac{(8k-17)^2}{4} + 8(4k-1) - 72 = 16k^2 - 36k - \frac{31}{4} > 0.$$

At $k = 2$, the transverse tangent cones from Proposition 8.28 have signatures $(2,5)$ and $(3,4)$ in $\mathbb{R}^7$. Each is a six-dimensional Lawson cone whose link has $|A|^2 = 5$, so its radial Hardy margin is $(6-2)^2/4 - 5 = -1$. Lemma 2.15, applied to the flat $\mathbb{R}^9$ factor and the smooth local model (8.77), transfers a compactly supported negative variation to the regular cone. Thus both $k = 2$ cones are unstable and not area minimizing. □

Proposition 8.29 (Distinction from the preceding families)**.** *For $k \geq 3$, the cones in (8.67) are not orthogonally congruent to the standard OT–FKM splitting cones or to pure full Clifford pullbacks of nondegenerate Lawson quadratic cones. For each fixed k, the two cones are not congruent to one another.*

Proof. The splitting cones in $\mathbb{R}^{2\ell}$ have singular set the two coordinate ℓ-planes, whereas (8.74) has dimension $\ell + 1$. For a full Clifford quadratic map $P : \mathbb{R}^{2\ell} \to \mathbb{R}^p$, the identity $DP\,DP^{\mathsf{T}} = 4R\,\mathrm{Id}_p$ shows that the nonvertex singular set of a nondegenerate Lawson pullback is $P^{-1}(0)$, of codimension p if nonempty. For the balanced target this is also recorded in [69, Remark 5.2.6(1)]. Congruence would force $p = \ell - 1$. The Clifford module restriction [32, Section 3.5], equivalently [69, Table 4.1], would then require $\delta(\ell - 2) = 2^{2k-1} \mid 4k = \ell$, which is impossible for $k \geq 3$ because $2^{2k-1} > 4k$. Finally the transverse signatures $(2, 4k-3)$ and $(3, 4k-4)$ are distinct even after interchanging their two entries, and hence distinguish the two sequences. □

Remark 8.30 (The weights $(2, 5, 10)$). Taking $\ell = 16$ and $m = 5$ in Proposition 8.28 gives the weights $(2, 5, 10)$ and $F = 13Q - 4W$ on $\mathbb{R}^{32}$. The required Clifford system exists since $\delta(6) = 8 \mid 16$. Here $h = 15$, so Theorem 8.15 gives the potential WF, strict area minimization, and strict stability.

8.5.2. *A complex rank-one singular set: weights* $(3, 7, 7)$. The balanced criterion Corollary 8.16 gives a harmonic example with a complex rank-one singular set. Let $E_i = L_{e_i}$, $1 \leq i \leq 7$, be left multiplication by an orthonormal imaginary basis of the octonions. Alternativity gives $E_i^T = -E_i$ and $E_iE_j + E_jE_i = -2\delta_{ij}\,\mathrm{Id}_8$. On $\mathbb{R}^{16} = \mathbb{R}^8 \oplus \mathbb{R}^8$ put

$$J = \begin{pmatrix} 0 & \mathrm{Id}_8 \\ -\mathrm{Id}_8 & 0 \end{pmatrix}, \qquad B_i = \begin{pmatrix} E_i & 0 \\ 0 & -E_i \end{pmatrix}. \tag{8.79}$$

These eight skew-symmetric matrices satisfy the negative Clifford relations. For $x, y \in \mathbb{R}^{16}$ define

$$\begin{aligned} R &= |x|^2 + |y|^2, \\ W &= |x|^2|y|^2 - \langle x, y\rangle^2 - \langle Jx, y\rangle^2, \\ F &= 2\sum_{i=1}^{7} \langle B_i x, y\rangle^2 - W. \end{aligned} \tag{8.80}$$

Proposition 8.31 (The complex rank-one example)**.** *The polynomial F in (8.80) is real irreducible and harmonic. Its cone $C = \{F = 0\} \subset \mathbb{R}^{32}$ has geometric singular set*

$$\operatorname{Sing} C = \Sigma = \{W = 0\} \cong \{(z, w) \in \mathbb{C}^8 \times \mathbb{C}^8 : \operatorname{rank}_{\mathbb{C}}[z\ w] \leq 1\}. \tag{8.81}$$

This set has real dimension 18 and is smooth away from the origin. At each nonzero singular point the tangent cone is orthogonally congruent to $\mathbb{R}^{18} \times \{|\xi|^2 = |\eta|^2\}$, with $\xi, \eta \in \mathbb{R}^7$. The potential $G = WF$ subcalibrates the full boundary current, which is strictly area minimizing. Its smallest Laplacian algebra is $\mathbb{R}[R, W, F]$, and (8.28) gives

$$Q_C(\varphi) \geq \frac{385}{4} \int_C r^{-2}\varphi^2, \qquad \varphi \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}). \tag{8.82}$$

Proof. The Clifford system associated with $\mathrm{Id}_{16}, J, B_1, \dots, B_7$ has quadratic coordinates $p_0 = |x|^2 - |y|^2$ and $p_A = 2\langle Ax, y\rangle$. Apply Lemma 8.6 with

$$\begin{aligned} E_0 &= p_0^2 + p_{\mathrm{Id}}^2 + p_J^2 = R^2 - 4W, \\ E_+ &= \sum_{i=1}^{7} p_{B_i}^2 = 2(W + F), \\ E_- &= R^2 - E_0 - E_+ = 2(W - F). \end{aligned} \tag{8.83}$$

Here $\ell = 16$, $m = 9$, and the weights are $(3, 7, 7)$, so Corollary 8.16 gives harmonicity, the smallest algebra, and the subcalibration. Irreducibility follows from Proposition 3.12; the Hardy coefficient uses $(N, h) = (32, 14)$.

Identify $(\mathbb{R}^{16}, J)$ with $\mathbb{C}^8$. The expression for W is the complex Gram determinant of the two columns x, y. Thus Lemma 8.14 and equality in Cauchy–Schwarz give (8.81). The rank-one matrices form a smooth complex manifold of dimension $8+2-1 = 9$, so $\dim_{\mathbb{R}} \Sigma = 18$ and $\Sigma \setminus \{0\}$ is smooth.

For $x \neq 0$, the vectors $x, Jx, B_1x, \dots, B_7x$ are orthogonal with length $|x|$. Decompose $y = y_0 + y_+ + y_-$ along $K_x = \operatorname{span}\{x, Jx\}$, $U_x^+ = \operatorname{span}\{B_1x, \dots, B_7x\}$, and $U_x^- = (K_x \oplus U_x^+)^\perp$. The last two spaces have dimension seven, and

$$W = |x|^2(|y_+|^2 + |y_-|^2), \qquad F = |x|^2(|y_+|^2 - |y_-|^2). \tag{8.84}$$

Local frames give transverse signature $(7, 7)$; interchanging x, y covers $\Sigma \setminus \{0\}$. The Hessian formulas in Lemma 8.14, with $T = W$ and $F_{\mathrm{gen}} = F/2$, give eigenvalues $\pm 2R(p)$, each sevenfold, and kernel dimension eighteen; Section 2.6 gives the asserted orthogonal tangent-cone model. □

The singular dimension 18 distinguishes this cone from the splitting and Hopf pairing cones in $\mathbb{R}^{32}$, whose singular sets have dimension 16. For a pure full Clifford pullback of a nondegenerate Lawson cone, (5.6) makes the nonvertex singular set the zero fiber of the Clifford map. Congruence would therefore require fourteen quadratic coordinates. Such a system on $\mathbb{R}^{32}$ would require $\delta(13) = 128 \mid 16$, contrary to the module restriction [32, Section 3.5]. In particular, the example is distinct from Riedler's family.

8.5.3. *An unequal Lawson pullback.*

Example 8.32 (An unequal quadratic target)**.** A Lawson cone in $\mathbb{R}^5 \oplus \mathbb{R}^6$ pulls back to a quartic in $\mathbb{R}^{64}$, illustrating Proposition 5.1. On the target, put

$$T = |u|^2 + |v|^2, \qquad f = 5|u|^2 - 4|v|^2, \qquad H = Tf.$$

The target potential satisfies (5.5) with $\ell = 4$ and $\kappa = 1/2$, strictly off $f = 0$; the calculation is recorded in Appendix C.5.

Choose eleven Clifford matrices on $\mathbb{R}^{64}$, and set $F = f \circ P$, $Q = T \circ P = |P|^2$, and $R = |x|^2$ on the source. Thus $G = H \circ P = QF$. The source blocks

$$E_0 = R^2 - Q, \qquad E_1 = \sum_{i=1}^{5} P_i^2, \qquad E_2 = \sum_{i=6}^{11} P_i^2$$

have weights $(22, 5, 6)$. Such matrices exist by [69, Table 4.1]. With $T_{\mathrm{gen}} = Q/4$, $F_{\mathrm{gen}} = F/36$, Theorem 8.15 gives the same potential QF, critical set $P^{-1}(0)$, full support, and strict area minimality. Here $Q > 0$ off $P^{-1}(0)$, so the multiplier has the required positive domain. The Hardy coefficient in (8.28) is $2273/4$.

8.6. **An example from complex Gram coordinates.** The following sextic is obtained from the nine quadratic coordinates of a complex Hermitian Gram matrix. Its profile uses the imaginary part of a cyclic product of three off-diagonal entries. It shows that the composition criterion also produces higher-degree minimal regular zero sets.

Example 8.33 (A complex Gram sextic)**.** For every $k \geq 2$, let $x_1, x_2, x_3 \in \mathbb{C}^k$ and $h_{ij} = x_i^* x_j$. Then

$$F = \operatorname{Im}(h_{12}h_{23}h_{31}) \tag{8.85}$$

is a real harmonic homogeneous polynomial of degree six on $\mathbb{R}^{6k}$ with nonempty minimal regular zero set.

More explicitly, set $r_i = h_{ii}$, $R = \sum_i r_i$,

$$a_1 = |h_{23}|^2, \qquad a_2 = |h_{31}|^2, \qquad a_3 = |h_{12}|^2, \qquad t = \operatorname{Re}(h_{12}h_{23}h_{31}).$$

With $\{i,j,l\}=\{1,2,3\}$ in each summand, define

$$S=\sum_{i=1}^{3} r_i a_i(a_j+a_l)-2t\sum_{i=1}^{3} a_i, \tag{8.86}$$

$$B=\sum_{i=1}^{3} a_i(r_i+r_j)(r_i+r_l)-\sum_{i=1}^{3} a_i^2-2Rt. \tag{8.87}$$

Then

$$\Delta F=0, \qquad |\nabla F|^2=S, \qquad \Gamma(F,S)=4BF, \qquad \mathcal{L}(F)=-2BF. \tag{8.88}$$

Proof. Use the complex bilinear extension of Γ in (8.17). For $Z=h_{12}h_{23}h_{31}$ and $A=a_1+a_2+a_3$, the product rule gives

$$\Gamma(Z,Z)=4AZ, \qquad \Gamma(Z,\overline{Z})=2\sum_i r_i a_i(a_j+a_l).$$

Taking real and imaginary parts proves $|\nabla F|^2=S$. The same table gives

$$\Gamma(F,r_i)=4F, \quad \Gamma(F,a_i)=2(R-r_i)F, \quad \Gamma(F,t)=2AF. \tag{8.89}$$

Applying these three identities to the expression for S gives $\Gamma(F,S)=4BF$. Indeed, writing $D=\sum_i r_i a_i$ and $E=\sum_i r_i a_i^2$, one has $S=AD-E-2tA$. After the A^2 terms cancel, the product rule gives

$$\Gamma(F,S)=4F\left[RD+\frac{A}{2}\sum_i r_i(R-r_i)-\sum_i a_i^2-\sum_i r_i a_i(R-r_i)-2Rt\right].$$

The coefficient of a_i in the brackets is $r_i^2+r_1r_2+r_2r_3+r_3r_1=(r_i+r_j)(r_i+r_l)$, so the brackets equal B. This is a polynomial identity, valid also when the Gram entries satisfy rank relations, as for $k=2$. For harmonicity, with x_2,x_3 fixed, F is the Hermitian quadratic form in x_1 with matrix

$$\frac{h_{23}x_2x_3^*-\overline{h_{23}}x_3x_2^*}{2i}.$$

Its trace is zero, so its block Laplacian vanishes. The same argument applies to the other two blocks. The point $x_1=x_3=e_1$, $x_2=e_1+e_2$ has $F=0$ and $S=2$, proving nonempty regularity; (8.88) then proves minimality. □

8.7. **Higher bidegrees and transverse instability.** The two-block equations also admit higher bidegrees. In the following octonionic examples, an unstable transverse tangent cone excludes area minimization.

Identify $\mathbb{R}^8$ with $\mathbb{O}$ and set

$$\mu(x,y)=x\overline{y}, \qquad x,y\in\mathbb{O}.$$

Then $|\mu|^2=T$ and

$$D\mu\, D\mu^{\mathsf{T}}=R\,\mathrm{Id}_8, \qquad \Delta\mu=0, \qquad D\mu(\nabla R)=4\mu. \tag{8.90}$$

These follow because multiplication by a fixed nonzero octonion is a linear similarity, with its squared norm as dilation squared; see [71, Chapter 1, Lemma 1.3(1) and Appendix, pp. 15–16 and 21–22] for the norm-multiplicative algebra and similarity identities.

We use T. Wang's octonionic pullback construction [89, Section 3.4]. For $g=3,4,6$, choose the normalized Cartan–Münzner polynomial Φ_g on $\mathbb{R}^8$ with multiplicities respectively $(2,2)$, $(1,2)$, and $(1,1)$. Set

$$P_g=\Phi_g\circ\mu, \qquad F_g=P_g-t_gT^{g/2}, \qquad (\kappa_g,t_g)=(0,0),(8,1/3),(0,0),$$

in the same order, where $\Delta\Phi_g=\kappa_g|z|^{g-2}$ and terms with zero coefficient are omitted. The value t_g is the minimal level from Section 6. These polynomials have bidegree (g,g).

Proposition 8.34. *Each $\mathbb{R}[u, v, P_g]$ is a graded Laplacian algebra with three algebraically independent generators. Each $C_g = \{F_g = 0\} \subset \mathbb{R}^{16}$ is a minimal hypercone, with singular set equal to the two coordinate $\mathbb{R}^8$ subspaces. For $g = 3, 4, 6$, these hypercones are unstable.*

Proof. The closure identities are

$$\begin{gathered}\Delta u = \Delta v = 16, \quad \Gamma(u, v) = 0, \quad \Gamma(u, u) = 4u, \quad \Gamma(v, v) = 4v, \\ \Gamma(u, P_g) = \Gamma(v, P_g) = 2gP_g, \quad \Gamma(P_g, P_g) = g^2 R T^{g-1}, \\ \Delta P_g = \kappa_g R T^{g/2-1}.\end{gathered} \tag{8.91}$$

The last expression is polynomial (zero for $g = 3$). The gradient determinant $16g^2R(T^g - P_g^2)$ is positive at intermediate levels. Thus Proposition 3.7 gives closure and (3.8) gives independence and spherical quotient dimension two.

Put $H_g(z) = \Phi_g(z) - t_g|z|^g$, with the second term zero when $t_g = 0$. This is a homogeneous polynomial in every case under consideration, and its zero set is the minimal isoparametric hypercone in $\mathbb{R}^8$. A useful consequence of (8.90) is, for any homogeneous H of degree h,

$$\mathcal{L}(H \circ \mu) = R^2(\mathcal{L}(H)) \circ \mu - 2h(H|\nabla H|^2) \circ \mu. \tag{8.92}$$

Indeed $|\nabla(H \circ \mu)|^2 = R(|\nabla H|^2) \circ \mu$, $\Delta(H \circ \mu) = R(\Delta H) \circ \mu$, and $\Gamma(H \circ \mu, R) = 4h(H \circ \mu)$. Substitution proves (8.92). Both terms vanish on $H_g \circ \mu = 0$, proving minimality. Off the coordinate subspaces $\mu \neq 0$, the target minimal level is regular, and (8.90) gives a nonzero source gradient. Along either coordinate subspace that gradient vanishes.

At $(x_0, 0)$ with $x_0 \neq 0$, the first nonzero term is $H_g(x_0\overline{y})$, so the tangent cone is a product of $\mathbb{R}^8$ with an orthogonal image of the target minimal cone. The target link has dimension six and constant squared curvature $6(g - 1)$. Its classical Hardy margin is

$$D_g = \frac{25}{4} - 6(g - 1) = \begin{cases} -23/4, & g = 3, \\ -47/4, & g = 4, \\ -95/4, & g = 6. \end{cases}$$

Each is negative, so Lemma 2.15 proves instability of C_g. The target cone is nonlinear, so these coordinate strata are genuine singularities. □

Remark 8.35 (The quadratic target and the sign obstruction). For a quadratic target with both multiplicities positive, let $Q = |\mathrm{pr}_W \mu|^2$, $m = \dim W$, and $c = (m - 1)/6$. This Q obeys (8.45) with $\ell = 8$. The curvature and cutoff proof above applies without a choice of Clifford splitting, so the full $(2, 2)$ hypercone $Q = cT$ is also unstable. Thus all four bidegrees in this $\mathbb{R}^8 \times \mathbb{R}^8$ construction give unstable hypercones. The minimizing question for a codimension-two balance section is separate.

9. Metrized Cubic Algebras and Differential Closure

For an essential irreducible harmonic cubic with minimal regular zero set, the smallest Laplacian algebra has at most four generators in the radial case and exactly five in the nonradial case (Theorem 9.13). Radial transport identities and the Clifford table prepare Section 10. The nonradial weight operator gives an intrinsic graded splitting, five differential invariants, and a reduction to Cartan cubics in dimensions $8, 14, 26$ (Corollary 9.26); compatible splittings still require classification. For the five-generator proof, pass from Section 9.2.1 to Sections 9.5 to 9.7; coordinate models appear in Section 11.

9.1. Hsiang's radial cubic problem and the classification work of Peng–Xiao and Tkachev.

Within his broader study of algebraic minimal cones, Hsiang [42] singled out the classification of cubic forms satisfying

$$\mathcal{L}(F) = -\gamma RF, \qquad R = |x|^2,$$

with a constant γ. These are now called *radial eigencubics*; radiality refers to the quadratic weight in this equation, not to radial dependence of F. This is the radial cubic classification problem considered here; see [84, Section 1.1] and [34, Section 1]. Hsiang's invariant-theoretic constructions recover Cartan cubics in $\mathbb{R}^5$ and $\mathbb{R}^8$ and give the $SO(4)$- and $SU(4)$-invariant models in $\mathbb{R}^9$ and $\mathbb{R}^{15}$; see [84, Section 6.1]. Tongyan Liu [51] later gave an explicit octonionic construction of a further cubic in $\mathbb{R}^{27}$, the $Sp(4)$-invariant model; see [66, Example 4 and Theorem C] and [41, Section 6, p. 1562].

Peng–Xiao [66] and, independently, Tkachev [83, 84] developed closely related approaches to radial minimal cubics.[2] Peng–Xiao assumed harmonicity; Tkachev proved it for irreducible radial cubics [84, Theorem 1]; the algebraic form of this result, with cubes of linear forms as the exception, is [64, Theorem 6.6.1].

Clifford and exceptional cubics. Following Tkachev, an irreducible radial cubic is of *Clifford type* if it is orthogonally equivalent, up to a nonzero constant factor, to

$$F_{\mathcal{A}}(y,z)=\sum_{i=0}^{q} z_i\langle A_i y,y\rangle, \qquad A_iA_j+A_jA_i=2\delta_{ij}\,\mathrm{Id},$$

where $y\in\mathbb{R}^{2m}$, $z\in\mathbb{R}^{q+1}$, and the A_i are symmetric matrices on $\mathbb{R}^{2m}$. The remaining irreducible radial cubics are called *exceptional*; see [84, Section 1.2] and [34, Definitions 2.2 and 2.8]. In particular, Cartan cubics belong to the exceptional branch. The Clifford family will be written explicitly in Section 9.4.2 to compute its differential closure.

Radial normal forms and the classical model families. Choosing a positive maximum direction on the unit sphere, with coordinate z, and normalizing F gives

$$F=z^3+zB+C, \qquad B=-3|\xi|^2-\tfrac{3}{2}|\eta|^2+\tfrac{3}{2}|\zeta|^2,$$
$$C=C_{111}+C_{102}+C_{012}+\sigma, \qquad \sigma=C_{030}.$$

The subscripts denote multidegrees in (ξ,η,ζ), whose dimensions are n_1,n_2,n_3, respectively. They satisfy $n_3=2n_1+n_2-2$ and $N=3n_1+2n_2-1$; see [66, Section 1, Theorem 1] and [84, Proposition 3.1].

The component σ satisfies $|\nabla\sigma|^2=\frac{9}{2}|\eta|^4$. The zero case means that the η-space is absent ($n_2=0$). Otherwise Peng–Xiao [66, Theorem 2] classify σ as reducible or a scaled Cartan cubic; a different proof of the eikonal classification is given in [82, Theorem 1.1]. The case $\sigma=0$ yields the Cartan cubics, whereas the case $\sigma\neq 0$ with σ reducible yields the Clifford cubics [66, Theorems A and B]; see also [84, Proposition 3.4 and Theorem 3]. In the remaining branch, $n_2\in\{5,8,14,26\}$. The cases $n_1=0$ and $n_1=1$ were treated in [66, Theorems C and D] and [84, Sections 6.2–6.3]: the former yields the three models invariant under $SO(4)$, $SU(4)$, and $Sp(4)$, and the latter yields four models interpreted through complexified rank-three Jordan norms [64, Section 6.11.6(iii)–(iv)].

Tkachev also identifies a hidden Clifford system encoded by $C_{111}+3^{-1/2}C_{102}$ when $n_1>0$, imposing Hurwitz–Radon restrictions on the multiplicities [84, Proposition 3.3]. His construction of a G_2-invariant cubic of type $(4,5)$ and his correspondence with isoparametric quartics provide further examples and restrictions [84, Section 6.4 and Theorem 5], using the isoparametric classification results of Cecil–Chi–Jensen [12] and Immervoll [43]. The exceptional branch has only finitely many possible multiplicity types, with $N\leq 72$ [64, Theorem 6.11.2]. This bounds the possible types rather than completing the classification of their realizations.

Subsequent work of Tkachev and his collaborators recast radial minimal cubics in the language of commutative nonassociative metrized algebras, giving intrinsic interpretations of their Clifford and Jordan structures [86, 64, 34]. We follow this terminology and recall the algebraic preliminaries needed to study their Laplacian algebras.

[2]Tkachev records that he learned of Peng–Xiao's 1993 paper from Zizhou Tang in November 2010; see [87, Remark 1, p. 40].

9.2. **Metrized multiplication and low-degree structure.**

9.2.1. *The multiplication associated with a cubic.* The radial equation above also defines the associated *radial eigencubic (REC) algebras* [64, Definition 6.4.1]. We allow an arbitrary quadratic weight in $\mathcal{L}(F) = -qF$.

We use the *Freudenthal–Springer algebra* $V(F)$ of a cubic [64, Definition 6.3.1 and equations (6.4.1)–(6.4.4)]. Let V be a Euclidean space and let $F \in \mathbb{R}[V]$ be homogeneous of degree three. Define a bilinear multiplication by

$$\langle xy, z\rangle = D^3F[x, y, z], \qquad x, y, z \in V. \tag{9.1}$$

The inner product determines xy; symmetry of D^3F gives

$$xy = yx, \qquad \langle xy, z\rangle = \langle x, yz\rangle.$$

Thus $V(F)$ is commutative with invariant Euclidean inner product. Its *Freudenthal multiplication* gives self-adjoint operators $L_x y = xy$. No unit, associativity, or Jordan identity is assumed; in particular, $L_{x^2}y = (xx)y$ and $L_x^2 y = x(xy)$ need not agree.

Proposition 9.1 (Differential quantities and multiplication)**.** *With the normalization* (9.1)*,*

$$F(x) = \frac{1}{6}\langle x^2, x\rangle, \qquad \nabla F(x) = \frac{1}{2}x^2, \qquad \operatorname{Hess} F(x) = L_x, \qquad \Delta F(x) = \operatorname{tr} L_x. \tag{9.2}$$

Here the Hessian is identified with a self-adjoint endomorphism using the Euclidean inner product. Conversely, a commutative bilinear multiplication with an invariant Euclidean inner product determines a cubic by the first formula, and its polarization recovers the multiplication.

Proof. Homogeneity gives $D^3F[x, x, x] = 6F(x)$. Differentiating the first formula and using invariance of the inner product gives

$$DF(x)[v] = \frac{1}{6}\big(2\langle xv, x\rangle + \langle x^2, v\rangle\big) = \frac{1}{2}\langle x^2, v\rangle.$$

Another derivative gives $D(\nabla F)(x)[v] = xv$; taking its trace proves the last assertion. For the converse, commutativity and invariance make $\langle xy, z\rangle$ fully symmetric, so the third derivative of $\langle x^2, x\rangle/6$ is precisely this trilinear form. □

Writing $x^3 = x(x^2)$, substitution in $\mathcal{L}(F) = |\nabla F|^2\Delta F - \operatorname{Hess} F(\nabla F, \nabla F)$ shows that, for a quadratic form q, the equation $\mathcal{L}(F) = -qF$ is equivalent to

$$|x^2|^2 \operatorname{tr} L_x - \langle x^2, x^3\rangle = -\frac{2}{3}q(x)\langle x^2, x\rangle.$$

If F is harmonic, then $\operatorname{tr} L_x = 0$ and the identity reduces to $\langle x^2, x^3\rangle = \frac{2}{3}q(x)\langle x^2, x\rangle$. The further specialization $q = \gamma R$ gives the radial REC identity in [64, Proposition 6.4.1], which uses the same cubic normalization and the parameter $\lambda = -\gamma$.

The metrized algebra on V encodes the derivatives of F, whereas $\mathcal{A}_R(F)$ contains scalar differential invariants such as $S = |\nabla F|^2 = |x^2|^2/4$. In particular, the critical points of F are exactly the square-zero elements of its metrized algebra.

The cylindrical directions are encoded by

$$D_v F(x) = \frac{1}{2}\langle x^2, v\rangle = \frac{1}{2}\langle L_v x, x\rangle,$$

which gives $V_{\mathrm{cyl}}(F) = \{v : L_v = 0\}$, the annihilator of its metrized algebra.

9.2.2. *The quadratic weight and linear generators.* An irreducible cubic with nonempty minimal regular zero set satisfies, by Proposition 3.14,

$$\mathcal{L}(F) = -qF, \tag{9.3}$$

where $q = -\lambda_F$ is a homogeneous quadratic polynomial, possibly zero. Irreducibility is used here to obtain the polynomial equation; its radial specialization was introduced in Section 9.1.

For brevity put $\mathcal{B} = \mathcal{A}_R(F)$ and $\mathcal{B}_d = \mathcal{B} \cap \mathbb{R}[V]_d$, the degree-$d$ homogeneous part.

Proposition 9.2 (Quadratic information in the closure)**.** *Suppose* $\dim V \geq 2$ *and* $F \not\equiv 0$ *is a homogeneous cubic satisfying* (9.3) *with a polynomial* q*. Then* $q \in \mathcal{B}_2$*. If* $\mathcal{B}_2 = \mathbb{R}R$*, then*

$$\Delta F = 0, \qquad q = \gamma R \quad \text{with } \gamma > 0.$$

In particular, a nonradial weight forces a quadratic element of $\mathcal{A}_R(F)$ *independent of* R*.*

Proof. The identity $\mathcal{L}(F) = S\Delta F - \frac{1}{2}\Gamma(F, S)$ shows that $\mathcal{L}(F) \in \mathcal{B}$. The polynomial-quotient property in Proposition 3.6 therefore gives $q = -\mathcal{L}(F)/F \in \mathcal{B}_2$. If $\mathcal{B}_2 = \mathbb{R}R$, the linear polynomial $\ell = \Delta F$ satisfies $\ell^2 \in \mathbb{R}R$. A nonzero square of a linear form has rank one, whereas R has rank $\dim V \geq 2$; hence $\ell = 0$. We already know $q = \gamma R$. At a positive maximum a of F on the unit sphere, attained at x, homogeneity gives $\nabla F(x) = 3ax$ and $\operatorname{Hess} F(x)x = 6ax$. Evaluating (9.3) there gives $-\gamma a = -54a^3$, so $\gamma = 54a^2 > 0$. □

Remark 9.3 (Additional quadratic generators). Since $\mathbb{R}[R, F, S]_2 = \mathbb{R}R$, Proposition 9.2 shows that its Laplacian closure forces radiality. Adjoining a nonradial q need not close the algebra. Conversely, the radial unbalanced Clifford cubics have several quadratic generators; see Theorem 9.12.

Remark 9.4 (Linear generators in the nonharmonic case). If $\Delta F \neq 0$, then $\mathcal{B}_1 \neq 0$. Choose orthonormal linear coordinates $t_1, \dots, t_r$ spanning $\mathcal{B}_1$, and put $W = \operatorname{span}\{\nabla t_i\}$. Then

$$\mathcal{B} = \mathbb{R}[t_1, \dots, t_r] \otimes \mathcal{B}', \qquad \mathcal{B}' = \mathcal{B} \cap \mathbb{R}[W^\perp], \qquad \mathcal{B}'_1 = 0.$$

Indeed, $\partial_{t_i} h = \Gamma(t_i, h) \in \mathcal{B}$, so the finite Taylor identity

$$h(0, y) = \sum_\alpha \frac{(-t)^\alpha}{\alpha!} \partial_t^\alpha h(t, y),$$

applied also to the derivatives of h, puts every coefficient in $\mathcal{B}'$. This algebra is Laplacian and contains $|y|^2 = R - \sum_i t_i^2$. The cubic F can still have mixed terms, and the splitting gives no uniform generator bound for $\mathcal{B}'$.

9.2.3. *One cubic generator and orthogonal blocks.* The quadratic part of a Laplacian algebra is a Jordan algebra of self-adjoint endomorphisms; see Mendes–Radeschi [59]. The following restriction on its cubic part forces its endomorphisms to commute.

Theorem 9.5 (Orthogonal blocks forced by the cubic part)**.** *Let* $F \not\equiv 0$ *be an essential homogeneous cubic on a Euclidean space* V *of dimension at least two, and suppose* $\mathcal{B}_3 = \mathbb{R}F$*. Then* $\mathcal{B}_1 = 0$ *and* $\Delta F = 0$*. There is an orthogonal decomposition into nonzero subspaces, unique up to permutation,*

$$V = V_1 \oplus \cdots \oplus V_r, \qquad \mathcal{B}_2 = \operatorname{span}\{R_1, \dots, R_r\}, \qquad R_i(x) = |\operatorname{pr}_{V_i} x|^2. \tag{9.4}$$

Moreover, F *is multihomogeneous of positive degrees* $(d_1, \dots, d_r)$*, with* $d_1 + \cdots + d_r = 3$*. Consequently* $r \leq 3$*, and, up to permutation, the possible degree patterns are*

$$(3), \qquad (2, 1), \qquad (1, 1, 1).$$

If F *also satisfies* (9.3)*, then* $q = \sum_i \gamma_i R_i$ *for constants* γ_i*.*

Proof. If $0 \neq \ell \in \mathcal{B}_1$, then both ℓR and ℓ^3 are nonzero elements of the one-dimensional space $\mathcal{B}_3$. Cancellation would give $\ell^2 = cR$, contradicting the rank argument above. Thus $\mathcal{B}_1 = 0$, and in particular $\Delta F = 0$.

Represent $\mathcal{B}_2$ by a subspace J of self-adjoint endomorphisms: $B \in J$ when $q_B(x) = \langle Bx, x\rangle$ belongs to $\mathcal{B}_2$. Since

$$\Gamma(q_B, q_C) = 4q_{B \circ C}, \qquad B \circ C = \tfrac{1}{2}(BC + CB),$$

J contains Id and is closed under the Jordan product. For $B \in J$, define $E_B h(x) = \langle Bx, \nabla h(x)\rangle$. Then $2E_B F = \Gamma(q_B, F) \in \mathcal{B}_3$, so $E_B F = \mu_B F$ for some real number μ_B.

We claim that all matrices in J commute. Fix $B, C \in J$ and put $K = [B, C]$. The identity $[E_B, E_C] = -E_K$ gives $E_K F = 0$. Since K is skew-adjoint, the orthogonal transformations

$\exp(tK)$ preserve F. Every orthogonal symmetry of F fixes every element of $\mathcal{B}$: its invariant polynomial algebra is Laplacian and contains R, F, and hence contains $\mathcal{A}_R(F)$. In particular, $q_B(\exp(tK)x) = q_B(x)$, which upon differentiation gives $[B, K] = 0$. In an orthonormal basis diagonalizing B, the (i, j) entry of $[B, [B, C]]$ is $(b_i - b_j)^2 C_{ij}$. Thus $[B, C] = 0$, as claimed.

Simultaneously diagonalize J, and group together the coordinates on which every member of J has the same diagonal value. These groups define the joint eigenspaces $V_1, \dots, V_r$. A generic element of J has distinct eigenvalues on distinct groups. Its spectral projections P_i are polynomials in this element and therefore belong to J, because on commuting matrices the Jordan product is ordinary multiplication. Hence $J = \operatorname{span}\{P_1, \dots, P_r\}$. The projections are precisely the primitive nonzero idempotents of this commutative matrix algebra, proving uniqueness in (9.4).

Finally, E_{P_i} is the Euler operator in the variables of V_i, and $E_{P_i} F = d_i F$ for a real scalar d_i. Expansion into monomials shows that each d_i is a nonnegative integer and that F has degree d_i in that block. Total homogeneity gives $\sum_i d_i = 3$. If $d_i = 0$, F would be independent of V_i, so $V_i \subset V_{\mathrm{cyl}}(F) = \{0\}$, contradicting $V_i \neq \{0\}$. This proves the degree list. The final assertion follows from Proposition 9.2. □

Clifford cubics can also require a quartic generator (Theorem 9.12). The assumption $\mathcal{B}_3 = \mathbb{R}F$ is absent from the subcalibration criteria below and fails for the nonradial models, whose two cubic components are separated by a block Euler operator.

9.3. **Critical Hessians and multiplier degeneracy.** By Proposition 9.1, critical points are square-zero elements $c^2 = 0$, and $L_c = \operatorname{Hess} F(c)$ governs the first transverse term of F. The Hessian identity below follows from (9.3) for an arbitrary quadratic weight q.

Proposition 9.6 (Critical Hessians and multiplier degeneracy)**.** *Let F be a nonzero homogeneous cubic on $\mathbb{R}^N$ satisfying (9.3), where q is a quadratic form. Let c be a critical point, and put $A = \operatorname{Hess} F(c)$ and $\tau = \operatorname{tr} A$. Then*

(i) *The Hessian satisfies*

$$A^3 - \tau A^2 = \frac{q(c)}{2} A. \tag{9.5}$$

In particular, every nonzero eigenvalue satisfies $\lambda^2 - \tau\lambda - q(c)/2 = 0$. If $q(c) \neq 0$, then $A \neq 0$. If F is harmonic and $A \neq 0$, then $q(c) > 0$ and the nonzero eigenvalues are $\pm\sqrt{q(c)/2}$, with equal multiplicities.

(ii) *If $q(c) > 0$ and a is C^2 near c, with $a(c) > 0$, then $G = aF$ cannot satisfy $F\mathcal{L}(G) \geq 0$ throughout a neighborhood of c.*

(iii) *If F is essential and harmonic, every nonzero critical point has $A \neq 0$ and $q(c) > 0$. Hence a nonnegative smooth multiplier giving the two-sided subcalibration sign must vanish at every such point across which it extends smoothly.*

Proof. For (i), Euler's identity gives $F(c) = 0$. Write

$$F(c + y) = h(y) + T(y), \qquad h(y) = \tfrac{1}{2}\langle Ay, y\rangle,$$

where T is homogeneous of degree three and is not identically zero. If $q(c) \neq 0$ and $A = 0$, the left side of (9.3), expanded at c, is homogeneous of degree five in y, whereas the right side has the nonzero term $-q(c)T(y)$ of degree three. This is impossible.

The terms of degree two in the same equation are

$$\tau\langle A^2 y, y\rangle - \langle A^3 y, y\rangle = -\frac{q(c)}{2}\langle Ay, y\rangle.$$

All matrices involved are symmetric, so this proves (9.5). If F is harmonic, $\tau = 0$. If in addition $A \neq 0$, a nonzero real eigenvalue gives $q(c)/2 = \lambda^2 > 0$. The zero trace then forces the multiplicities of the two nonzero eigenvalues to agree.

For (ii), Lemma 4.3 gives $\mathcal{K}_F(a)(c) = -a(c)^3 q(c) < 0$. Since (i) gives $A \neq 0$, choose y with $h(y) \neq 0$. Continuity and $F(c+ry) = r^2 h(y) + O(r^3)$ yield

$$F(c+ry)\mathcal{L}(aF)(c+ry) = -a(c)^3 q(c) r^4 h(y)^2 + o(r^4) < 0 \tag{9.6}$$

for small $r > 0$. The negative product implies $\nabla(aF) \neq 0$, so the sign fails where the field is defined; harmonicity is not used.

For (iii), $L_c = 0$ would put c in the annihilator $V_{\mathrm{cyl}}(F) = \{0\}$. Thus (i) gives $q(c) > 0$. The multiplier conclusion is also the harmonic obstruction of Proposition 4.24. □

For essential harmonic F as above, Section 2.6 identifies the tangent at every nonzero critical point as $\{(u,v) \in \mathbb{R}^s \oplus \mathbb{R}^s : |u| = |v|\} \times \mathbb{R}^{N-2s}$, where $2s$ is the Hessian rank, with smooth convergence on compact regular subsets. Radial powers R^β fail as smooth positive multipliers there, whereas S has the required vanishing. The following trace bound supplies removability for Theorem 10.2.

Lemma 9.7 (A trace bound for the critical set)**.** *Let $F \not\equiv 0$ be a harmonic cubic on $\mathbb{R}^N$, $N \geq 2$, satisfying $\mathcal{L}(F) = -2RF$, and put $Z = \{\nabla F = 0\}$. If $\Delta S \geq 16R$, then every $c \in Z \setminus \{0\}$ satisfies* $\operatorname{rank} \operatorname{Hess} F(c) \geq 8$*, and Z is locally contained near c in a smooth submanifold of codimension eight. In particular, Z has locally zero $\mathcal{H}^{N-1}$-measure.*

Proof. At $c \in Z \setminus \{0\}$, Proposition 9.6 gives $A^3 = |c|^2 A$, where $A = \operatorname{Hess} F(c)$. Its nonzero eigenvalues are $\pm|c|$, and Bochner's identity yields

$$|c|^2 \operatorname{rank} A = \operatorname{tr}(A^2) = \tfrac{1}{2}\Delta S(c) \geq 8|c|^2.$$

Choose eight components of ∇F with independent differentials. The implicit function theorem places Z locally in their smooth common zero set of codimension eight. This proves the local codimension claim and, by a countable cover, nullity off the origin. The origin is also null since $N \geq 2$. If $N < 8$, the rank inequality simply excludes nonzero critical points. □

9.4. **Radial transport and Clifford closure.**

9.4.1. *Intrinsic transport and the scalar trace form.* For harmonic F, the identity $\Gamma(F,S) = 4RF$ is the normalization $\lambda = -2$ for REC algebras. We use this normalization and its polarized identities from [64, Section 6.7.1, equations (6.7.6)–(6.7.7)].

Lemma 9.8 (Intrinsic radial transport identities)**.** *Let F be a homogeneous cubic on a Euclidean space V, with $\Gamma(F,S) = 4RF$, where $R = |x|^2$ and $S = |\nabla F|^2$. Define*

$$D = \frac{\Gamma(S,S) - 4RS}{12} = \frac{|x^3|^2 - R|x^2|^2}{12}. \tag{9.7}$$

Then

$$\Gamma(F,D) = 2SF, \qquad \Gamma(S,D) = 8RD, \qquad \Gamma(D,D) = 4SD. \tag{9.8}$$

Neither harmonicity nor a scalar quadratic trace form is needed.

Proof. Use the multiplication (9.1), and put

$$y = x^2, \quad z = xy, \quad u = y^2, \quad w = yz,$$
$$r = |x|^2, \quad c = \langle x, y\rangle, \quad s = |y|^2, \quad t = |z|^2, \quad \delta = t - rs.$$

Thus $F = c/6$, $S = s/4$, $D = \delta/12$, and the hypothesis is $P(x) := \langle y, z\rangle - 4rc/3 = 0$. Its gradient gives the polarized identity

$$Q(x) := 4xz + u - \tfrac{8}{3}cx - 4ry = 0. \tag{9.9}$$

Pairing with y gives

$$\eta := \langle y, u\rangle = \tfrac{8}{3}c^2 - 4\delta. \tag{9.10}$$

The coefficient of τ^2 in $P(x+\tau y)=0$, with y held fixed, and the pairing $\langle Q, z\rangle = 0$ give, respectively,

$$4\langle z, xz\rangle + 6\langle u, z\rangle = \tfrac{16}{3}r^2c + \tfrac{28}{3}cs,$$
$$4\langle z, xz\rangle + \langle u, z\rangle = \tfrac{16}{3}r^2c + \tfrac{8}{3}cs.$$

Subtracting, and then using $DQ(x)[y]/2 - xQ(x) = 0$, yields

$$\langle u, z\rangle = \tfrac{4}{3}cs, \qquad xu + 4w = 4sx + \tfrac{8}{3}cy. \tag{9.11}$$

Equation (9.10) gives $D = 2F^2 - F(\nabla F)$. Differentiating and applying the second identity in (9.11), we obtain

$$\nabla D = \tfrac{c}{3}y - \tfrac{1}{8}xu = \tfrac{1}{2}(w - sx). \tag{9.12}$$

Write $b = \langle z, w\rangle$ and $U = |u|^2$. Pairing (9.9) with u, and using $\langle z, xu\rangle = 4s^2 + 32rc^2/9 - 4b$, gives

$$b = s^2 + r\delta + \frac{U}{16}.$$

The coefficient of τ^3 in $P(x+\tau y)=0$ is

$$2U + 4\langle z, xu\rangle + 4b = \tfrac{4}{3}\big(r\eta + 6c\langle y, z\rangle + 3s^2\big).$$

Substitution of (9.10) and $\langle Q, u\rangle = 0$ reduces this to $b = s^2 - U/4 + 8r\delta/3$. The two expressions for b imply

$$U = \tfrac{16}{3}r\delta, \qquad b = s^2 + \tfrac{4}{3}r\delta. \tag{9.13}$$

Differentiate $\langle u, z\rangle = 4cs/3$ in the direction u:

$$4\langle w, xu\rangle + \langle u, yu\rangle + 2|xu|^2 = 4s\eta + \tfrac{16}{3}c\langle z, u\rangle.$$

Here $P(y) = 0$ gives $\langle u, yu\rangle = 4s\eta/3$, while (9.11) gives

$$\langle w, xu\rangle = 4st + \tfrac{32}{9}c^2 s - 4|w|^2, \qquad |xu|^2 = 16rs^2 - 32st + 16|w|^2.$$

Substitution yields

$$|w|^2 = \tfrac{7}{3}st - \tfrac{4}{3}rs^2. \tag{9.14}$$

Finally, $\nabla F = y/2$, $\nabla S = z$, and (9.12) give

$$\Gamma(F, D) = \tfrac{1}{4}(\langle u, z\rangle - sc) = cs/12 = 2SF,$$
$$\Gamma(S, D) = \tfrac{1}{2}(b - s^2) = 2r\delta/3 = 8RD,$$
$$\Gamma(D, D) = \tfrac{1}{4}(|w|^2 - 2st + rs^2) = s\delta/12 = 4SD.$$

□

Proposition 9.9 (Closure under a scalar quadratic trace form)**.** *Suppose, in addition to the hypotheses of Lemma 9.8, that F is harmonic and $\operatorname{tr}(L_x^2) = kR$ for a constant k. Writing $N = \dim V$, one has*

$$\Delta S = 2kR, \qquad \Delta D = \left(2k - \frac{2N}{3}\right) S, \qquad \mathcal{A}_R(F) = \mathbb{R}[R, F, S, D]. \tag{9.15}$$

In particular, this algebra has at most four generators.

Proof. Polarization gives $\operatorname{tr}(L_x L_v) = k\langle x, v\rangle$. Differentiating (9.9) and dividing by four yields the operator identity

$$L_z + L_x L_y + L_y L_x + 2L_x^3 - 2rL_x - \tfrac{2}{3}c\,\mathrm{Id} - 2x \otimes y - 2y \otimes x = 0,$$

where $(a \otimes b)v = \langle b, v\rangle a$. Taking traces, using harmonicity, gives

$$\operatorname{tr} L_x^3 = \alpha c, \qquad \alpha = N/3 + 2 - k.$$

Its derivative in the direction y is $\operatorname{tr}(L_y L_x^2) = \alpha s$. Therefore

$$\operatorname{div}(yz) = 2\operatorname{tr}(L_z L_x) + \operatorname{tr} L_y^2 + 2\operatorname{tr}(L_y L_x^2) = (3k + 2\alpha)s,$$
$$\operatorname{div}(sx) = (N+4)s.$$

Taking the divergence in (9.12) proves the formula for ΔD; Bochner's identity gives that for ΔS. Together with $\Delta R = 2N$, $\Delta F = 0$, homogeneity, and Lemma 9.8, these formulas show that $\mathbb{R}[R, F, S, D]$ is Laplacian. Conversely, $S = \Gamma(F, F)$ and (9.7) put S, D in every Laplacian algebra containing R, F, proving equality. □

Remark 9.10 (The radial bound and its trace input). An irreducible harmonic radial eigencubic of exceptional type, in the terminology of Section 9.1, has $\operatorname{tr}(L_x^2) = k|x|^2$ for a constant k, in the multiplication convention (9.1). This is [64, Proposition 6.11.1]; see also [84, Proposition 5.2, equation (5.11)]. The scalar trace condition is not a consequence of radiality alone: in the Clifford family, (9.21) is scalar precisely when $m = q + 1$.

Consequently every irreducible radial eigencubic has at most four ordinary algebra generators in $\mathcal{A}_R(F)$. After normalizing F, the exceptional branch follows from this trace theorem and Proposition 9.9. In the Clifford branch, the same proposition applies when $m = q + 1$; when $m \neq q + 1$, Theorem 9.12 recovers the block radii and identifies the smallest algebra directly.

9.4.2. *The Clifford family and its differential table.* We write the Clifford family of [66, Theorem B] in the symmetric-system notation of [83].

Let $q \geq 1$, and let $\mathcal{A} = (A_0, \dots, A_q)$ be a symmetric Clifford system on $\mathbb{R}^{2m}$, so that

$$A_i A_j + A_j A_i = 2\delta_{ij} \operatorname{Id}_{\mathbb{R}^{2m}}. \tag{9.16}$$

For $(y, z) \in \mathbb{R}^{2m} \oplus \mathbb{R}^{q+1}$, put $A_z = \sum_{i=0}^{q} z_i A_i$. The corresponding cubic and its zero cone are

$$F = F_{\mathcal{A}}(y, z) = \sum_{i=0}^{q} z_i \langle A_i y, y\rangle = y^T A_z y, \qquad \mathcal{C}_{\mathcal{A}} = \{F = 0\}. \tag{9.17}$$

The Freudenthal–Springer algebra of $F/2$ is a polar algebra in the normalization of [64, Definition 6.5.1 and Propositions 6.5.1–6.5.2]. We call the pair (m, q) admissible when such a Clifford system exists.

Set

$$R = |y|^2 + |z|^2, \qquad S = |\nabla F|^2, \qquad N = 2m + q + 1.$$

The matrices are trace free, so F is harmonic. The Clifford relations give $\mathcal{L}(F) = -8RF$, and hence the regular zero set is minimal; the calculations are given below. We orient the cone as $\partial[\![\{F < 0\}]\!]$ and use the local mass-minimizing convention for integral-current competitors with compactly supported difference that bounds.

Differential identities. Introduce

$$\tau_i = y^T A_i y, \qquad \tau = (\tau_0, \dots, \tau_q), \qquad A_\tau = \sum_{i=0}^{q} \tau_i A_i,$$

and the scalar functions

$$u = |y|^2, \qquad v = |z|^2, \qquad R = u + v, \qquad T = |\tau|^2.$$

The Clifford relations imply

$$A_z^2 = v \operatorname{Id}, \qquad A_\tau^2 = T \operatorname{Id}, \qquad A_z A_\tau + A_\tau A_z = 2F \operatorname{Id}.$$

Consequently,

$$|A_z y|^2 = uv, \quad |A_\tau y|^2 = uT, \quad \langle A_z y, A_\tau y\rangle = uF. \tag{9.18}$$

The vectors $A_0 y, \dots, A_q y$ are orthogonal, each of length $|y|$. Bessel's inequality, followed by Cauchy–Schwarz, gives

$$0 \leq T \leq u^2, \qquad 0 \leq F^2 \leq vT. \tag{9.19}$$

We shall also use

$$\langle A_i y, A_z y\rangle = u z_i, \qquad \langle A_i y, A_\tau y\rangle = u\tau_i.$$

Set

$$U = \nabla F, \qquad S = |U|^2, \qquad Z = \nabla S, \qquad J = |Z|^2, \qquad K = \Delta S.$$

The four functions $I = (u, v, T, F)$ have the following gradient inner products and Laplacians:

$$\big(\langle \nabla I_i, \nabla I_j\rangle\big) = \begin{pmatrix} 4u & 0 & 8T & 4F \\ 0 & 4v & 0 & 2F \\ 8T & 0 & 16uT & 8uF \\ 4F & 2F & 8uF & 4uv+T \end{pmatrix}, \tag{9.20}$$

$$\Delta I = (4m,\, 2(q+1),\, 8(q+1)u,\, 0).$$

For example, $\nabla_y T = 4A_\tau y$, so $|\nabla T|^2 = 16uT$ and $\langle \nabla T, \nabla F\rangle = 8uF$. Euler's identity and the Clifford relations give the remaining entries. Since $R = u + v$, Proposition 3.7 shows that $\mathbb{R}[u, v, T, F]$ is a Laplacian algebra. Hence Theorem 3.10 applies to potentials $G(u, v, T, F)$.

Proposition 9.11 (Gradients and Laplacians)**.** *One has*

$$\begin{aligned} U &= (2A_z y, \tau), & S &= 4uv + T, \\ Z &= (8vy + 4A_\tau y, 8uz), & \Delta F &= 0, \\ K &= 16mv + 16(q+1)u, & \langle U, Z\rangle &= 16RF, \\ J &= 16\big(4uvR + (u+4v)T\big). \end{aligned} \tag{9.21}$$

Proof. Differentiate to obtain U, Z; their squared norms follow from (9.18). Conjugation by A_j, $j \neq i$, sends A_i to $-A_i$, proving trace-freeness and $\Delta F = 0$. The remaining formulas are $K = \Delta(4uv + T)$ and $\langle U, Z\rangle = \Gamma(F, 4uv + T)$ in (9.20). □

Theorem 9.12 (The smallest algebra of an unbalanced Clifford cubic)**.** *If $m \neq q + 1$, then*

$$\mathcal{A}_R(F) = \mathbb{R}[u, v, T, F], \qquad (\mathcal{A}_R(F))_2 = \operatorname{span}\{u, v\}, \qquad (\mathcal{A}_R(F))_3 = \mathbb{R}F.$$

The two coordinate blocks are therefore the intrinsic quadratic blocks of Theorem 9.5, with degree pattern $(2, 1)$.

Proof. Every Laplacian algebra containing R, F contains $S = \Gamma(F, F)$ and ΔS. Equation (9.21) recovers the two block radii by

$$u = \frac{\Delta S/16 - mR}{q + 1 - m}, \qquad v = R - u.$$

It then recovers $T = S - 4uv$. Thus $\mathbb{R}[u, v, T, F]$ is contained in $\mathcal{A}_R(F)$, and the reverse inclusion follows from its closure table (9.20). The generators have degrees $2, 2, 4, 3$, giving the two homogeneous components stated above; u, v are linearly independent. Finally, if a constant vector (ξ, ζ) is a translation direction of F, its derivative has terms $2\langle A_z y, \xi\rangle$ and $y^T A_\zeta y$ of different bidegrees, so both vanish. Invertibility of A_0 gives $\xi = 0$, and $A_\zeta^2 = |\zeta|^2 \operatorname{Id}$ gives $\zeta = 0$. The block theorem therefore applies. □

When $m = q + 1$, one has $\Delta S = 16mR$. Applying Proposition 9.9 to the normalized cubic $F/2$ therefore gives

$$\mathcal{A}_R(F) = \mathbb{R}[R, F, S, vT] \subsetneq \mathbb{R}[u, v, T, F].$$

Indeed, $|\nabla(F/2)|^2 = S/4$, and its generator D from (9.7) is $vT/4$. In the present normalization,

$$vT = \frac{\Gamma(S, S) - 16RS}{48} \in \mathcal{A}_R(F).$$

The generators R, F, S, vT have degrees $2, 3, 4, 6$, so the quadratic component of this smallest algebra is $\mathbb{R}R$. Since u, v are linearly independent, the inclusion is strict. The larger algebra remains useful for the coordinate calculations below.

Since $Z = 2\operatorname{Hess} F(U)$, (9.21) and harmonicity give

$$\mathcal{L}(F) = -\tfrac{1}{2}\Gamma(F, S) = -8RF,$$

recovering the radial minimality identity for this family directly from the closure table.

9.5. **The weight operator and nonradial spectral rigidity.**

Theorem 9.13 (The five-generator bound)**.** *Let F be a nonzero irreducible homogeneous harmonic cubic on a Euclidean space V, with $V = V_{\mathrm{ess}}(F)$, and suppose its regular zero set is minimal. Then the smallest Laplacian algebra $\mathcal{A}_R(F)$ is generated by at most five polynomials. More precisely:*

(i) *If F is radial, $\mathcal{A}_R(F)$ has at most four generators.*
(ii) *If F is nonradial, $\mathcal{A}_R(F)$ is a polynomial algebra on five algebraically independent homogeneous generators of degrees $(2, 2, 3, 3, 4)$. Its spherical quotient dimension is four.*

The upper bound five is attained by the examples of Section 11.1.

Extending the Peirce framework of [64, Sections 6.6–6.8] to the weight operator gives spectral rigidity and a graded splitting. Irreducibility forces an exact eikonal core. The five resulting invariants close under the differential operations, are recovered from R, F, and have rank five. Spectral exclusion and core exactness are proved in Appendices B.3 and B.4.

9.5.1. *Maximal idempotents and the weight operator.* By Lemma 3.11, an irreducible harmonic F has a nonsingular real zero. Thus Proposition 3.14 gives $\mathcal{L}(F) = -qF$, where homogeneity makes q quadratic.

Use the commutative Euclidean metrized algebra and normalization of Proposition 9.1. Harmonicity is the condition $\operatorname{tr} L_x = 0$ for every x; we call such an algebra *exact*, following [34, Section 3.2 and Definition 4.1]. For a self-adjoint operator T, the equation $\mathcal{L}(F) = -\frac{3}{2}\langle Tx, x\rangle F$ is then equivalent to

$$\langle x^2, x^3\rangle = \langle Tx, x\rangle\langle x^2, x\rangle. \tag{9.22}$$

This defines an *exact Hsiang algebra*, with symmetric bilinear weight $b(x, y) = \langle Tx, y\rangle$. The radial case is the special case in which T is scalar. Here $x^3 = x(x^2)$ and $x^4 = x(x^3)$; no associativity is assumed. Differentiating (9.22) gives

$$4x^4 + (x^2)^2 - 2\langle x^2, x\rangle Tx - 3\langle Tx, x\rangle x^2 = 0. \tag{9.23}$$

As shown in Section 9.2.1, $\operatorname{Ann}(A) = \{v : vA = 0\}$ is the cylindrical subspace of F. Metric invariance makes $A_{\mathrm{ess}} = \operatorname{Ann}(A)^{\perp}$ a subalgebra; its orthogonal splitting from the annihilator has zero mixed products.

We use the spherical maximum construction of an idempotent [64, Lemma 6.6.1] and derive the spectral bounds below.

Lemma 9.14 (Maximal normalization)**.** *Let A be a commutative metrized algebra with a Euclidean metric and nonzero multiplication. After multiplying its product by a positive constant, there is a unit idempotent c such that*

$$\max_{|x|=1}\langle x^2, x\rangle = \langle c^2, c\rangle = 1. \tag{9.24}$$

Every eigenvalue of L_c on $c^{\perp}$ lies in $[-1, \frac{1}{2}]$.

Proof. Put $h(x) = \langle x^2, x\rangle$. Polarization shows that h is a nonzero odd cubic, so its maximum M on the unit sphere is positive. At a maximizer c, the Lagrange multiplier equation is $c^2 = Mc$. Replacing the product by xy/M proves (9.24). It preserves exactness and replaces

T in (9.22) by T/M^2. The constrained Hessian of h at c is $6L_c - 3\,\mathrm{Id}$ on $c^\perp$, giving the upper bound. For a unit eigenvector $u \perp c$, with $cu = ru$, maximality gives

$$h\left(-\frac{c}{2} \pm \frac{\sqrt{3}}{2}u\right) = -\frac{1}{8} - \frac{9}{8}r \pm \frac{3\sqrt{3}}{8}h(u) \le 1.$$

Averaging gives $r \ge -1$. □

Fix this normalization, which rescales the cubic by a positive constant.

Lemma 9.15 (The weight operator at a unit idempotent)**.** *Every unit idempotent e in an algebra satisfying (9.22) satisfies*

$$Te = e, \qquad T = p(L_e) - 6P_e, \qquad p(t) = 4t^3 + 4t^2 - t, \tag{9.25}$$

where $P_e x = \langle e, x\rangle e$. In particular, $[T, L_e] = 0$.

Proof. At e, the scalar identity gives $\langle Te, e\rangle = 1$, and the vector identity gives $Te = e$. Its derivative at e is

$$8L_e^3 y + 8L_e^2 y - 2L_e y - 2Ty - 12\langle e, y\rangle e = 0.$$

This proves (9.25). Since L_e is self-adjoint and $L_e e = e$, it commutes with P_e. □

Write $E_r = \ker(L_c - r\,\mathrm{Id})$. By maximality $E_1 = \mathbb{R}c$, and

$$Tu = p(r)u \qquad (u \in E_r \subset c^\perp). \tag{9.26}$$

9.5.2. *Fusion identities and spectral rigidity.* We adapt the radial polarization argument in [64, Lemma 6.6.2], specifically equation (6.6.8). For nonradial T, we derive the fusion rule directly from (9.23), using

$$\mathscr{K}(r, s, t) = 4(r^2 + s^2 + t^2 + rs + rt + st) + 2(r + s + t) - 3. \tag{9.27}$$

Lemma 9.16 (Fusion and the square identity)**.** *For $u \in E_r$, $v \in E_s$, $w \in E_t$, all perpendicular to c, one has*

$$\mathscr{K}(r, s, t)\langle uv, w\rangle = 0. \tag{9.28}$$

For $u \in E_r \subset c^\perp$, set $Q_r(t) = \mathscr{K}(r, r, t)$. Then

$$Q_r(L_c)u^2 = 3\big(2r + p(r)\big)|u|^2 c. \tag{9.29}$$

Writing $u^2 = r|u|^2 c + \sum_t q_t$, with $q_t \in E_t \cap c^\perp$, gives

$$\sum_t (t + 4r)|q_t|^2 = 4r^2(3r - 1)(r + 1)|u|^4. \tag{9.30}$$

Proof. The mixed coefficient of ab in (9.23) at $c + au + bv$, projected to $c^\perp$ and divided by two, is the projection of

$$\big[4L_c^2 + (4r + 4s + 2)L_c + 4(r^2 + rs + s^2) + 2(r + s) - 3\big]uv.$$

Pairing with w proves (9.28). The coefficient of a^2 at $c + au$ gives (9.29); thus $Q_r(t)q_t = 0$. For the norm identity, pair the vector identity at u with c. Metric invariance gives

$$\langle u^4, c\rangle = r|u^2|^2, \qquad \langle (u^2)^2, c\rangle = r^2|u|^4 + \sum_t t|q_t|^2.$$

The term containing Tu pairs to zero, and the last term is $-3p(r)r|u|^4$. Rearranging yields (9.30), since $3p(r)r - 4r^3 - r^2 = 4r^2(3r - 1)(r + 1)$. □

Proposition 9.17 (A uniform spectral interval)**.** *Put* $r_0 = -(2+\sqrt{7})/6$. *Then*

$$\operatorname{Spec}(L_c|_{c^\perp}) \subset \{-1,0\} \cup [r_0, -1/6] \cup [1/3, 1/2]. \tag{9.31}$$

Moreover,

$$\operatorname{Spec} T \subset \{0\} \cup \left[\frac{7}{27}, \frac{17+7\sqrt{7}}{27}\right], \qquad \ker T = \operatorname{Ann}(A). \tag{9.32}$$

Proof. All possible output eigenvalues t in (9.30) lie in $[-1, 1/2]$. If $-1 < r < r_0$, then $Q_r(t) = 4t^2 + (8r+2)t + 12r^2 + 4r - 3$ is decreasing on that interval and $Q_r(1/2) = 12r^2 + 8r - 1 > 0$. All q_t therefore vanish, contradicting the negative right side of (9.30). If $-1/6 < r < 0$, both endpoint values

$$Q_r(-1) = (6r+1)(2r-1), \qquad Q_r(1/2) = 12r^2 + 8r - 1$$

are negative. Convexity makes Q_r negative throughout the interval, giving the same contradiction. If $0 < r < 1/3$, then $Q_r(-1) < 0$ and $Q_r(0) < 0$; hence every admissible root is positive. The left side of the norm identity is nonnegative and the right side is negative. This proves (9.31). At $r = 1/3$, the norm identity also shows that every q_t vanishes, because $t + 4/3 > 0$.

Applying p to these intervals gives the spectral inclusion in (9.32). The only zero of p allowed by (9.31) is 0. Thus $u \in \ker T$ implies $cu = 0$, and (9.29) gives

$$(4L_c^2 + 2L_c - 3\,\mathrm{Id})u^2 = 0.$$

Its two roots $(-1 \pm \sqrt{13})/4$ lie outside $[-1, 1/2]$, and it is nonzero at 1. Consequently $u^2 = 0$. The coefficient of t^3 in the vector identity at $x + tu$ is then $8L_u^3 x$: all other terms have degree at most two, since $Tu = u^2 = 0$. Hence $L_u^3 = 0$, and self-adjointness gives $L_u = 0$. Conversely, an annihilator element is orthogonal to $c = c^2$ and is killed by L_c, so (9.25) kills it as well. □

By Proposition 9.17, restriction to A_{ess} preserves exactness and the defining identity, and $T|_{A_{\mathrm{ess}}}$ is positive definite.

Let

$$B = E_T(1) = \mathbb{R}c \oplus Z \oplus H \oplus W, \qquad Z = E_{-1}, \quad H = E_{-1/2}, \quad W = E_{1/2}, \tag{9.33}$$

since $p(t) - 1 = (t+1)(2t-1)(2t+1)$. The fusion polynomial gives the following possible outputs on $c^\perp$:

(r,s)	$(-1,-1)$	$(-1,-\frac{1}{2})$	$(-1,\frac{1}{2})$	$(-\frac{1}{2},-\frac{1}{2})$	$(-\frac{1}{2},\frac{1}{2})$	$(\frac{1}{2},\frac{1}{2})$
t	none	$\frac{1}{2}$	$-\frac{1}{2},\frac{1}{2}$	$-\frac{1}{2}$	$-1,\frac{1}{2}$	$-1,-\frac{1}{2}$

(The additional root 1 in the fourth column is absent on $c^\perp$.) In particular,

$$B^2 \subset B, \qquad BB^\perp \subset B^\perp. \tag{9.34}$$

A Euclidean commutative metrized algebra is *eikonal* if $x^3 = |x|^2 x$, equivalently $|x^2|^2 = |x|^4$. Equivalently, the associated cubic $6F$ is a cubic eiconal in the sense of [64, Definition 3.1.1]; exactness is a separate condition.

Lemma 9.18 (The negative Peirce spaces)**.** (i) *If* $H \neq 0$, *then* $\operatorname{Spec} T \subset \{1, 0\}$.

(ii) *If* $Z \neq 0$, *then* $\operatorname{Spec} T \subset \{1, 7/27, 0\}$. *For* $U = E_T(7/27)$, *one has*

$$U = P \oplus Q, \quad P = E_{1/3}, \quad Q = E_{-1/6}, \qquad L_z|_P = 0, \quad (L_z|_Q)^2 = \frac{1}{12}|z|^2\,\mathrm{Id}_Q\,. \tag{9.35}$$

The proof is given in Appendix B.3.

Exactness excludes $Z = H = W = 0$.

Lemma 9.19. *In an exact Hsiang algebra whose multiplication is not identically zero,* $E_T(1) \neq \mathbb{R}c$.

The proof is given in Appendix B.3.

Theorem 9.20 (Spectral rigidity)**.** *Every exact Euclidean Hsiang algebra whose multiplication is not identically zero admits a positive rescaling for which*

$$\operatorname{Spec} T \subset \{0, 7/27, 1\}, \qquad \ker T = \operatorname{Ann}(A). \tag{9.36}$$

On A_{ess}*, either* $T|_{A_{\mathrm{ess}}} = \mathrm{Id}$ *or*

$$A_{\mathrm{ess}} = B \oplus U, \qquad T|_B = \mathrm{Id}, \qquad T|_U = \frac{7}{27}\,\mathrm{Id}\,. \tag{9.37}$$

Proof. The kernel assertion is Proposition 9.17. If $H \neq 0$ or $Z \neq 0$, use Lemma 9.18. Otherwise the fusion table gives $w^2 = |w|^2 c/2$ for $w \in W$. The norm identity with $r = 1/2$ then reads $0 = 3|w|^4/4$, so $W = 0$. This would give $B = \mathbb{R}c$, contrary to Lemma 9.19. □

9.6. **The eikonal core and its representation.** Work on A_{ess}, denoted by A, with nonradial weight operator. By Theorem 9.20 and (9.34),

$$A = B \oplus U, \qquad T|_B = \mathrm{Id}, \qquad T|_U = \tfrac{7}{27}\,\mathrm{Id}, \qquad B^2 \subset B, \quad BU \subset U. \tag{9.38}$$

Proposition 9.21 (The eikonal core and its grading)**.** *Assume* $T|_{A_{\mathrm{ess}}}$ *is not scalar. Then* $H = 0$*,* $Z \neq 0$*, and the subalgebra* $B = E_T(1)$ *is eikonal. Moreover,* $U^2 \subset B$*, so* $A = B \oplus U$ *is a* $\mathbb{Z}_2$*-grading.*

Proof. Lemma 9.18(i) and Theorem 9.20 give $H = 0$. If $Z = 0$, the proof of Theorem 9.20 would give $B = \mathbb{R}c$, so $Z \neq 0$. Fusion and invariance give

$$zz' = -\langle z, z'\rangle c, \qquad ZW \subset W, \qquad w^2 = \frac{1}{2}|w|^2 c + z_w, \quad z_w \in Z.$$

For a unit $z \in Z$, the self-adjoint map L_z preserves W. If $w \in W$ is an eigenvector with eigenvalue y, then $L_{e_\pm} w = (-1/4 \pm \sqrt{3}y/2)w$, where $e_\pm = (-c \pm \sqrt{3}z)/2$. Since $w \perp e_\pm$ and $Tw = w$, the quotient formula gives

$$p(-1/4 + \sqrt{3}y/2) = p(-1/4 - \sqrt{3}y/2) = 1.$$

The roots of $p(t) - 1$ are $-1, -1/2, 1/2$. Two such roots with sum $-1/2$ must be $-1, 1/2$, so $y^2 = 3/4$. Consequently $(L_z|_W)^2 = 3\,\mathrm{Id}\,/4$. Scaling and the norm identity at $w \in W$ yield

$$|zw|^2 = \frac{3}{4}|z|^2|w|^2, \qquad |z_w|^2 = \frac{3}{4}|w|^4.$$

For $b = tc + z + w$, its square has the orthogonal components

$$b^2 = \left(t^2 - |z|^2 + \frac{1}{2}|w|^2\right) c + (-2tz + z_w) + (tw + 2zw).$$

On taking norms, the terms linear in t cancel because $\langle z, z_w\rangle = \langle zw, w\rangle$. The displayed norm identities then give $|b^2|^2 = |b|^4$. Differentiating within the metrized subalgebra B proves $b^3 = |b|^2 b$.

For the grading, use $U = P \oplus Q$ from (9.35). For the input eigenvalue pairs $(1/3, 1/3)$, $(-1/6, -1/6)$, and $(1/3, -1/6)$, the fusion rule (9.28) permits, among the eigenvalues on $c^\perp$, respectively no output, only -1, and only $1/2$. Metric invariance therefore gives

$$p^2 = \tfrac{1}{3}|p|^2 c, \qquad q^2 \in \mathbb{R}c \oplus Z, \qquad pq \in W \qquad (p \in P,\ q \in Q).$$

Polarization yields $U^2 \subset B$; the other two grading relations are already contained in (9.34). □

At the maximal unit idempotent c, the decomposition is therefore

$$B = \mathbb{R}c \oplus Z \oplus W, \qquad U = P \oplus Q,$$

with L_c-eigenvalues $1, -1, 1/2, 1/3, -1/6$, respectively, and $Z \neq 0$. To determine the action of B on U, set $M_b = L_b|_U$ and $\{A_1, A_2\} = A_1A_2 + A_2A_1$.

Proposition 9.22. *For $b, d \in B$,*

$$\begin{aligned} M_{b^2} &= 6M_b^2 - \tfrac{1}{3}|b|^2 \operatorname{Id}_U, \\ M_{bd} &= 3\{M_b, M_d\} - \tfrac{1}{3}\langle b, d\rangle \operatorname{Id}_U . \end{aligned} \tag{9.39}$$

$$108M_b^3 - 9|b|^2 M_b - \langle b^2, b\rangle \operatorname{Id}_U = 0. \tag{9.40}$$

Proof. Every unit idempotent $e \in B$ satisfies $\langle e^2, e\rangle = 1$, so it attains the global spherical maximum in (9.24). The maximality bounds and Lemma 9.15, applied on $U \subset e^\perp$, give

$$\operatorname{Spec} M_e \subset \{1/3, -1/6\}.$$

Indeed, the third root $-7/6$ of $p(t) = 7/27$ is below the lower bound -1. Hence $M_e = 6M_e^2 - \operatorname{Id}_U/3$.

Consider the homogeneous quadratic operator polynomial $\mathscr{Q}(b) = M_{b^2} - 6M_b^2 + \frac{1}{3}|b|^2 \operatorname{Id}_U$. Every nonzero $b \in B$ lies in an eikonal subalgebra of dimension at most two. Indeed, differentiating $b^3 = |b|^2 b$ in direction b^2 gives

$$(b^2)^2 = 2\langle b^2, b\rangle\, b - |b|^2 b^2,$$

so $\operatorname{span}\{b, b^2\}$ is such a subalgebra. In dimension one it is generated by a unit idempotent, and hence $\mathscr{Q}$ vanishes there. In dimension two choose a positive spherical maximum e for its cubic. Since $|e^2| = 1$, the multiplier equation gives $e^2 = e$ without rescaling. For a unit vector $z \perp e$, write $ez = \alpha z$, $z^2 = \alpha e + \beta z$. Linearization at e gives $2\alpha^2 + \alpha = 1$, while $z^3 = z$ gives $\alpha\beta = 0$ and $\alpha^2 + \beta^2 = 1$. Thus $\alpha = -1$, $\beta = 0$, and

$$e, \qquad \frac{-e + \sqrt{3}z}{2}, \qquad \frac{-e - \sqrt{3}z}{2}$$

are three pairwise nonproportional unit idempotents. A homogeneous quadratic polynomial on a plane vanishing on three distinct lines is zero. Applying this to each matrix entry of $\mathscr{Q}$ proves $\mathscr{Q}(b) = 0$. Polarization gives the second quadratic identity. Finally, apply it to $b, d = b^2$. Since $b^3 = |b|^2 b$,

$$|b|^2 M_b = 3\{M_b, 6M_b^2 - \tfrac{1}{3}|b|^2 \operatorname{Id}_U\} - \tfrac{1}{3}\langle b^2, b\rangle \operatorname{Id}_U = 36M_b^3 - 2|b|^2 M_b - \tfrac{1}{3}\langle b^2, b\rangle \operatorname{Id}_U,$$

which gives the cubic identity. □

The remaining identities on U are as follows.

Lemma 9.23. *For $u \in U$, set $s = |u|^2$, $w = u^2 \in B$, and $v = wu \in U$. Then*

$$v = \tfrac{s}{9}u, \qquad |w|^2 = \tfrac{s^2}{9}, \qquad w^2 = \tfrac{s}{3}w, \qquad uv = \tfrac{s}{9}w. \tag{9.41}$$

Moreover, for every $b \in B$,

$$-3wb + 18u(bu) = sb. \tag{9.42}$$

Proof. Pair the first identity in (9.39) with u and differentiate in b to obtain (9.42). At u, (9.23) reads $4uv + w^2 = 7sw/9$. Together with (9.42) at $b = w$, this gives $uv = sw/9$ and $w^2 = sw/3$. Since B is eikonal, $|w^2|^2 = |w|^4$, and therefore

$$|u^2|^2\left(|u^2|^2 - \tfrac{1}{9}|u|^4\right) = 0 \qquad (u \in U).$$

The first factor is not the zero polynomial: $U \neq 0$ and the only possible eigenvalues $1/3, -1/6$ of M_c are nonzero, so $\langle u^2, c\rangle = \langle M_c u, u\rangle$ is not identically zero. The polynomial ring on U is an integral domain, whence $|u^2|^2 = |u|^4/9$ identically. Differentiating this identity within U gives $4wu = 4su/9$. □

Proposition 9.24 (Exactness of the core)**.** *If the original cubic is irreducible over $\mathbb{R}$, then B is exact,*

$$\operatorname{tr}_B L_b = \operatorname{tr}_U M_b = 0 \qquad (b \in B),$$

and $\dim B = \dim U + 2$.

The proof is given in Appendix B.4.

9.7. **The intrinsic five-generator closure.** Use the normalization in Lemma 9.14.

Theorem 9.25 (Five generators in the nonradial branch)**.** *Let F be a nonzero irreducible homogeneous harmonic cubic on a Euclidean space $V = V_{\mathrm{ess}}(F)$, with minimal regular zero set and nonradial quadratic weight. After a constant rescaling of F, let $V = B \oplus U$ be the orthogonal splitting of Theorem 9.20. For $x = b + u$, define*

$$r = |b|^2, \qquad s = |u|^2, \qquad a = \langle b^2, b\rangle, \qquad p = \langle bu, u\rangle, \qquad h = |bu|^2. \tag{9.43}$$

Then

$$\mathcal{A}_R(F) = \mathbb{R}[r, s, a, p, h], \qquad R = r + s, \qquad F = \frac{a + 3p}{6}. \tag{9.44}$$

These five polynomials are algebraically independent. In particular, the smallest Laplacian algebra of F requires exactly five ordinary algebra generators, of degrees $2, 2, 3, 3, 4$.

Proof. By Theorem 9.20 and Proposition 9.21, both B and U are nonzero, B is eikonal, and $B^2 \subset B$, $BU \subset U$, $U^2 \subset B$. By Proposition 9.24, irreducibility makes B exact, and the symmetric operators $M_b = L_b|_U$ satisfy

$$\operatorname{tr}_B L_b = 0, \qquad \operatorname{tr}_U M_b = 0 \qquad (b \in B). \tag{9.45}$$

Write $m = \dim U$, $n = \dim B = m + 2$, and $w = u^2 \in B$. The identities of Proposition 9.22 and Lemma 9.23, together with (9.42), give

$$M_{b^2} = 6M_b^2 - \frac{r}{3}\operatorname{Id}_U, \qquad 108M_b^3 = 9rM_b + a\operatorname{Id}_U, \tag{9.46}$$

$$|w|^2 = \frac{s^2}{9}, \qquad w^2 = \frac{s}{3}w,$$

$$M_w u = \frac{s}{9}u, \qquad 18u(bu) = 3bw + sb. \tag{9.47}$$

The eikonal identity on B gives $b(b^2) = rb$ and $|b^2|^2 = r^2$.

Differential closure. The five bidegrees with respect to $B \oplus U$ are, respectively,

$$(2,0), \quad (0,2), \quad (3,0), \quad (1,2), \quad (2,2).$$

Thus, for any of the five generators f of bidegree (i, j),

$$\Gamma(r, f) = 2if, \qquad \Gamma(s, f) = 2jf.$$

All remaining gradient products are

$$\begin{aligned}
\Gamma(a, a) &= 9r^2,\\
\Gamma(a, p) &= 18h - rs,\\
\Gamma(p, p) &= \frac{s^2}{9} + 4h,\\
\Gamma(a, h) &= rp + \frac{sa}{3},\\
\Gamma(p, h) &= \frac{rp}{3} + \frac{2sp}{9} + \frac{sa}{27},\\
\Gamma(h, h) &= \frac{r + s}{3}h + \frac{p^2}{9} + \frac{ap}{27}.
\end{aligned} \tag{9.48}$$

To verify the table intrinsically, metric invariance and (9.47) give the gradients, written in their $B \oplus U$ components,

$$\nabla a = (3b^2, 0), \qquad \nabla p = (w, 2M_b u), \qquad \nabla h = \left(\frac{bw}{3} + \frac{sb}{9}, 2M_b^2 u\right). \tag{9.49}$$

Their pairings, using (9.46), yield the first five lines of (9.48). For the last line, polarize the eikonal identity on B at w to obtain

$$2w(wb) + w^2 b = |w|^2 b + 2\langle w, b\rangle w.$$

Pairing with b and using $\langle b^2, w\rangle = 6h - rs/3$ gives

$$|bw|^2 = \frac{s^2 r}{9} + p^2 - sh.$$

Consequently

$$|\nabla_B h|^2 = \frac{sh}{3} + \frac{p^2}{9}, \qquad |\nabla_U h|^2 = 4\langle M_b^4 u, u\rangle = \frac{rh}{3} + \frac{ap}{27},$$

which proves the final pairing.

The Laplacians are

$$\Delta r = 2(m+2), \qquad \Delta s = 2m, \qquad \Delta a = \Delta p = 0, \qquad \Delta h = \frac{mr + (m+2)s}{9}. \tag{9.50}$$

To verify these formulas, exactness of B implies $\sum_{\alpha=1}^n e_\alpha^2 = 0$ for an orthonormal basis of B. Summing the quadratic identity in (9.46), and separately taking its trace, gives

$$\sum_{\alpha=1}^n M_{e_\alpha}^2 = \frac{n}{18}\,\mathrm{Id}_U, \qquad \operatorname{tr} M_b^2 = \frac{m}{18} r.$$

Consequently,

$$\Delta_B h = 2\sum_{\alpha=1}^n |M_{e_\alpha} u|^2 = \frac{n}{9} s, \qquad \Delta_U h = 2\operatorname{tr} M_b^2 = \frac{m}{9} r.$$

The other formulas in (9.50) follow from (9.45). The differential chain rule now proves that $\mathbb{R}[r, s, a, p, h]$ is a Laplacian algebra.

Recovery from R and F. The grading gives $F = (a + 3p)/6$. Applying (9.48) and (9.50) gives

$$S = \frac{1}{4}(r - s/3)^2 + 4h, \qquad \Delta S = \frac{(10m + 36)r + (2m + 4)s}{9}. \tag{9.51}$$

Since $R = r + s$ and $m > 0$, we recover the two radii directly:

$$r = \frac{9\Delta S - (2m + 4)R}{8(m + 4)}, \qquad s = R - r. \tag{9.52}$$

Both belong to $\mathcal{A}_R(F)$, because $S = \Gamma(F, F)$ and ΔS do. Setting $D = \Gamma(r, F) = a + p$ then gives

$$p = 3F - \frac{D}{2}, \qquad a = \frac{3D}{2} - 3F, \qquad h = \frac{\Gamma(a, p) + rs}{18}.$$

Thus all five generators belong to $\mathcal{A}_R(F)$. Since their algebra is Laplacian and contains R, F, equality follows.

Algebraic independence. At the maximal unit idempotent c, Proposition 9.21 supplies a nonzero (-1)-Peirce space $Z \subset B$. Choose a unit vector $z \in Z$. Then $z^2 = -c$, so at $b = z$ one has $r = 1$, $a = 0$, and

$$\nabla_B r = 2z, \qquad \nabla_B a = -3c.$$

In particular, the differential of (r, a) has rank two. By (9.46), every eigenvalue of M_z lies in

$$\{\lambda_1, \lambda_2, \lambda_3\} = \left\{-\frac{1}{2\sqrt{3}}, 0, \frac{1}{2\sqrt{3}}\right\}.$$

Let their multiplicities be k_1, k_2, k_3, allowing zero provisionally. The identities $\operatorname{tr} M_z = 0$, $\operatorname{tr} M_z^2 = m/18$, and $k_1 + k_2 + k_3 = m$ imply $k_1 = k_2 = k_3 = m/3 > 0$. Choose $u = u_1 + u_2 + u_3$ with a nonzero component in each eigenspace. For fixed $b = z$, the map from $t_i = |u_i|^2$ to the remaining invariants is

$$\begin{pmatrix} s \\ p \\ h \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \\ t_3 \end{pmatrix}.$$

The Vandermonde matrix is invertible, and the differentials dt_i are independent because $u_i \neq 0$. Thus (s, p, h) has rank three in the U directions, whereas (r, a) depends only on b and has rank two. The full five-invariant map has rank five at this point. Its components are therefore algebraically independent. Their polynomial algebra has transcendence degree five and cannot be generated by fewer than five elements. □

Corollary 9.26 (Cartan reduction and ambient dimensions)**.** *Under the hypotheses and normalization of Theorem 9.25, the cubic*

$$H = a - 9p = 6\big(\Gamma(r, F) - 5F\big) \tag{9.53}$$

satisfies

$$\Delta H = 0, \qquad |\nabla H|^2 = 9R^2.$$

It is orthogonally equivalent to a normalized Cartan cubic, and

$$\dim V \in \{8, 14, 26\}.$$

Writing $C = H/6$ and $C_B = C \circ \mathrm{pr}_B$, one has

$$F = \frac{4C_B - C}{3}. \tag{9.54}$$

Proof. The formula for H follows from $6F = a + 3p$ and $\Gamma(r, F) = a + p$. By (9.49),

$$\nabla H = (3b^2 - 9u^2, -18bu).$$

The identities $|b^2|^2 = r^2$, $|u^2|^2 = s^2/9$, and $\langle b^2, u^2\rangle = 6h - rs/3$ therefore give

$$|\nabla H|^2 = 9r^2 + 9s^2 - 54(6h - rs/3) + 324h = 9(r + s)^2.$$

Also $\Delta H = 0$ by (9.50).

Put $m = \dim U$. The multiplicity calculation in the proof of Theorem 9.25 gives $m/3 \in \mathbb{N}$ and $m > 0$. Since $\dim B = m + 2$, it follows that $\dim V = 2m + 2 \geq 8$. Cartan's classification of harmonic cubic eikonals [11], or [82, Theorem 1.1], now identifies H with a Cartan cubic in dimension $5, 8, 14$, or 26. In the latter theorem, the additional reducible possibility $x_N^3 - 3x_N \sum_{i<N} x_i^2$ has Laplacian $6(2 - N)x_N$, so harmonicity and $N \geq 8$ exclude it. The same dimension bound excludes $N = 5$. Finally, $C_B = a/6$ and $C = (a - 9p)/6$ give (9.54). □

The reduction recovers F from a Cartan cubic and a compatible splitting $B \oplus U$. Classifying F up to equivalence therefore also requires classifying these splittings.

The models in Section 11.1 realize the sharp bound and express r, s and the other invariants in coordinates. The programs check their closure, recovery, and rank-five minor; the universal spectral and trace arguments are proved above and in Appendix B.

Proof of Theorem 9.13. The divisibility argument in Section 9.5.1 and the cubic correspondence associate an exact Hsiang algebra to F. In the radial case, write $\mathcal{L}(F) = -\gamma RF$, where $\gamma > 0$ by the maximal normalization above. Harmonicity gives $\Gamma(F, S) = 2\gamma RF$. Replacing F by $\widetilde{F} = \sqrt{2/\gamma}\, F$ gives $\Gamma(\widetilde{F}, |\nabla \widetilde{F}|^2) = 4R\widetilde{F}$, as required by Proposition 9.9; this rescaling does not change $\mathcal{A}_R(F)$. The branch argument in Remark 9.10 then gives the radial four-generator bound.

In the nonradial case, Theorem 9.25 gives the polynomial algebra with the stated generator degrees. Restriction to $R = 1$ reduces its quotient dimension from five to four. The examples in Proposition 11.1 establish sharpness. □

10. Cubic Subcalibration Criteria and Radial Models

The radial table of Section 9 gives a degree-six subcalibration from two invariant inequalities or one sufficient trace bound. An additional identity yields a degree-seven polynomial alternative.

10.1. **Existence from the radial cubic table.** The common defect formula also applies to the nonradial models of Section 11, using their five-generator table.

Lemma 10.1 (The common harmonic potential)**.** *Let $F \not\equiv 0$ be a homogeneous harmonic cubic on $\mathbb{R}^N$, $N \geq 2$, satisfying $\mathcal{L}(F) = -qF$ for a homogeneous quadratic polynomial q. Put $K = \Delta S$ and $T = \Gamma(S,S)$. For $G = F\sqrt{RS}$, there is a homogeneous polynomial $\mathcal{P}_F$ of degree eighteen such that*

$$\mathcal{L}(G) = \frac{F\mathcal{P}_F}{16\sqrt{R}\,S^{3/2}} \qquad \text{on } \{S > 0\}. \tag{10.1}$$

If $\mathcal{P}_F \geq 0$ there and $Z = \{S = 0\}$ has locally zero $\mathcal{H}^{N-1}$-measure, then G generates a two-sided subcalibration. The phase $\{F < 0\}$ is locally perimeter minimizing and its boundary current is locally mass minimizing, with support $\{F = 0\}$.

Proof. Using $\Gamma(F,S) = 2qF$ and homogeneity gives, on $S > 0$,

$$Q_G := |\nabla G|^2 = RS^2 + F^2\left[R\left(2q + \frac{T}{4S}\right) + 11S\right], \tag{10.2}$$

$$\Delta G = G\left[\frac{N+9}{R} + \frac{2q + K/2}{S} - \frac{T}{4S^2}\right]. \tag{10.3}$$

Substitution into $\mathcal{L}(G) = Q_G\Delta G - \Gamma(G, Q_G)/2$ gives (10.1) with a polynomial $\mathcal{P}_F$. Its explicit coefficient formula is $\mathcal{P}_F = C_0 + FC_1 + F^2C_2$, with (C.14) specialized to $t = 0$ and $H_2 = |\operatorname{Hess} F|^2 = K/2$. Appendix C.3 gives the full expansion and derivation. Since $Z \subset \{F = 0\}$ by Euler's identity, G extends continuously with the signs of F. Its positive multiplier and degree six give $\nabla G \neq 0$ off Z by the homogeneous-multiplier verification following Corollary 2.8. That corollary and Theorem 2.5 give comparison; harmonicity identifies the full support by Remark 2.7. □

By Proposition 9.2 and closure, $\mathcal{P}_F \in \mathcal{A}_R(F)$; its sign is tested on the actual generator image. No separate condition (C.15) is needed.

For the radial existence criterion, use D from (9.7) and put $K = \Delta S = 2\operatorname{tr}(L_x^2)$. By Lemma 9.8, the criterion depends only on the radial table and K, R, S, D.

Theorem 10.2 (Subcalibrations from the radial cubic table)**.** *Let $F \not\equiv 0$ be a homogeneous harmonic cubic on a Euclidean space of dimension $N \geq 8$, with $\Gamma(F,S) = 4RF$. Define D by (9.7) and put $K = \Delta S$. Assume, at every point,*

$$K \geq 16R \tag{10.4}$$

and

$$\mathscr{H} := (N-12)S^2 + \frac{1}{2}R(K - 16R)S - 12RD \geq 0. \tag{10.5}$$

Then the phase $\{F < 0\}$ is locally perimeter minimizing, and its oriented boundary current is strictly area minimizing with support $\{F = 0\}$. These conclusions are realized by the degree-six potential

$$G = F\sqrt{RS} = \frac{1}{12}|x|\,|x^2|\langle x^2, x\rangle$$

which is continuous on V, smooth on $V \setminus Z$, and has $\{G < 0\} = \{F < 0\}$, where $Z = \{S = 0\}$. Its gradient is nonzero off Z, and its normalized gradient is a two-sided subcalibration whose divergence has the strict sign of F when $F \neq 0$.

The polynomial $\mathscr{H}$ has a direct boundary interpretation. Apply Proposition 4.5 with $e = 3$, $\sigma = 1$, $\gamma = 1/2$, $q = 2R$, and $T = 4RS + 12D$. Since $w = 1/(rS)$, this gives

$$\mathscr{H}|_{C_{\mathrm{reg}}} = RS^2\left(-\frac{J_C w}{w}\right). \tag{10.6}$$

Since G has degree six, Theorem 4.6 also gives

$$Q_C(u) \geq \left(6 - \frac{N-1}{2}\right)^2 \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}).$$

This proves strict stability whenever $N \neq 13$, including when the link is singular.

Lemma 10.3 (Bounds for radial cubics)**.** *Assume the structural hypotheses of Theorem 10.2, without its dimension and inequality assumptions. Then*

$$D \geq 0, \qquad S \leq \frac{R^2}{3}, \qquad 3RD \leq S^2, \qquad 9D \leq RS, \qquad F^2 \leq D. \tag{10.7}$$

Proof. By Lemma 9.8, the last identity in (9.8) gives $D \geq 0$ on $S > 0$. At $S = 0$, the nonnegative function S has zero gradient, and (9.7) gives $D = 0$. Take a positive maximum of S on $R = 1$. Homogeneity gives $\nabla S = 4Sx$ there. Pairing with ∇D and using Euler's identity gives $24SD = 8D$. Thus $S = 1/3$ if $D > 0$. If $D = 0$, then $16S^2 = \Gamma(S,S) = 4S$, so $S = 1/4$. Homogeneity proves the bound for S.

Cauchy–Schwarz gives $(8RD)^2 \leq (4RS + 12D)(4SD)$. When $D > 0$, this is $D(4R^2 - 3S) \leq RS^2$. Since $3S \leq R^2$, it implies $3RD \leq S^2$. The assertion is immediate when $D = 0$. Similarly, Euler's identity and (9.8) give

$$36D^2 = \langle x, \nabla D\rangle^2 \leq 4RSD, \qquad 4S^2F^2 = \Gamma(F,D)^2 \leq 4S^2D.$$

These prove the final two bounds wherever cancellation is possible. At $S = 0$, Euler's identity for F gives $F = 0$, so the remaining zero cases also follow. □

Proof of Theorem 10.2. Specializing Lemma 10.1 to $q = 2R$ and $T = 4RS + 12D$ gives, on $S > 0$,

$$Q := |\nabla G|^2 = RS^2 + F^2\left(\frac{3DR}{S} + 5R^2 + 11S\right), \tag{10.8}$$

$$\Delta G = G\left(\frac{N+9}{R} + \frac{K}{2S} + \frac{3R}{S} - \frac{3D}{S^2}\right). \tag{10.9}$$

In particular, $Q \geq RS^2 > 0$. Substituting these formulas into $\mathcal{L}(G) = Q\Delta G - \Gamma(G,Q)/2$ yields

$$\mathcal{L}(G) = \frac{G}{RS^2}\left(RS^2\mathscr{H} + F^2\mathscr{J}\right), \tag{10.10}$$

where

$$\begin{aligned}\mathscr{J} = {} & RD\left[\tfrac{3}{2}R(K - 16R) + 15(R^2 - 3S) + 3(N-8)S\right] \\ & + \tfrac{5}{2}(K - 16R)R^3S + 45R^4S + \tfrac{11}{2}(K - 16R)RS^2 \\ & + (5N + 33)R^2S^2 + (11N + 44)S^3.\end{aligned} \tag{10.11}$$

This calculation uses only Lemma 9.8 and homogeneity; no derivative of K is involved. Every term in $\mathscr{J}$ is nonnegative by Lemma 10.3, (10.4), and $N \geq 8$. The term $45R^4S$ is positive. Hence $F\mathcal{L}(G) > 0$ wherever $F \neq 0$.

The critical set is removable by Lemma 9.7. Since $\mathcal{P}_F = 16(RS^2\mathscr{H} + F^2\mathscr{J})$, Lemma 10.1 gives the minimizing and support conclusions. Both open phases are nonempty because F is odd, and the strict divergence sign makes both integrals in (2.18) positive. The conical sets $E = \{F < 0\}, Z$ and zero-homogeneous field X_G therefore satisfy Theorem 2.11, proving strict area minimality. □

Corollary 10.4 (Existence from a trace bound)**.** *Under the structural hypotheses of Theorem 10.2, with $N \geq 8$, the single bound*

$$\mathrm{tr}(L_x^2) \geq \max\left\{8, \frac{40 - N}{3}\right\}|x|^2 \tag{10.12}$$

implies its two inequality assumptions and all its conclusions.

Proof. The bound implies $K \geq 16R$. By Lemma 10.3,

$$\mathscr{H} \geq (N-16)S^2 + \tfrac{1}{2}R(K-16R)S.$$

For $N \geq 16$ this is nonnegative. For $8 \leq N < 16$, the bound $S \leq R^2/3$, with the negative coefficient $N-16$, gives

$$\mathscr{H} \geq RS\left[\frac{K}{2} + \frac{N-40}{3}R\right] \geq 0.$$

□

The trace test allows nonscalar K, without a closure condition on ΔD. The low-dimensional Clifford endpoint instead uses (10.5) directly.

10.2. **Polynomial alternatives of degree seven.** To obtain polynomial potentials in place of $F\sqrt{RS}$, we use the degree-seven potential SF. Its defect identity holds on the full radial cubic table. We first impose the additional relation $D = F^2$; Proposition 10.13 instead assumes a scalar trace coefficient and allows an independent D.

Lemma 10.5 (The gradient-square defect)**.** *Let F be a nonzero homogeneous harmonic cubic satisfying $\Gamma(F,S) = 4RF$, put $K = \Delta S$, and define D by (9.7). For $G = SF$, one has*

$$Q := |\nabla G|^2 = S^3 + 12F^2(RS+D), \qquad \Delta G = (K+8R)F, \tag{10.13}$$

and

$$\begin{aligned}\frac{\mathcal{L}(G)}{F} &= (K-16R)S^3 - 30DS^2 \\ &\quad + F^2\big[(12K-72R)D + 12RKS - 96S^2\big].\end{aligned} \tag{10.14}$$

The quotient denotes its polynomial extension across $F=0$. No scalar-trace condition or identity for ΔD is required.

Proof. Apply the product rule and Lemma 9.8 to $Q = |\nabla(SF)|^2$, then substitute in $\mathcal{L}(SF) = Q\Delta(SF) - \Gamma(SF,Q)/2$. The individual pairings are recorded in Appendix C.5. □

Corollary 10.6 (A polynomial gradient-square certificate)**.** *Let V be a Euclidean commutative metrized algebra of dimension $N \geq 3$. Suppose that, for every $x \in V$,*

$$\operatorname{tr} L_x = 0, \tag{10.15}$$

$$\langle x^2, x^3\rangle = \frac{4}{3}|x|^2\langle x^2, x\rangle, \tag{10.16}$$

$$|x^3|^2 = |x|^2|x^2|^2 + \frac{1}{3}\langle x^2, x\rangle^2, \tag{10.17}$$

$$\operatorname{tr}(L_x^2) \geq 8|x|^2. \tag{10.18}$$

Set

$$F(x) = \frac{1}{6}\langle x^2, x\rangle, \qquad R = |x|^2, \qquad S = |\nabla F|^2, \qquad G = SF.$$

Then G is a polynomial potential of degree seven with

$$\{G < 0\} = \{F < 0\}, \qquad \{\nabla G = 0\} = \{\nabla F = 0\}.$$

Its normalized gradient is a two-sided subcalibration away from this common critical set, which has locally zero $\mathcal{H}^{N-1}$-measure. Hence $\partial[\![\{F<0\}]\!]$ is locally mass minimizing. The divergence has a strict sign at every point with $F \neq 0$.

Proof. The multiplication dictionary turns the three identities into $\Delta F = 0$, $\Gamma(F,S) = 4RF$, and $\Gamma(S,S) = 4RS + 12F^2$, so $D = F^2$. Bochner's identity gives $K = \Delta S = 2\operatorname{tr}(L_x^2) \geq 16R$, and Lemma 10.3 gives $S \leq R^2/3$. Substituting these facts in Lemma 10.5 yields the nonnegative decomposition (C.29), which is strict wherever $F \neq 0$. The same gradient identity identifies the exceptional set with $\{\nabla F = 0\}$, and Lemma 9.7 and Corollary 2.8 give the minimizing conclusion. The coefficient decomposition is proved in Appendix C.5. □

Remark 10.7 (Relation to gradient-square closure). The norm identity (10.17) is exactly $D = F^2$. In the three-generator closure of Section 10.3, Proposition 7.4 gives $K = 4kR$, so the trace bound becomes $k \geq 4$. The quaternionic rank-four trace model has $k = 5$ and satisfies these hypotheses (Corollary 10.9). Unlike this certificate, Theorem 10.2 allows an independent D; SF provides a polynomial alternative when the displayed defect is nonnegative.

The cubic–algebra dictionary and the multiplier obstruction do not require harmonicity. For a general quadratic weight in $\mathcal{L}(F) = -qF$, Appendix C.3 retains the trace terms in the defect of $F\sqrt{RS}$. Proposition C.2 gives a sufficient polynomial positivity test.
Applications to the classical cubic models. The normal forms in Section 9.1 give three-generator rank-four trace cubics and four-generator complexified rank-three Jordan norms and G_2 trilinear cubics. The table collects their $F\sqrt{RS}$ data and the nonradial model of Section 11.

For the Cartan–Münzner cubics Φ in dimensions $14, 26$, set $F = \Phi/(3\sqrt{3})$. Then $S = R^2/3$, $D = R^3/27$, and $K = 4(N+2)R/3$. The common potential is proportional to $R^{3/2}F$; the polynomial alternative RF is given in Theorem 6.5 and Example 4.23.

model	N	verification data
Cartan, quaternionic	14	$K/R = 64/3$
Cartan, octonionic	26	$K/R = 112/3$
rank-four quaternionic trace	27	$K/R = 20$
complexified rank-three Jordan norm, $d = 2$	18	$K/R = 16$
complexified rank-three Jordan norm, $d = 4$	30	$K/R = 24$
complexified rank-three Jordan norm, $d = 8$	54	$K/R = 40$
G_2 trilinear	21	$K/R = 24$
nonradial quaternionic model	26	five-generator closure

The first seven rows satisfy the radial trace bound $K/R \geq \max\{16, 2(40-N)/3\}$. The final row is verified in Theorem 11.5. In every row the boundary current is strictly area minimizing.

10.3. **Three-generator closure and rank-four trace cubics.** For the closure and realizations in Propositions 7.4 and 7.5, we determine area minimization, trace models, and the sharp stability constant. The degree-seven polynomial $G = SF$ complements the common degree-six potential $F\sqrt{RS}$.

10.3.1. *The tangent obstruction and the existence theorem.* The following closure criterion retains its displayed generality: for example, the reducible polynomial $F = xyz$ on $\mathbb{R}^3$ satisfies (7.2) with $k = 1$, by direct differentiation. This comparison endpoint is not subject to the introductory irreducibility convention.

Theorem 10.8 (Area minimization from gradient-square closure)**.** *Let F be a nonzero homogeneous cubic on $\mathbb{R}^N$, $N \geq 3$, and let $S = |\nabla F|^2$. Suppose that $\mathbb{R}[R, F, S]$ is a Laplacian algebra and that F has a nonzero critical point. Write $N = 6k - 3$ as in Proposition 7.4. Then*

$$\partial[\![\{F < 0\}]\!] \text{ is locally mass minimizing} \quad \Longleftrightarrow \quad k \geq 4. \tag{10.19}$$

Every minimizing member is strictly area minimizing in the sense of Definition 2.1, with polynomial potential $G = SF$ of degree seven. It is also strictly stable, with the optimal Hardy constant determined in Theorem 10.10.

Proof. Normalize as in (7.2) and suppose $k \geq 4$. The dictionary in Proposition 9.1 translates the normalized gradient identities into the product identities of Corollary 10.6. Moreover, Bochner's identity gives

$$K = \Delta S = 4kR, \qquad \operatorname{tr}(L_x^2) = \tfrac{1}{2}K = 2kR \geq 8R.$$

That theorem therefore gives the polynomial subcalibration potential $G = SF$, including the required sign and the removability of its critical set. Its divergence is strictly signed

wherever $F \neq 0$. Since G is homogeneous and polynomial and both phases are nonempty, Corollary 2.13 gives the strict mass gap.

The sharp stability constant is computed in Theorem 10.10. Harmonicity and Proposition 3.12 also show that every minimizing member is irreducible.

For the converse, Proposition 7.5 identifies $k = 1$ with the reducible harmonic cubic xyz, whose phases cannot minimize perimeter by Proposition 3.12. For $k = 2, 3$, take a nonzero critical point p. The Taylor expansion in Proposition 7.4 gives

$$r^{-2}F(p + rz) \longrightarrow \frac{|p|}{2}\big(|u|^2 - |v|^2\big), \qquad z = (u, v, w), \quad u, v \in \mathbb{R}^k.$$

The transverse quadratic cone is unstable, since its link has $|A|^2 = 2k - 2$ and

$$\frac{(2k-3)^2}{4} - (2k-2) < 0.$$

By Section 2.6, the Taylor expansion gives smooth regular graphical convergence. Lemma 2.15 therefore makes these cases unstable and excludes area minimization. □

10.3.2. *The rank-four trace realizations.* The trace cubics on $\mathrm{Herm}_4(\mathbb{F}_d)_0$, $d = 1, 2, 4$, realize the three exceptional cases in dimensions $9, 15, 27$ [64, Section 6.11.6(iii)]. They admit the following uniform calculation.

Corollary 10.9 (The rank-four trace cubics)**.** *Let $\mathbb{F}_d = \mathbb{R}, \mathbb{C}, \mathbb{H}$, with real dimension $d = 1, 2, 4$, and equip*

$$V = \mathrm{Herm}_4(\mathbb{F}_d)_0, \qquad \dim_{\mathbb{R}} V = 3 + 6d,$$

with $\langle A, B\rangle = \mathrm{Re}\,\mathrm{tr}(AB)$. For $F(A) = \frac{1}{3}\,\mathrm{Re}\,\mathrm{tr}(A^3)$, $R = \mathrm{tr}(A^2)$, and $S = |\nabla F|^2$, one has

$$\mathcal{A}_R(F) = \mathbb{R}[R, F, S], \qquad \Delta F = 0, \quad \Delta S = 4(d+1)R,$$

$$\Gamma(F, S) = 4RF, \qquad \Gamma(S, S) = 4RS + 12F^2.$$

The generators are algebraically independent. The oriented zero cone is area minimizing if and only if $d = 4$; in that case $G = SF$ proves strict area minimality. The common potential $F\sqrt{RS}$ also gives a subcalibration in this case.

Proof. Projection onto the trace-free subspace gives

$$\nabla F = A^2 - \tfrac{R}{4}\,\mathrm{Id}, \qquad S = \mathrm{tr}(A^4) - \tfrac{R^2}{4},$$
$$(\mathrm{Hess}\,F)_A(B) = AB + BA - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{tr}(AB)\,\mathrm{Id}\,.$$

Write $p_j = \sum_i \lambda_i^j$ after diagonalizing A, where $p_1 = 0$. If P is the orthogonal projection onto the sum-zero diagonal vectors, the diagonal Hessian is $2P\,\mathrm{diag}(\lambda_i)P$, with trace zero and squared Hilbert–Schmidt norm $2R$. Each off-diagonal entry contributes d eigenvalues $\lambda_i + \lambda_j$. Thus

$$\Delta F = 0, \qquad |\mathrm{Hess}\,F|^2 = 2R + d\sum_{i<j}(\lambda_i + \lambda_j)^2 = 2(d+1)R,$$

and $\Delta S = 4(d+1)R$. Also, $\nabla S = 4A^3 - RA - p_3\,\mathrm{Id}$. Newton's identities give

$$p_5 = \tfrac{5}{6}Rp_3, \qquad p_6 = \tfrac{3}{4}Rp_4 - \tfrac{1}{8}R^3 + \tfrac{1}{3}p_3^2.$$

Substitution proves the two gradient identities. Together with Euler's identities, Proposition 3.7 gives Laplacian closure. Since $S = \Gamma(F, F)$ belongs to $\mathcal{A}_R(F)$, this is the smallest algebra containing R, F. At distinct eigenvalues, the Vandermonde determinant shows that p_2, p_3, p_4 have independent differentials on $p_1 = 0$; hence R, F, S are algebraically independent.

The gradient formula and the transverse model in Proposition 7.4 identify the singular set as

$$\mathrm{Sing}\{F = 0\} = \{A \in V : A^2 = \tfrac{1}{4}\,\mathrm{tr}(A^2)\,\mathrm{Id}\}.$$

Its nonzero matrices have eigenvalues $(\lambda,\lambda,-\lambda,-\lambda)$, with $\lambda>0$. In particular, $\operatorname{diag}(1,1,-1,-1)$ is a nonzero critical point. Since $\Delta S=4kR$ with $k=d+1$, Theorem 10.8 proves all the minimizing and nonminimizing assertions. For the common potential, put $D=F^2$. The displayed identities imply

$$\Gamma(F,D)=2SF,\qquad \Gamma(S,D)=8RD,\qquad \Gamma(D,D)=4SD.$$

For $d=4$, $N=27$ and $\operatorname{tr}L_x^2=10R$, so Corollary 10.4 applies. □

10.3.3. *Exact link geometry and the sharp Hardy constant.* On the sphere write $f=F|_{\mathbb{S}^{N-1}}$ and $s=S|_{\mathbb{S}^{N-1}}$. Let $M=Z^*\cap\mathbb{S}^{N-1}$. Since Z^* is conical and radial vectors are transverse to the sphere, M is a compact smooth submanifold of dimension $4k-4$. The zero link $\Sigma=\{F=0\}\cap\mathbb{S}^{N-1}$ has regular part of dimension $6k-5$. Its transverse dimension along M is thus $2k-1$. If ρ denotes spherical distance to M, then, near M,

$$s\asymp\rho^2,\qquad \mathcal{H}^{N-2}\big(\Sigma_{\mathrm{reg}}\cap\{\rho<\varepsilon\}\big)\le C\varepsilon^{2k-1}. \tag{10.20}$$

The proof of these estimates is given in Appendix B.5.

Two geometric identities. With $\nu=\nabla F/\sqrt{S}$, the curvature identity proved in Proposition 4.5 is $|A_C|^2=(\Delta S)/(2S)-|\nabla S|^2/(2S^2)$. Substituting $\Delta S=4kR$ and $|\nabla S|^2=4RS$ on the regular cone gives

$$|A_C|^2=\frac{2(k-1)R}{S},\qquad |A_\Sigma|^2=\frac{2(k-1)}{s}. \tag{10.21}$$

The spherical cross term $\Gamma_{\mathbb{S}}(f,s)=4f(1-3s)$ vanishes on the zero link. Therefore

$$|\nabla_\Sigma s|^2=4s-16s^2. \tag{10.22}$$

In particular $0<s\le 1/4$ on Σ_{reg}, improving the global bound $s\le 1/3$ in (C.27).

To compute the link Laplacian, differentiate the closure table:

$$\Gamma(F,\Gamma(F,S))=4RS+24F^2,$$
$$\operatorname{Hess}F(\nabla F,\nabla S)=\tfrac12|\nabla S|^2=2RS+6F^2.$$

Their difference is $\operatorname{Hess}S(\nabla F,\nabla F)$; hence $\operatorname{Hess}S(\nu,\nu)=2R$ on the cone. Since S has degree four, its spherical normal Hessian at $r=1$ equals $2-4s$. Subtract this from $\Delta_{\mathbb{S}}s=4k-4(N+2)s$. The link is minimal, so there is no mean-curvature term, and

$$\Delta_\Sigma s=4k-2-4(N+1)s=4k-2-(24k-8)s. \tag{10.23}$$

A positive eigenfunction and its domain. Let $J_\Sigma=\Delta_\Sigma+|A_\Sigma|^2$. For any real p, equations (10.21)–(10.23) give

$$J_\Sigma(s^{-p})=\left\{\frac{4p^2+(6-4k)p+2k-2}{s}+4p(6k-6-4p)\right\}s^{-p}. \tag{10.24}$$

For $k\ge 4$, define

$$\delta_k=\sqrt{4k^2-20k+17}>0,\qquad p_k=\frac{2k-3-\delta_k}{4}>0,\qquad B_k=4p_k(6k-6-4p_k). \tag{10.25}$$

The coefficient of s^{-1} in (10.24) vanishes, so

$$\phi=s^{-p_k}>0,\qquad J_\Sigma\phi=B_k\phi. \tag{10.26}$$

Theorem 10.10 (Optimal Hardy stability constant)**.** *Under (7.2), with a nonvertex critical point and $k\ge 4$, the Friedrichs bottom of $-J_\Sigma$, initially defined on $C_c^\infty(\Sigma_{\mathrm{reg}})$, is $-B_k$. The optimal constant in*

$$Q_C(u)\ge H_k\int_C r^{-2}u^2,\qquad u\in C_c^\infty(C_{\mathrm{reg}}\setminus\{0\}),$$

is

$$\boxed{H_k=\big(k+\sqrt{4k^2-20k+17}\big)^2.} \tag{10.27}$$

In particular, for $k = 5$, $N = 27$, $H_5 = 42 + 10\sqrt{17}$.

Proof. For every compactly supported smooth v on the regular link, the positive-solution identity applied to (10.26) gives

$$\int_\Sigma \big(|\nabla v|^2 - |A_\Sigma|^2 v^2\big) = \int_\Sigma \phi^2\, |\nabla(v/\phi)|^2 - B_k \int_\Sigma v^2. \tag{10.28}$$

The bottom is therefore at least $-B_k$.

By (10.20), $\phi \asymp \rho^{-2p_k}$, and $2k - 3 - 4p_k = \delta_k > 0$. Thus $\phi \in L^2(\Sigma)$, with finite gradient and potential energy near M. Choose smooth cutoffs η_ϵ, zero when $\rho < \epsilon$, one when $\rho > 2\epsilon$, with $|\nabla \eta_\epsilon| \le C/\epsilon$. The local volume bound gives

$$\int_\Sigma \phi^2 |\nabla \eta_\epsilon|^2 = O(\epsilon^{2k-3-4p_k}) = O(\epsilon^{\delta_k}) \longrightarrow 0.$$

Use $v = \eta_\epsilon \phi$ in (10.28). Its Rayleigh quotient tends to $-B_k$, proving the claimed bottom. The same estimates give convergence to ϕ in the Friedrichs form norm. Testing (10.26) first against compactly supported smooth functions and then using form density shows that ϕ belongs to the Friedrichs operator domain, with eigenvalue $-B_k$ for $-J_\Sigma$.

The cone dimension is $n = N - 1 = 6k - 4$. By (4.16), the optimal cone constant is

$$\frac{(n-2)^2}{4} + \inf \operatorname{spec}(-J_\Sigma) = 9(k-1)^2 - B_k.$$

Finally,

$$9(k-1)^2 - B_k = \big(3(k-1) - 4p_k\big)^2 = (k + \delta_k)^2,$$

which proves (10.27). □

Remark 10.11. For the larger root in (10.25), $2k - 3 - 4p = -\delta_k$, so the cutoff energy does not tend to zero. The radicand δ_k^2 equals four times the Hardy stability constant of the transverse balanced quadratic cone:

$$\delta_k^2 = 4\left[\frac{(2k-3)^2}{4} - (2k-2)\right].$$

10.4. **Four-generator radial models.**

10.4.1. *Complexified Jordan norms.* The sextic D is now independent: $D = (\Gamma(S,S) - 4RS)/12$. In the preceding trace family, $D = F^2$.

Let $\mathbb{K}_d = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ for $d = 1, 2, 4, 8$, respectively, and set $J_d = \mathrm{Herm}_3(\mathbb{K}_d)$, writing $J = J_d$ locally. Its Jordan product is $x \bullet y = (xy + yx)/2$, its unit is e, and its Euclidean inner product is $\operatorname{tr}(x \bullet y)$. Thus $(J_d)_0 = e^\perp = \{x \in J_d : \operatorname{tr} x = 0\}$ denotes its trace-free part. On the real vector space underlying $V = J \otimes_{\mathbb{R}} \mathbb{C}$, use the real part of the Hermitian extension of this inner product. Thus $|x + iy|^2 = |x|^2 + |y|^2$ and $\dim_{\mathbb{R}} V = 6(d+1)$. Let N be the complexification of the Jordan determinant. A Jordan frame consists of three idempotents c_i with $c_i \bullet c_j = \delta_{ij} c_i$, $\operatorname{tr} c_i = 1$, and $\sum_i c_i = e$. The normalization is

$$N(z_1c_1 + z_2c_2 + z_3c_3) = z_1z_2z_3, \qquad |z_1c_1 + z_2c_2 + z_3c_3|^2 = \sum_i |z_i|^2.$$

These are the complexified rank-three models in [64, Section 6.11.6(iv)]; the polynomial there is $2 \operatorname{Re} N$, with the same zero set. Their REC characteristic dimensions are $(n_1, n_2) = (1, 3d+2)$.

The determinant has the coordinate formula

$$z = \begin{pmatrix} a & \bar{u} & \bar{v} \\ u & b & w \\ v & \bar{w} & h \end{pmatrix}, \qquad N(z) = abh - a\, n(w) - b\, n(v) - h\, n(u) + 2\,\mathrm{Sc}((\bar{u}w)v). \tag{10.29}$$

Here $a, b, h \in \mathbb{C}$ and $u, v, w \in \mathbb{K}_d \otimes_{\mathbb{R}} \mathbb{C}$; the bar is algebra conjugation fixing complex scalars, $n(u) = u\bar{u}$, and Sc is the complex-linear extension of the scalar part. Thus n is a complex quadratic form, distinct from the Hermitian norm $|u|^2$.

The quadratic adjoint $z^{\#}$ satisfies $(dN)_z(v) = \operatorname{tr}(z^{\#} \bullet v)$, with the complex bilinear trace form. The rank-three identities are

$$z \bullet z^{\#} = N(z)e, \qquad (z^{\#})^{\#} = N(z)z, \qquad \operatorname{tr}(z \bullet z^{\#}) = 3N(z). \tag{10.30}$$

They extend from J to V as polynomial identities; see [64, Section 3.2.7] for the Jordan algebra background. Put

$$F = \operatorname{Re} N, \qquad R = |z|^2, \qquad S = |z^{\#}|^2, \qquad D = |N(z)|^2. \tag{10.31}$$

Holomorphicity gives $\Delta F = 0$ and $|\nabla F|^2 = S$. The Jordan product on J is distinct from the metrized product obtained by polarizing F on V.

Theorem 10.12 (The smallest closure of the complexified models)**.** *For each $d \in \{1, 2, 4, 8\}$, the four polynomials in (10.31) are algebraically independent and*

$$\mathcal{A}_R(F) = \mathbb{R}[R, F, S, D].$$

Their degrees are $2, 3, 4, 6$; in particular, $(\mathcal{A}_R(F))_2 = \mathbb{R}R$ and $(\mathcal{A}_R(F))_3 = \mathbb{R}F$. The Laplacian identities are

$$\Delta R = 12(d+1), \qquad \Delta F = 0, \qquad \Delta S = 4(d+2)R, \qquad \Delta D = 4S. \tag{10.32}$$

The gradient pairings, apart from those determined by $\Gamma(R, P) = 2\deg(P)P$, are

$$\begin{aligned} &\Gamma(F, F) = S, && \Gamma(F, S) = 4RF, \quad \Gamma(F, D) = 2SF, \\ &\Gamma(S, S) = 4RS + 12D, && \Gamma(S, D) = 8RD, \quad \Gamma(D, D) = 4SD. \end{aligned} \tag{10.33}$$

In particular, $\mathcal{L}(F) = -2RF$, so the regular zero set is minimal. Throughout V,

$$R^2 \geq 3S, \qquad RS \geq 9D, \qquad S^2 \geq 3RD, \qquad 0 \leq F^2 \leq D. \tag{10.34}$$

Proof. Differential identities. For real $a \in J$, $T_a : z \mapsto a \bullet z$ is self-adjoint for the Hermitian trace metric, and (10.30) gives $(dN)_z(T_a z) = (\operatorname{tr} a)N(z)$. Consequently $\exp(itT_a)$ is unitary and multiplies N by $\exp(it \operatorname{tr} a)$. Real Jordan automorphisms are also unitary and preserve N. These transformations preserve R, S, D and transport identities involving N with its phase factor.

The linearized Jordan identity implies that the commutators $[T_a, T_b]$ are derivations, whose exponentials are automorphisms. Take the standard diagonal frame (c_i). At $x = \sum_i r_i c_i$, with distinct positive r_i, the Peirce decomposition [64, Section 3.2.6] is

$$J = \bigoplus_i \mathbb{R}c_i \ \oplus \bigoplus_{i<j} J_{ij}, \qquad \dim_{\mathbb{R}} J_{ij} = d.$$

Here J_{ij} consists of matrices supported in the (i, j) and (j, i) entries, and $c_i \bullet a = c_j \bullet a = a/2$ for $a \in J_{ij}$. Varying the r_i supplies the real diagonal directions, while the derivations and unitary infinitesimal transformations give

$$[T_{c_i}, T_a]x = \tfrac{1}{4}(r_j - r_i)a, \qquad iT_a x = \tfrac{1}{2}(r_i + r_j)ia, \qquad iT_{c_i} x = ir_i c_i.$$

These directions span the underlying real space of V. The map obtained by varying the diagonal entries and the transformation parameters is therefore a submersion at x, so its image contains a real open subset of V.

On the diagonal, with $t_i = r_i^2$, $R = e_1(t), S = e_2(t), D = e_3(t)$, where $e_1(t) = \sum_i t_i$, $e_2(t) = \sum_{i<j} t_i t_j$, and $e_3(t) = t_1 t_2 t_3$. The first derivatives of these three invariants in the off-diagonal directions vanish. Differentiation gives the last row of (10.33). Extending Γ complex bilinearly, diagonal differentiation gives $\Gamma(N, S) = 4RN$, whose real part is the desired identity for F. Holomorphicity gives the remaining pairings with F. The complex Hessian of N has diagonal-block squared norm $2R$; on J_{ij} its block is $-r_k \operatorname{Id}$, where $\{i, j, k\} = \{1, 2, 3\}$, contributing dR in total. Therefore

$$|\operatorname{Hess}_{\mathbb{C}} N|^2 = (d+2)R, \qquad |\operatorname{Hess}_{\mathbb{R}} F|^2 = 2(d+2)R,$$
$$\Delta S = 2|\operatorname{Hess}_{\mathbb{R}} F|^2 = 4(d+2)R.$$

Here $|\operatorname{Hess}_{\mathbb{R}} \operatorname{Re}(e^{i\theta}N)|^2 = 2|\operatorname{Hess}_{\mathbb{C}} N|^2$, so the real Hessian norm is unchanged by the determinant phase. Also $\Delta|N|^2 = 4|z^{\#}|^2$. Phase covariance transports these identities to the open subset just obtained, and polynomiality then proves them on all of V.

The smallest closure. The differential identities satisfy Proposition 9.9, which gives the claimed smallest algebra. At distinct positive t_i, the Jacobian of (e_1, e_2, e_3) is nonzero. Changing the overall complex phase preserves R, S, D and changes F whenever N is nonreal. Hence the four differentials are independent at such a point.

Global inequalities. The gradient table and Lemma 10.3 give (10.34). □

10.4.2. *Polynomial potentials and the Jordan minimizing range.* The following polynomial alternative also applies to the model in Section 10.4.3.

Proposition 10.13 (An algebraic sign criterion for the multiplier S)**.** *Let F be a harmonic cubic on a Euclidean space V, set $R = |x|^2$ and $S = |\nabla F|^2$, and let D be a sextic satisfying (10.33) and (10.34). Suppose $\Delta S = cR$. For $G = SF$, one has $\mathcal{L}(G) = F\mathcal{P}_c$, where*

$$\begin{aligned}\mathcal{P}_c = {} & (c - \tfrac{58}{3})RS^3 + \tfrac{10}{3}S^2(RS - 9D) \\ & + F^2\big[(12c - 72)RD + 12cS(R^2 - 3S) + (36c - 96)S^2\big].\end{aligned} \tag{10.35}$$

If $c \geq 58/3$, then $G\mathcal{L}(G) \geq 0$, and $\nabla G \neq 0$ wherever $S > 0$.

Proof. Set $K = cR$ in (10.14) and regroup to obtain (10.35). Every summand is nonnegative by (10.34) when $c \geq 58/3$. The gradient formula (10.13) gives $|\nabla G|^2 \geq S^3$. □

Theorem 10.14 (Area minimization for the complexified Jordan models)**.** *For the four models in Section 10.4.1, $\{F < 0\}$ is locally perimeter minimizing if and only if $d \in \{2, 4, 8\}$. In all three minimizing dimensions, $G = F\sqrt{RS}$ generates a subcalibration. For $d = 4, 8$, the polynomial potential SF does so as well.*

Proof. For $d = 2, 4, 8$, the dimension and trace coefficient are

$$(N, K/R) = (18, 16),\ (30, 24),\ (54, 40), \qquad K = \Delta S.$$

Since $\operatorname{tr} L_x^2 = K/2$, each pair satisfies Corollary 10.4; this proves the minimizing assertions for the common potential. For $d = 4, 8$, Proposition 10.13 additionally gives the polynomial potential SF.

The singular set is the Jordan rank-at-most-one cone

$$\operatorname{Sing}\{F = 0\} = \Sigma = \{z \in V : z^{\#} = 0\}.$$

To describe its nonzero stratum, take $a \neq 0$ in (10.29) and set

$$B = b - \frac{n(u)}{a}, \quad C = h - \frac{n(v)}{a}, \quad W = w - \frac{u\bar{v}}{a}.$$

The determinant formula gives

$$N(z) = a\big(BC - n(W)\big). \tag{10.36}$$

This identity also holds for $d = 8$. Indeed, multiplicativity of n and associativity of the scalar part give

$$n\Big(w - \frac{u\bar{v}}{a}\Big) = n(w) - \frac{2}{a}\operatorname{Sc}((\bar{u}w)v) + \frac{n(u)n(v)}{a^2},$$

which proves (10.36) after expanding aBC. The coordinate change is holomorphically invertible, so $dN = 0$ is exactly $B = C = W = 0$ in this chart. Hence these points are geometrically singular: the transverse real quadratic form $\operatorname{Re}(a(BC - n(W)))$ is nondegenerate and indefinite. Moreover, $\Sigma \setminus \{0\}$ is smooth of complex dimension $1 + 2d$, and real codimension $2(d + 2)$. To see that these charts cover the nonzero set, recall that the spectral theorem makes primitive idempotents span J, and that real Jordan automorphisms act transitively on them [28, Theorem III.1.2 and Corollary IV.2.7]. Some primitive idempotent therefore has nonzero trace pairing with any nonzero z, and an automorphism takes it to c_1.

For $d = 4, 8$, the codimension bound and (10.13) allow Corollary 2.8 to be applied to SF. Harmonicity identifies the boundary-current support with $\{F = 0\}$ by Remark 2.7.

For $d = 1$, take $p \in \Sigma \setminus \{0\}$ and put $H = \operatorname{Hess} F(p)$. The chart (10.36) shows that H has real rank $2(d+2)$. Proposition 9.6, with $q = 2R$, gives $H^3 = R(p)H$. Since $\operatorname{tr} H = 0$, its nonzero eigenvalues are $\pm\sqrt{R(p)}$, each of multiplicity $d+2$. Thus the tangent cone, with the induced Euclidean metric, is

$$C_{d+1,d+1} \times \mathbb{R}^{4d+2}, \qquad C_{q,q} = \{(x, y) \in \mathbb{R}^{q+1} \oplus \mathbb{R}^{q+1} : |x| = |y|\}. \tag{10.37}$$

For $d = 1$, the normal factor $C_{2,2}$ is unstable by the Hardy criterion in Section 2.6. Lemma 2.15 therefore rules out local perimeter minimization. □

For $d = 2$, complexification identifies V isometrically with $M_3(\mathbb{C})$ with its Frobenius metric, and N with the determinant. In the notation of Section 14, $R = E_1$, $S = E_2$, and $D = E_3$. Thus both the closure table and the potential above are the three-by-three cases of the construction there.

10.4.3. *The G_2 trilinear model.* Let $\times$ be the standard cross product on $\operatorname{Im} \mathbb{O} = \mathbb{R}^7$, and set

$$F(a, b, c) = \langle a \times b, c\rangle, \qquad (a, b, c) \in (\mathbb{R}^7)^3.$$

Equivalently, $F(a, b, c) = \varphi(a, b, c)$, where φ is the associative calibration on $\mathbb{R}^7$ of Harvey and Lawson [40, Section IV.1.A]. This is the classical model in [84, Section 6.4, equation (6.26), Proposition 6.3], whose defining polynomial $\operatorname{Re}(a(bc))$ is $-F$. Up to similarity, its subtracted Jordan model is the cubic norm restricted to the trace-orthogonal complement of $\operatorname{Herm}_3(\mathbb{R})$ in $\operatorname{Herm}_3(\mathbb{O})$; see [64, Proposition 6.4.4 and Table 6.4.1]. Let A be the real 7×3 matrix with columns a, b, c, and put

$$B = A^t A, \qquad R = \operatorname{tr} B, \qquad S = e_2(B), \qquad D = \det B,$$

where $e_2(B)$ is the second elementary symmetric polynomial of the eigenvalues of B.

The compact real group $G_2 = \operatorname{Aut}(\mathbb{O})$ acts diagonally on $V = (\mathbb{R}^7)^3$. Schwarz's invariant-generation theorem [70, Theorem 4 and equations (5.1)–(5.3)] gives

$$\mathbb{R}[V]^{G_2} = \mathbb{R}[B_{ij}, F].$$

The commuting $SO(3)$ action on the columns fixes F and conjugates B. Averaging polynomial expressions over $SO(3)$, whose conjugation invariants on symmetric matrices are generated by the elementary symmetric polynomials, therefore gives

$$\mathbb{R}[V]^{G_2 \times SO(3)} = \mathbb{R}[R, F, S, D].$$

This invariant ring equals $\mathcal{A}_R(F)$:

Theorem 10.15 (The trilinear model)**.** *The four polynomials R, F, S, D are algebraically independent, and $\mathcal{A}_R(F) = \mathbb{R}[R, F, S, D]$. In particular, $(\mathcal{A}_R(F))_2 = \mathbb{R}R$ and $(\mathcal{A}_R(F))_3 = \mathbb{R}F$. They satisfy (10.33) and (10.34), with*

$$\Delta R = 42, \qquad \Delta F = 0, \qquad \Delta S = 24R, \qquad \Delta D = 10S. \tag{10.38}$$

Both $G = F\sqrt{RS}$ and the polynomial potential SF give subcalibrations of $\{F < 0\} \subset \mathbb{R}^{21}$, which is locally perimeter minimizing.

Proof. The cross-product identity $|a \times b|^2 = |a|^2|b|^2 - \langle a, b\rangle^2$ gives $|\nabla F|^2 = S$, while trilinearity gives $\Delta F = 0$. Alternation gives $\langle \nabla_{a_i} F, a_j\rangle = \delta_{ij} F$, with a_i denoting the columns. In matrix notation,

$$\nabla_A S = 2A(RI - B), \qquad \nabla_A D = 2A \operatorname{adj} B.$$

These formulas and the three-by-three Newton identities give (10.33). Direct differentiation of the Gram invariants gives $\Delta S = 24R$ and $\Delta D = 10S$. The global inequalities (10.34) follow from Lemma 10.3.

The smallest-algebra assertion now follows from Proposition 9.9. For independence, choose orthonormal e_1, e_2, put $e_3 = e_1 \times e_2$, and choose a unit e_4 orthogonal to their span. At

$$a = r_1 e_1, \qquad b = r_2 e_2, \qquad c = r_3(\cos\theta\, e_3 + \sin\theta\, e_4),$$

the three invariants R, S, D are the elementary symmetric functions of r_1^2, r_2^2, r_3^2, while $F = r_1 r_2 r_3 \cos\theta$. Distinct positive r_i^2 and $0 < \theta < \pi$ give four independent differentials.

Here $N = 21$ and $K = 24R$, so Corollary 10.4 gives the common potential. For the polynomial potential, Proposition 10.13 applies with $c = 24$. The singular set of the zero cone is

$$\operatorname{Sing}\{F = 0\} = \Sigma = \{S = 0\} = \{A \in M_{7,3}(\mathbb{R}) : \operatorname{rank} A \le 1\}.$$

At a nonzero rank-one matrix, a column rotation reduces to $(a, 0, 0)$, whose transverse leading form $\langle a \times \delta b, \delta c\rangle$ is indefinite of rank twelve. The nonzero stratum has dimension $7+3-1 = 9$, hence codimension 12. The gradient bound (10.13) and Corollary 2.8 therefore give the same minimizing phase. □

Remark 10.16. For both families, Bochner's identity gives $2|L_x|^2_{\mathrm{HS}} = c|x|^2$, where $c > 0$. Hence $L_v = 0$ implies $v = 0$, and their smallest closures have the intrinsic degree pattern (3) of Theorem 9.5. For the G_2 model this differs from its trilinear coordinate presentation: the larger invariant algebra $\mathbb{R}[V]^{G_2}$ contains their separate radii, whereas the smallest algebra is also invariant under column rotations and does not contain $|a|^2, |b|^2, |c|^2$.

10.5. **Area-minimizing Clifford cubic cones.**

Theorem 10.17 (Area-minimizing classification)**.** *For every admissible symmetric Clifford system with $q \ge 1$, the oriented multiplicity-one cone $\mathcal{C}_{\mathcal{A}}$ is area minimizing if and only if*

$$\boxed{m \ge 4, \qquad q \ge 3, \qquad (m, q) \ne (4, 3).} \tag{10.39}$$

Every area-minimizing member in this family is strictly area minimizing.

The positive range uses $G = F\sqrt{RS}$ and Theorem 10.2; singular-stratum obstructions exclude the low-dimensional cases. The remaining pair $(4, 3)$ requires a link variation:

Theorem 10.18 (Instability of the pair $(4, 3)$)**.** *For every Clifford system with $(m, q) = (4, 3)$, the cone $\mathcal{C}_{\mathcal{A}} \subset \mathbb{R}^{12}$ is unstable. More precisely, if $\Lambda = \mathcal{C}_{\mathcal{A}} \cap \mathbb{S}^{11}$, then $\phi = S^{3/4}(1 - S/2)$ belongs to the energy space $\mathcal{D}(\Lambda)$ defined in Section 10.8 and satisfies*

$$\int_{\Lambda_{\mathrm{reg}}} \left(|\nabla_\Lambda \phi|^2 - |A_\Lambda|^2 \phi^2 + \frac{81}{4}\phi^2 \right) \mathrm{d}\mathcal{H}^{10} < 0,$$

where A_Λ is the second fundamental form of $\Lambda \subset \mathbb{S}^{11}$. Every tangent cone of $\mathcal{C}_{\mathcal{A}}$ at a nonzero point is area minimizing. At such a singular point it is the product of the seven-dimensional Simons cone and $\mathbb{R}^4$.

10.6. **The Clifford positivity range.**

Proposition 10.19 (The common potential for Clifford cubics)**.** *Suppose either $m \ge 8$, $q \ge 3$, or $m \ge 4$, $q \ge 4$. Then*

$$G = F\sqrt{RS}$$

generates a two-sided subcalibration for $\{F < 0\}$, and $\mathcal{C}_{\mathcal{A}}$ is strictly area minimizing with the prescribed orientation. The gradient of G is nonzero on $\{S > 0\}$, and $F\mathcal{L}(G) > 0$ wherever $F \ne 0$.

Proof. For Theorem 10.2, use the normalization

$$f = \frac{F}{2}, \qquad s = |\nabla f|^2 = \frac{S}{4} = uv + \frac{T}{4}, \qquad D = \frac{vT}{4},$$
$$\kappa = \Delta s = \frac{K}{4} = 4mv + 4(q+1)u. \tag{10.40}$$

The closure table (9.20) gives

$$\Gamma(f,s) = 4Rf, \qquad \Gamma(s,s) = 4Rs + 12D. \tag{10.41}$$

Thus $\mathcal{L}(f) = -2Rf$, and the remaining transport identities are (9.8), with (F,S) replaced by (f,s). If $m \geq 8$ and $q \geq 3$, then $N = 2m+q+1 \geq 20$ and $\kappa \geq 16R$. The trace criterion Corollary 10.4 applies directly.

For the second range, write the remaining sign expression as

$$\mathscr{H}_{m,q} = (2m+q-11)s^2 + \frac{1}{2}R\big(4mv + 4(q+1)u - 16R\big)s - 12RD.$$

At $(m,q) = (4,4)$, direct substitution gives

$$\mathscr{H}_{4,4} = \frac{T(8u^2+T)}{16} + (u^2 - T)v(2u+3v) \geq 0, \tag{10.42}$$

by (9.19) and $u, v \geq 0$. For every $m \geq 4$, $q \geq 4$,

$$\begin{aligned}\mathscr{H}_{m,q} = \mathscr{H}_{4,4} &+ \big(2(m-4)+q-4\big)s^2 \\ &+ 2R\big((m-4)v + (q-4)u\big)s \geq 0,\end{aligned}$$

and $\kappa \geq 16R$. Hence Theorem 10.2 applies in this range as well. Finally,

$$f\sqrt{Rs} = \frac{1}{4}F\sqrt{RS}.$$

This positive factor preserves the normalized gradient and the strict divergence sign, giving the result for the original F. □

In the normalized variables of (10.40), the sextic is generally independent of the cubic:

$$D - f^2 = \frac{vT - F^2}{4} \geq 0.$$

This Cauchy–Schwarz defect need not vanish, so the polynomial gradient-square criterion requiring $D = f^2$ does not apply in general.

Dimensions in which Clifford systems exist. For a symmetric Clifford system $(A_0, \dots, A_q)$ on $\mathbb{R}^{2m}$, the Hurwitz–Radon admissibility condition [71, Chapter 0, pp. 3–4] is $q \leq \rho(m)$, with the Clifford-system convention of [32, 83]. Here, if $m = 2^{4a+b}c$ with c odd and $0 \leq b \leq 3$, then $\rho(m) = 8a + 2^b$. In particular, $\rho(4) = 4$, $\rho(5) = 1$, $\rho(6) = 2$, $\rho(7) = 1$. Thus an admissible pair with $q \geq 3$ has either $m = 4$ or $m \geq 8$. If $m = 4$, then $q \in \{3, 4\}$.

Sufficiency in Theorem 10.17. By admissibility, every pair in (10.39) has either $m \geq 8$, $q \geq 3$, or $(m,q) = (4,4)$. Both cases are covered by Proposition 10.19. □

10.7. **Singular strata and tangent cones.** From (9.21), the critical set is

$$\Sigma := \{\nabla F = 0\} = \Sigma_0 \cup \Sigma_+,$$

where

$$\Sigma_0 = \{(0,z) : z \in \mathbb{R}^{q+1}\}, \qquad \Sigma_+ = \{(y,0) : \tau(y) = 0\}.$$

Indeed, $A_z^2 = v\,\mathrm{Id}$ shows that $A_z y = 0$ forces $z = 0$ or $y = 0$.

At a nonzero point of Σ_+, the gradients $\nabla\tau_i = 2A_i y$ are mutually orthogonal, so this stratum is smooth there and has ambient codimension $2q+2$. The stratum Σ_0 has ambient codimension $2m$. In the sufficient range $m \geq 4$, $q \geq 3$, and with $N = 2m+q+1$,

$$\mathcal{H}^{N-1}(\Sigma \cap \mathcal{K}) = 0 \quad \text{for every compact } \mathcal{K}. \tag{10.43}$$

Define the generalized Simons cone

$$\mathcal{S}_k = \{(a,b) \in \mathbb{R}^k \oplus \mathbb{R}^k : |a| = |b|\}.$$

Let $E = \{F < 0\}$.

Proposition 10.20 (Unique tangents with multiplicity one)**.**

(i) *At a nonzero point $p = (0, z_0)$ of Σ_0, choose orthogonal coordinates*

$$\mathbb{R}^{2m+q+1} = \mathbb{R}^m \oplus \mathbb{R}^m \oplus \mathbb{R}^{q+1}.$$

The unique set blow-up is

$$E_p^{(0)} = \{(a, b, \eta) : |a| < |b|\}, \qquad \partial E_p^{(0)} = \mathcal{S}_m \times \mathbb{R}^{q+1}. \tag{10.44}$$

(ii) *At $p = (y_0, 0) \in \Sigma_+ \setminus \{0\}$, after an orthogonal identification $\mathbb{R}^{2m+q+1} = \mathbb{R}^{q+1} \oplus \mathbb{R}^{q+1} \oplus \mathbb{R}^{2m-q-1}$, the unique set blow-up is*

$$E_p^{(+)} = \{(x, x', \xi^\perp) : |x| < |x'|\}, \qquad \partial E_p^{(+)} = \mathcal{S}_{q+1} \times \mathbb{R}^{2m-q-1}. \tag{10.45}$$

(iii) *At the vertex, every dilation fixes E and $\mathcal{C}_\mathcal{A}$, so the tangent cone is $\mathcal{C}_\mathcal{A}$ itself.*

In each case, the oriented current tangent is the boundary of the stated set, and the varifold tangent has multiplicity one on its regular part. The perimeter measures converge under dilation. These statements hold for every admissible Clifford system, without an area-minimizing assumption.

Proof. The Taylor argument in Section 2.6 gives the set tangents. We use exact local coordinates also to determine the current and varifold limits. Near $p = (0, z_0)$, $z_0 \neq 0$, the identity $A_z^2 = |z|^2 \,\mathrm{Id}$ gives smooth positive and negative eigenspaces. Their dimensions are both m, since A_w interchanges them whenever $0 \neq w \perp z$. Choose smooth orthonormal frames and write $y = P(z)(a, b)$. Then

$$F(y, z) = |z|\big(|a|^2 - |b|^2\big).$$

The coordinate map has orthogonal derivative at $(0, z_0)$, since its derivatives in z are multiplied by $y = 0$. This gives (10.44), including the stated set side.

Near $p = (y_0, 0) \in \Sigma_+$, $y_0 \neq 0$, the vectors $e_i = A_i y_0/|y_0|$, $0 \leq i \leq q$, are orthonormal. Let $P_\perp$ be the orthogonal projection onto their common orthogonal complement. The map

$$y \longmapsto \left(a_i(y) = \frac{\tau_i(y)}{2|y_0|} \ (0 \leq i \leq q), \quad \xi^\perp = P_\perp(y - y_0)\right)$$

has orthogonal derivative at y_0, because $da_i|_{y_0}(\xi) = \langle e_i, \xi\rangle$, and is a smooth local coordinate map. With $z = \eta$ and

$$x = \frac{a + \eta}{\sqrt{2}}, \qquad x' = \frac{a - \eta}{\sqrt{2}},$$

we have exactly

$$F(y, \eta) = 2|y_0| a \cdot \eta = |y_0|\big(|x|^2 - |x'|^2\big).$$

This proves (10.45) with its stated side.

In either case, after the stated orthogonal identification, a smooth local diffeomorphism Φ takes the quadratic sublevel model onto E, with $\Phi(0) = p$ and $D\Phi(0) = \mathrm{Id}$. The rescaled maps $\Phi_r(w) = (\Phi(rw) - p)/r$ converge to the identity in C^1 on compact sets. Each quadratic model has locally finite perimeter and multiplicity one on its regular boundary. The area formula for Φ_r, whose tangential Jacobians converge uniformly to one, gives convergence of the perimeter measures and the associated varifolds. The C^1 convergence also gives L^1_{loc} convergence of the sublevel sets, whose model boundaries are Lebesgue null, and identifies the oriented current limit with the boundary of the stated model. At the vertex, all rescalings agree exactly by homogeneity. □

Remark 10.21. All nonzero algebraic singular points are genuine geometric singularities: neither (10.44) nor (10.45) is a hyperplane. Thus $\operatorname{Sing} \mathcal{C}_\mathcal{A} = \Sigma_0 \cup \Sigma_+$. The transverse Simons cones are area minimizing exactly when $m \geq 4$ and $q + 1 \geq 4$, respectively [6, 25]. For a transverse parameter at least two, this is also exactly the stable range [77, Section 6.1]; the parameter-one planar crossing is stable on its regular part but is not minimizing. In particular, for $(m, q) = (4, 3)$ both strata have the same tangent model $\mathcal{S}_4 \times \mathbb{R}^4$. The corresponding

multiplicity-one current tangents are area minimizing; at regular points the tangent is a multiplicity-one hyperplane.

For the Hurwitz cubics $F(x,y,z) = \mathrm{Re}((xy)z)$, with $x,y,z \in \mathbb{K}$ and $\mathbb{K} = \mathbb{C}, \mathbb{H}, \mathbb{O}$, the singular set is the union of the three mutually orthogonal coordinate subspaces:

$$\mathrm{Sing}\{F=0\} = (\mathbb{K}\oplus 0\oplus 0)\cup(0\oplus\mathbb{K}\oplus 0)\cup(0\oplus 0\oplus\mathbb{K}).$$

Indeed, $|\nabla F|^2 = |x|^2|y|^2+|y|^2|z|^2+|z|^2|x|^2$. The complex case gives the explicit cubic tangent in Section 2.6.

Lemma 10.22. *If $m > q$, then $\Sigma_+ \setminus \{0\}$ is nonempty.*

Proof. After an orthogonal change of coordinates, the first two Clifford matrices have the standard form

$$A_0 = \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & -\mathrm{Id} \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 0 & \mathrm{Id} \\ \mathrm{Id} & 0 \end{pmatrix}, \tag{10.46}$$

and, for $i \geq 2$,

$$A_i = \begin{pmatrix} 0 & E_i \\ -E_i & 0 \end{pmatrix}, \tag{10.47}$$

where the E_i are skew-symmetric orthogonal matrices. Fix $b \neq 0$ in $\mathbb{R}^m$. The q vectors $b, E_2 b, \dots, E_q b$ are mutually orthogonal. Since $m > q$, choose $a \neq 0$ of the same length as b and orthogonal to their span. For $y = (a,b)$, all components of $\tau(y)$ vanish. □

Necessity in Theorem 10.17. Suppose first that $m \leq 3$. The stratum $\Sigma_0 \setminus \{0\}$ has dimension $q+1$ and consists of genuine singular points by Proposition 10.20. Its ambient codimension is $2m < 8$. Federer's codimension-seven bound for the singular set of an area-minimizing hypersurface [30] therefore excludes area minimization.

Now suppose $q \leq 2$ and $m \geq 4$. Then $m > q$, so Lemma 10.22 applies. The smooth stratum $\Sigma_+ \setminus \{0\}$ has ambient codimension $2q+2 < 8$ and again consists of genuine singular points. The same dimension bound excludes area minimization in this case as well. □

10.8. **The exceptional pair $(4,3)$.**
Second variation and the radial Hardy inequality. Fix $(m,q) = (4,3)$ and write

$$\Lambda = \mathcal{C}_{\mathcal{A}} \cap \mathbb{S}^{11}, \qquad \mathscr{S} = \mathrm{Sing}(\mathcal{C}_{\mathcal{A}}) \cap \mathbb{S}^{11}, \qquad \mathscr{S}_0 = \Sigma_0 \cap \mathbb{S}^{11}, \qquad \mathscr{S}_+ = \Sigma_+ \cap \mathbb{S}^{11}.$$

Thus $\dim \Lambda = 10$, while the two singular strata $\mathscr{S}_0$ and $\mathscr{S}_+$ each have dimension 3. Let $\mathcal{D}(\Lambda)$ be the closure of $C_c^\infty(\Lambda_{\mathrm{reg}})$ in the norm with square $\int_{\Lambda_{\mathrm{reg}}}(|\nabla\psi|^2 + (1+|A_\Lambda|^2)\psi^2)$. The radial Hardy inequality in cone dimension $n = 11$ has optimal constant $(n-2)^2/4 = 81/4$. Separating the radial variable in the second variation of area therefore gives the quadratic form

$$\mathcal{B}(\varphi,\psi) = \int_{\Lambda_{\mathrm{reg}}} \left(\langle \nabla_\Lambda \varphi, \nabla_\Lambda \psi \rangle - |A_\Lambda|^2 \varphi\psi + \frac{81}{4}\varphi\psi \right).$$

It remains to find $\phi \in \mathcal{D}(\Lambda)$ with $\mathcal{B}(\phi,\phi) < 0$.

For the present link, the smooth compact singular strata and their nondegenerate transverse quadratic models give

$$\mathcal{H}^{10}(N_r(\mathscr{S}) \cap \Lambda) \leq C r^7, \qquad |A_\Lambda| \leq C\rho^{-1}, \qquad \rho = \mathrm{dist}(\cdot, \mathscr{S}). \tag{10.48}$$

Indeed, the local coordinates in Proposition 10.20 identify the transverse zero sets with seven-dimensional quadratic cones in eight normal variables. Their volume element has radial order $\rho^6\,\mathrm{d}\rho$, and their second fundamental forms have size $O(\rho^{-1})$. Restricting to the unit sphere removes the radial direction along the singular stratum. The coordinate maps and their inverses have bounded first and second derivatives on a finite cover of the compact singular strata. The same estimates therefore hold on Λ, which proves (10.48).

Curvature identities on the link. Use u, v, S, J from Section 9.4.2. On Λ one has $R = u+v = 1$ and $F = 0$.

Lemma 10.23 (Link curvature)**.** *On* Λ_{reg}*,*

$$|A_\Lambda|^2 = \frac{32}{S} - \frac{J}{2S^2}, \tag{10.49}$$

$$|\nabla_\Lambda S|^2 = J - 16S^2. \tag{10.50}$$

Proof. The curvature identity in the proof of Proposition 4.5 is $|A_C|^2 = K/(2S) - J/(2S^2)$. Here (9.21) gives $K = 64$ at $(m, q) = (4, 3)$, $R = 1$, proving (10.49): the radial principal curvature is zero, so $|A_C| = |A_\Lambda|$ at radius one. Moreover, $Z \perp \nabla F$ on $F = 0$ by (9.21), and homogeneity gives $\langle Z, (y, z)\rangle = 4S$. Hence $\nabla_\Lambda S = Z - 4S(y, z)$, proving (10.50). □

Integration along the Hopf fibers. For $(4, 3)$, the standard block form allows us to extend the Clifford system by one generator. In the notation of (10.46)–(10.47), put $E_4 = E_2E_3$ and

$$A_4 = \begin{pmatrix} 0 & E_4 \\ -E_4 & 0 \end{pmatrix}.$$

Then $(A_0, \dots, A_4)$ is a Clifford system on $\mathbb{R}^8$. For $\xi \in \mathbb{S}^7$, the map

$$\Pi(\xi) = \big(\langle A_0\xi, \xi\rangle, \dots, \langle A_4\xi, \xi\rangle\big) \tag{10.51}$$

is the quaternionic Hopf map $\mathbb{S}^7 \to \mathbb{S}^4$; see, for example, [32]. To verify its measure normalization, write $p_i(\xi) = \langle A_i\xi, \xi\rangle$. If $\xi = (a, b) \in \mathbb{R}^4 \oplus \mathbb{R}^4$, then b, E_2b, E_3b, E_4b are mutually orthogonal with the same length; this proves $|\Pi(\xi)| = 1$. The Clifford relations also give

$$\nabla_{\mathbb{S}^7} p_i = 2(A_i\xi - p_i\xi), \qquad \langle \nabla_{\mathbb{S}^7} p_i, \nabla_{\mathbb{S}^7} p_j\rangle = 4(\delta_{ij} - p_ip_j). \tag{10.52}$$

Thus $D\Pi$ multiplies lengths by 2 on the horizontal space $(\ker D\Pi)^\perp$, and its four-dimensional Jacobian is the constant 2^4. The quaternionic Hopf fibers are congruent round 3-spheres. Integration over the fibers gives, for every integrable function h,

$$\int_{\mathbb{S}^7} h(\Pi(\xi))\,\mathrm{d}\xi = c_H \int_{\mathbb{S}^4} h(p)\,\mathrm{d}p, \qquad c_H > 0.$$

The Hopf map also describes the second singular stratum: $\mathscr{S}_+$ is the inverse image under Π of the two poles $(0, 0, 0, 0, \pm 1) \in \mathbb{S}^4$. Hence $\mathscr{S}_+$ has two connected components, each a Hopf fiber diffeomorphic to S^3.

Set

$$\theta = \frac{T}{u^2} \in [0, 1].$$

If $s = \langle A_4\xi, \xi\rangle$, then $\theta = 1 - s^2$. The following lemma expresses integration on the link in terms of u and θ.

Lemma 10.24 (Integration on the link)**.** *There is a constant* $c_0 > 0$ *such that every integrable function* $h = h(u, \theta)$ *satisfies*

$$\int_{\Lambda_{\mathrm{reg}}} h\ \mathrm{d}\mathcal{H}^{10} = c_0 \int_0^1 \int_0^1 h(u, \theta)\, u^2(1-u)^{1/2} \left(\frac{\theta}{1-\theta}\right)^{1/2} S^{1/2}\ \mathrm{d}\theta\ \mathrm{d}u, \tag{10.53}$$

$$S = 4u(1-u) + u^2\theta,$$

$$J = 16\left[4u(1-u) + \big(u + 4(1-u)\big)u^2\theta\right]. \tag{10.54}$$

Proof. Write $y = \sqrt{u}\,\xi$ and $z = \sqrt{1-u}\,\zeta$, with $\xi \in \mathbb{S}^7$ and $\zeta \in \mathbb{S}^3$. Spherical measure on $\mathbb{S}^{11}$ decomposes, up to a positive constant, as

$$u^3(1-u)\ \mathrm{d}u\ \mathrm{d}\xi\ \mathrm{d}\zeta.$$

If $\bar\tau = (\langle A_0\xi,\xi\rangle,\dots,\langle A_3\xi,\xi\rangle)$, then $|\bar\tau| = \sqrt{\theta}$ and

$$F = u\sqrt{1-u}\,\langle\zeta,\bar\tau\rangle.$$

For fixed $0<u<1$ and ξ with $\theta>0$, the equation $F=0$ defines an equatorial $\mathbb{S}^2$ in the ζ-sphere. Along this equator,

$$|\nabla_{\mathbb{S}^3,\zeta}F| = u\sqrt{1-u}\sqrt{\theta}. \tag{10.55}$$

The coordinate exceptional sets have zero $\mathcal{H}^{10}$-measure in Λ. Indeed, $u=0$ and $u=1$ have dimensions three and seven. For $0<u<1$, the set $\theta=0$ has dimension $1+3+3=7$, since ξ lies in two Hopf fibers and ζ is unrestricted; the set $\theta=1$ has dimension $1+6+2=9$, since ξ lies above an equatorial $\mathbb{S}^3\subset\mathbb{S}^4$ and ζ lies in an equatorial $\mathbb{S}^2$. We may therefore work on $0<u<1$, $0<\theta<1$ without changing the link integral.

On $\{F=0\}\cap\mathbb{S}^{11}$, Euler's identity gives $|\nabla_{S^{11}}F| = |\nabla F| = \sqrt{S}$. Applying the coarea formula first on $\mathbb{S}^{11}$ and then on the ζ-sphere gives, for a constant $c_1>0$,

$$\int_{\Lambda_{\mathrm{reg}}} h\,\mathrm{d}\mathcal{H}^{10} = c_1\int_0^1 u^2(1-u)^{1/2}\int_{\mathbb{S}^7} h(u,\theta)\frac{S^{1/2}}{\sqrt{\theta}}\,\mathrm{d}\xi\,\mathrm{d}u.$$

It remains to integrate in ξ. The Hopf map (10.51) has congruent fibers and constant Jacobian. For the last coordinate s on $\mathbb{S}^4$, spherical measure has density proportional to $(1-s^2)\,\mathrm{d}s$. Since $\theta = 1-s^2$, the two branches of s contribute

$$\frac{\theta}{\sqrt{1-\theta}}\,\mathrm{d}\theta.$$

Substitution into the preceding integral proves (10.53); (10.54) follows from the formulas for S and J in (9.21) with $R=1$ and $T=u^2\theta$. □

A test function proving instability. For $a\in\mathbb{R}$, put $\phi_a = S^a$ on Λ_{reg}. Combining Lemmas 10.23 and 10.24 gives, whenever the integral is finite,

$$\begin{aligned}\frac{1}{c_0}\mathcal{B}(\phi_a,\phi_b) = \int_0^1\int_0^1 u^2(1-u)^{1/2}\left(\frac{\theta}{1-\theta}\right)^{1/2}\Bigg[\left(ab+\frac12\right)JS^{a+b-3/2}\\ -32S^{a+b-1/2}+\left(\frac{81}{4}-16ab\right)S^{a+b+1/2}\Bigg]\mathrm{d}\theta\,\mathrm{d}u.\end{aligned} \tag{10.56}$$

For $a,b\in\{3/4,7/4\}$, every power of S in (10.56) is a nonnegative integer. Expanding (10.54) and using only

$$\int_0^1 u^{i+2}(1-u)^{1/2}\,\mathrm{d}u = \mathrm{B}\left(i+3,\frac32\right),$$
$$\int_0^1 \theta^{j+1/2}(1-\theta)^{-1/2}\,\mathrm{d}\theta = \mathrm{B}\left(j+\frac32,\frac12\right),$$

we obtain

$$\frac{1}{c_0\pi}\begin{pmatrix}\mathcal{B}(\phi_{3/4},\phi_{3/4}) & \mathcal{B}(\phi_{3/4},\phi_{7/4})\\ \mathcal{B}(\phi_{7/4},\phi_{3/4}) & \mathcal{B}(\phi_{7/4},\phi_{7/4})\end{pmatrix} = \begin{pmatrix}\frac{96}{715} & \frac{471936}{1616615}\\ \frac{471936}{1616615} & \frac{571904}{969969}\end{pmatrix}.$$

In particular, for

$$\phi = \phi_{3/4} - \frac12\phi_{7/4} = S^{3/4}\left(1-\frac12 S\right),$$

$$\mathcal{B}(\phi,\phi) = -c_0\pi\,\frac{9952}{969969} < 0. \tag{10.57}$$

Near either singular stratum of $\mathscr{S}$, the nondegenerate transverse quadratic models give

$$S\asymp\rho^2,\qquad J=O(\rho^2),\qquad |\nabla_\Lambda S| = O(\rho),$$

where $\rho = \operatorname{dist}(\cdot, \mathscr{S})$; the last estimate also follows from (10.50). Therefore

$$|\phi| = O(\rho^{3/2}), \qquad |\nabla_\Lambda \phi| = O(\rho^{1/2}).$$

In particular, both $|\nabla_\Lambda \phi|^2$ and $|A_\Lambda|^2\phi^2$ are $O(\rho)$. Since the transverse volume element has order $\rho^6\,\mathrm{d}\rho$, all terms in the form energy are locally integrable. Choose smooth cutoffs χ_ε that vanish for $\rho \le \varepsilon$, equal one for $\rho \ge 2\varepsilon$, and satisfy $|\nabla_\Lambda \chi_\varepsilon| \le C/\varepsilon$. The estimate (10.48) and the preceding pointwise bounds give

$$\|(1-\chi_\varepsilon)\phi\|_{\mathcal{D}}^2 \le C\varepsilon^8 \longrightarrow 0.$$

Indeed, the gradient and curvature terms are bounded by $C\int_0^{2\varepsilon} \rho\,\rho^6\,\mathrm{d}\rho = O(\varepsilon^8)$; the cutoff derivative contributes at most $C\varepsilon^{-2}\varepsilon^3\varepsilon^7 = O(\varepsilon^8)$. Thus $\chi_\varepsilon\phi \in C_c^\infty(\Lambda_{\mathrm{reg}})$ approximates ϕ in $\mathcal{D}(\Lambda)$. The cone stability formula (4.16) and (10.57) prove Theorem 10.18. Together with the sufficient and dimension-obstruction arguments above, this also completes the proof of Theorem 10.17.

Remark 10.25 (Power multipliers at $(4,3)$). Instability already excludes every positive smooth multiplier with the two-sided sign near the regular cone: Theorem 4.4 and (4.9) would otherwise imply stability. In particular, no choice of exponents in

$$G_{\alpha,\gamma} = R^\alpha S^\gamma F, \qquad (\alpha,\gamma) \in \mathbb{R}^2,$$

can give a subcalibration for $(m,q) = (4,3)$. Appendix C.2 additionally locates explicit points in the actual invariant image with $R, S > 0$, $F \neq 0$, and $\nabla G_{\alpha,\gamma} \neq 0$, where $F\,\operatorname{div}(\nabla G_{\alpha,\gamma}/|\nabla G_{\alpha,\gamma}|) < 0$.

10.9. **Unequal dimensions: Hopf incidence cubics.** Let $d \in \{2,4,8\}$ and let $h_d : \mathbb{R}^{2d} \to \mathbb{R}^{d+1}$ be the standard quadratic Hopf map, with $|h_d(y)|^2 = |y|^4$. Set

$$F_d(x,y) = \langle x, h_d(y)\rangle, \qquad x \in \mathbb{R}^{d+1}, \quad y \in \mathbb{R}^{2d}, \quad u = |y|^2, \quad v = |x|^2.$$

The bidegree is $(1,2)$, and the Hopf identities give

$$(\Gamma(I_i,I_j))_{I=(u,v,F_d)} = \begin{pmatrix} 4u & 0 & 4F_d \\ 0 & 4v & 2F_d \\ 4F_d & 2F_d & u^2+4uv \end{pmatrix}, \qquad \Delta I = (4d, 2(d+1), 0). \tag{10.58}$$

Its determinant is $16(u+4v)(u^2v - F_d^2)$, positive on an open set. Thus the algebra has three independent generators of degrees $(2,2,3)$. Furthermore $\mathcal{L}(F_d) = -8(u+v)F_d$, and the singular set of the minimal zero hypercone is $\{y=0\}$. At a nonzero point of this stratum the transverse quadratic form has signature (d,d).

These are the Clifford cubics with parameters $(m,q) = (d,d)$. By Theorem 10.17, the cases $d = 4, 8$ are area minimizing, in ambient dimensions $13, 25$. For $d = 2$, the transverse Lawson cone in $\mathbb{R}^4$ is unstable, and the localization argument following (8.48) gives instability in $\mathbb{R}^7$. For $d = 4$, the degree-six potential is critical: $6 = (13-1)/2$. Substituting $T = u^2$ into (10.42) and (10.6) gives

$$\beta_G = \frac{9u^2}{(u+4v)^2} > 0 \quad \text{on } C_{\mathrm{reg}}, \qquad \inf_{C_{\mathrm{reg}} \cap \mathbb{S}^{12}} \beta_G = 0.$$

As $u \to 0$ near the singular stratum, the bound degenerates. It proves stability but gives no uniform positive Hardy gap. The three-generator description uses $|h_d(y)|^2 = u^2$. For a general Clifford system, $|\tau(y)|^2$ is a separate quartic invariant. The computations in Section 9.4.2 use the four-generator algebra $\mathbb{R}[u, v, |\tau|^2, F]$.

The corresponding regular levels on $\mathbb{S}^d \times \mathbb{S}^{2d-1}$ are isoparametric hypersurfaces. Indeed, on the product of unit round spheres $P_d = \mathbb{S}^d \times \mathbb{S}^{2d-1}$, put $f_d = F_d|_{P_d}$. The partial degrees $(1,2)$ and (10.58) give

$$|\nabla_{P_d} f_d|^2 = 5(1-f_d^2), \qquad \Delta_{P_d} f_d = -5d f_d.$$

Thus the levels $f_d = t$, $-1 < t < 1$, are isoparametric, and $f_d = 0$ is minimal in P_d.

11. Nonradial Cubic Examples and Cubic Pullbacks

Cartan deformations attain the five-generator bound of Theorem 9.13 in the dimensions allowed by Corollary 9.26. Their two weight eigenvalues in (11.7) distinguish them from the radial irreducible cubics known to Fox and Tkachev [34, Introduction and Section 2.1]. The models in $\mathbb{R}^8, \mathbb{R}^{14}$ are unstable; five-generator inequalities prove strict area minimality in $\mathbb{R}^{26}$. We then pull back the minimizing cubic targets.

11.1. **Nonradial harmonic models.**

11.1.1. *One-parameter deformations of Cartan cubics.* Use the notation $J_d = \mathrm{Herm}_3(\mathbb{K}_d)$ from Section 10.4. For the three Cayley–Dickson inclusions $\mathbb{K}_d \subset \mathbb{K}_{2d}$, $d = 1, 2, 4$, the involution $a + b\ell \mapsto a - b\ell$ preserves multiplication and conjugation by the Cayley–Dickson formula. Its entrywise extension therefore preserves the Jordan product and defines an automorphism σ of J_{2d} with fixed algebra J_d. Relative to the common diagonal Jordan frame, it refines each non-diagonal Peirce space as

$$J_{ij} \cong \mathbb{K}_{2d} = \mathbb{K}_d \oplus \mathbb{K}_d\ell, \qquad J_{ij}^{+} \cong \mathbb{K}_d, \quad J_{ij}^{-} \cong \mathbb{K}_d\ell.$$

Thus the fixed algebra remains simple of rank three, and its non-diagonal Peirce spaces have half the dimensions of those of J_{2d}. On $V = (J_{2d})_0$, with inner product $\langle x, y\rangle = \frac{1}{2}\operatorname{tr}(x \bullet y)$, this gives the orthogonal splitting

$$V = V_+ \perp V_-, \qquad V_+ = (J_d)_0, \qquad V_- = U \cong \mathbb{K}_d^3.$$

The Cartan cubic $N = \det_{J_{2d}}|_V$ is invariant under σ. Hence $N(b+u) = N(b-u)$ for $b \in V_+$, $u \in V_-$. Since N is cubic and even in u, it has the unique bidegree decomposition

$$N(b+u) = A(b) + \mathsf{B}(b,u), \qquad \deg A = (3,0), \quad \deg \mathsf{B} = (1,2).$$

We deform the mixed component by setting

$$F_t = A + t\mathsf{B}, \qquad t \in \mathbb{R}. \tag{11.1}$$

The choice of these pairs makes $A = \det_{J_d}|_{(J_d)_0}$ itself a Cartan cubic; see [64, Theorem 3.1.1 and Sections 3.2.6, 6.4.4–6.4.5] and [86]. Both N and A are therefore harmonic. Since $\Delta = \Delta_b + \Delta_u$ and B is linear in b,

$$0 = \Delta N = \Delta_b A + \Delta_u \mathsf{B}, \qquad \Delta F_t = (1-t)\Delta_b A = 0.$$

Thus the entire family is harmonic. With $\mathbb{K} = \mathbb{K}_d$, this identifies V with

$$V_d = \mathrm{Herm}_3(\mathbb{K})_0 \oplus \mathbb{K}^3, \qquad \langle (Z,u), (Z',u')\rangle = \tfrac{1}{2}\operatorname{Re}\operatorname{tr}(ZZ') + \operatorname{Re}(u^*u').$$

Thus $\dim V_d = 6d + 2$. The half-trace metric here differs from the trace metric on the complexification in Section 10.4. Set

$$\rho = \tfrac{1}{2}\operatorname{tr} Z^2, \quad \nu = |u|^2, \quad A = \det Z, \quad \mathsf{B} = -u^*Zu, \quad W = 4u^*Z^2u. \tag{11.2}$$

Then $R = \rho + \nu$. For the nonradial member $F_{-1/3}$, the dictionary with the intrinsic invariants (9.43) is

$$(r,s,a,p,h) = \left(\rho, \nu, \frac{3\sqrt{3}}{2}A, -\frac{\mathsf{B}}{2\sqrt{3}}, \frac{W}{48}\right).$$

The intrinsic normalization is $\sqrt{3}F_{-1/3}/4$, with $3/16$ times the quadratic weight below. We verify the model by its coordinate table before invoking the intrinsic theorem.

To verify the mixed term and the isometry, write

$$Z = \begin{pmatrix} r_1 & c & \bar{b} \\ \bar{c} & r_2 & a \\ b & \bar{a} & r_3 \end{pmatrix}, \qquad r_1 + r_2 + r_3 = 0.$$

In the determinant formula $r_1r_2r_3 - r_1|a|^2 - r_2|b|^2 - r_3|c|^2 + 2\,\mathrm{Re}(abc)$, replace (a,b,c) by $(a+v_1\ell, b+v_2\ell, c+v_3\ell)$. The quadratic part in v, with $u=(\bar v_1,\bar v_2,\bar v_3)$, is $-u^*Zu = \mathsf{B}$. This verifies the isometry and the signs, including the inclusion $\mathbb{H}\subset\mathbb{O}$. In particular,

$$F_{-1/3} = \frac{4A-N}{3} = \frac{1}{3}\,\mathrm{tr}\big(\sigma(x)\bullet x^{\bullet 2}\big),$$

where $x\in(J_{2d})_0$ corresponds to (Z,u). After the common scaling above, this is the Cartan reconstruction in Corollary 9.26. These models realize all three ambient dimensions allowed there; a classification of compatible splittings would still be needed to classify all nonradial cubics.

Proposition 11.1 (The nonradial differential algebra)**.** *The algebra $\mathbb{R}[\rho,\nu,A,\mathsf{B},W]$ is Laplacian. For $F=F_{-1/3}$, it is the smallest Laplacian algebra containing R,F:*

$$\mathcal{A}_R(F) = \mathbb{R}[\rho,\nu,A,\mathsf{B},W].$$

Its five generators are algebraically independent, and its spherical quotient dimension is four. Their Laplacians and gradient products are

$$\begin{gathered}\Delta\rho = 6d+4, \qquad \Delta\nu = 6d, \qquad \Delta A = \Delta\mathsf{B} = 0,\\ \Delta W = 16d\rho + \tfrac{16}{3}(3d+2)\nu.\end{gathered} \tag{11.3}$$

and, using symmetry, the following relations:

$$\begin{aligned}&\Gamma(\rho,\rho)=4\rho, && \Gamma(\rho,\nu)=0, && \Gamma(\nu,\nu)=4\nu,\\ &\Gamma(\rho,A)=6A, && \Gamma(\nu,A)=0, && \Gamma(\rho,\mathsf{B})=2\mathsf{B},\\ &\Gamma(\nu,\mathsf{B})=4\mathsf{B}, && \Gamma(\rho,W)=4W, && \Gamma(\nu,W)=4W.\end{aligned} \tag{11.4}$$

$$\begin{aligned}\Gamma(A,A) &= \tfrac{4}{3}\rho^2, & \Gamma(\mathsf{B},\mathsf{B}) &= \tfrac{4}{3}\nu^2 + W,\\ \Gamma(A,\mathsf{B}) &= \tfrac{4}{3}\rho\nu - \tfrac{1}{2}W, & \Gamma(A,W) &= 16\nu A - \tfrac{16}{3}\rho\mathsf{B},\\ \Gamma(\mathsf{B},W) &= 16\rho\mathsf{B} - 16\nu A + \tfrac{32}{3}\nu\mathsf{B},\\ \Gamma(W,W) &= 16(\rho+\nu)W + \tfrac{64}{3}\mathsf{B}^2 - 64A\mathsf{B}.\end{aligned} \tag{11.5}$$

Proof. Put $Q=uu^*$ and $m_j = u^*Z^ju$. The trace-free characteristic identity and the relevant gradients are

$$\begin{aligned}Z^3 &= \rho Z + AI, & \nabla_Z A &= 2(Z^2 - \tfrac{2}{3}\rho I),\\ \nabla_Z\mathsf{B} &= -2(Q - \tfrac{1}{3}\nu I), & \nabla_u\mathsf{B} &= -2Zu,\\ \nabla_Z W &= 8(ZQ + QZ - \tfrac{2}{3}m_1 I), & \nabla_u W &= 8Z^2u.\end{aligned}$$

Their pairings give (11.5), including

$$|\nabla_Z W|^2 = 16\nu W + \tfrac{64}{3}\mathsf{B}^2, \qquad |\nabla_u W|^2 = 64m_4 = 16\rho W - 64A\mathsf{B}.$$

An orthonormal basis E_α of $\mathrm{Herm}_3(\mathbb{K})_0$, with the indicated trace metric, satisfies $\sum_\alpha E_\alpha^2 = \frac{2}{3}(3d+2)I$. Taking diagonal and off-diagonal basis elements proves this identity; it gives the last formula in (11.3). The other Laplacians follow directly by taking traces. The chain rule therefore proves that the displayed algebra is Laplacian.

For $F=F_{-1/3}$ and $S=|\nabla F|^2$, the table gives

$$S = \tfrac{4}{3}(\rho-\nu/3)^2 + \tfrac{4}{9}W, \qquad \Delta S = \tfrac{32}{27}\big((15d+18)\rho + (3d+2)\nu\big). \tag{11.6}$$

It also gives

$$\Gamma(F,S) = 16(\rho + 7\nu/27)F.$$

Thus $\mathcal{L}(F) = -qF$, where $q = 8(\rho+7\nu/27)$ is positive and nonradial. We check the remaining hypotheses needed for Theorem 9.25.

The Cartan cubic A is irreducible by its harmonic eikonal identity $|\nabla A|^2 = 4\rho^2/3$. Indeed, if a harmonic cubic P in $n>2$ variables satisfying $|\nabla P|^2 = \kappa|x|^4$, $\kappa>0$, had a real linear

factor, write it as $P(t,y) = t(at^2 + t\ell(y) + Q(y))$. Restriction to $t = 0$ gives $Q = b|y|^2$, $b^2 = \kappa$. Harmonicity gives $\ell = 0$ and $6a + 2b(n-1) = 0$. The coefficient of $t^2|y|^2$ in the eikonal equation then gives $6ab + 4b^2 = 2\kappa$, forcing $n = 2$, a contradiction. Any real factorization of F would have a linear factor; restriction to $u = 0$ would factor A. Thus F is irreducible.

If $D_vF \equiv 0$ for a constant vector v, differentiating $\mathcal{L}(F) = -qF$ gives $(D_vq)F = 0$. Since q is positive definite, this forces $v = 0$. The point $(Z,u) = (\mathrm{diag}(1,-1,0),0)$ is a regular zero, and the two distinct eigenvalues of q establish nonradiality.

The coordinate dictionary above now transfers Theorem 9.25, with $m = 3d$, to these models, proving the smallest-algebra identity, algebraic independence, and spherical quotient dimension four. □

Theorem 11.2 (Minimal members of the Cartan deformation)**.** *For $d = 1, 2, 4$, every cubic F_t in (11.1) is harmonic. Its regular zero set is minimal if and only if*

$$t \in \{0, 1, -\tfrac{1}{3}\}.$$

The members $F_0 = A$ and $F_1 = N$ are, respectively, a cylindrical extension of a Cartan cubic and the original Cartan cubic. The cubic $F = F_{-1/3}$ is irreducible over $\mathbb{R}$, has no cylindrical directions, and satisfies

$$\Delta F = 0, \qquad \Gamma(F,S) = 2qF, \qquad \mathcal{L}(F) = -qF, \qquad q = 8(\rho + 7\nu/27). \tag{11.7}$$

Its regular zero set is a minimal hypersurface in ambient dimension $8, 14, 26$, respectively. Its quadratic weight has two distinct positive eigenvalues, with ratio $7/27$.

Proof. Harmonicity was proved above. The closure table gives

$$S_t := |\nabla F_t|^2 = \tfrac{4}{3}(\rho + t\nu)^2 + t(t-1)W. \tag{11.8}$$

Pairing its gradient with ∇F_t yields

$$\mathcal{L}(F_t) = -8\big(\rho + t^2(2-t)\nu\big)F_t - \tfrac{8}{3}t(t-1)(3t+1)(\rho + t\nu)\mathsf{B}. \tag{11.9}$$

This proves minimality for the three stated values and gives (11.7) at $t = -1/3$. For necessity, take $Z = \mathrm{diag}(1,1,-2)$ and, for $t \neq 0$, choose

$$u = t^{-1/2}e_3 \quad (t > 0), \qquad u = \sqrt{-2/t}\,e_1 \quad (t < 0).$$

Here e_i are the standard coordinate vectors of $\mathbb{K}^3$. Then $A = -2$, $\mathsf{B} = 2/t$, and $F_t = 0$, whereas $\nabla_u F_t = -2tZu \neq 0$. Moreover, $\rho + t\nu$ equals 4 or 1, respectively. Thus (11.9) cannot vanish at this regular zero unless $t = 1$ or $t = -1/3$.

The remaining assertions for $F = F_{-1/3}$ were established in Proposition 11.1 while verifying the hypotheses of the intrinsic five-generator theorem. □

These cones are not congruent to radial cubic cones. Indeed, if $P \circ Q$ has the same zero set as F, with Q orthogonal and P a radial cubic, the smooth-patch divisibility fact proved in Proposition 3.12 gives $P \circ Q = cF$. Here F is irreducible and has regular real zeros by Lemma 3.11. Orthogonal equivariance of $\mathcal{L}$ and $\mathcal{L}(cF) = c^3\mathcal{L}(F)$ would then make c^2q a multiple of R, contradicting (11.7).

For the rest of this section, put $F = F_{-1/3}$ and $S = |\nabla F|^2$. The generator degrees give

$$(\mathcal{A}_R(F))_2 = \mathrm{span}\{\rho, \nu\}, \qquad (\mathcal{A}_R(F))_3 = \mathrm{span}\{A, \mathsf{B}\}.$$

Thus the cubic part is two-dimensional, unlike the hypothesis of Theorem 9.5.

11.1.2. *Singularities and the two unstable models.*

Lemma 11.3. *The singular set of $\{F = 0\}$, equivalently the critical set of F, is*

$$\mathrm{Sing}\{F = 0\} = \Sigma = \{(Z,u) : \nu = 3\rho,\ Zu = 0\}. \tag{11.10}$$

The punctured set $\Sigma \setminus \{0\}$ is smooth of dimension $4d$. At

$$c = (\mathrm{diag}(1,-1,0), (0,0,\sqrt{3})),$$

the Hessian of F has eigenvalues $8/3, -8/3, 0$, with multiplicities $d+1, d+1, 4d$, respectively, and its kernel is $T_c\Sigma$. Every nonzero point of Σ is obtained from c by a Hermitian change of basis and a positive dilation. The zero sets, rescaled about such a point, converge smoothly on compact subsets of the regular part of

$$\mathbb{R}^{4d} \times \{(v_+, v_-) \in \mathbb{R}^{d+1} \oplus \mathbb{R}^{d+1} : |v_+| = |v_-|\}. \tag{11.11}$$

Proof. Equation (11.6) proves the description of Σ. For a nonzero u, the restriction of Z to $u^\perp$ is a trace-free Hermitian 2×2 matrix of a fixed positive norm. These matrices form a sphere of dimension d, so $\Sigma \setminus \{0\}$ is a smooth sphere bundle over $\mathbb{K}^3 \setminus \{0\}$, of dimension $4d$. Its eigenvalues are $(\lambda, -\lambda, 0)$, $\lambda > 0$, which also proves the assertion about changes of basis and dilations.

At c, let the diagonal variation of Z be $(\xi, \eta, -\xi - \eta)$, let h_i be the variations of u_i, and use a, b, c' for the off-diagonal variations in the matrix displayed above. The quadratic term of F is

$$Q_2 = \xi^2 - \eta^2 - \tfrac{2}{\sqrt{3}}(\xi + \eta) \operatorname{Re} h_3 + |b + \bar{h}_1/\sqrt{3}|^2 - |a - h_2/\sqrt{3}|^2.$$

The first three terms become $8st'/3$ under the orthonormal changes

$$s = \tfrac{\sqrt{3}}{2}(\xi + \eta), \qquad t = \tfrac{1}{2}(\xi - \eta), \qquad t' = \tfrac{1}{2}(\sqrt{3}t - \operatorname{Re} h_3).$$

Completing to orthonormal coordinates gives $Q_2 = \frac{4}{3}(|v_+|^2 - |v_-|^2)$ and the stated spectrum. Differentiating $\nabla F = 0$ along Σ, and comparing dimensions, identifies the kernel. Finally, $r^{-2}F(c + rx) = Q_2(x) + rF(x)$, so Section 2.6 gives the tangent and smooth regular graphical convergence. The transverse cone is not a hyperplane, so every nonzero point of Σ is geometrically singular; the vertex is singular by homogeneity. □

By Proposition 4.24, a smooth nonnegative subcalibrating multiplier must vanish on $\Sigma \setminus \{0\}$; in particular, radial powers fail.

Proposition 11.4. *The nonradial cones in ambient dimensions 8 and 14 are unstable on their regular parts and are not area minimizing.*

Proof. The transverse cone in (11.11) has dimension $2d + 1$ and squared second fundamental form $2d/r^2$. Its radial second variation, up to a positive factor, is

$$\int_0^\infty \left(|\phi'|^2 - \frac{2d}{r^2}\phi^2 \right) r^{2d}\, dr.$$

The sharp Hardy constant $(2d-1)^2/4$ is smaller than $2d$ for $d = 1, 2$, so a compactly supported test function on a radial annulus gives a negative value. Lemma 2.15, using the regular graphical convergence in Lemma 11.3, proves instability of the original cone. □

The boundary formula suggests a gradient factor. Put $K = \Delta S$ and $T = |\nabla S|^2$. For $H = F\sqrt{S}$, the boundary speed is $w = S^{-1}$. On C_{reg}, the closure table gives $T/S = \frac{64}{27}(9\rho + \nu)$. Using (11.6) and (11.7) in Proposition 4.5 yields

$$-\frac{J_C(S^{-1})}{S^{-1}} = \frac{K/2 - 2q - T/S}{S} = \frac{16(d-3)(5\rho + \nu)}{9S}. \tag{11.12}$$

For $d = 1, 2$, this candidate fails the boundary condition. For $d = 4$, its trace is strictly positive; we now prove the global subcalibration sign.

11.1.3. *The minimizing nonradial cone in dimension twenty-six.*

Theorem 11.5. *For $d = 4$, both homogeneous potentials*

$$H = F\sqrt{S}, \qquad G = F\sqrt{RS}$$

generate two-sided subcalibrations for $E = \{F < 0\}$. Their divergence signs are strict on $\{F \neq 0\}$. Consequently E is locally perimeter minimizing and $\partial[\![E]\!]$ is strictly area minimizing, with support exactly $\{F = 0\} \subset \mathbb{R}^{26}$.

Proof. We first prove the assertion for the degree-five potential H. The closure table gives

$$T = \tfrac{64}{27}(9\rho+\nu)S - \tfrac{1024}{81}F\mathsf{B}. \tag{11.13}$$

On $S>0$, differentiation yields

$$\begin{aligned}\Delta H &= \frac{F}{\sqrt{S}}\left(2q + \tfrac{1}{2}K - \frac{T}{4S}\right),\\ |\nabla H|^2 &= S^2 + 2qF^2 + \frac{F^2T}{4S} \geq S^2.\end{aligned} \tag{11.14}$$

The resulting defect is

$$\begin{aligned}\frac{\mathcal{L}(H)}{S^{3/2}F} &= \tfrac{1}{2}K - 2q - \frac{T}{S} - \frac{F\Gamma(F,q)}{S}\\ &\quad + \frac{F^2}{S^2}\left(qK + 2q^2 - \tfrac{1}{2}\Gamma(S,q)\right) - \frac{F\Gamma(F,T)}{8S^2}\\ &\quad + \frac{F^2}{8S^3}\left(TK - \tfrac{1}{2}\Gamma(S,T)\right).\end{aligned} \tag{11.15}$$

The quotient is initially written for $F\neq 0$; multiplying through gives an identity on all of $\{S>0\}$. Substituting the table gives

$$\mathcal{L}(H) = \frac{16F}{729S^{3/2}}\mathcal{N}, \tag{11.16}$$

where

$$\begin{aligned}\mathcal{N} &= (405\rho+81\nu)S^3 + 144F\mathsf{B}S^2\\ &\quad + F^2S(46692\rho^2 + 20040\rho\nu + 2212\nu^2 - 4239S)\\ &\quad - (4992\rho+896\nu)F^3\mathsf{B} - 256\nu F^4.\end{aligned} \tag{11.17}$$

To prove its sign, define

$$\begin{aligned}S_* &= \tfrac{4}{3}\rho^2 + \tfrac{40}{27}\rho\nu + \tfrac{4}{27}\nu^2,\\ K_0 &= 4992\rho + 896\nu,\\ H_0 &= \tfrac{8}{9}(36639\rho^2 + 12634\rho\nu + 1555\nu^2).\end{aligned}$$

The invariant image satisfies

$$S\leq S_*, \qquad 9F^2\leq RS, \qquad 16\mathsf{B}^2\leq 9\nu S, \qquad F\mathsf{B}\leq \tfrac{3}{16}(9\rho+\nu)S. \tag{11.18}$$

Indeed, the trace-free eigenvalues of Z obey $\lambda_i^2\leq 4\rho/3$, whence $W\leq 16\rho\nu/3$ and the first inequality follows from (11.6). The second is Euler's identity and Cauchy–Schwarz. The third follows from $\mathsf{B}^2\leq \nu u^{*}Z^2u = \nu W/4\leq 9\nu S/16$. The last is $T\geq 0$ in (11.13). There is the exact decomposition

$$\begin{aligned}\mathcal{N} &= 405\rho S^3 + 36S^2(F+2\mathsf{B})^2 + S^2(81\nu S - 144\mathsf{B}^2) + F^2SH_0\\ &\quad + K_0F^2\left(\tfrac{3}{16}(9\rho+\nu)S - F\mathsf{B}\right)\\ &\quad + 256\nu F^2\left(\tfrac{1}{9}RS - F^2\right) + 4275F^2S(S_*-S).\end{aligned} \tag{11.19}$$

Expansion recovers (11.17); every summand is nonnegative by (11.18). Since $F\neq 0$ implies $\rho>0$ and $S>0$, the first summand proves the required strict sign for $\mathcal{L}(H)$ off $\{F=0\}$.

For the degree-six potential $G=\sqrt{R}\,H$, we use (11.14) and (11.13), which give

$$\Delta H = \frac{F}{\sqrt{S}}\left(\frac{512}{9}\rho + \frac{320}{27}\nu + \frac{256F\mathsf{B}}{81S}\right).$$

The second and third bounds in (11.18) imply

$$\frac{|F\mathsf{B}|}{S}\leq \frac{1}{4}\sqrt{\nu(\rho+\nu)}\leq \frac{\rho+2\nu}{8}.$$

Thus the parenthesis is at least $(4576\rho + 896\nu)/81 > 0$ for $R > 0$, and $H\Delta H \geq 0$. On $\{S > 0\}$, the degree-five potential H is smooth, satisfies $H\mathcal{L}(H) \geq 0$, and has nonvanishing gradient by (11.14). Proposition A.5, with degree 5 and ambient dimension $26 \geq 2 \cdot 5 + 2$, therefore gives $G\mathcal{L}(G) \geq 0$, strictly where $F \neq 0$, as well as $\nabla G \neq 0$ on $\{S > 0\}$.

The exceptional set has codimension ten by Lemma 11.3. Both potentials extend continuously by zero across it with the signs of F, and their normalized gradients agree with the outward normal at regular zeros. Thus Theorem 2.5 applies; harmonicity gives the full zero-set support. For both fields the strict sign on the two nonempty phases gives (2.18), so Theorem 2.11 proves strict area minimality. □

The degree-five potential H also proves strict stability directly. Indeed, the cone has dimension $n = 25$, and its subcalibration sign gives a nonnegative Jacobi boundary defect. Hence Theorem 4.6 yields

$$Q_C(u) \geq \left(5 - \frac{25}{2}\right)^2 \int_C r^{-2}u^2 = \frac{225}{4} \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}).$$

11.2. **Cubic targets.** The common cubic potential gives the following pullback construction, including for targets with nonisolated singularities.

Corollary 11.6 (Pullback of the common cubic potential)**.** *Let $f \not\equiv 0$ be a harmonic cubic on $\mathbb{R}^m$, and put $q = |y|^2$, $S = |\nabla_y f|^2$, and $H = f\sqrt{qS}$. Suppose, on $S > 0$, that*

$$f\Gamma_y(f, S) \geq 0, \qquad H\mathcal{L}_y(H) \geq 0.$$

For every quadratic Clifford map $P : \mathbb{R}^N \to \mathbb{R}^m$ with $N \geq 32$, set $F = f(P)$, $Q = q(P)$, and $T = S(P)$. Then

$$G = R^2 F\sqrt{QT}$$

is a subcalibration potential for $\{F < 0\}$. Its boundary current is locally mass minimizing and strictly area minimizing, with support $\{F = 0\}$. The defining polynomial has degree six and the potential has homogeneity degree sixteen.

Proof. Put $H_0 = f\sqrt{S}$. Bochner's identity and $\Delta_y f = 0$ give

$$H_0\Delta_y H_0 = f^2\left(|\operatorname{Hess} f|^2 - \frac{|\operatorname{Hess} f(\nabla f)|^2}{S}\right) + f\Gamma_y(f, S) \geq 0.$$

Since H_0 has degree five, differentiating $H = \sqrt{q}\, H_0$ yields

$$H\Delta_y H = qH_0\Delta_y H_0 + (m + 9)H_0^2 \geq 0.$$

Thus the ratio condition holds with $M = 0$. Euler's identity gives $\nabla H \neq 0$ where $H \neq 0$; at a zero with $S > 0$, $\nabla H = \sqrt{qS}\nabla f \neq 0$. The exceptional set $Z = \{\nabla f = 0\}$ is conical and algebraic, and Euler's identity gives $Z \subset \{f = 0\}$. Its codimension is at least two by harmonicity and Lemma 3.11. Now apply Theorem 5.4 with $\ell = 6$, $a = 2$, and $M = 0$: its bound is $N \geq 1 + 24 + 4 + 3 = 32$. Harmonicity gives both signs near every target zero and hence the support and strict-minimality assertions. □

The first hypothesis holds for the radial cubic models, where $\Gamma(f, S)$ is a positive multiple of qf, and for the nonradial models of Theorem 11.2, whose quadratic minimality weight is positive. The second holds in the minimizing ranges of Theorems 10.2 and 11.5. Since $q, f, S \in \mathcal{A}_q(f)$, this potential is a function of the smallest target algebra. For an essential target with r minimal homogeneous generators and a pullback with $Q < R^2$ somewhere, Theorem 5.5 gives exactly $r + 1$ source generators. Thus the three-, four-, and five-generator cubic models yield four-, five-, and six-generator pullback algebras whenever the two displayed inequalities hold.

For the minimizing nonradial target in Theorem 11.5, the degree-five potential $H_0 = f\sqrt{S}$ already satisfies $H_0\mathcal{L}(H_0) \geq 0$, and the Bochner identity above gives $H_0\Delta H_0 \geq 0$. Thus

Theorem 5.4, with $\ell = 5$, $a = 1$, and $M = 0$, gives the degree-twelve potential $G = RF\sqrt{T}$ for every such Clifford map with $N \geq 28$.

For Cartan targets, a weighted inequality gives the lower-degree pullback $G = H(P)$.

11.2.1. *Cartan targets with a polynomial pullback potential.* Cartan's harmonic cubics on $\mathbb{R}^m$, for $m = 14, 26$, can be normalized so that

$$\Delta f = 0, \qquad |\nabla f|^2 = 9q^2, \qquad |f|^2 \leq q^3, \qquad q = |y|^2. \tag{11.20}$$

Their multiplicities are four and eight, respectively; see [17, Theorems 2.5 and 4.1]. The algebra $\mathbb{R}[q, f]$ is Laplacian, with $\Gamma(q, f) = 6f$.

Theorem 11.7 (Cartan cubic pullbacks)**.** *Let f be either normalized Cartan cubic in (11.20), and let $P : \mathbb{R}^N \to \mathbb{R}^m$ be a quadratic Clifford map. Then*

$$F = f \circ P, \qquad Q = |P|^2, \qquad G = QF$$

give a perimeter-minimizing phase $\{F < 0\}$. Its boundary current C is strictly area minimizing and is supported on $\{F = 0\}$. It is strictly stable, with the explicit bound

$$Q_C(u) \geq \left[\left(10 - \frac{N-1}{2}\right)^2 + 8m - 106\right] \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}). \tag{11.21}$$

The polynomial F has degree six and the subcalibration potential G has degree ten. The construction exists for $(m, N) = (14, 256)$ and $(26, 8192)$, and for direct sums of either Clifford system.

Proof. Use the degree-five target potential $H = qf$ from Example 4.23. The target closure gives

$$|\nabla H|^2 = 9q^4 + 16qf^2, \qquad \Delta H = 2(m+6)f, \tag{11.22}$$

$$\mathcal{L}(H) = 9(2m-24)q^4 f + 16(2m+1)qf^3. \tag{11.23}$$

Substituting the target closure in $\mathcal{L}(H) = |\nabla H|^2 \Delta H - \Gamma(H, |\nabla H|^2)/2$ gives the second identity. Hence

$$2q\mathcal{L}(H) - 5H|\nabla H|^2 = q^2 f\big(9(4m-53)q^3 + 16(4m-3)f^2\big). \tag{11.24}$$

Both coefficients in parentheses are positive for $m = 14, 26$. After multiplication by H, this is precisely (5.5) for $\kappa = 1/2$.

The smallest source algebra is $\mathcal{A}_R(F) = \mathbb{R}[R, Q, F]$ by Theorem 5.5. Indeed, the target algebra is $\mathcal{A}_q(f) = \mathbb{R}[q, f]$, and $|\nabla f|^2 = 9q^2$ excludes a nonzero cylindrical direction. For example,

$$\Delta Q = 8mR, \quad \Delta F = 0, \quad \Gamma(Q, F) = 24RF, \quad |\nabla F|^2 = 36RQ^2.$$

The target inequality therefore gives the required source sign by Proposition 5.1; the expanded gradient and defect are recorded in Appendix C.5.

By (11.22), $\nabla H \neq 0$ off the target origin. The pullback results Proposition 5.1 and Lemma 5.3 therefore give comparison across $P^{-1}(0)$, of codimension m away from the origin. Surjectivity makes $F \neq 0$; harmonicity identifies the full support and gives both phases by Remark 2.7. The strict sign in (C.31) then invokes Corollary 2.13.

For the improved Hardy bound, compute the boundary term. On the regular cone, $a = Q$ and $|\nabla F|^2 = 36RQ^2$, so (4.11) gives

$$\beta_G = \left.\frac{R\mathcal{K}_F(Q)}{Q^3|\nabla F|^2}\right|_C = (8m - 96)\frac{R^2}{Q} - 10 \geq 8m - 106 > 0.$$

Apply Theorem 4.6 with $\ell = 10$ to obtain (11.21).

Finally, the standard Clifford module dimensions give systems with fourteen matrices on $\mathbb{R}^{256}$ and twenty-six matrices on $\mathbb{R}^{8192}$; see [69, Table 4.1]. Direct sums preserve the Clifford relations and all the identities used above. □

12. Spectral Laplacian Algebras and a Uniform Positivity Criterion

From the spectral gradient-pairing table below and one Laplacian coefficient bound, Theorem 12.3 constructs $G = F|\nabla F|$ for octonionic products and Pfaffians (Section 13); complex determinants require another multiplier (Section 14).

12.1. **Common gradient pairings and the smallest algebra.** Fix $d \geq 2$. For $r = (r_1, \dots, r_d)$, let $e_k(r)$ be the elementary symmetric polynomial of degree k, with $e_0 = 1$ and $e_k = 0$ for $k < 0$ or $k > d$. Consider real homogeneous polynomials $F, E_1, \dots, E_d$ on $\mathbb{R}^N$, of degrees $d, 2, \dots, 2d$, such that $F \not\equiv 0$, $E_1 = R = |x|^2$, and at each point

$$(E_1(x), \dots, E_d(x)) = (e_1(r), \dots, e_d(r)) \quad \text{for some } r_1, \dots, r_d \geq 0. \tag{12.1}$$

No differentiable choice or ordering of these roots is required. Assume the identities

$$\Delta F = 0, \qquad \Gamma(F, F) = E_{d-1}, \tag{12.2}$$

$$\Gamma(F, E_k) = 2(d - k + 1)E_{k-1}F, \qquad 1 \leq k \leq d, \tag{12.3}$$

$$\Gamma(E_i, E_j) = 4\sum_{\alpha=1}^{d} r_\alpha \partial_\alpha e_i(r) \partial_\alpha e_j(r). \tag{12.4}$$

The right-hand side of (12.4) is a symmetric polynomial in the roots. Its explicit polynomial form is

$$\Gamma(E_i, E_j) = 4\sum_{\ell=0}^{i-1} (j - i + 1 + 2\ell) E_{i-1-\ell} E_{j+\ell}, \qquad 1 \leq i \leq j \leq d. \tag{12.5}$$

To verify this formula, let $p(t) = \prod_\alpha (1 + r_\alpha t)$. The generating function for one quarter of the pairings is

$$\sum_\alpha r_\alpha \partial_\alpha p(s) \partial_\alpha p(t) = \frac{st}{s - t}\big(sp'(s)p(t) - tp'(t)p(s)\big).$$

The numerator vanishes at $s = t$. Polynomial division and comparison of the coefficients of $s^i t^j$ give (12.5).

Proposition 12.1 (Finite closure and exact generation)**.** *Suppose* (12.2)–(12.4) *hold and*

$$\Delta E_k = b_k E_{k-1} \qquad (1 \leq k \leq d) \tag{12.6}$$

for real constants b_k. Then $\mathbb{R}[E_1, \dots, E_d, F]$ is a Laplacian algebra. If $d \geq 3$ and $b_k \neq 0$ for $2 \leq k \leq d - 1$, then

$$\mathcal{A}_R(F) = \mathbb{R}[E_1, \dots, E_d, F]. \tag{12.7}$$

For $d = 2$, one instead has $\mathcal{A}_R(F) = \mathbb{R}[R, F]$. If the $d + 1$ generators in (12.7) have independent differentials at some point, the spherical quotient has dimension d.

Proof. The generator criterion Proposition 3.7 gives closure. Conversely, $E_{d-1} = \Gamma(F, F) \in \mathcal{A}_R(F)$, and successive Laplacians recover $E_{d-2}, \dots, E_1$ by (12.6). For $d \geq 3$, (12.5) gives

$$\Gamma(E_2, E_{d-1}) = 4\big((d - 2)E_1 E_{d-1} + dE_d\big), \tag{12.8}$$

which recovers E_d. This proves equality of the algebras. For $d = 2$, the identities $\Gamma(F, F) = R$, $\Gamma(R, F) = 4F$, $\Gamma(R, R) = 4R$, $\Delta F = 0$, and $\Delta R = 2N$ already close on R, F. Finally, independence gives quotient dimension $d + 1$ in the Euclidean space; fixing the nonzero radial generator E_1 decreases this by one. □

The positivity theorem below requires only a lower bound for ΔE_{d-1}, rather than the full drift formula (12.6).

12.2. **Two elementary inequalities.** Set

$$S = E_{d-1}, \qquad B = E_{d-2}, \qquad C = E_{d-3}, \qquad D = E_d. \tag{12.9}$$

In this subsection the letter C denotes this invariant, not the zero cone. All formulas will be used on $S > 0$, where $B > 0$ and every E_k with $0 \le k \le d-1$ is positive.

Lemma 12.2 (A Newton inequality and an antitone coefficient)**.** *For nonnegative roots and $d \ge 3$,*

$$BS \ge \frac{3d}{d-2} CD. \tag{12.10}$$

For every $d \ge 2$, the function

$$Q_* = \frac{5B}{S} + \frac{3CD}{S^2} \tag{12.11}$$

is positive and nonincreasing in each root r_i wherever $S > 0$.

Proof. First suppose all roots are positive and put $t_i = r_i^{-1}$. Writing $p_k = e_k(t_1, \dots, t_d)$, one has $S = Dp_1$, $B = Dp_2$, and $C = Dp_3$. For each triple of nonnegative numbers, the sum of the six monomials $t_i^2 t_j$, with distinct indices in the triple, is at least $6t_it_jt_k$, by the arithmetic–geometric mean inequality. Summing over triples counts each ordered pair $d-2$ times. Consequently

$$p_1 p_2 = \sum_{i \ne j} t_i^2 t_j + 3p_3 \ge \frac{3d}{d-2} p_3,$$

which proves (12.10); continuity includes zero roots.

In the reciprocal variables, $Q_* = 5p_2/p_1 + 3p_3/p_1^2$. Fix $t_i = z$, and let u, v, w be the elementary symmetric polynomials of orders $1, 2, 3$ in the remaining variables. Then $p_1 = z+u$, $p_2 = zu + v$, and $p_3 = zv + w$, so

$$\frac{\partial Q_*}{\partial z} = \frac{z(5u^2 - 8v) + 5u^3 - 2uv - 6w}{(z+u)^3} \ge 0. \tag{12.12}$$

Indeed $u^2 \ge 2v$ and $u^3 \ge 6w$; the numerator is at least $zu^2 + 3u^3$. Since $\partial_{r_i} = -t_i^2 \partial_{t_i}$, this proves the asserted sign. Formula (12.11) is smooth in the original roots wherever $S > 0$, so its derivative sign extends to zero roots by continuity. Positivity follows from $B > 0$. For $d = 2$, the same argument has $v = w = 0$, or one can use $Q_* = 5/E_1$ directly. □

12.3. **Subcalibration existence from a Laplacian bound.**

Theorem 12.3 (Spectral subcalibrations)**.** *Assume (12.1)–(12.4). Suppose on $U = \{S > 0\}$ that*

$$\Delta S \ge \beta B, \qquad \beta \ge 24 - \frac{16}{d}. \tag{12.13}$$

Then $N \ge 4d$, and there is a potential G, homogeneous of degree $2d-1$, with nonzero gradient on U and satisfying $F\mathcal{L}(G) \ge 0$. It is given explicitly by

$$G = F\sqrt{S} = F|\nabla F|. \tag{12.14}$$

Writing $X = \nabla G / |\nabla G|$, one moreover has

$$\operatorname{sgn}(F) \operatorname{div} X > 0 \qquad \text{on } \{F \ne 0\}. \tag{12.15}$$

The field X is a two-sided subcalibration of $\{F < 0\}$ across $Z = \{S = 0\}$, and the perimeter and current conclusions of Theorem 2.5 hold. The cone $\Sigma = \{F = 0\}$ is strictly area minimizing and strictly stable, with

$$Q_\Sigma(u) \ge \frac{(N - 4d + 1)^2}{4} \int_\Sigma r^{-2} u^2, \qquad u \in C_c^\infty(\Sigma_{\mathrm{reg}} \setminus \{0\}). \tag{12.16}$$

The constant in this estimate is at least $1/4$.

Proof. For positive roots, the decreasing ratios of the normalized elementary symmetric functions $e_j/\binom{d}{j}$, as given by Newton's inequalities [7, Section 3.1], imply

$$RB \geq \frac{d(d-1)}{2} S. \tag{12.17}$$

This also holds at zero roots by continuity, and is an equality for $d = 2$. Since $S = |\nabla F|^2$ is a nonzero polynomial, U is dense, so (12.13) extends to all points by continuity. Integrating on the unit sphere and using the homogeneous degree $2d-2$ of S gives

$$(2d-2)(N+2d-4) \int_{\mathbb{S}^{N-1}} S = \int_{\mathbb{S}^{N-1}} \Delta S \geq (12d-8)(d-1) \int_{\mathbb{S}^{N-1}} S.$$

The integral of S is positive. Division proves $N \geq 4d$.

The mixed pairing at $k = d-1$ gives $\Gamma(F, S) = 4BF$. Thus the regular zero set is minimal and (3.12) holds with $q = 2B$. By (12.5), $T = \Gamma(S, S) = 4(BS + 3CD)$. Take $h = \frac{1}{2} \log S$, and write $\delta = \beta/4$. The auxiliary function in (4.26) is exactly Q_*. Moreover,

$$\Delta h \geq 2(\delta - 1)\frac{B}{S} - 6\frac{CD}{S^2}, \tag{12.18}$$

$$D_h \geq 2(\delta - 4)B - 12\frac{CD}{S}. \tag{12.19}$$

For $d \geq 3$, Lemma 12.2 implies $BS \geq 3CD$; hence $\Delta h \geq 2(\delta - 2)B/S > 0$. The sharper constant gives

$$D_h \geq 2\left(\delta - 6 + \frac{4}{d}\right) B \geq 0.$$

For $d = 2$, $C = 0$ and $\delta \geq 4$, so $\Delta h \geq 2(\delta - 1)/S > 0$ and $D_h \geq 0$ follow directly from (12.18)–(12.19).

For every smooth symmetric function $u(r)$ expressed in the generators, the mixed pairings give

$$\Gamma(F, u) = 2F \sum_i \partial_{r_i} u, \tag{12.20}$$

because $\sum_i \partial_{r_i} e_k = (d - k + 1)e_{k-1}$. The root pairing similarly gives

$$\Gamma(h, Q_*) = \frac{2}{S} \sum_i r_i (\partial_{r_i} S)(\partial_{r_i} Q_*) \leq 0, \qquad F\Gamma(F, Q_*) = 2F^2 \sum_i \partial_{r_i} Q_* \leq 0.$$

Here $\partial_{r_i} S \geq 0$, and the other derivative has the opposite sign by Lemma 12.2. The invariant identities extend to repeated and zero roots, verifying Theorem 4.12. In particular,

$$|\nabla G|^2 = S^2 + 5BF^2 + \frac{3CD}{S} F^2 \geq S^2 > 0. \tag{12.21}$$

Since $Q_* > 0$ and $\Delta h > 0$, the term $F^2 Q_* \Delta h$ in (4.22) is strictly positive when $F \neq 0$, while all the other terms in that bracket are nonnegative. Thus $F\mathcal{L}(G) > 0$ there. Euler's identity shows that $F \neq 0$ implies $S > 0$, so this proves (12.15) on both entire phases.

Harmonicity makes Z removable by Lemma 3.11; the monotonicity theorem gives comparison with full support by Remark 2.7. Both phases are nonempty, and (12.15) verifies the flux hypothesis of Theorem 2.11. Homogeneity therefore gives strict area minimality.

For strict stability, we use the boundary Jacobi estimate. At a smooth point of the full harmonic nodal hypersurface, the two local phases have opposite signs, and the Hopf boundary lemma gives $\nabla F \neq 0$. Thus the regular cone lies in U, where (4.23) gives $V_G = D_h/S \geq 0$. Apply Theorem 4.6 with $n = N - 1$ and $\ell = 2d - 1$ to obtain (12.16), since

$$\left(\ell - \frac{n}{2}\right)^2 = \frac{(N - 4d + 1)^2}{4}.$$

The bound $N \geq 4d$ makes this constant at least $1/4$. □

Remark 12.4 (An explicit coefficient identity). If $\Delta S = 4\delta B$ exactly, put $A = E_{d-4}$ and $J = E_{d-5}$, retaining the convention for negative indices. The same closure table gives the checkable identity

$$\mathcal{L}(F\sqrt{S}) = \frac{F}{2S^{3/2}}\left\{4S^2\left[(\delta-4)BS - 6CD\right] + F^2\mathscr{P}_\delta\right\}, \tag{12.22}$$

where

$$\mathscr{P}_\delta = (20\delta+10)B^2S + (12\delta-18)BCD - 56CS^2 - 82ASD - 30JD^2.$$

Theorem 12.3 proves the sign of this identity by monotonicity, uniformly in d.

12.4. **The threshold and the role of the Laplacian.** The threshold is sharp for $G = F\sqrt{S}$ when $\Delta S = 4\delta B$ and the invariant image contains $r_1 = \cdots = r_d = 1$ with arbitrarily small nonzero F. Indeed, at these roots,

$$D_h = d(d-1)\left(\delta - 6 + \frac{4}{d}\right). \tag{12.23}$$

For $\delta < 6 - 4/d$, the multiplier sign fails near $F = 0$. Complex determinants realize these values with $\delta = 4$, so $F\sqrt{S}$ fails in every order $d \geq 3$. Their product multiplier in Section 14 succeeds: the threshold is not necessary for area minimization.

The next proposition compares reduced operators with different Laplacian coefficients; their ambient spaces may differ.

Proposition 12.5 (Monotonicity under an increase of the drift)**.** *Suppose two reduced operators have the same gradient pairing $\widehat{\Gamma}$, annihilate F, and have generator drifts $\ell_i = \widehat{\Delta}E_i$ and $\widetilde{\ell}_i = \widetilde{\Delta}E_i$. These are the full drift functions, including any factors E_{i-1}. For a positive smooth multiplier $a = a(E_1, \dots, E_d)$, put $G = aF$ and $W = \widehat{\Gamma}(G, G)$. Their curvature operators satisfy*

$$G\widetilde{\mathcal{L}}(G) - G\mathcal{L}(G) = F^2 aW \sum_i (\widetilde{\ell}_i - \ell_i)\partial_{E_i} a. \tag{12.24}$$

On any common invariant domain where the reference certificate holds, a nonnegative sign certificate is preserved if $\widetilde{\ell}_i \geq \ell_i$ for every generator used by a and $\partial_{E_i} a \geq 0$.

Proof. The common gradient pairing gives the same W and the same term $-\widehat{\Gamma}(G, W)/2$. The difference of the two Laplacians on G is $F\sum_i(\widetilde{\ell}_i - \ell_i)\partial_{E_i}a$. Multiplication by GW proves the formula. □

For example, consider drifts of the form

$$\Delta E_k = 2j[\alpha + \eta(j-1)]E_{k-1}, \qquad j = d - k + 1. \tag{12.25}$$

The simple criterion requires $\alpha + \eta \geq 6 - 4/d$. The determinant drift $(\alpha, \eta) = (2, 2)$ instead admits $a = (E_1 \cdots E_{d-1})^{1/2}$ by Section 14. Its nonnegative partial derivatives allow Proposition 12.5 to transfer the certificate whenever

$$\alpha + \eta \geq 4, \qquad \alpha + (d-1)\eta \geq 2d. \tag{12.26}$$

Indeed these endpoint conditions are equivalent to $\alpha + \eta(j-1) \geq 2j$ for $2 \leq j \leq d$. To apply this comparison geometrically, the field must be defined off a removable exceptional set.

13. Normed Products and Complex Pfaffians

We apply Theorem 12.3 to octonionic products and complex Pfaffians in every integer degree. The multiplication maps over normed division algebras are harmonic morphisms; see [69, Section 4.5.2]. The minimality of the real-part Pfaffian cones also follows from the holomorphic construction of Hoppe–Tkachev [41, Corollary 5.2 and Remark 5.5]. Their smallest algebras have the spectral table, with Laplacian coefficients 32 and 24, respectively. Both exceed $24 - 16/d$, giving the potential $F|\nabla F|$.

13.1. **Products in a normed division algebra.** Let $\mathbb{K} = \mathbb{H}$ or $\mathbb{O}$, put $\delta = \dim_{\mathbb{R}} \mathbb{K}$, and fix an integer $d \geq 2$. Choose a full parenthesization of the ordered product of d variables, and write

$$P : \mathbb{K}^d \longrightarrow \mathbb{K}, \qquad P(x_1, \dots, x_d) = x_1 \cdots x_d, \qquad F = \operatorname{Re} P. \tag{13.1}$$

The fixed parentheses allow for the nonassociativity of $\mathbb{O}$. Set

$$r_a = |x_a|^2, \qquad E_k = e_k(r_1, \dots, r_d), \qquad R = E_1 = \sum_a |x_a|^2, \tag{13.2}$$

where e_k denotes the k-th elementary symmetric polynomial. We use $E_0 = 1$ and $E_k = 0$ for $k > d$.

Proposition 13.1 (Differential closure of normed products)**.** *For every parenthesization in* (13.1)*, the polynomial F is real irreducible, homogeneous of degree d, and satisfies*

$$\begin{aligned} &\Delta F = 0, \qquad \Gamma(F, F) = E_{d-1}, \\ &\Gamma(F, E_k) = 2(d - k + 1) E_{k-1} F, \end{aligned} \tag{13.3}$$

$$\begin{aligned} \Delta E_k &= 2\delta(d - k + 1) E_{k-1}, \\ \Gamma(E_i, E_j) &= 4 \sum_{a=1}^{d} r_a \frac{\partial e_i}{\partial r_a} \frac{\partial e_j}{\partial r_a}. \end{aligned} \tag{13.4}$$

Consequently

$$\mathcal{L}(F) = -2E_{d-2} F. \tag{13.5}$$

The polynomial critical set is

$$Z_d = \{\nabla F = 0\} = \bigcup_{1 \leq a < b \leq d} \{x_a = x_b = 0\}, \tag{13.6}$$

and has real codimension 2δ.

Proof. Multiplication in $\mathbb{K}$ is real bilinear and satisfies $|uv| = |u||v|$; see [71, Chapter 1, Appendix, pp. 21–22]. A parenthesized product is therefore real multilinear, and induction on its multiplication tree gives

$$|P(x)|^2 = \prod_{a=1}^{d} r_a.$$

With all variables except x_a fixed, the derivative $A_a = D_{x_a} P : \mathbb{K} \to \mathbb{K}$ is a linear map satisfying

$$|A_a v|^2 = \left(\prod_{b \neq a} r_b \right) |v|^2.$$

By polarization, $A_a^* A_a = (\prod_{b \neq a} r_b) I$, the usual characterization of a linear similarity [71, Chapter 1, Lemma 1.3(1), pp. 15–16]. The source and target have the same dimension, so also $A_a A_a^* = (\prod_{b \neq a} r_b) I$; this remains true when the coefficient is zero. Since $F = \langle P, 1 \rangle$, it follows that

$$|\nabla_{x_a} F|^2 = |A_a^* 1|^2 = \prod_{b \neq a} r_b.$$

Summing over a proves $\Gamma(F, F) = E_{d-1}$. Every second derivative within a single variable block vanishes, so $\Delta F = 0$. The choice $x_1 = \cdots = x_d = 1$ shows that F is nonzero.

For irreducibility, a factorization $F = AB$ must be homogeneous in every variable block, by the lowest- and highest-degree argument in Lemma 3.11. Since each block degree of F is one, A and B depend on disjoint groups of blocks. If both are nonconstant, retain one variable from each group and set all remaining variables equal to 1. Every parenthesization then reduces F to $\operatorname{Re}(uv) = \langle u, \bar{v} \rangle$, a nondegenerate bilinear form of rank $\delta > 1$ [71, Chapter 1,

Appendix, pp. 21–22]. The specialized factorization would express it as a product of two linear forms in separate variables, of rank one, a contradiction.

Euler's identity in each variable block gives $\Gamma(F, r_a) = 2F$. Hence

$$\Gamma(F, E_k) = 2F \sum_a \frac{\partial e_k}{\partial r_a} = 2(d-k+1)E_{k-1}F.$$

Moreover, $\Delta r_a = 2\delta$, $\Gamma(r_a, r_b) = 4r_a\delta_{ab}$, and each e_k is affine in each individual r_a. These facts prove (13.4) by the chain rule. Harmonicity and the identity for $\Gamma(F, E_{d-1})$ now give (13.5).

Finally, the nonnegative sum $E_{d-1} = \sum_a \prod_{b\neq a} r_b$ vanishes precisely when at least two blocks vanish. Each largest linear component of this set has dimension $\delta(d-2)$, proving the codimension claim. □

The vector-valued map itself satisfies

$$\Delta P = 0, \qquad DP\,DP^* = E_{d-1}I_\delta, \qquad DP(\nabla E_{d-1}) = 4E_{d-2}P. \tag{13.7}$$

Indeed, the first two identities were established blockwise in the proof, while multilinearity gives $DP(\nabla r_a) = 2P$, whence the third. Thus the target of the pullback defining our hypersurface is the hyperplane $\{\operatorname{Re} y = 0\} \subset \mathbb{K}$. The umbilic condition in (13.7) places its regular pullback within Riedler's umbilic framework [69, Theorem 5.1.4]. The direct identity (13.5) additionally covers regular zeros lying over $P = 0$, for instance points with exactly one zero block.

Proposition 13.2 (Exact algebra and quotient image)**.** *For $d \geq 3$,*

$$\mathcal{A}_R(F) = \mathbb{R}[E_1, \dots, E_d, F], \tag{13.8}$$

and its $d+1$ displayed generators are algebraically independent. Their joint image is exactly

$$\left\{(e_1(r), \dots, e_d(r), f) : r \in [0,\infty)^d, \quad f^2 \leq e_d(r)\right\}. \tag{13.9}$$

The spherical quotient dimension is d. For $d = 2$, the smaller algebra $\mathcal{A}_R(F) = \mathbb{R}[R, F]$ is already closed and its spherical quotient dimension is one.

Proof. Apply Proposition 12.1 with $r_a = |x_a|^2$ and $b_k = 2\delta(d-k+1) > 0$ from Proposition 13.1.

The norm identity implies $F^2 \leq E_d$. Conversely, choose an imaginary unit $u \in \mathbb{K}$, and put

$$x_1 = \sqrt{r_1}(\cos\theta + u\sin\theta), \qquad x_a = \sqrt{r_a} \quad (a \geq 2).$$

All bracketings then give $F = \sqrt{e_d(r)}\cos\theta$. This realizes every value in (13.9), including its boundary. At distinct positive r_a, the Jacobian of $(e_1, \dots, e_d)$ is a nonzero Vandermonde determinant. Taking $\sin\theta \neq 0$ adds a phase direction changing F while fixing all E_k. The joint map thus has rank $d+1$ on a nonempty open set, which proves algebraic independence. Fixing $E_1 = 1$ lowers the rank by one. In the quadratic case, R, F have independent differentials at generic points and give the asserted one-dimensional spherical quotient. □

In particular, different octonionic parenthesizations have the same abstract differential closure and the same quotient image. These statements do not identify the corresponding embedded hypersurfaces up to orthogonal congruence.

Theorem 13.3 (Octonionic product cones in every degree)**.** *For every $d \geq 2$ and every fixed parenthesization, let $F = \operatorname{Re}(x_1 \cdots x_d)$ on $\mathbb{O}^d$. Then the cone $C_d = \{F = 0\} \subset \mathbb{R}^{8d}$ is the support of the locally mass-minimizing boundary $\partial[\![\{F < 0\}]\!]$. A global two-sided subcalibration off the set Z_d in (13.6) is*

$$G = F\sqrt{E_{d-1}} = F|\nabla F|, \qquad X = \frac{\nabla G}{|\nabla G|}. \tag{13.10}$$

Here Z_d has real codimension sixteen; it contains nonzero points whenever $d \geq 3$. The cone is strictly area minimizing and strictly stable, with

$$Q_{C_d}(u) \geq \frac{(4d+1)^2}{4} \int_{C_d} r^{-2} u^2, \qquad u \in C_c^\infty((C_d)_{\mathrm{reg}} \setminus \{0\}). \tag{13.11}$$

Proof. By Proposition 13.1, $\Delta E_{d-1} = 32 E_{d-2}$ and Z_d has codimension sixteen. Since $32 > 24 - 16/d$, Theorem 12.3, with $N = 8d$, gives all assertions, including (13.11). □

Remark 13.4 (The quaternionic closure). Quaternionic products have the same algebra and minimality identities, but $\Delta S = 16B$. The potential $F\sqrt{S}$ works for $d = 2$ and fails for $d \geq 3$: at equal positive roots its boundary coefficient is $(-4 + 8/d)B < 0$, and the phase construction above realizes arbitrarily small nonzero F with those roots fixed.

13.2. **Complex Pfaffians and their minor norms.** Let W_d be the space of complex skew-symmetric $2d$-by-$2d$ matrices, with the Euclidean metric

$$|Z|^2 = \sum_{i<j} |z_{ij}|^2 = \tfrac{1}{2} \operatorname{tr}(ZZ^*), \qquad \dim_{\mathbb{R}} W_d = 2d(2d-1). \tag{13.12}$$

The independent coordinates are the complex entries z_{ij}, $i < j$. For $d \geq 2$, set

$$f(Z) = \operatorname{Pf}(Z), \qquad F = \operatorname{Re} f, \qquad E_k(Z) = \sum_{\substack{I \subset \{1,\dots,2d\} \\ |I| = 2k}} |\operatorname{Pf}(Z_I)|^2. \tag{13.13}$$

Here Z_I is the principal submatrix on I, and $\operatorname{Pf}(Z_\varnothing) = 1$. In particular, $E_1 = R = |Z|^2$ and $E_d = |f|^2$.

Proposition 13.5 (Pfaffian differential closure)**.** *For F, E_k defined by (13.13), F is real irreducible and*

$$\begin{aligned} \Delta F &= 0, \qquad \Gamma(F,F) = E_{d-1}, \\ \Gamma(F, E_k) &= 2(d-k+1) E_{k-1} F, \end{aligned} \tag{13.14}$$

$$\Delta E_k = 4(d-k+1)(2d-2k+1) E_{k-1}. \tag{13.15}$$

At every point there exist nonnegative numbers $r_1, \dots, r_d$ such that $E_k = e_k(r)$, and

$$\Gamma(E_i, E_j) = 4 \sum_{a=1}^{d} r_a \frac{\partial e_i}{\partial r_a} \frac{\partial e_j}{\partial r_a}. \tag{13.16}$$

In particular, $\mathcal{L}(F) = -2E_{d-2}F$. For $d \geq 3$,

$$\mathcal{A}_R(F) = \mathbb{R}[E_1, \dots, E_d, F], \tag{13.17}$$

with algebraically independent generators and spherical quotient dimension d. For $d = 2$, $\mathcal{A}_R(F) = \mathbb{R}[R, F]$.

Proof. Complexification gives $2F = \operatorname{Pf}(Z) + \operatorname{Pf}(W)$ for independent skew-symmetric matrices Z, W. It is linear in z_{12}, with coefficient $\operatorname{Pf}(Z_{\{3,\dots,2d\}})$. That coefficient is coprime to the constant term, since the latter contains $w_{12}w_{34}\cdots w_{2d-1,2d}$ with coefficient one. Gauss' lemma proves irreducibility [44, Chapter VIII, Section 5, pp. 395–398].

Write $\omega = \sum_{i<j} z_{ij} e_i \wedge e_j$ in an orthonormal basis of $\mathbb{C}^{2d}$. The coefficient of e_I in $\omega^k/k!$ is $\operatorname{Pf}(Z_I)$. Consequently

$$E_k = \left| \frac{\omega^k}{k!} \right|^2, \tag{13.18}$$

which proves that all E_k are invariant under unitary congruence $Z \mapsto UZU^T$. The skew-symmetric unitary normal form is

$$Z = U \operatorname{diag}(\sigma_1 J, \dots, \sigma_d J) U^T, \qquad \sigma_a \geq 0, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a direct proof, let $T(v) = Z\bar{v}$. Skew-symmetry gives $T^2 = -ZZ^*$ and $\langle v, Tw\rangle = -\langle w, Tv\rangle$, where the Hermitian inner product is conjugate linear in its first argument. Choose a unit eigenvector v of ZZ^* with positive eigenvalue σ^2, and put $w = -Tv/\sigma$. These identities imply that v, w are orthonormal, $Tv = -\sigma w$, and $Tw = \sigma v$. Their orthogonal complement is T-invariant. Induction gives the displayed normal form, with zero blocks at the last step. At this normal form only principal minors which are unions of complete two-by-two blocks can have nonzero Pfaffian. Thus $E_k = e_k(\sigma_1^2, \dots, \sigma_d^2)$.

To compute first derivatives, more generally use the block diagonal matrix with complex entries $z_a J$. A nonzero principal Pfaffian is a product of its block entries; its derivative in any off-block coordinate vanishes. A principal Pfaffian which vanishes contributes zero to the first derivative of its squared modulus. Therefore the gradient of E_k has only block components at this matrix, and

$$\frac{\partial E_k}{\partial z_a} = \bar{z}_a \, e_{k-1}(r_1, \dots, \widehat{r}_a, \dots, r_d), \qquad r_a = |z_a|^2.$$

The real Euclidean inner product of these gradients gives (13.16). Unitary congruence preserves the metric (13.12), so this calculation holds everywhere.

The polynomial f is holomorphic, hence $\Delta F = 0$. Differentiating its Pfaffian expansion gives

$$\frac{\partial f}{\partial z_{ij}} = \pm \operatorname{Pf}(Z_{\{1,\dots,2d\}\setminus\{i,j\}}).$$

For a holomorphic polynomial on complex Euclidean space,

$$|\nabla \operatorname{Re} f|^2 = \sum_{i<j} |\partial f/\partial z_{ij}|^2.$$

This proves $\Gamma(F, F) = E_{d-1}$. At the block diagonal matrix, complex bilinear extension of Γ gives

$$\Gamma(f, E_k) = 2 \sum_a \frac{\partial f}{\partial z_a} \frac{\partial E_k}{\partial \bar{z}_a} = 2(d-k+1) E_{k-1} f.$$

Unitary congruence preserves every E_k, and $f(UZU^T) = \det(U) f(Z)$. Hence the identity is covariant under this action. Taking real parts proves the remaining identity in (13.14); polynomial continuity includes all degenerate normal forms.

For the Laplacian, apply $\Delta |p|^2 = 4 \sum_{i<j} |\partial p/\partial z_{ij}|^2$ to each holomorphic principal Pfaffian in (13.13). Each derivative deletes a pair of indices. A fixed set of $2k-2$ indices is obtained in exactly $\binom{2d-2k+2}{2} = (d-k+1)(2d-2k+1)$ ways, proving (13.15). The minimality identity follows from $-\frac{1}{2}\Gamma(F, E_{d-1}) = -2E_{d-2}F$.

Now apply Proposition 12.1 with $r_a = \sigma_a^2$ and $b_k = 4(d-k+1)(2d-2k+1) > 0$. To verify independence, take distinct positive r_a, with $z_1 = \sqrt{r_1} e^{i\theta}$ and $z_a = \sqrt{r_a}$ for $a \geq 2$. Then $E_k = e_k(r)$ and $F = \sqrt{e_d(r)} \cos\theta$. The Vandermonde and phase argument in Proposition 13.2 applies without change. It also proves that the joint image is exactly (13.9). □

Theorem 13.6 (Area minimization for complex Pfaffians)**.** *For every $d \geq 2$, the cone*

$$C_d = \{\operatorname{Re} \operatorname{Pf}(Z) = 0\} \subset W_d \cong \mathbb{R}^{2d(2d-1)}$$

is the support of the locally mass-minimizing boundary of $\{\operatorname{Re} \operatorname{Pf}(Z) < 0\}$. A two-sided subcalibration is the normalized gradient of

$$G = F\sqrt{E_{d-1}} = F|\nabla F|, \tag{13.19}$$

defined off

$$Z_d = \{Z : \operatorname{rank}_{\mathbb{C}} Z \leq 2d-4\} = \{\nabla F = 0\}. \tag{13.20}$$

This exceptional set has real codimension twelve. The cone is strictly area minimizing and strictly stable, with

$$Q_{C_d}(u) \geq \frac{(4d^2-6d+1)^2}{4}\int_{C_d} r^{-2}u^2, \qquad u \in C_c^\infty((C_d)_{\mathrm{reg}} \setminus \{0\}). \tag{13.21}$$

Proof. By Proposition 13.5, the spectral table holds with $S = E_{d-1}$, $B = E_{d-2}$, $\Delta S = 24B$, and $24 \geq 24 - 16/d$.

The normal form shows that $S = 0$ precisely when at most $d-2$ of the σ_a are nonzero, proving (13.20). The stratum of complex skew-symmetric n-by-n matrices of rank $2q$ has complex dimension $2q(n-2q) + \binom{2q}{2}$. Indeed, choose its kernel, of dimension $n-2q$, and then choose a nondegenerate alternating form on the $2q$-dimensional quotient. Its complex codimension is therefore $\binom{n-2q}{2}$. For $n = 2d$ and $q = d-2$, this is six; lower rank strata have larger codimension. Hence Z_d is a closed algebraic set of real codimension twelve and has locally zero $\mathcal{H}^{2d(2d-1)-1}$-measure.

Theorem 12.3, with $N = 2d(2d-1)$, gives the minimizing assertions and (13.21). Here $4d^2 - 6d + 1 > 0$ for $d \geq 2$. □

14. Real Parts of Complex Determinants in All Orders

The determinant has the gradient table of Section 12, but $\Delta E_{m-1} = 16E_{m-2}$ excludes $\sqrt{S}$ in orders at least three; see (12.23). We prove Theorem I using the product of proper minor norms and two coefficient inequalities uniform in matrix order. We use the Frobenius Euclidean metric $\langle U, V\rangle = \operatorname{Re}\operatorname{tr}(UV^*)$ on $M_m(\mathbb{C})$, so that $|U|^2 = \sum_{i,j}|U_{ij}|^2$, and write

$$f(Z) = \det_{\mathbb{C}} Z, \qquad F = \operatorname{Re} f, \qquad C_m = \{F = 0\}, \qquad E_m^- = \{F < 0\}.$$

For $0 \leq k \leq m$, let

$$E_0 = 1, \qquad E_k(Z) = \sum_{|I|=|J|=k} |\det Z_{I,J}|^2. \tag{14.1}$$

Let $x_1, \dots, x_m \geq 0$ be the squared singular values of Z, and retain the elementary symmetric polynomials e_k from Section 12. Cauchy–Binet gives

$$E_k(Z) = \sum_{|I|=k} \det(ZZ^*)_{I,I}, \qquad \sum_{k=0}^{m} E_k(Z)t^k = \det(\mathrm{Id} + tZZ^*) = \prod_{i=1}^{m}(1+tx_i).$$

Comparing coefficients yields $E_k = e_k(x_1, \dots, x_m)$; in particular, $E_1 = |Z|^2$ and $E_m = |f|^2$. The minor invariants $E_1, \dots, E_{m-1}$ are positive at rank at least $m-1$. Their product vanishes precisely at rank at most $m-2$, the singular set of the cone, and is positive along its regular part. We take the product candidate

$$a = \left(\prod_{k=1}^{m-1} E_k\right)^{1/2}, \qquad G = aF. \tag{14.2}$$

Logarithmic derivatives add over the minor orders. Lemma 14.5 reduces the defect sign to a boundary coefficient controlled by Newton inequalities and a second coefficient controlled by a covariance decomposition. The homogeneous degree of G is $m(m+1)/2$. The case $m = 1$ is a hyperplane; the coefficient calculations assume $m \geq 2$.

For $m = 2, 3$, the potential is

$$G_2 = F_2 E_1^{1/2}, \qquad G_3 = F_3(E_1E_2)^{1/2}.$$

The first multiplier is the radius $E_1^{1/2} = |Z|$; the second also includes the norm $E_2^{1/2}$ of the second-order minors and agrees with the $d = 2$ Jordan potential in Theorem 10.14. The low-order geometry and determinant phase are discussed in Remark 14.6; the coefficient signs are proved in Sections 14.3 and 14.4.

14.1. **Invariant calculus and the multiplier identity.**

Lemma 14.1. *For $1 \le j, k \le m$, one has*

$$\Delta F = 0, \qquad \Gamma(F, F) = E_{m-1}, \tag{14.3}$$

$$\Gamma(F, E_k) = 2(m-k+1)E_{k-1}F, \qquad \Delta E_k = 4(m-k+1)^2 E_{k-1}, \tag{14.4}$$

$$\Gamma(E_j, E_k) = 4\sum_{i=1}^{m} x_i \frac{\partial e_j}{\partial x_i}\frac{\partial e_k}{\partial x_i}. \tag{14.5}$$

Proof. The real part of a holomorphic polynomial is harmonic. The complex first derivatives of f are its cofactors, so their squared norms sum to E_{m-1}. This proves (14.3).

For the mixed identity, use complex matrix coordinates z_α and observe that

$$\Gamma(F, E_k) = 2\,\mathrm{Re}\sum_{\alpha} \frac{\partial f}{\partial z_\alpha}\frac{\partial E_k}{\partial \overline{z}_\alpha}.$$

The complex expression in this sum transforms by the same determinant phase as f under the left and right unitary actions. Thus singular value decomposition reduces the identity to a diagonal matrix, with its determinant phase retained. At such a matrix,

$$\sum_{\alpha} \frac{\partial f}{\partial z_\alpha}\frac{\partial E_k}{\partial \overline{z}_\alpha} = f\sum_{i} e_{k-1}(x_1, \dots, \widehat{x}_i, \dots, x_m) = (m-k+1)E_{k-1}f.$$

The polynomial identity extends to singular matrices by continuity.

For a holomorphic polynomial q, $\Delta|q|^2 = 4\sum_\alpha |\partial q/\partial z_\alpha|^2$. Apply this to the minors in (14.1). Each fixed $(k-1)$-minor occurs $(m-k+1)^2$ times, which gives the Laplacian formula.

The same unitary invariance reduces (14.5) to $D = \mathrm{diag}(\sigma_1, \dots, \sigma_m)$, where $x_i = \sigma_i^2$. At D, nonprincipal minors vanish, and

$$\mathrm{d}(\det Z_{I,I})_D[H] = \sum_{i\in I}\left(\prod_{\ell\in I\setminus\{i\}} \sigma_\ell\right) H_{ii}.$$

Differentiating their squared moduli and summing gives

$$\mathrm{d}E_k(D)[H] = 2\sum_i \sigma_i \frac{\partial e_k}{\partial x_i}\,\mathrm{Re}\,H_{ii}.$$

Thus all other first derivatives vanish and

$$\nabla E_k(D) = \mathrm{diag}\left(2\sigma_i\frac{\partial e_k}{\partial x_i}\right)_{i=1}^{m}.$$

Pairing these gradients proves (14.5), including repeated or zero singular values. □

Apply Proposition 12.1 with $d = m$, roots x_i, and $b_k = 4(m-k+1)^2 > 0$ from Lemma 14.1. The algebra is Laplacian and, for $m \ge 3$,

$$\mathcal{A}_R(F) = \mathbb{R}[E_1, \dots, E_m, F]. \tag{14.6}$$

For $m = 2$, the same proposition gives $\mathcal{A}_R(F) = \mathbb{R}[R, F]$.

The joint image of $(E_1, \dots, E_m, F)$ is exactly

$$\left\{(e_1(x), \dots, e_m(x), t) : x \in [0,\infty)^m, \quad t \in \mathbb{R}, \quad t^2 \le e_m(x)\right\}. \tag{14.7}$$

Indeed, $F^2 \le |\det Z|^2 = E_m$, and the diagonal matrices

$$Z = \mathrm{diag}\left(\sqrt{x_1}e^{i\vartheta}, \sqrt{x_2}, \dots, \sqrt{x_m}\right)$$

have $E_k = e_k(x)$ and $F = \sqrt{e_m(x)}\cos\vartheta$. The phase and Vandermonde argument of Proposition 13.2 therefore gives the full image, including zero spectra, and generic rank $m+1$. Thus for $m \ge 3$ the generators are independent and the spherical quotient dimension is m; for $m = 2$, the algebra $\mathbb{R}[R, F]$ has spherical quotient dimension one.

The identities (14.3)–(14.4) also give

$$\mathcal{L}F = -\tfrac{1}{2}\Gamma(F, E_{m-1}) = -2E_{m-2}F. \tag{14.8}$$

Thus the regular part of C_m is minimal, as in [41, Corollary 5.2 and Example 5.4].

Example 14.2 (Generator degrees for harmonic determinant cones)**.** For $m \geq 3$, the polynomial $F = \operatorname{Re}\det Z$ is an irreducible homogeneous harmonic equation of degree m on $\mathbb{R}^{2m^2}$, with nonempty regular minimal zero set. Its Laplacian algebra has $m+1$ algebraically independent generators, and the least possible maximal degree of homogeneous generators is $2m$.

Harmonicity and minimality follow from (14.3) and (14.8). For irreducibility, complexify the real coordinates to independent matrices Z, W, so that $2F = \det Z + \det W$. As a polynomial in z_{11}, this is $Dz_{11} + C$, where D is the corresponding cofactor. Any factor of D is independent of W, whereas C has coefficient 1 on $w_{11}\cdots w_{mm}$. Thus D and C are coprime, so $Dz_{11} + C$ is primitive and linear over the fraction field. Gauss' lemma proves irreducibility [44, Chapter VIII, Section 5, pp. 395–398]. The zero $Z = \operatorname{diag}(i, 1, \dots, 1)$ is regular by (14.3).

By (14.6),

$$\mathcal{A}_R(F) = \mathbb{R}[E_1, \dots, E_m, F], \qquad \deg E_k = 2k.$$

Algebraic independence gives the lower bound $m+1$ for the number of generators. Every homogeneous element of degree less than $2m$ belongs to the proper subalgebra $\mathbb{R}[E_1, \dots, E_{m-1}, F]$, and such elements cannot generate E_m. Conversely, the displayed generators have maximal degree $2m$. Thus the least possible maximal degree is exactly $2m$, and the uniform bound in Proposition 3.8 must satisfy $B(2m^2, m) \geq 2m$.

The singular set has real codimension eight and is given by

$$\mathcal{S}_m = \{Z : \operatorname{rank}_{\mathbb{C}} Z \leq m-2\} = \{\nabla F = 0\}. \tag{14.9}$$

The cofactor identity gives the second equality. To see that every point of this set is geometrically singular, take a matrix of rank $q \leq m-2$. Schur-complement coordinates, absorbing a nonvanishing determinant factor into one row, identify its zero set locally with a smooth factor times $\{\operatorname{Re}\det W = 0\}$ at $W = 0$, where $W \in M_{m-q}(\mathbb{C})$. Since $m - q \geq 2$, this transverse cone contains every complex rank-one line. Their real tangent directions span the whole matrix space, so the zero set cannot be a smooth hypersurface there. The rank-q stratum has real dimension $2q(2m-q)$, so the largest stratum in $\mathcal{S}_m$ has dimension $2m^2 - 8$. The multiplier in (14.2) is smooth and strictly positive on $M_m(\mathbb{C}) \setminus \mathcal{S}_m$: at every matrix of rank at least $m-1$, all $E_1, \dots, E_{m-1}$ are positive. This includes the regular zeros of F, even when the matrix is not invertible. On $\mathcal{S}_m$ the displayed product gives $a = 0$ and $G = 0$. For $m \geq 3$, this set contains nonzero points, where Proposition 4.24 excludes multipliers extending as strictly positive C^2 functions. For $m = 2$, $\mathcal{S}_2 = \{0\}$.

Proposition 14.3 (Tangent cones along the determinant rank strata)**.** *Let $p \in C_m$ have complex rank $q \leq m-2$, and put $r = m - q$. The rescalings of C_m about p have the unique tangent cone*

$$\mathbb{R}^{2(m^2-r^2)} \times C_r, \qquad C_r = \{\operatorname{Re}\det W = 0 : W \in M_r(\mathbb{C})\}, \tag{14.10}$$

up to an orthogonal transformation. The corresponding oriented boundary currents and their associated varifolds also converge, with multiplicity one. The first nonzero Taylor term of F_m at p has degree r.

Proof. The assertion at $p = 0$ is homogeneity. Otherwise, unitary left and right changes of coordinates put p in the form $\operatorname{diag}(A_0, 0)$, where A_0 is a positive diagonal q-by-q matrix. These changes are Euclidean isometries; they replace F_m by $\operatorname{Re}(\chi \det Z)$ for some $|\chi| = 1$. Write nearby matrices as

$$Z = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad V = D - CA^{-1}B.$$

Block elimination, with determinant one, gives $\det Z = (\det A)\det V$. Put

$$\zeta(A) = \frac{\chi \det A}{|\det A|}, \qquad U(A) = \operatorname{diag}(\zeta(A), 1, \dots, 1), \qquad W = U(A)V.$$

Then

$$\operatorname{Re}(\chi \det Z) = |\det A| \operatorname{Re}\det W.$$

Thus the smooth local coordinate map $\Psi(Z) = (A - A_0, B, C, W)$ identifies both the zero set and its negative phase with a product of a smooth coordinate space and the corresponding determinant zero set or phase. At p, the differential of the Schur complement is $dV = dD$, and the derivative of $U(A)$ contributes nothing because $V(p) = 0$. Consequently $D\Psi_p$ is the identity on the free blocks and the unitary row transformation $D \mapsto U(A_0)D$ on the last block. In particular, it is an isometry.

For $s \downarrow 0$, the maps $s^{-1}\Psi(p + sH)$ converge in C^1 on compact sets to $D\Psi_p$. Since the target zero set and phase are products of cones, this proves local Hausdorff and local L^1 convergence of their inverse images. The same local diffeomorphisms push forward the multiplicity-one boundary current. Their C^1 convergence and the area formula give local current and mass convergence, hence convergence of the associated varifolds. This also proves uniqueness. Finally, the determinant identity gives the leading term

$$\operatorname{Re}\big(\chi \det(A_0) \det H_{22}\big),$$

which has degree r and the zero set in (14.10). □

Example 14.4 (Cubic transverse tangents at nonvertex points)**.** For $F_4 = \operatorname{Re}\det Z$ on $M_4(\mathbb{C})$, take $p = \operatorname{diag}(1, 0, 0, 0)$. Then

$$\operatorname{Tan}_p C_4 = \mathbb{R}^{14} \times \{\operatorname{Re}\det W = 0 : W \in M_3(\mathbb{C})\}.$$

The transverse equation is cubic. In contrast, the rank-two stratum has quadratic transverse tangents. Both the original cone and the cubic factor are area minimizing by Theorem I. Their regular minimality is the determinant construction of [41, Corollary 5.2 and Example 5.4].

The normed-product family gives an equally direct example. Let $P_d(x_1, \dots, x_d)$ denote the left-associated octonionic product, and let $\mathcal{O}_d = \{\operatorname{Re} P_d = 0\}$. At $p = (1, 0, 0, 0) \in \mathcal{O}_4$, the coordinate change

$$(a, b, c, d) \longmapsto (a, (1 + a)b, c, d)$$

is a local diffeomorphism whose derivative at zero is the identity. It transforms $\operatorname{Re} P_4(1 + a, b, c, d)$ into $\operatorname{Re}((b'c)d)$. Hence the unique set and boundary-current tangent is $\mathbb{R}^8 \times \mathcal{O}_3$. The same argument gives $\mathbb{R}^8 \times \mathcal{O}_r$ at $(1, 0, \dots, 0) \in \mathcal{O}_{r+1}$ for every $r \geq 2$. These cones are irreducible and area minimizing by Proposition 13.1 and Theorem 13.3.

These are not cylindrical extensions of their defining equations. For determinants, evaluation of directional derivatives on rank-$(m-1)$ matrices detects every matrix entry of the direction. For normed products, the directional derivative terms have distinct block multidegrees, and $\operatorname{Re}(uv)$ is a nondegenerate bilinear form. In both cases $D_v F \equiv 0$ forces $v = 0$. The Euclidean factor appears in the tangent at the singular point; the radial line splitting also follows from Section 2.6.

For comparison, if a homogeneous cubic F is defined on its essential space, every nonzero critical point p has $\operatorname{Hess} F(p) \neq 0$, since $D_p F(x) = \frac{1}{2}\operatorname{Hess} F(p)[x, x]$ and essentiality excludes a nonzero direction with $D_p F \equiv 0$. Thus its first nonzero Taylor term is quadratic; for the harmonic minimal case, see Proposition 9.6 and Section 2.6. The higher-order transverse cones above occur in defining degree four and higher.

We now specialize Theorem 4.2 to a positive smooth invariant multiplier $a = a(E_1, \dots, E_m)$ on the full-rank locus. By (14.4), the chain rule gives

$$\Gamma(F, a) = F \sum_{k=1}^{m} 2(m - k + 1) E_{k-1} \frac{\partial a}{\partial E_k}.$$

Thus division by F below gives smooth invariant functions. Put $Q = E_{m-1}$, $P = a^{-2}|\nabla G|^2$, and

$$c = \frac{\Gamma(F,a)}{aF}, \qquad t = \frac{\Gamma(a,a)}{a^2}, \qquad K = 2c + t, \qquad \delta = \frac{\Delta a}{a} + c - t. \tag{14.11}$$

For an invariant coefficient $q = q(E_1, \dots, E_m)$, write

$$\mathcal{T}q = \frac{\Gamma(F,q)}{F} = \sum_{k=1}^{m} 2(m-k+1)E_{k-1}\frac{\partial q}{\partial E_k}. \tag{14.12}$$

Lemma 14.5 (The multiplier identity)**.** *With the notation just introduced, one has*

$$P = Q + F^2 K, \tag{14.13}$$

$$\mathcal{L}(G) = a^3 F(D_0 + F^2 D_1), \tag{14.14}$$

$$D_0 = Q\delta - 2E_{m-2} - QK - \frac{\Gamma(a,Q)}{2a}, \tag{14.15}$$

$$D_1 = K\delta - \tfrac{1}{2}\left(\mathcal{T}K + 2cK + \frac{\Gamma(a,K)}{a}\right). \tag{14.16}$$

Proof. Apply Lemma 4.11 with $h = \log a$. Its notation becomes

$$S = Q, \quad q = 2E_{m-2}, \quad \tau = c, \quad H_h = t,$$
$$Q_h = K, \quad \Delta h = \delta - c, \quad \kappa = \mathcal{T}K.$$

The last identity follows from (14.12); all these coefficients are smooth across $F = 0$. Substituting in (4.21) and (4.24), with $\Gamma(h,Q) = \Gamma(a,Q)/a$ and $\Gamma(h,K) = \Gamma(a,K)/a$, gives the stated formulas. □

In the notation of Theorem 4.2, the lemma states

$$\mathcal{K}_F(a) = a^3(D_0 + F^2 D_1), \qquad G\mathcal{L}(G) = aF^2\mathcal{K}_F(a) = a^4F^2(D_0 + F^2 D_1).$$

The coefficient D_0 determines the boundary Jacobi sign, while $D_1 > 0$ supplies strict divergence off the cone once $D_0 \geq 0$. We now take the uniform product multiplier

$$a = \prod_{k=1}^{m-1} E_k^{\lambda}. \tag{14.17}$$

The value $\lambda = 1/2$ gives (14.2).

14.2. **From the coefficient signs to area minimization.** We first deduce the geometric conclusions from the coefficient signs proved in Sections 14.3 and 14.4.

Proof of Theorem I. The case $m = 1$ is an oriented hyperplane.

The sign of $G\mathcal{L}(G)$. For $m \geq 2$, take $\lambda = 1/2$ in (14.17). Then $G = aF$ is the potential in (14.2), and Lemmas 14.7 and 14.11 and Theorem 14.8 give $D_0, D_1 \geq 0$. Therefore (14.14) yields

$$G\mathcal{L}(G) = a^4F^2(D_0 + F^2 D_1) \geq 0 \qquad \text{on the full-rank locus.} \tag{14.18}$$

Extension and comparison. Off $\mathcal{S}_m$, the proper minor norms are positive, so a, G are smooth and (14.18) extends by continuity to rank $m-1$. Positive homogeneity gives $\nabla G \neq 0$ there by the verification following Corollary 2.8. The continuous potential has the phases of F, and the exceptional set has real codimension eight by (14.9). Thus Corollary 2.8 and Theorem 2.5 give perimeter and mass comparison; harmonicity gives full support by Remark 2.7.

Strict minimality. For $\lambda = 1/2$, Lemma 14.11 gives $D_1 > 0$. Hence $q_{X_G} = \operatorname{sgn}(G)\operatorname{div} X_G > 0$ at full-rank points with $F \neq 0$. Each phase contains such points, giving positive flux integrals. Homogeneity and Theorem 2.11 prove strict area minimality.

Stability. Off $\mathcal{S}_m$, all of $E_1, \dots, E_{m-1}$ are positive, so the invariant expression (14.15) is smooth. Thus $D_0 \geq 0$ extends from full-rank matrices to rank $m-1$ by continuity. On the regular cone, the Jacobi potential in (4.11) is consequently

$$V_G = \frac{\mathcal{K}_F(a)}{a^3|\nabla F|^2} = \frac{D_0}{E_{m-1}} \geq 0.$$

Apply Theorem 4.6 with $n = 2m^2 - 1$ and $\ell = m(m+1)/2$. This gives (1.6):

$$Q_{C_m}(u) \geq \Big(\ell - \frac{n}{2}\Big)^2 \int_{C_m} r^{-2}u^2 = \frac{(m^2-m-1)^2}{4} \int_{C_m} r^{-2}u^2. \qquad \square$$

Remark 14.6 (Low orders and determinant phase). For $m = 2$, the cone C_2 is orthogonally congruent to the classical Simons cone in $\mathbb{R}^4 \oplus \mathbb{R}^4$. Indeed, for $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_3 & z_4 \end{smallmatrix}\right)$, the unitary change of coordinates

$$w = \frac{1}{\sqrt{2}}\big(z_1 + z_4,\ i(z_1 - z_4),\ i(z_2 + z_3),\ z_2 - z_3\big)$$

gives $\sum_{j=1}^4 w_j^2 = 2 \det Z$. Writing $w = u + iv$, with $u, v \in \mathbb{R}^4$, we obtain $2F_2 = |u|^2 - |v|^2$. Thus this order recovers the standard quadratic cone.

For every real θ, the isometry $Z \mapsto e^{-i\theta/m}Z$ preserves all the E_k and sends F to $\mathrm{Re}(e^{-i\theta} \det_{\mathbb{C}} Z)$. Consequently the phase cone $\{\mathrm{Re}(e^{-i\theta} \det_{\mathbb{C}} Z) = 0\}$ has the corresponding subcalibration and is locally area minimizing for every $m \geq 1$, with the same strict minimality and stability conclusions when $m \geq 2$. In particular, the real and imaginary determinant zero cones are orthogonally congruent.

14.3. **Nonnegativity of the first coefficient.** We prove $D_0 \geq 0$ for the choice $\lambda = 1/2$ in (14.17). On the full-rank locus, set $y_i = x_i^{-1}$. For the remainder of this section, an unadorned e_s denotes the elementary symmetric polynomial in $y_1, \dots, y_m$. For $1 \leq s \leq m-1$, define

$$a_{si} = \frac{e_s(y_1, \dots, \widehat{y_i}, \dots, y_m)}{e_s(y)}, \qquad P_i = \sum_{s=1}^{m-1} a_{si}, \tag{14.19}$$

and introduce four scalar sums for the coefficient calculation:

$$A = \sum_i y_i \sum_{s=1}^{m-1} (s a_{si} - a_{si}^2), \qquad B = \sum_i y_i P_i, \qquad V = \sum_i y_i P_i^2, \qquad Y = \sum_i y_i^2 P_i. \tag{14.20}$$

Lemma 14.7. *For the weight* (14.17),

$$\frac{D_0}{Q} = 4\lambda A - 4\lambda^2 V - 2\frac{e_2}{e_1} + 2\lambda \frac{Y}{e_1}. \tag{14.21}$$

Consequently, at $\lambda = 1/2$, *the inequality* $D_0 \geq 0$ *is equivalent to*

$$2A + \frac{Y}{e_1} - V - 2\frac{e_2}{e_1} \geq 0. \tag{14.22}$$

Proof. We first explain the reciprocal substitution. Put $p = \prod_i x_i$. For a k-element index set I, $\prod_{i \in I} x_i = p \prod_{j \notin I} y_j$. Summing over I gives

$$E_k(x) = p\, e_{m-k}(y), \qquad \frac{E_{k-1}(x)}{E_k(x)} = \frac{e_{m-k+1}(y)}{e_{m-k}(y)}.$$

Write $y_{\widehat{i}}$ for the tuple with y_i omitted. Since $\partial_{y_i} e_{s+1}(y) = e_s(y_{\widehat{i}})$, Euler's identity for the homogeneous polynomial e_{s+1} gives

$$\sum_i y_i e_s(y_{\widehat{i}}) = (s+1) e_{s+1}(y), \qquad \sum_i y_i a_{si} = (s+1)\frac{e_{s+1}}{e_s}.$$

For $s = m-k$, applying the complement identity in the $m-1$ variables with x_i omitted also gives

$$e_{k-1}(x_{\widehat{i}}) = p\, y_i e_s(y_{\widehat{i}}), \qquad \frac{e_{k-1}(x_{\widehat{i}})}{E_k(x)} = y_i a_{si}.$$

Consequently, the gradient formula (14.5) yields

$$\frac{\Gamma(E_k,E_k)}{E_k^2} = 4\sum_i x_i \left(\frac{e_{k-1}(x_{\widehat{i}})}{E_k(x)}\right)^2 = 4\sum_i y_i a_{si}^2,$$

where $x_i y_i^2 = y_i$. Put $h = \log a = \lambda \sum_{k=1}^{m-1} \log E_k$. The chain rule in (4.4) gives

$$\frac{\nabla a}{a} = \lambda \sum_{k=1}^{m-1} \frac{\nabla E_k}{E_k}, \qquad \frac{\Delta a}{a} - t = \lambda \sum_{k=1}^{m-1} \left(\frac{\Delta E_k}{E_k} - \frac{\Gamma(E_k,E_k)}{E_k^2}\right).$$

Using Lemma 14.1 and the identities just proved, we obtain the quantities defined in (14.11):

$$\begin{aligned} c = 2\lambda B, \qquad t = 4\lambda^2 V, \qquad \frac{\Delta a}{a} - t = 4\lambda(A+B), \\ \delta = 4\lambda A + 6\lambda B, \qquad K = 4\lambda B + 4\lambda^2 V. \end{aligned} \tag{14.23}$$

Finally,

$$\frac{E_{m-2}}{Q} = \frac{e_2}{e_1}, \qquad \frac{\Gamma(a,Q)}{aQ} = 4\lambda\left(B - \frac{Y}{e_1}\right).$$

Insert these identities in (14.15). The three terms involving B cancel, leaving (14.21). □

Theorem 14.8 (First-coefficient inequality)**.** *Let $m \geq 2$ and let $y_1, \dots, y_m$ be positive. Then the quantities in (14.19)–(14.20) satisfy (14.22).*

We prove the theorem by a recurrence for symmetric polynomials. Put

$$D_m^{\mathrm{sym}} = \prod_{s=1}^{m-1} e_s, \qquad \mathcal{C}_m = e_1 (D_m^{\mathrm{sym}})^2 \left(2A + \frac{Y}{e_1} - V - 2\frac{e_2}{e_1}\right). \tag{14.24}$$

This is a symmetric polynomial, viewed as a polynomial in $e_1, \dots, e_m$. In the first two orders,

$$\mathcal{C}_2 = 0, \qquad \mathcal{C}_3 = 6 e_1 e_2 e_3 (e_1^2 - 2e_2).$$

Thus the first coefficient vanishes identically for $m = 2$; for $m = 3$, the normalization $e_1(y) = 1$ gives

$$2A + Y - V - 2e_2 = 6e_3 \frac{\sum_i y_i^2}{e_2} > 0.$$

For $N \geq m$, let $\mathcal{C}_m^{[N]}$ denote this same formal polynomial evaluated at the elementary symmetric functions of N positive variables. Its coefficients and summation limits remain those defining $\mathcal{C}_m$; they are not recomputed by replacing m with N in (14.24). In the induction, N is fixed and r increases, so $\mathcal{C}_r^{[N]}$ is evaluated on a fixed N-tuple. The formal recurrence below remains valid under these evaluations.

Lemma 14.9 (Recurrence for the symmetric polynomials)**.** *One has $\mathcal{C}_2 = 0$. For every $r \geq 2$,*

$$\mathcal{C}_{r+1} = e_r^2 \mathcal{C}_r + \mathcal{R}_r. \tag{14.25}$$

With

$$\mathcal{P}_r = e_1^3 e_r e_{r+1} \prod_{j=2}^{r-1} e_j^2,$$

the remainder is given by

$$\frac{\mathcal{R}_r}{\mathcal{P}_r} = r(r+1) - \sum_{p=1}^{\lfloor r/2 \rfloor} (r+1)(2 + \mathbf{1}_{p=1} + \mathbf{1}_{2p=r}) \frac{e_r}{e_p e_{r-p}}$$

$$-\sum_{d=1}^{\lfloor (r-1)/2\rfloor}\sum_{k=1}^{r-2d}(r-k+1)(2+\mathbf{1}_{k=r-2d})\frac{e_k e_r}{e_{k+d}e_{r-d}}. \tag{14.26}$$

Every denominator on the right cancels against $\mathcal{P}_r$.

The proof is given in Appendix C.6.

Newton's inequalities bound the sums in (14.26) by the following binomial sums, whose total is at most $r(r+1)$. This will give $\mathcal{R}_r^{[N]} \geq 0$.

Lemma 14.10 (Scalar binomial inequality)**.** *For $r \geq 2$, set*

$$I_r = \sum_{p=1}^{\lfloor r/2\rfloor}\frac{(r+1)(2+\mathbf{1}_{p=1}+\mathbf{1}_{2p=r})}{\binom{r}{p}},$$

$$J_r = \sum_{d=1}^{\lfloor (r-1)/2\rfloor}\sum_{k=1}^{r-2d}(r-k+1)(2+\mathbf{1}_{k=r-2d})\frac{\binom{k+d}{d}}{\binom{r}{d}}.$$

Then

$$I_r + J_r \leq r(r+1). \tag{14.27}$$

The proof is given in Appendix C.6.

Proof of Theorem 14.8. Fix $N \geq r+1$ positive variables. Newton's inequalities state that the normalized sequence $\overline{e}_j = e_j/\binom{N}{j}$ satisfies $\overline{e}_j^2 \geq \overline{e}_{j-1}\overline{e}_{j+1}$. This is its log concavity. Hence

$$\frac{e_r}{e_p e_{r-p}} \leq \frac{\binom{N}{r}}{\binom{N}{p}\binom{N}{r-p}} \leq \frac{1}{\binom{r}{p}}, \tag{14.28}$$

$$\frac{e_k e_r}{e_{k+d}e_{r-d}} \leq \frac{\binom{N}{k}\binom{N}{r}}{\binom{N}{k+d}\binom{N}{r-d}} \leq \frac{\binom{k+d}{d}}{\binom{r}{d}}. \tag{14.29}$$

For the last inequality, the quotient of the middle expression by the last expression is

$$\prod_{j=1}^{d}\frac{N-r+j}{N-k-d+j} \leq 1,$$

since $k+d \leq r$. Thus Lemma 14.10 and (14.26) give $\mathcal{R}_r^{[N]} \geq 0$ for every $N \geq r+1$.

We induct in r with the assertion $\mathcal{C}_r^{[N]} \geq 0$ for every $N \geq r$. The base is $\mathcal{C}_2 = 0$, and (14.25) proves the next step. Taking $N = m$ and dividing by the positive factor in (14.24) proves (14.22). Equivalently, under $e_1(y) = 1$, this is $2A + Y - V - 2e_2 \geq 0$. □

14.4. **Nonnegativity of the second coefficient.** We prove $D_1 \geq 0$ by a covariance formula valid for every $\lambda \geq 0$, including $\lambda = 1/2$. We retain the variables $y_i = x_i^{-1}$.

Lemma 14.11. *For the uniform weight (14.17), $D_1 \geq 0$ on the full-rank locus whenever $\lambda \geq 0$. If $\lambda > 0$, then $D_1 > 0$.*

Proof. To differentiate the ratios a_{si}, we use a finite probability space. For each $1 \leq s \leq m-1$, consider the probability measure on s-element subsets of $\{1, \ldots, m\}$ given by

$$\mathbb{P}_s(S) = \frac{\prod_{i\in S} y_i}{e_s(y)}.$$

This is the conditional Bernoulli distribution of [14, pp. 876–877, equations (3) and (6); Section 3.2, p. 883], with weights y_i and fixed sample size s. For random variables ξ, η on this finite probability space, their covariance is

$$\mathrm{Cov}_s(\xi, \eta) = \mathbb{E}_s(\xi\eta) - \mathbb{E}_s(\xi)\mathbb{E}_s(\eta).$$

Let $\xi_i(S)$ be the indicator of $i \in S$, equal to one when i belongs to S and zero otherwise. Define

$$\pi_{si} = \mathbb{P}_s(i \in S) = 1 - a_{si}, \qquad w_{sij} = \pi_{si}\pi_{sj} - \mathbb{P}_s(i, j \in S) \quad (i \neq j).$$

Thus $w_{sij} = -\operatorname{Cov}_s(\xi_i, \xi_j)$. If e_q^{ij} denotes the elementary symmetric polynomial with y_i, y_j omitted, then

$$w_{sij} = \frac{y_i y_j}{e_s^2}\left[(e_{s-1}^{ij})^2 - e_{s-2}^{ij} e_s^{ij}\right] \geq 0. \tag{14.30}$$

This is Newton's inequality, with the convention $e_{-1}^{ij} = 0$. It is also the Rayleigh inequality for e_s, expressing pairwise negative correlation; see [7, Definitions 2.3 and 2.5]. Since every subset has exactly s elements, $\sum_i \xi_i = s$. This gives

$$\sum_{j \neq i} w_{sij} = \pi_{si}(1 - \pi_{si}) = \pi_{si} a_{si}. \tag{14.31}$$

Indeed, the sum of the covariances of the indicator of $i \in S$ with all the indicators is its covariance with the constant $|S| = s$.

In addition to (14.20), put

$$L = \sum_i y_i \sum_{s=1}^{m-1} s a_{si}, \qquad Z_2 = \sum_i y_i^2 P_i^2, \qquad U = \sum_i y_i^2 P_i^3,$$

and define

$$H_s = \sum_i y_i^2 a_{si} - \sum_{i<j} w_{sij}(y_i - y_j)^2, \tag{14.32}$$

$$\mathcal{Q}_1 = \sum_{s,i<j} w_{sij}\left[B(y_i + y_j) - (y_i - y_j)(y_i P_i - y_j P_j)\right], \tag{14.33}$$

$$\mathcal{Q}_2 = \sum_{s,i<j} w_{sij}\left[V(y_i + y_j) - (y_i P_i - y_j P_j)^2\right]. \tag{14.34}$$

We claim the decomposition

$$D_1 = \lambda R_1 + \lambda^2 R_2 + \lambda^3 R_3, \tag{14.35}$$

where

$$R_1 = 4\sum_s H_s, \tag{14.36}$$

$$R_2 = 16LB + 12Z_2 + 16\mathcal{Q}_1, \tag{14.37}$$

$$R_3 = 16LV + 8U + 16\mathcal{Q}_2. \tag{14.38}$$

We derive it first, then prove the coefficient signs. Write $\mathscr{D}_j = y_j \partial_{y_j}$. Differentiating the probabilities gives

$$\mathscr{D}_j a_{si} = \begin{cases} w_{sij}, & i \neq j, \\ -\sum_{\ell \neq i} w_{si\ell}, & i = j. \end{cases} \tag{14.39}$$

Indeed, direct differentiation gives $\mathscr{D}_j \mathbb{P}_s(S) = (\xi_j(S) - \pi_{sj})\mathbb{P}_s(S)$. Thus $-\mathscr{D}_j a_{si} = \operatorname{Cov}_s(\xi_i, \xi_j)$, the usual covariance identity in the logarithmic parameters $\log y_j$. Consequently,

$$\mathscr{D}_j B = y_j P_j + \sum_s \sum_{i \neq j} w_{sij}(y_i - y_j), \tag{14.40}$$

$$\mathscr{D}_j V = y_j P_j^2 + 2\sum_s \sum_{i \neq j} w_{sij}(y_i P_i - y_j P_j). \tag{14.41}$$

For any smooth invariant function q written in the y variables, the change of variables gives $\partial_{x_j} q = -y_j \mathscr{D}_j q$. The product multiplier (14.17) satisfies $a^{-1}\partial_{x_j} a = \lambda y_j P_j$, by the complement identity in the proof of Lemma 14.7. Thus (14.12) and the gradient pairing (14.5) give

$$\mathcal{T} q = -2 \sum_j y_j \mathscr{D}_j q, \qquad \frac{\Gamma(a,q)}{a} = -4\lambda \sum_j y_j P_j \mathscr{D}_j q. \tag{14.42}$$

Insert (14.23) and (14.42) in (14.16). The coefficients of $\lambda, \lambda^2, \lambda^3$ are, respectively,

$$\begin{aligned} R_1 &= 4 \sum_j y_j \mathscr{D}_j B, \\ R_2 &= 16B(A+B) + 4 \sum_j y_j \mathscr{D}_j V + 8 \sum_j y_j P_j \mathscr{D}_j B, \\ R_3 &= 16V(A+B) + 8 \sum_j y_j P_j \mathscr{D}_j V. \end{aligned}$$

Pairing the terms with indices i, j in (14.40)–(14.41) gives

$$\begin{aligned} \sum_j y_j \mathscr{D}_j B &= \sum_s H_s, \\ \sum_j y_j \mathscr{D}_j V &= Z_2 - 2 \sum_{s, i<j} w_{sij} (y_i - y_j)(y_i P_i - y_j P_j), \\ \sum_j y_j P_j \mathscr{D}_j B &= Z_2 - \sum_{s, i<j} w_{sij} (y_i - y_j)(y_i P_i - y_j P_j), \\ \sum_j y_j P_j \mathscr{D}_j V &= U - 2 \sum_{s, i<j} w_{sij} (y_i P_i - y_j P_j)^2. \end{aligned}$$

The row-sum identity (14.31) also gives

$$A + B = L + \sum_{s, i<j} w_{sij} (y_i + y_j). \tag{14.43}$$

These identities yield (14.36)–(14.38).

To prove the signs, use (14.31) to obtain

$$\sum_{i<j} w_{sij} (y_i - y_j)^2 \le \sum_i y_i^2 \pi_{si} a_{si} \le \sum_i y_i^2 a_{si}.$$

Thus $H_s \ge 0$. Also,

$$|(y_i - y_j)(y_i P_i - y_j P_j)| \le (y_i + y_j)(y_i P_i + y_j P_j) \le B(y_i + y_j),$$

so $\mathcal{Q}_1 \ge 0$. Cauchy–Schwarz gives

$$(y_i P_i - y_j P_j)^2 \le (y_i + y_j)(y_i P_i^2 + y_j P_j^2) \le V(y_i + y_j),$$

which proves $\mathcal{Q}_2 \ge 0$. The other terms in (14.37)–(14.38) are nonnegative by definition. Finally, $P_i > 0$ and $y_i > 0$ imply $Z_2 > 0$. Thus $R_2 \ge 12 Z_2$ and $D_1 \ge 12\lambda^2 Z_2 > 0$ when $\lambda > 0$. □

Despite a common gradient-pairing table, the different Laplacian coefficients of the spectral families in Section 13 and the determinants lead to the multipliers $\sqrt{S}$ and the product of proper minor norms, respectively. Algebraic closure and spectral inequalities certify both in every order. For determinants, $m+1$ independent generators when $m \ge 3$ give unbounded defining degree and generator count.

Appendix A. Power Multipliers and Radial Perturbations

All identities involving negative or nonintegral powers are used on $R, S > 0$. In the power calculations, F is homogeneous of degree $e \geq 1$ and $\mathcal{L}(F) = F\lambda_F$, with λ_F smooth on the calculation domain. It is a polynomial whenever the divisibility hypothesis of Theorem 4.2 holds.

A.1. **Notation and power factors.** Consider the power family

$$G_{\alpha,\gamma} = R^\alpha S^\gamma F, \qquad R = |x|^2, \quad S = |\nabla F|^2, \tag{A.1}$$

with real exponents. Write $e = \deg F$, and use the contractions

$$U = \Gamma(F, S), \quad V = \Gamma(S, S), \quad D = \Delta F, \quad B = \Delta S.$$

Here U, V, D, B are ambient functions; in a finite Laplacian closure they are functions of the generators. Define

$$\mathcal{V}_\gamma = S + \frac{2\gamma FU}{S} + \frac{\gamma^2 F^2 V}{S^2}, \tag{A.2}$$

$$\mathcal{D}_\gamma = D + \frac{2\gamma U}{S} + \frac{\gamma FB}{S} + \frac{\gamma(\gamma-1)FV}{S^2}. \tag{A.3}$$

Let $\mathcal{Q}_\gamma$ be the explicit expression in (A.4). It satisfies $\mathcal{L}(S^\gamma F) = S^{3\gamma} F \mathcal{Q}_\gamma$.

A.2. **Gradient and radial identities.**

Proposition A.1 (The gradient-power defect)**.** *For $H = S^\gamma F$,*

$$|\nabla H|^2 = S^{2\gamma}\mathcal{V}_\gamma, \qquad \Delta H = S^\gamma \mathcal{D}_\gamma, \qquad \mathcal{L}(H) = S^{3\gamma} F \mathcal{Q}_\gamma,$$

where $\mathcal{V}_\gamma, \mathcal{D}_\gamma$ are given in (A.2)–(A.3), and

$$\begin{aligned} \mathcal{Q}_\gamma = \lambda_F + \gamma B &+ \frac{2\gamma UD - \gamma\Gamma(F,U) - (\gamma^2 + \frac{3}{2}\gamma)V}{S} + \frac{\gamma(\gamma+1)U^2}{S^2} \\ &+ \frac{\gamma^2 F}{S^2}\left(2UB + VD - \Gamma(S,U) - \tfrac{1}{2}\Gamma(F,V)\right) \\ &+ \frac{\gamma^3 F^2}{S^3}\left(VB - \tfrac{1}{2}\Gamma(S,V)\right). \end{aligned} \tag{A.4}$$

At every regular zero of F, one has $S > 0$. Under the assumed smoothness of λ_F, the displayed formula is therefore well-defined there; no division by F remains.

Proof. The gradient and Laplacian follow from the product rule. For the defect, insert $a = S^\gamma$ in Theorem 4.2. The derivative identities are

$$\nabla a = \gamma S^{\gamma-1}\nabla S, \qquad \operatorname{Hess} a = \gamma S^{\gamma-1}\operatorname{Hess} S + \gamma(\gamma-1)S^{\gamma-2}\, dS \otimes dS.$$

Differentiating $S = |\nabla F|^2$, $U = \Gamma(F, S)$, and $V = |\nabla S|^2$ gives

$$\begin{aligned} \operatorname{Hess} F(\nabla F, \nabla S) &= V/2, \\ \operatorname{Hess} S(\nabla F, \nabla F) &= \Gamma(F, U) - V/2, \\ \operatorname{Hess} F(\nabla S, \nabla S) &= \Gamma(S, U) - \Gamma(F, V)/2, \\ \operatorname{Hess} S(\nabla F, \nabla S) &= \Gamma(F, V)/2, \\ \operatorname{Hess} S(\nabla S, \nabla S) &= \Gamma(S, V)/2. \end{aligned}$$

Substitution and collection of the powers of F yield (A.4). □

Proposition A.2 (The radial master identity)**.** *Let H be smooth and homogeneous of degree m, and set $S_H = |\nabla H|^2$, $D_H = \Delta H$, and $c_\alpha = 4\alpha(m+\alpha)$. Then*

$$|\nabla(R^\alpha H)|^2 = R^{2\alpha}\left(S_H + \frac{c_\alpha H^2}{R}\right), \tag{A.5}$$

$$\mathcal{L}(R^\alpha H) = R^{3\alpha}\Bigg[\mathcal{L}(H) + \frac{c_\alpha H^2 D_H}{R} + \frac{\big(2\alpha(N-1) - c_\alpha\big) H S_H}{R} + \frac{c_\alpha\big(m + 2\alpha(N-1)\big)H^3}{R^2}\Bigg]. \tag{A.6}$$

Proof. Euler's identity gives $\Gamma(R, H) = 2mH$. Hence

$$\Delta(R^\alpha H) = R^\alpha\left[D_H + \frac{2\alpha(2m + N + 2\alpha - 2)H}{R}\right],$$

and the product rule gives (A.5). Writing $W = S_H + c_\alpha H^2/R$, of degree $2m - 2$, yields

$$R^{-3\alpha}\mathcal{L}(R^\alpha H) = W D_H + \frac{2\alpha(N-1)HW}{R} - \tfrac{1}{2}\Gamma(H, W).$$

Use $\Gamma(H, H^2/R) = 2HS_H/R - 2mH^3/R^2$ to expand the last term. This proves (A.6). □

A.3. **The combined multiplier.**

Proposition A.3 (The combined power multiplier)**.** *Suppose $\mathcal{L}(F) = F\lambda_F$. Set*

$$m = e + 2\gamma(e-1), \quad \ell = m + 2\alpha, \quad c_\alpha = 4\alpha(m+\alpha), \quad b_\alpha = 2\alpha(N-1) - c_\alpha.$$

For $G = G_{\alpha,\gamma}$, one has

$$|\nabla G|^2 = R^{2\alpha} S^{2\gamma} \widehat{V}_{\alpha,\gamma}, \qquad \widehat{V}_{\alpha,\gamma} = \mathcal{V}_\gamma + \frac{c_\alpha F^2}{R}, \tag{A.7}$$

$$\mathcal{L}(G) = R^{3\alpha} S^{3\gamma} F \widehat{Q}_{\alpha,\gamma}, \tag{A.8}$$

where

$$\widehat{Q}_{\alpha,\gamma} = \mathcal{Q}_\gamma + \frac{c_\alpha F \mathcal{D}_\gamma}{R} + \frac{b_\alpha \mathcal{V}_\gamma}{R} + \frac{c_\alpha\big(m + 2\alpha(N-1)\big)F^2}{R^2}. \tag{A.9}$$

Thus $\widehat{V}_{\alpha,\gamma} > 0$ and $\widehat{Q}_{\alpha,\gamma} \geq 0$ give the normalized-gradient sign on the region where they hold. If that region is one phase, they give its one-sided sign. Comparison across the omitted set requires the hypotheses on the exceptional set and boundary flux in Theorem 2.9.

Proof. Apply Proposition A.2 to $H = S^\gamma F$, of degree m, using the three identities of Proposition A.1. This gives (A.7)–(A.9). Since $R^\alpha S^\gamma > 0$, $G\mathcal{L}(G)$ has the sign of $F^2\widehat{Q}_{\alpha,\gamma}$. □

A.4. **Radial perturbations giving strict divergence estimates.**

Proposition A.4 (Moving the degree toward $n/2$)**.** *Let $G \in C^\infty(\mathbb{R}^{n+1}\setminus\{0\})$ be homogeneous of degree $\ell \neq n/2$, with $\nabla G \neq 0$ throughout the punctured space. Suppose $C = \{G = 0\}$ has a smooth compact minimal link, G takes both signs, and*

$$\operatorname{sgn}(G)\operatorname{div} X_G > 0 \quad \text{off } C, \qquad \beta_G \geq 0 \quad \text{on } C \cap \mathbb{S}^n,$$

where β_G is defined in (4.8). Put $\sigma = \operatorname{sgn}(\ell - n/2)$. For all sufficiently small $\varepsilon > 0$, the potential

$$G_\varepsilon = r^{-\sigma\varepsilon} G, \qquad \ell_\varepsilon = \ell - \sigma\varepsilon, \tag{A.10}$$

has nonvanishing gradient on the punctured space and generates a global subcalibration satisfying

$$\operatorname{sgn}(G_\varepsilon)\operatorname{div} X_{G_\varepsilon} \geq c_\varepsilon \frac{\operatorname{dist}(x, C)}{|x|^2} \qquad (x \neq 0) \tag{A.11}$$

for a constant $c_\varepsilon > 0$. Writing $\beta_\varepsilon = \beta_{G_\varepsilon}$, its boundary coefficient is

$$\beta_\varepsilon = \beta_G + \left(\ell - \frac{n}{2}\right)^2 - \left(\ell_\varepsilon - \frac{n}{2}\right)^2 = \beta_G + 2\left|\ell - \frac{n}{2}\right|\varepsilon - \varepsilon^2. \tag{A.12}$$

The new potential is generally not polynomial. If G is polynomial and ε is positive rational, then G_ε is semialgebraic on $r > 0$: the radial factor is a rational power of the positive polynomial R, hence is described by a polynomial equation and the choice of its positive root.

Proof. Nonvanishing of ∇G_ε for small ε follows from smooth parameter dependence and compactness of the unit sphere. On C, its reciprocal gradient length is $w_\varepsilon = r^{\sigma\varepsilon} w = r^{1-\ell_\varepsilon}\phi$. For $w = r^\alpha \phi$, separation of variables gives

$$-r^2 \frac{J_C w}{w} = \frac{(-\Delta_\Sigma - |A_\Sigma|^2)\phi}{\phi} - \alpha(\alpha + n - 2).$$

Use $\alpha = 1 - \ell$ and $\alpha_\varepsilon = 1 - \ell_\varepsilon$ to obtain (A.12).

For the global sign, let s be signed spherical distance to the link in a fixed tubular neighborhood, positive on $G > 0$. On $r = 1$, put $f_\varepsilon = \operatorname{div} X_{G_\varepsilon}$. Minimality gives $f_\varepsilon = 0$ on the link. The quotient $h_\varepsilon = f_\varepsilon / s$ extends smoothly across it, with $h_\varepsilon|_\Sigma = \beta_\varepsilon$. By (A.12), $\partial_\varepsilon h_\varepsilon|_{\varepsilon=0,\Sigma} = 2|\ell - n/2| > 0$. Compactness and continuity give a smaller fixed tube and a parameter interval on which $\partial_\varepsilon h_\varepsilon \geq |\ell - n/2|$. Since $h_0 \geq 0$, integration in the parameter yields $h_\varepsilon \geq \varepsilon|\ell - n/2|$ throughout that tube. On the complement of a still smaller tube the original strict sign has a positive minimum, which persists for small ε. Spherical normal distance is comparable to $\operatorname{dist}(\theta, C)$ near the link; away from it the latter is bounded. This proves the uniform unit-sphere estimate, and homogeneity gives (A.11). □

For a singular link, (A.12) still holds on the regular part. The global sign requires additional estimates near the singular strata.

A.5. **Radial multiplication preserving the divergence sign.**

Proposition A.5 (Radial multiplication of a homogeneous potential)**.** *Let h be smooth and homogeneous of degree $d > 0$ on a conical open subset of $\mathbb{R}^N \setminus \{0\}$. If*

$$h\mathcal{L}(h) \geq 0, \qquad h\Delta h \geq 0, \qquad N \geq 2d + 2,$$

then $g = \sqrt{R}\, h$ satisfies $g\mathcal{L}(g) \geq 0$, strictly where $h \neq 0$. If $\nabla h \neq 0$ throughout the domain, then $\nabla g \neq 0$ there as well.

Proof. The radial master identity, Proposition A.2 with $\alpha = 1/2$, reads

$$\mathcal{L}(\sqrt{R}\, h) = R^{3/2}\Bigg[\mathcal{L}(h) + (2d+1)\frac{h^2\Delta h}{R} + (N - 2d - 2)\frac{h|\nabla h|^2}{R} + (2d+1)(d+N-1)\frac{h^3}{R^2}\Bigg]. \tag{A.13}$$

Multiplication by $g = \sqrt{R}\, h$ makes every term nonnegative; the last one is positive when $h \neq 0$. The same master identity gives

$$|\nabla g|^2 = R|\nabla h|^2 + (2d+1)h^2,$$

proving the final assertion. □

Appendix B. Odd-Degree Rigidity, Cubic Structure and Local Geometry

B.1. **Odd-degree rigidity by second-order expansions.**

We prove Theorem 7.1 and determine the normalized cubic closure.

B.1.1. *The closure identities and the degree.* Throughout this subsection, $e \geq 3$ is odd and $S = |\nabla F|^2$. Counting homogeneous monomials in the generators of degrees $2, e, 2e-2$ gives

$$\begin{aligned} \Delta F &= 0, & \Delta S &= \alpha R^{e-2}, \\ \Gamma(F,S) &= \beta R^{e-2} F, & \Gamma(S,S) &= \gamma R^{2e-3} + \delta R^{e-3} F^2 + \varepsilon R^{e-2} S. \end{aligned} \tag{B.1}$$

Bochner's identity gives $2|\operatorname{Hess} F|^2 = \alpha R^{e-2}$, with $\alpha > 0$.

Let $p \neq 0$ be a critical point. Euler's identity gives $F(p) = 0$, and $\nabla S(p) = 0$, so $\gamma = 0$. Put $H = \operatorname{Hess} F(p)$ and $R_0 = |p|^2$. The quadratic terms at p in the last two identities of (B.1) give

$$H^3 = \frac{\beta}{4} R_0^{e-2} H, \qquad H^4 = \frac{\varepsilon}{4} R_0^{e-2} H^2.$$

Since $H \neq 0$, these identities imply $\beta > 0$ and $\varepsilon = \beta$.

At a positive maximum of F on the unit sphere, $\nabla F = eFx$ and $S = e^2 F^2 > 0$. Moreover, $\nabla S = 2 \operatorname{Hess} F \nabla F$ and the differentiated Euler identity $\operatorname{Hess} F\, x = (e-1)\nabla F$ give $\nabla S = 2(e-1)Sx$. Substitution gives

$$S = \frac{\beta}{2e(e-1)}, \qquad 4(e-1)^2 S^2 = \delta \frac{S}{e^2} + \beta S,$$

whence $\delta = e(e-2)\beta$. Multiplying F by $2/\sqrt{\beta}$ and reusing F, S, α for the normalized quantities, we obtain

$$\begin{aligned} \Delta F &= 0, & \Delta S &= \alpha R^{e-2}, \\ \Gamma(F,S) &= 4R^{e-2} F, & \Gamma(S,S) &= 4R^{e-2} S + 4e(e-2) R^{e-3} F^2. \end{aligned} \tag{B.2}$$

Write $f = F|_{\mathbb{S}^{N-1}}$ and $s = S|_{\mathbb{S}^{N-1}}$. Removing the radial derivatives from (B.2) gives

$$\begin{aligned} |\nabla_{\mathbb{S}} f|^2 &= s - e^2 f^2, \\ \langle \nabla_{\mathbb{S}} f, \nabla_{\mathbb{S}} s \rangle &= f\big(4 - 2e(e-1)s\big), \\ |\nabla_{\mathbb{S}} s|^2 &= 4s + 4e(e-2) f^2 - 4(e-1)^2 s^2. \end{aligned} \tag{B.3}$$

Let $a > 0$ be the maximum of f, attained at x_0. Both spherical gradients vanish at x_0, and

$$s(x_0) = \frac{2}{e(e-1)}, \qquad a^2 = \frac{2}{e^3(e-1)}.$$

Let $B = \operatorname{Hess}_{\mathbb{S}} f(x_0)$ and $C = \operatorname{Hess}_{\mathbb{S}} s(x_0)$. Comparing second-order terms in the three identities of (B.3), in geodesic normal coordinates at x_0, gives

$$\begin{aligned} C &= 2B(B + e^2 a I), \\ BC &= -e(e-1) a C, \\ C^2 &= \left(\frac{8}{e} - 6\right) C + 4e(e-2) a B. \end{aligned} \tag{B.4}$$

The first equation shows that B and C commute. The first two equations restrict every eigenvalue of B to $0, -e^2 a, -e(e-1)a$. The value $-e^2 a$ would give a zero eigenvalue of C but a nonzero right-hand side in the last equation, so it is impossible. Since $\operatorname{tr} B = -e(N+e-2)a < 0$, some eigenvalue of B is nonzero. On the corresponding eigenvector, $B = -e(e-1)a$ and $C = -4/e$. The last equation now gives

$$0 = \frac{16}{e^2} - \left(\frac{8}{e} - 6\right)\left(-\frac{4}{e}\right) + 4e^2(e-1)(e-2)a^2 = \frac{8(e-2)(e-3)}{e^2}.$$

Thus $e = 3$, proving Theorem 7.1.

B.1.2. *The remaining cubic coefficient.* For $e = 3$, the spherical Laplacians are

$$\Delta_{\mathbb{S}} f = -3(N+1)f, \qquad \Delta_{\mathbb{S}} s = \alpha - 4(N+2)s.$$

Let brackets denote normalized spherical integration, and set

$$M_0 = \langle f^2 \rangle > 0, \quad M_1 = \langle s \rangle, \quad M_2 = \langle s^2 \rangle, \quad M_3 = \langle f^2 s \rangle.$$

Using (B.3) with $e = 3$, integration by parts gives

$$M_1 = 3(N+4)M_0, \tag{B.5}$$

$$(\alpha+8)M_0 = 4(N+8)M_3, \tag{B.6}$$

$$M_2 + 4M_0 = 3(N+8)M_3, \tag{B.7}$$

$$4(N+6)M_2 = (\alpha+4)M_1 + 12M_0. \tag{B.8}$$

These follow, respectively, by integrating $\Delta_{\mathbb{S}}(f^2)$, by testing the equation for $\Delta_{\mathbb{S}} s$ against f^2, by testing the equation for $\Delta_{\mathbb{S}} f$ against fs, and by integrating $\Delta_{\mathbb{S}}(s^2)$. The middle two identities give $4M_2 = (3\alpha+8)M_0$. Substituting this and (B.5) into (B.8), and cancelling $M_0 > 0$, yields $(N+6)(3\alpha+8) = 3(N+4)(\alpha+4) + 12$. Consequently,

$$\alpha = \frac{2(N+3)}{3}. \tag{B.9}$$

B.2. **Odd gradient-square closure with trivial critical set.**

Proof of Theorem 7.2. The count in (B.1) applies without a nonzero critical point. We retain its constants $\alpha, \beta, \gamma, \delta, \varepsilon$, without imposing the normalization (B.2). Write $f = F|_{\mathbb{S}^{N-1}}$, $s = S|_{\mathbb{S}^{N-1}}$, and put

$$K = 2e(e-1), \qquad L = 4(e-1)^2, \qquad D = 2(e-1)(N+2e-4).$$

The spherical identities are

$$\begin{aligned} |\nabla f|^2 &= s - e^2 f^2, & \langle \nabla f, \nabla s \rangle &= f(\beta - Ks), \\ |\nabla s|^2 &= Q(s) + \delta f^2, & Q(t) &= \gamma + \varepsilon t - Lt^2, \\ \Delta f &= -e(N+e-2)f, & \Delta s &= \alpha - Ds. \end{aligned} \tag{B.10}$$

Here and throughout this proof, gradients, Hessians, and Laplacians without subscripts are spherical. These formulas follow by subtracting the radial derivatives from (B.1); the Laplacian formula uses $\Delta(r^j v) = r^{j-2}[\Delta_{\mathbb{S}} v + j(j+N-2)v]$.

Step 1: the upper bound for s. The assumption on the critical set gives $s > 0$. At a critical point of f, Euler's identity gives $\nabla_{\mathbb{R}^N} F = efx$, and differentiation of Euler's identity gives $\nabla_{\mathbb{R}^N} S = 2(e-1)Sx$. Since $s = e^2 f^2 > 0$ there, the identity for $\Gamma(F, S)$ yields $\beta = 2e^3(e-1)f^2 > 0$. Rescale F so that $\beta = 1$, and write

$$s_0 = \frac{1}{K}, \qquad a^2 = \frac{s_0}{e^2}.$$

Thus the only critical values of f are a and $-a$, and $s = s_0$ at every such point. Oddness ensures that both values occur and that $-a \le f \le a$.

We claim that $s \le s_0$ everywhere. Otherwise follow the complete gradient flow of f from a point where $g = s - s_0 > 0$. Along this trajectory, with parameter t,

$$\dot{f} = s - e^2 f^2, \qquad \dot{g} = -Kfg.$$

The second equation preserves $g > 0$. The function f increases and is bounded, and every accumulation point is critical: indeed, $\int_0^\infty |\nabla f|^2 \, dt < \infty$, while $|\nabla f|^2$ is uniformly continuous along a smooth flow on the compact sphere. Since the initial value is larger than $-a$, it follows that $f \to a$ and $g \to 0$. Direct differentiation of the two flow equations shows that

$$\big(f^2 - a^2 - g/e\big) g^{-e/(e-1)}$$

is constant. Consequently

$$f^2 - a^2 = \frac{g}{e} + c_1 g^{e/(e-1)}.$$

The right-hand side is positive for sufficiently small positive g, contradicting $|f| \le a$. This proves the claim.

Step 2: the minimum and maximum Hessians. Suppose that s is nonconstant. Its minimum m satisfies $0 < m < s_0$. At a minimum point, the second identity in (B.10) gives $f = 0$, so $v = \nabla f$ has $|v|^2 = m$. Let $B_m = \operatorname{Hess} s$ there. Differentiating that identity once, and the identity for $|\nabla s|^2$ twice, gives

$$B_m v = \eta v, \qquad B_m^2 = \frac{Q'(m)}{2} B_m + \delta v \otimes v, \qquad \eta = 1 - Km > 0. \tag{B.11}$$

In particular,

$$Q(m) = 0, \qquad \eta^2 = \frac{Q'(m)}{2}\eta + \delta m. \tag{B.12}$$

For example, the second matrix identity follows because the Hessian of $|\nabla s|^2$ at a critical point of s is $2B_m^2$, whereas the Hessian of δf^2 at $f = 0$ is $2\delta v \otimes v$.

At a point where $f = a$, put $B = \operatorname{Hess} f$ and $C = \operatorname{Hess} s$. Differentiating (B.10) twice gives

$$C = 2B(B + e^2 a I), \qquad BC = -e(e-1)aC, \qquad C^2 = \Big(\frac{\varepsilon}{2} - Ls_0\Big) C + \delta a B. \tag{B.13}$$

In the middle identity we used the first one to commute B and C. It follows that the eigenvalues of B belong to

$$0, \quad -e^2 a, \quad -e(e-1)a,$$

and the corresponding eigenvalues of C are $0, 0, -1/e$.

Step 3: exclusion of a nonconstant s. If $\delta = 0$, the equations at the two extrema give $Q(m) = Q(s_0) = 0$. Hence $Q(t) = -L(t-m)(t-s_0)$. The scalar identity in (B.12), together with $\eta > 0$, gives

$$K(s_0 - m) = \eta = \frac{Q'(m)}{2} = \frac{L}{2}(s_0 - m).$$

This is impossible because $K = 2e(e-1) \ne 2(e-1)^2 = L/2$.

Now suppose $\delta \ne 0$. In (B.13), the eigenvalue $-e^2 a$ of B is impossible, because its C-eigenvalue is zero. Since $\operatorname{tr} B = -e(N + e - 2)a < 0$, the eigenvalue $-e(e-1)a$ occurs. Substitution in the last identity yields

$$\delta = 4e - 6 - e\varepsilon. \tag{B.14}$$

The value of the gradient-square identity at this same point is

$$Q(s_0) + \delta s_0 / e^2 = 0. \tag{B.15}$$

Set $x = m/s_0 \in (0, 1)$. Eliminating γ with (B.15), and substituting (B.14), the two scalar equations (B.12), after multiplication by their nonzero denominators, become

$$\begin{aligned} 0 &= e\varepsilon[ex - (e-1)] + 2(e^2 - 3e + 3) - 2e(e-1)x^2, \\ 0 &= e\varepsilon[ex - (e-1)] + 2e(e-1) + (10 - 8e)x - 2(e-1)(e-2)x^2. \end{aligned}$$

Their difference is

$$2(x-1)\big(2(e-1)x - (2e-3)\big) = 0.$$

Since $x < 1$, this gives

$$\begin{aligned} x &= \frac{2e-3}{2(e-1)}, & \varepsilon &= \frac{4e^2 - 15e + 12}{e(e-2)}, \\ \delta &= \frac{e}{e-2}, & Q'(m) &= -\frac{1}{e-2}, & \eta &= \frac{1}{2(e-1)}. \end{aligned} \tag{B.16}$$

The nonnegative Hessian B_m preserves $v^\perp$. By (B.11), its eigenvalues on $v^\perp$ are either zero or $Q'(m)/2 < 0$, so they are all zero. Thus $\Delta s = \eta$ at the minimum.

At the maximum point of f, let r be the multiplicity of $-e(e-1)a$ in B. Taking traces gives

$$r = \frac{N+e-2}{e-1}, \qquad \Delta s = -\frac{r}{e}.$$

On the other hand, subtracting the two values of $\Delta s = \alpha - Ds$ gives

$$\frac{1}{2(e-1)} + \frac{r}{e} = D(s_0 - m) = \frac{N+2e-4}{2e(e-1)}.$$

Consequently $r = (N+e-4)/(2(e-1))$, in contradiction with the preceding trace formula. Hence s is constant, proving $S = cR^{e-1}$ with $c > 0$.

Step 4: the harmonic eikonal classification. Rescale once more so that $S = e^2R^{e-1}$. Tkachev's polynomial eikonal theorem [85, Theorem 1] gives, up to sign,

$$f = T_j(g), \qquad e = jp, \qquad p \in \{1,2,3,4,6\},$$

where g is the spherical restriction of a Cartan–Münzner polynomial of degree p. Since e is odd, $p \in \{1,3\}$ and that polynomial is harmonic. In particular,

$$|\nabla g|^2 = p^2(1-g^2), \qquad \Delta g = -p(p+N-2)g.$$

The Chebyshev equation $(1-t^2)T_j'' - tT_j' + j^2T_j = 0$ and the chain rule give

$$\Delta f = -e^2T_j(g) - p(N-2)gT_j'(g).$$

Comparing this with harmonicity of F, namely $\Delta f = -e(e+N-2)f$, and using $N > 2$, yields

$$tT_j'(t) = jT_j(t) \qquad (-1 < t < 1).$$

This identity would make T_j a monomial. For $j > 1$, its coefficient of t^{j-2} is nonzero, so $j = 1$. Thus $e = p = 3$. Cartan's classification of harmonic cubic eikonal polynomials [11], also recovered in [82, Theorem 1.1], gives the four stated dimensions and models. □

B.3. **Spectral exclusions for the weight operator.** In the notation of Sections 9.5.1 and 9.5.2, A is an exact Euclidean Hsiang algebra, c is its maximal unit idempotent, $E_r = \ker(L_c - r\,\mathrm{Id})$, and $T = p(L_c) - 6P_c$, where $p(t) = 4t^3 + 4t^2 - t$. In particular, $Z = E_{-1}$, $H = E_{-1/2}$, $W = E_{1/2}$, and $B = E_T(1) = \mathbb{R}c \oplus Z \oplus H \oplus W$. We use Lemma 9.16 and Proposition 9.17.

B.3.1. *The negative Peirce spaces.*

Proof of Lemma 9.18. (i) For $h \in H$, fusion gives $h^2 = -|h|^2c/2 + q(h)$, with $q(h) \in H$. The norm identity at $r = -1/2$ gives $|q(h)|^2 = |h|^4/2$. Thus $h \diamond k = \sqrt{2}\,\pi_H(hk)$ makes H a Euclidean eikonal algebra. A positive spherical maximum for its cubic gives a unit idempotent a: the multiplier equation is $a \diamond a = Ma$, and $|a \diamond a| = 1$ forces $M = 1$. In the original algebra,

$$a^2 = -\frac{1}{2}c + \frac{1}{\sqrt{2}}a, \qquad e = -\frac{1}{3}c + \frac{2\sqrt{2}}{3}a$$

gives a unit idempotent e of A. Lemma 9.15, applied to c and e, yields $[T, L_a] = 0$.

Fix $U_m = E_T(m)$ with $m \neq 1$. A nonzero block of $L_a|_{U_m}$ from E_r to E_s requires

$$4(r^2 + rs + s^2) - 3 = 0, \qquad p(r) = p(s) = m.$$

If $r = s$, then $r = \pm 1/2$, giving $m = 1$. If $r \neq s$, the divided difference of p, together with this equality, gives $r + s = -1/2$. The third root of $p(t) - m$ is then $-1/2$, again giving $m = 1$. Thus $L_a|_{U_m} = 0$. For $u \in U_m \cap E_r$, it follows that $L_e u = -ru/3$. Since $u \perp e$, the quotient formula at e gives

$$0 = p(r) - p(-r/3) = \frac{4}{27}r(28r^2 + 24r - 9).$$

The quadratic factor has one root below -1 and the other in $(0, 1/3)$, both excluded by (9.31). Hence $r = m = 0$.

(ii) For a unit $z \in Z$, fusion and invariance give $z^2 = -c$ and $cz = -z$. The vectors $e_\pm = (-c \pm \sqrt{3}z)/2$ are unit idempotents. Together with c, they form a triality system in the terminology of [64, Definition 6.8.1]. Thus Lemma 9.15 gives $[T, L_z] = 0$. Fix $U_m = E_T(m)$, $m \neq 1$, and put $X = L_c|_{U_m}$, $Y = L_z|_{U_m}$. For a nonzero block of Y between distinct roots r, s of $p(t) = m$, fusion and the divided difference of p require

$$4(r^2 + rs + s^2) - 2(r + s) - 1 = 0, \qquad 4(r^2 + rs + s^2) + 4(r + s) - 1 = 0.$$

Thus $r + s = 0$ and $\{r, s\} = \{1/2, -1/2\}$, which would give $m = 1$. Hence X, Y commute and have a common orthonormal eigenbasis.

On a common eigenvector with eigenvalues r, y, the quotient formula at c, e_+, e_- gives

$$p(r) = p\left(\frac{-r + \sqrt{3}y}{2}\right) = p\left(\frac{-r - \sqrt{3}y}{2}\right) = m.$$

These three arguments sum to zero, whereas the three roots of $p(t) - m$ sum to -1. Therefore two of the arguments coincide. If $y = 0$, then

$$0 = p(r) - p(-r/2) = \frac{3}{2}r(3r - 1)(r + 1),$$

so $r = 0$ or $r = 1/3$; the root -1 would give $m = 1$. If $y \neq 0$, one of the other arguments equals r, and the remaining argument is $-2r$. Thus $y^2 = 3r^2$ and

$$0 = p(r) - p(-2r) = 3r(6r + 1)(2r - 1).$$

Here $r \neq 0$, and $r = 1/2$ again gives $m = 1$. Hence $r = -1/6$, $m = 7/27$, and $y^2 = 1/12$. This proves the spectral assertion and the actions on P, Q. Scaling z proves (9.35). □

B.3.2. *Excluding a one-dimensional weight-one space.*

Proof of Lemma 9.19. Suppose $E_T(1) = \mathbb{R}c$. On A_{ess} the Peirce eigenvalues $-1, -1/2, 1/2, 0$ are absent. Fix another eigenvalue $r \neq 1/3$. The right side of (9.30) is nonzero for every nonzero $u \in E_r$, so u^2 has a nonzero component in $c^\perp$. There is at most one root of Q_r in $[-1, 1/2]$. Indeed, if $r \leq -3/4$, then Q_r is strictly decreasing there. If $-3/4 < r \leq -1/6$, then $Q_r(1/2) < 0$, so $1/2$ lies strictly between its two roots. If $1/3 < r < 1/2$, then $Q_r(-1) < 0$, and -1 lies strictly between the roots. These cases cover every remaining r. The nonzero square component must therefore belong to a single eigenspace E_s. This root is simple: in the decreasing case, $s < 1/2$ gives $Q_r'(s) < 0$, and the other cases have two distinct roots. Thus the other root lies outside the interval. In particular, the endpoint $r = r_0$ cannot occur: its unique root in the interval is $1/2$, which is absent. We have

$$E_r^2 \subset \mathbb{R}c \oplus E_s, \qquad u^2 = r|u|^2 c + q(u), \qquad 0 \neq q(u) \in E_s. \tag{B.17}$$

The equality $r = s$ would imply $\mathscr{K}(r, r, r) = 3(4r - 1)(2r + 1) = 0$, contrary to (9.31) and the absence of $E_{-1/2}$.

Let $d = -2r - s - 1/2$ be the other root of Q_r. Symmetry of $\mathscr{K}$ gives

$$\mathscr{K}(r, r, t) = 4(t - s)(t - d), \qquad \mathscr{K}(r, s, t) = 4(t - r)(t - d).$$

Since d is inadmissible and $r \neq s$, fusion and orthogonality imply $E_r E_s \subset E_r$. Hence

$$\langle u^2, u\rangle = 0, \qquad u^3 \in E_r, \qquad u^4 \in \mathbb{R}c \oplus E_s.$$

The full vector identity, evaluated at u, now implies $(u^2)^2 \in \mathbb{R}c \oplus E_s$. Expanding (B.17) shows that $q(u)^2 \in \mathbb{R}c \oplus E_s$. But $\mathscr{K}(s, s, s) \neq 0$, since $s = -1/2, 1/4$ are excluded. Its E_s component vanishes, and therefore $q(u)^2 \in \mathbb{R}c$. Applying (9.30) to this nonzero vector gives $s^2(3s - 1)(s + 1) = 0$. Since $s = 0, -1$ are absent, $s = 1/3$.

Every eigenvalue $r \neq 1/3$ thus satisfies

$$\mathscr{K}(r, r, 1/3) = 0 \quad \Longleftrightarrow \quad 108r^2 + 60r - 17 = 0.$$

The positive root $(-5+2\sqrt{19})/18$ belongs to $(0,1/3)$ and is excluded. Only $r_* = (-5-2\sqrt{19})/18$ remains. If $a = \dim E_{r_*}$ and $b = \dim E_{1/3}$, exactness gives

$$0 = \operatorname{tr} L_c = 1 + \frac{-5-2\sqrt{19}}{18}a + \frac{1}{3}b.$$

For $a = 0$ this is positive, while for $a > 0$ its irrational part cannot cancel. Both cases are impossible. □

B.4. **The trace argument for exactness of the core.** Here $A = B \oplus U$ is the nonradial grading of Proposition 9.21, B is eikonal, and $M_b = L_b|_U$. We use the representation identities of Proposition 9.22 and the square-norm identity $|u^2|^2 = |u|^4/9$ from Lemma 9.23.

Proof of Proposition 9.24. Put $n = \dim B$, $m = \dim U$, and let $\tau = \sum_{\alpha=1}^n e_\alpha^2$ for an orthonormal basis (e_α) of B. Thus $\langle \tau, b\rangle = \operatorname{tr}_B L_b$. Exactness of A and the grading give

$$\operatorname{tr}_U M_b = -\langle \tau, b\rangle, \qquad \Delta_U(u^2) = -2\tau. \tag{B.18}$$

Summing (9.39) over the basis yields

$$K := \sum_{\alpha=1}^n M_{e_\alpha}^2 = \frac{n}{18}\operatorname{Id}_U + \frac{1}{6}M_\tau.$$

Apply Δ_U to $|u^2|^2 = |u|^4/9$. Using (B.18), this gives

$$8\langle Ku, u\rangle - 4\langle \tau, u^2\rangle = \frac{4(m+2)}{9}|u|^2.$$

Consequently

$$M_\tau = \kappa \operatorname{Id}_U, \qquad \kappa = \frac{n-m-2}{6}, \qquad m\kappa = -|\tau|^2. \tag{B.19}$$

The last equality is its trace, using (B.18).

Suppose $\tau \neq 0$, so $\kappa < 0$. The coefficient of t^2 in (9.40) at $b + t\tau$ is

$$3\big(36\kappa^2 - |\tau|^2\big)M_b = \langle \tau^2 + 6\kappa\tau, b\rangle \operatorname{Id}_U. \tag{B.20}$$

If $36\kappa^2 \neq |\tau|^2$, every M_b is scalar: $M_b = \ell(b)\operatorname{Id}_U$ for a linear form ℓ, which is nonzero because $\ell(\tau) = \kappa$. The cubic representation identity then gives, with $r = |b|^2$ and $s = |u|^2$,

$$\langle b^2, b\rangle = 108\ell(b)^3 - 9r\ell(b), \qquad \langle bu, u\rangle = \ell(b)s.$$

The grading therefore factors the original cubic as

$$F(b+u) = \frac{\ell(b)}{2}\big(36\ell(b)^2 - 3r + s\big),$$

contrary to irreducibility.

It remains that $36\kappa^2 = |\tau|^2$. From Equations (B.19) and (B.20), we obtain

$$\kappa = -\frac{m}{36}, \qquad \tau^2 = -6\kappa\tau.$$

Set $k = -6\kappa = m/6 > 0$ and $e = \tau/k$. Then e is a unit idempotent of B and $M_e = -\operatorname{Id}_U/6$. Linearizing the eikonal identity at e gives the orthogonal decomposition

$$B = \mathbb{R}e \oplus Z' \oplus W', \qquad L_e|_{Z'} = -\operatorname{Id}, \qquad L_e|_{W'} = \tfrac{1}{2}\operatorname{Id}.$$

For $\eta \in W'$, the polarized quadratic identity gives $M_\eta/2 = 3\{M_e, M_\eta\} = -M_\eta$, hence $M_\eta = 0$. Taking the trace of the quadratic identity at η now gives

$$-\frac{m}{3}|\eta|^2 = \operatorname{tr}_U M_{\eta^2} = -\langle \tau, \eta^2\rangle = -\frac{k}{2}|\eta|^2 = -\frac{m}{12}|\eta|^2.$$

Thus $W' = 0$. Since $\dim B \geq 2$ by $Z \neq 0$ in the original Peirce decomposition, $\dim Z' \geq 1$. But then

$$0 < k = \langle \tau, e\rangle = \operatorname{tr}_B L_e = 1 - \dim Z' \leq 0,$$

a contradiction. Therefore $\tau = 0$, proving both trace assertions; (B.19) then gives $n = m + 2$. □

B.5. **Local geometry near the cubic critical set.** The local geometry in Proposition 7.4 and (10.20) follows from (7.2) and the Hessian eigenvalues established in that proposition.

B.5.1. *The critical set.* Fix $p \in Z^*$. Translate p to the coordinate origin and write orthonormal coordinates as (x, y), with $x \in \mathbb{R}^{2k}$ in the nonzero Hessian eigenspaces and $y \in \mathbb{R}^{N-2k}$ in its kernel. In these translated coordinates, R is still the original squared radius, so its constant term is $|p|^2 > 0$. The matrix $F_{xx}(0,0)$ is invertible, while $F_{xy}(0,0) = F_{yy}(0,0) = 0$.

The analytic implicit function theorem solves $F_x = 0$ locally and uniquely as $x = \psi(y)$, with $\psi(0) = 0$ and $\psi'(0) = 0$. Set $g(y) = F(\psi(y), y)$. Since $F(p) = 0$, $\nabla F(p) = 0$, and $F_{yy}(p) = 0$, the function g vanishes to order at least three. We claim that it vanishes identically.

Suppose instead that its first nonzero homogeneous term is g_j, of degree $j \geq 3$. On the graph of ψ, the identity $F_x = 0$ gives

$$\nabla F = (0, \nabla g), \qquad S = |\nabla g|^2, \qquad F = g.$$

Differentiating $F_x(\psi(y), y) = 0$ and $g_y = F_y$ gives

$$F_{xy} = -F_{xx}\psi', \qquad F_{yy} = g_{yy} + (\psi')^t F_{xx}\psi'.$$

Consequently,

$$F_{xy} = O(|y|), \qquad F_{yy} = O(|y|^{j-2}) + O(|y|^2), \qquad \nabla g = O(|y|^{j-1}).$$

Since $j \geq 3$, these estimates imply $\nabla S = 2\operatorname{Hess} F \nabla F = O(|y|^j)$ on the graph. The last identity in (7.2) therefore becomes

$$|\nabla S|^2 = 4R|\nabla g|^2 + 12g^2.$$

Its left side vanishes to order at least $2j$, but its right side has the nonzero term $4|p|^2|\nabla g_j|^2$ of degree $2j - 2$. This contradiction proves $g \equiv 0$, by analyticity.

Every point of the graph $x = \psi(y)$ is now critical, and every nearby critical point lies on that graph. Thus Z^* is real analytic of dimension $N - 2k$. Its tangent space is the Hessian kernel: differentiating $\nabla F = 0$ along the graph gives inclusion in that kernel, and the dimensions agree.

B.5.2. *Local coordinates and the link volume.* Subtract the graph by setting $x = \psi(y) + \xi$. The resulting function $\widetilde{F}(\xi, y)$ and its first ξ-derivatives vanish at $\xi = 0$. Taylor's integral formula gives

$$\widetilde{F}(\xi, y) = \xi^t A(\xi, y)\xi, \qquad A(\xi, y) = \int_0^1 (1 - t)\, F_{xx}(\psi(y) + t\xi, y)\, dt.$$

The matrix A is analytic and symmetric. After shrinking the neighborhood it is invertible with k positive and k negative eigenvalues. Successive completion of squares gives an analytic invertible matrix $B(\xi, y)$ such that

$$A = B^t D B, \qquad D = \operatorname{diag}(\mathrm{Id}_k, -\mathrm{Id}_k).$$

Indeed, at the coordinate origin A is diagonal with nonzero entries. All pivots remain nonzero on a sufficiently small neighborhood, and the positive square roots used to normalize their absolute values are analytic there. The map $(\xi, y) \mapsto (B(\xi, y)\xi, y)$ has invertible derivative at the origin. The analytic inverse function theorem therefore gives coordinates (u, v, y) in which

$$F = |u|^2 - |v|^2, \qquad Z^* = \{u = v = 0\}.$$

The coordinate derivatives and inverse derivatives are bounded on smaller neighborhoods, so Euclidean lengths and submanifold areas are comparable to those in the coordinate model.

The radial vector lies in $T_p Z^*$, because Z^* is a cone. It is transverse to the unit sphere. Hence $M = Z^* \cap \mathbb{S}^{N-1}$ is a smooth compact submanifold of dimension $N - 2k - 1$. On the

sphere, $f = F|_{\mathbb{S}^{N-1}}$ and its first derivatives vanish along M. Its Hessian has the same $2k$ nonzero transverse eigenvalues as the ambient Hessian; the removed radial direction belongs to the kernel. Applying the preceding Taylor formula and completion of squares in a sphere chart gives local coordinates (u, v, y), now with $y \in \mathbb{R}^{N-2k-1}$, for which

$$f = |u|^2 - |v|^2, \qquad M = \{u = v = 0\}.$$

Put $t = (|u|^2 + |v|^2)^{1/2}$. The smooth positive-definite spherical metric and the bounded coordinate derivatives give $\rho \asymp t$, where ρ is spherical distance to M. They also give $|\nabla_{\mathbb{S}} f|^2 \asymp t^2$. Since $s = |\nabla_{\mathbb{S}} f|^2 + 9f^2$ and $|f| \le t^2$, it follows that $s \asymp \rho^2$.

In these coordinates the regular zero link is the product, in a bounded chart, of the y-variables with

$$\{(u, v) : |u| = |v| > 0\} \subset \mathbb{R}^{2k}.$$

This quadratic cone has dimension $2k - 1$. Its link is the compact product of two spheres of radius $1/\sqrt{2}$, and its area inside a ball of radius a is a constant times a^{2k-1}. Bounded coordinate distortion thus bounds the area in each chart with $\rho < \varepsilon$ by $C\varepsilon^{2k-1}$. A finite cover of M gives

$$\mathcal{H}^{N-2}\big(\Sigma_{\mathrm{reg}} \cap \{\rho < \varepsilon\}\big) \le C\varepsilon^{2k-1},$$

which completes the proof of (10.20).

Appendix C. Exact Positivity and Obstruction Certificates

C.1. **Exact positivity certificates.** Our use of Bernstein coefficients to certify polynomial nonnegativity was inspired by [49, Section 6.3 and Appendix A]. For a polynomial P of degree at most m on $[a, b]$, put $z = a + (b - a)t$ and write $P(a + (b - a)t) = \sum_{j=0}^{m} c_j t^j$. Its Bernstein expansion is

$$P(a + (b - a)t) = \sum_{i=0}^{m} b_i \binom{m}{i} t^i (1 - t)^{m-i}, \tag{C.1}$$

where

$$b_i = \sum_{j=0}^{i} c_j \frac{\binom{i}{j}}{\binom{m}{j}}. \tag{C.2}$$

Indeed,

$$\sum_{i=j}^{m} \frac{\binom{i}{j}}{\binom{m}{j}} \binom{m}{i} t^i (1 - t)^{m-i} = t^j \sum_{i=j}^{m} \binom{m - j}{i - j} t^{i-j} (1 - t)^{m-i} = t^j.$$

The Bernstein basis functions are nonnegative and sum to one on $[0, 1]$, so $P \ge \min_i b_i$ on $[a, b]$.

Example C.1 (An exact check for the $(4, 1, 7)$ correction)**.** For the polynomial P in (6.38), (C.2), at degree five on each interval, gives the following minimum coefficients:

interval	$[-7/4, -1]$	$[-1, -1/2]$	$[-1/2, 0]$	$[0, 1/8]$	$[1/8, 1/4]$
$\min b_i$	$372925/1024$	$33735/32$	42	$10973/1280$	$1099085/32768$

All five numbers are positive. These intervals cover the entire realizable interval, so this proves the positivity assertion in Proposition 6.9.

The accompanying file `LSC-INTERVAL-POSITIVITY.wl` checks these positivity certificates with exact rational arithmetic. The related isoparametric identities are checked in `LSC-ISOPARAMETRIC.wl`.

C.2. **The Clifford power-multiplier obstruction.** Use the Clifford closure variables from Section 9.4.2, with $(m, q) = (4, 3)$, $J = |\nabla S|^2$, and $K = \Delta S = 64R$. The instability theorem already rules out these multipliers; the calculation below supplies the explicit wrong-sign points described in Remark 10.25.

Construction of the wrong-sign points. Put $a = R^\alpha S^\gamma$. The multiplier identity in Proposition A.3, together with Proposition 9.11, gives

$$\mathcal{L}(G_{\alpha,\gamma}) = a^3 F Q_{\alpha,\gamma},$$

where $Q_{\alpha,\gamma}$ extends smoothly across $F = 0$ on $\{R > 0, S > 0\}$. We need only its restriction to $R = 1$, $F = 0$, $T = u^2$. Writing $u = \lambda$, $v = 1 - \lambda$, set

$$\begin{aligned} s_\lambda &= \lambda(4 - 3\lambda), \\ j_\lambda &= 16\big(4\lambda(1-\lambda) + (4 - 3\lambda)\lambda^2\big). \end{aligned}$$

These are respectively S, J at the stated boundary values. The closure table gives $\Gamma(F, \Gamma(F, S)) = 16RS + 96F^2$. Using this identity and $K = \Delta S = 64$ in (A.4)–(A.9) gives

$$\begin{aligned} B_{\alpha,\gamma}(\lambda) = 48\gamma - 8 - \gamma(\gamma + \tfrac{3}{2})\frac{j_\lambda}{s_\lambda} \\ + 2\alpha(5 - 8\gamma - 2\alpha)s_\lambda. \end{aligned} \tag{C.3}$$

In particular,

$$\lim_{\lambda \downarrow 0} B_{\alpha,\gamma}(\lambda) = -8(\gamma - 1)(2\gamma - 1), \tag{C.4}$$

$$B_{\alpha,\gamma}\left(\frac{2}{3}\right) = -\frac{8}{3}\left[2\alpha^2 + (8\gamma - 5)\alpha + 8\gamma^2 - 6\gamma + 3\right]. \tag{C.5}$$

If the required sign $Q_{\alpha,\gamma} \geq 0$ held, then (C.4) would force

$$\frac{1}{2} \leq \gamma \leq 1. \tag{C.6}$$

But the bracket in (C.5) satisfies

$$\begin{aligned} &2\alpha^2 + (8\gamma - 5)\alpha + 8\gamma^2 - 6\gamma + 3 \\ &\quad = 2\left(\alpha + \frac{8\gamma - 5}{4}\right)^2 + 4\gamma - \frac{1}{8} \geq \frac{15}{8} \quad \text{under (C.6).} \end{aligned} \tag{C.7}$$

Therefore the value in (C.5) is at most -5.

To realize and perturb these boundary values, put the first two matrices in (10.46). For prescribed $\lambda \in (0, 1)$, set $u = \lambda$, $v = 1 - \lambda$, choose $x \in \mathbb{R}^4$ with $|x|^2 = u/2$, and let

$$y = (x, x). \tag{C.8}$$

Then

$$\tau(y) = (0, u, 0, \ldots, 0), \qquad T = u^2.$$

For $0 < \varepsilon < \sqrt{v}$, set

$$z_\varepsilon = (\sqrt{v - \varepsilon^2}, \varepsilon, 0, \ldots, 0).$$

Now $R = 1$, $T = u^2$, and

$$F(y, z_\varepsilon) = u\varepsilon \longrightarrow 0, \qquad Q_{\alpha,\gamma}(y, z_\varepsilon) \longrightarrow B_{\alpha,\gamma}(\lambda). \tag{C.9}$$

For fixed α, γ, the product rule on $\{F = 0, R > 0, S > 0\}$ gives

$$|\nabla(R^\alpha S^\gamma F)|^2 = R^{2\alpha} S^{2\gamma+1} > 0.$$

Hence the gradient remains nonzero for sufficiently small ε. If $\gamma \notin [1/2, 1]$, first fix a sufficiently small $\lambda > 0$ using (C.4); otherwise take $\lambda = 2/3$. The function $Q_{\alpha,\gamma}$ remains negative for all sufficiently small $\varepsilon > 0$, so the normalized gradient is defined and has the wrong divergence sign at the nearby points. □

C.3. **The cubic certificate without harmonicity.** For a cubic F, put $R = |x|^2$, $S = |\nabla F|^2$. We specialize Appendix A to $G = F\sqrt{RS}$, retaining the quadratic weight and trace terms.

For $\mathcal{L}(F) = -qF$, put

$$t = \Delta F, \qquad H_2 = |\operatorname{Hess} F|^2, \qquad K = \Delta S, \qquad T = \Gamma(S,S).$$

Then

$$\Gamma(F,S) = 2St + 2qF, \qquad K = 2H_2 + 2\Gamma(F,t). \tag{C.10}$$

In the associated metrized algebra this means

$$t(x) = \operatorname{tr} L_x, \qquad H_2(x) = \operatorname{tr}(L_x^2), \qquad K(x) = 2\operatorname{tr}(L_x^2) + \langle x^2, b\rangle, \quad \langle b, x\rangle = \operatorname{tr} L_x.$$

The fixed vector b is the trace vector.

Polynomial coefficients. Define

$$B_0 = 16S^2\big[S(H_2 - 2q + 3t^2) - T\big], \tag{C.11}$$

$$\begin{aligned} B_1 = {} & -16S^2\Gamma(F,q) - 8S^2\Gamma(S,t) + 16KS^2t \\ & + 80S^2qt - 4STt - 2S\Gamma(F,T), \end{aligned} \tag{C.12}$$

$$B_2 = -8S\Gamma(S,q) + 16KqS + 2KT + 32q^2S - \Gamma(S,T). \tag{C.13}$$

Then put

$$\begin{aligned} C_0 &= R^2B_0 + 16(N-12)RS^4, \\ C_1 &= R^2B_1 + 16(2N+9)RS^3t, \\ C_2 &= R^2B_2 + R\big[88KS^2 + 32(N-1)qS^2 + 4(N-23)ST\big] \\ &\quad + 176(N+4)S^3. \end{aligned} \tag{C.14}$$

The degrees of (B_0, B_1, B_2) are $(14, 11, 8)$, and those of (C_0, C_1, C_2) are $(18, 15, 12)$. In particular, all terms of $C_0 + FC_1 + F^2C_2$ have degree eighteen.

The conditional certificate.

Proposition C.2 (A polynomial certificate without harmonicity)**.** *Let F be a nonzero square-free homogeneous cubic on $\mathbb{R}^N$, $N \geq 2$, satisfying $\mathcal{L}(F) = -qF$ for a homogeneous quadratic polynomial q. Let C_0, C_1, C_2 be the explicit homogeneous polynomials in (C.14). If*

$$C_0 \geq 0, \qquad C_2 \geq 0, \qquad C_1^2 \leq 4C_0C_2 \tag{C.15}$$

at every point, then $G = F\sqrt{RS}$ generates a two-sided subcalibration off $Z = \{\nabla F = 0\}$. Consequently $E = \{F < 0\}$ is locally perimeter minimizing and $\partial[\![E]\!]$ is locally mass minimizing. No harmonicity or radiality assumption is required.

Proof. The derivation below gives

$$\mathcal{L}(G) = \frac{F}{16\sqrt{R}\,S^{3/2}}(C_0 + FC_1 + F^2C_2) \qquad (S > 0). \tag{C.16}$$

The positive semidefinite matrix in (C.15), evaluated on $(1, F)$, gives $F\mathcal{L}(G) \geq 0$. The potential is continuous with the phases of F; its positive multiplier and degree six give $\nabla G \neq 0$ off Z by Corollary 2.8. Square-freeness makes Z removable by Lemma 3.11, so Theorem 2.5 applies. □

The leading coefficient is

$$C_0 = 16RS^2\mathscr{H}_{\mathrm{gen}}, \qquad \mathscr{H}_{\mathrm{gen}} = (N-12)S^2 + RS(H_2 - 2q + 3t^2) - RT. \tag{C.17}$$

When $t = 0$, $q = 2R$, and $T = 4RS + 12D$, this is precisely $\mathscr{H}$ in (10.5). For harmonic models, the transport identities of Theorem 10.2 give sharper conditions than (C.15).

The minimizing current has support $\{F = 0\}$ if both phases have positive volume in every ball centered at a zero; square-freeness alone does not ensure this.

Derivation of the defect. First let $H = F\sqrt{S}$, of degree five. Product differentiation gives

$$Q_H := |\nabla H|^2 = S^2 + 2FSt + F^2\left(2q + \frac{T}{4S}\right), \tag{C.18}$$

$$\Delta H = 3\sqrt{S}\,t + \frac{F}{\sqrt{S}}\left(2q + \frac{K}{2} - \frac{T}{4S}\right). \tag{C.19}$$

To compute $\mathcal{L}(H)$, regard Q_H as a function of F, S, t, q, T. Its partial derivatives are

$$(Q_H)_F = 2St + 4qF + \frac{FT}{2S}, \quad (Q_H)_S = 2S + 2Ft - \frac{F^2T}{4S^2},$$
$$(Q_H)_t = 2FS, \qquad (Q_H)_q = 2F^2, \qquad (Q_H)_T = \frac{F^2}{4S}.$$

Use the chain rule with $\Gamma(F,F) = S$, $\Gamma(F,S) = 2St + 2qF$, $\Gamma(S,S) = T$, and $\Gamma(F,t) = K/2 - H_2$. Since

$$\Gamma(H, Q_H) = \sqrt{S}\,\Gamma(F, Q_H) + \frac{F}{2\sqrt{S}}\Gamma(S, Q_H),$$

substitution into $\mathcal{L}(H) = Q_H\Delta H - \Gamma(H, Q_H)/2$ and collection of the displayed powers of F yield

$$\mathcal{L}(H) = \frac{F}{16S^{3/2}}(B_0 + FB_1 + F^2B_2). \tag{C.20}$$

The term independent of these displayed factors uses the Bochner relation $K/2 - \Gamma(F,t) = H_2$; this accounts for the H_2 term in (C.11).

Finally $G = \sqrt{R}\,H$. The radial identity (A.13), with $d = 5$, gives

$$\mathcal{L}(G) = R^{3/2}\left[\mathcal{L}(H) + \frac{11H^2\Delta H}{R} + \frac{(N-12)H|\nabla H|^2}{R} + \frac{11(N+4)H^3}{R^2}\right].$$

Inserting (C.18)–(C.20) and factoring out $F/(16\sqrt{R}\,S^{3/2})$ gives (C.14) and (C.16), with polynomial extension across the zeros of F.

C.4. **Quartic potential identities.** For the normalization of Example 8.25, take $G = RQ^{3/4}F$ and use the differential table (8.63). The quotient chain rule gives

$$|\nabla G|^2 = RQ^{1/2}\big(16R^2Q^2 + 33R^2F^2 + 32QF^2\big), \tag{C.21}$$

$$\begin{aligned}\frac{\mathcal{L}(G)}{F} = RQ^{1/4}\Big[&16(12k-47)R^4Q^2 + 16(2N-41)R^2Q^3 \\ &+ F^2\big((396k+66)R^4 + (384k+66N-681)R^2Q \\ &\qquad + (64N+160)Q^2\big)\Big].\end{aligned} \tag{C.22}$$

Every coefficient is positive for $N \geq 32$ and $k \geq 4$. Equivalently, apply Proposition A.3 with (8.64).

On the regular zero set, these formulas yield (8.65).

The $(3,7,7)$ specialization. The differential table and certificate used for Proposition 8.31 are Equations (8.43) and (8.44) at $N = 32$ and $\nu = 7$:

$$\begin{aligned}\Delta(R,W,F) &= (64, 28R, 0), \qquad \Gamma(F,W) = 4RF, \\ \Gamma(F,F) &= \Gamma(W,W) = 4RW.\end{aligned} \tag{C.23}$$

$$\mathcal{L}(WF) = 16WF\left[(3R^2 - 2W)W^2 + (21R^2 - 6W)F^2\right]. \tag{C.24}$$

The certificate file is `LSC-WEIGHTED-CLIFFORD-QUARTIC.wl`.

C.4.1. *Trace reductions for rectangular quartics.* We complete the trace calculations for Proposition 7.9, using $B = X^{\mathsf{T}}X$, $P_j = \operatorname{tr}(B^j)$, and $R = P_1$. In (7.15), summing coordinate derivatives of XB gives $(p+3+1)P_1$. For XB^2, the derivative of the first factor contributes pP_2; the two derivatives of the first B contribute P_2 and P_1^2, and those of the second contribute P_2 and $3P_2$. Thus

$$\Delta P_2 = 4(p+4)P_1, \qquad \Delta P_3 = 6\big((p+5)P_2 + P_1^2\big).$$

These identities hold without any regular-eigenvalue assumption.

For the three eigenvalues of B, put

$$e_2 = (R^2 - P_2)/2, \qquad e_3 = (P_3 - RP_2 + Re_2)/3.$$

Their characteristic equation gives $P_j = RP_{j-1} - e_2P_{j-2} + e_3P_{j-3}$ for $j \geq 4$. Together with (7.16), this yields

$$\begin{aligned} P_2 &= (F + R^2)/2, \\ P_3 &= (S + 32RF + 16R^3)/64, \\ P_4 &= RS/48 + 5R^2F/12 + R^4/8 + F^2/8, \\ P_5 &= 5R^2S/256 + 5R^3F/16 + R^5/16 + 5RF^2/24 + 5FS/768. \end{aligned}$$

Differentiation of S gives

$$\nabla S = X\big(384B^2 - 256RB + (32R^2 - 64F)\,\mathrm{Id}\big).$$

Pairing this with $\nabla F = 4X(2B - R\,\mathrm{Id})$ and with itself, and then substituting the displayed power traces, proves the last two entries of (7.13).

C.5. **Polynomial cubic alternatives and pullback calculations.** We detail the polynomial potentials of Section 10.2, Cartan pullbacks of Theorem 11.7, and unequal Lawson pullback of Example 8.32.

The degree-seven defect.

Proof of Lemma 10.5. The product rule gives (10.13). Lemma 9.8 and Euler's identities give

$$\begin{aligned} \Gamma(G,R) &= 14SF, & \Gamma(G,F) &= S^2 + 4RF^2, \\ \Gamma(G,S) &= 8RSF + 12DF, & \Gamma(G,D) &= 2S^2F + 8RDF. \end{aligned}$$

Differentiating the displayed polynomial Q in R, F, S, D and substituting in $\mathcal{L}(G) = Q\Delta G - \Gamma(G,Q)/2$ proves (10.14). □

The positive decomposition for the polynomial certificate. Under the hypotheses of Corollary 10.6, the multiplication dictionary and Bochner identity give

$$\Delta F = 0, \qquad \Gamma(F,S) = 4RF, \qquad \Gamma(S,S) = 4RS + 12F^2, \tag{C.25}$$

and

$$K := \Delta S = 2\operatorname{tr}(L_x^2) \geq 16R. \tag{C.26}$$

The trace bound gives $F \not\equiv 0$, and Lemma 10.3 gives

$$0 \leq S \leq R^2/3. \tag{C.27}$$

Set $D = F^2$. Then $\Gamma(F,D) = 2SF$ and $\Gamma(S,D) = 8RD$, so Lemma 10.5 gives

$$Q := |\nabla G|^2 = S^3 + 12RSF^2 + 12F^4, \qquad \Delta G = (K + 8R)F. \tag{C.28}$$

The defect (10.14) specializes to

$$\begin{aligned} \frac{\mathcal{L}(G)}{F} &= (K - 16R)S^3 + F^2S(12RK - 126S) + 12F^4(K - 6R) \\ &= (K - 16R)Q + 150R^2SF^2 + 126F^2S\left(\frac{R^2}{3} - S\right) + 120RF^4. \end{aligned} \tag{C.29}$$

The quotient is understood by polynomial extension across $F = 0$. Every term on the last line is nonnegative by (C.26) and (C.27). If $F \neq 0$, Euler's identity implies $S > 0$ and $R > 0$, so the term $150R^2SF^2$ is strictly positive. Thus $F\mathcal{L}(G) > 0$ away from $F = 0$.

The geometric comparison is proved in Corollary 10.6.

The Cartan source certificate. For F, Q, G in Theorem 11.7, the pullback identities give

$$|\nabla G|^2 = 4RQ(9Q^3 + 16F^2), \tag{C.30}$$

$$\mathcal{L}(G) = 8QF\Big[9\big((4m-48)R^2 - 5Q\big)Q^3 + 16\big((4m+2)R^2 - 5Q\big)F^2\Big]. \tag{C.31}$$

Since $Q \leq R^2$, the bracket is positive when $Q > 0$. Thus $G\mathcal{L}(G) \geq 0$ throughout the regular set.

The unequal Lawson pullback. For the target data T, f, H in Example 8.32,

$$\Delta f = 2, \qquad |\nabla f|^2 = 80T + 4f, \qquad \Gamma(T, f) = 4f.$$

In particular $\mathcal{L}(f) = -160f$, so its regular zero set is minimal. The product rule gives

$$\begin{aligned} |\nabla H|^2 &= 4T(20T^2 + Tf + 3f^2), \\ \mathcal{L}(H) &= 8Tf(60T^2 + 7Tf + 33f^2), \\ 2T\mathcal{L}(H) - 4H|\nabla H|^2 &= 32T^2 f(20T^2 + 3Tf + 15f^2). \end{aligned}$$

The last quadratic factor is positive for $T > 0$: its discriminant as a polynomial in f is $-1191T^2$ and its leading coefficient is positive. Hence (5.5) holds with $\ell = 4$ and $\kappa = 1/2$, strictly off $f = 0$. For the source weights $(22, 5, 6)$, one has $N = 64$ and $h = 11$. Substitution in (8.28) gives

$$Q_C(u) \geq \frac{2273}{4} \int_C r^{-2}u^2, \qquad u \in C_c^\infty(C_{\mathrm{reg}} \setminus \{0\}), \quad C = \{F = 0\},$$

since $(64 - 17)^2/4 + 8 \cdot 11 - 72 = 2273/4$.

C.6. **Recurrence and scalar bounds for determinant potentials.** We prove the algebraic lemmas for the all-degree induction in Section 14.3, using $\mathcal{C}_m$, $\mathcal{R}_r$, and the binomial sums of (14.24)–(14.27).

Coefficient extraction for the recurrence.

Proof of Lemma 14.9. For $q \geq 1$, write

$$H_{st}^{(q)} = \sum_i y_i^q e_s(\widehat{y}_i) e_t(\widehat{y}_i).$$

Here and below $e_j = 0$ for $j < 0$. Direct substitution in (14.24) gives

$$\mathcal{C}_m = e_1(D_m^{\mathrm{sym}})^2\Big[2\sum_{s=1}^{m-1} s(s+1)\frac{e_{s+1}}{e_s} - 2\sum_{s=1}^{m-1}\frac{H_{ss}^{(1)}}{e_s^2} - \sum_{s,t=1}^{m-1}\frac{H_{st}^{(1)}}{e_s e_t} + \frac{1}{e_1}\sum_{s=1}^{m-1}\frac{H_{s0}^{(2)}}{e_s} - \frac{2e_2}{e_1}\Big]. \tag{C.32}$$

This also shows that the denominators cancel. The case $m = 2$ gives $\mathcal{C}_2 = 0$.

We calculate the coefficient of e_{r+1} in (C.32) for $m = r + 1$. Let $\mathscr{E}(z) = \sum_{j=0}^{r+1} e_j z^j = \prod_i (1 + y_i z)$. Summing $H_{st}^{(1)} z^s w^t$ gives the formal power series identity

$$\sum_{s,t\geq 0} H_{st}^{(1)} z^s w^t = \mathscr{E}(z)\mathscr{E}(w) \sum_i \frac{y_i}{(1 + y_i z)(1 + y_i w)}$$

$$= \frac{z\mathscr{E}'(z)\mathscr{E}(w) - w\mathscr{E}'(w)\mathscr{E}(z)}{z-w}. \tag{C.33}$$

Extracting the coefficient of $z^s w^t$ in (C.33) gives, for $0 \le s,t \le r$,

$$H^{(1)}_{st} = \sum_{j=0}^{\min(s,t)} (s+t+1-2j) e_j e_{s+t+1-j},$$

where $e_j = 0$ for $j > r+1$ in this calculation with $r+1$ variables. Since $j \le r$ in every summand, this expression is affine in e_{r+1}. For $1 \le s,t \le r$, differentiation with respect to that formal variable therefore yields

$$\frac{\partial H^{(1)}_{st}}{\partial e_{r+1}} = \begin{cases} (2r+1-s-t)e_{s+t-r}, & s+t \ge r, \\ 0, & s+t < r. \end{cases} \tag{C.34}$$

Similarly,

$$\sum_{s\ge 0} H^{(2)}_{s0} z^s = \frac{e_1\mathscr{E}(z) - \mathscr{E}'(z)}{z}, \qquad H^{(2)}_{s0} = e_1 e_{s+1} - (s+2)e_{s+2}.$$

It follows that

$$\frac{\partial H^{(2)}_{s0}}{\partial e_{r+1}} = \begin{cases} 0, & s \le r-2, \\ -(r+1), & s = r-1, \\ e_1, & s = r. \end{cases} \tag{C.35}$$

These formulas show that $\mathcal{C}_{r+1}$ depends affinely on e_{r+1}. Setting $e_{r+1} = 0$ leaves $e_r^2\mathcal{C}_r$. Indeed, append a zero variable to r positive variables. The terms with $s = r$ vanish at every nonzero coordinate, and every term from the zero coordinate has a factor y_i or y_i^2. This proves the claimed restriction as a polynomial identity.

Since the ratio of $e_1(D^{\mathrm{sym}}_{r+1})^2$ to $\mathcal{P}_r/e_{r+1}$ is e_r, equations (C.34)–(C.35) give

$$\begin{aligned} \frac{\mathcal{R}_r}{\mathcal{P}_r} &= 2r(r+1) - 2e_r \sum_{\substack{1\le s\le r \\ 2s\ge r}} \frac{(2r+1-2s)e_{2s-r}}{e_s^2} \\ &\quad - e_r \sum_{\substack{1\le s,t\le r \\ s+t\ge r}} \frac{(2r+1-s-t)e_{s+t-r}}{e_s e_t} + 1 - (r+1)\frac{e_r}{e_1 e_{r-1}}. \end{aligned} \tag{C.36}$$

The diagonal term $s = r$ contributes -2. In the double sum, the terms with $s = r$ or $t = r$ contribute $-(r^2+r-1)$. Together with $2r(r+1)+1$, these terms give the constant $r(r+1)$.

The remaining terms with $s+t = r$ have the form $e_r/(e_p e_{r-p})$. The double sum contributes the factor $2 - \mathbf{1}_{2p=r}$, and the diagonal sum contributes $2\mathbf{1}_{2p=r}$. The last term of (C.36) adds $\mathbf{1}_{p=1}$. Their total coefficient is the first coefficient in (14.26).

For the terms with $s+t = r+k > r$ and $s,t < r$, choose $s \le t$ and write $s = k+d$, $t = r-d$. Then $1 \le d \le \lfloor (r-1)/2 \rfloor$ and $1 \le k \le r-2d$. The two orders in the double sum and the diagonal contribution give $2 + \mathbf{1}_{k=r-2d}$, while $2r+1-s-t = r-k+1$. This is the second sum in (14.26), and proves the recurrence. □

The scalar binomial estimate.

Proof of Lemma 14.10. Let $S_{r,d}$ denote the inner sum in J_r, and put

$$t_{r,d} = \frac{\binom{r-d}{d}}{\binom{r}{d}}, \qquad t_{r,d} = 0 \quad \text{if } 2d > r.$$

For $\ell = r-2d$, the identities

$$\sum_{k=1}^{\ell} \binom{k+d}{d} = \binom{r-d+1}{d+1} - 1, \qquad \sum_{k=1}^{\ell} k\binom{k+d}{d} = (d+1)\binom{r-d+1}{d+2}$$

give

$$S_{r,d} = \frac{2}{\binom{r}{d}}\left[(r+1)\left(\binom{r-d+1}{d+1}-1\right)-(d+1)\binom{r-d+1}{d+2}\right] + (2d+1)t_{r,d}. \tag{C.37}$$

Also,

$$\frac{t_{r,d+1}}{t_{r,d}} = \frac{(r-2d)(r-2d-1)}{(r-d)^2}.$$

We claim that

$$S_{r,d} \le \frac{2r^3}{r-1}\left(\frac{t_{r,d}}{d+1}-\frac{t_{r,d+1}}{d+2}\right). \tag{C.38}$$

To check the sign explicitly, divide (C.37) by $t_{r,d}$. After omitting the negative term $-2(r+1)/\binom{r-d}{d}$, the result is

$$B_{r,d} = \frac{2(r-d+1)(r+2d^2+3d+2)}{(d+1)(d+2)} + 2d + 1.$$

Put $u = d-1 \ge 0$ and $v = r-2d-1 \ge 0$. Over the positive denominator $(r-1)(d+1)(d+2)(r-d)^2$, the numerator of

$$\frac{2r^3}{r-1}\left(\frac{1}{d+1}-\frac{(r-2d)(r-2d-1)}{(d+2)(r-d)^2}\right) - B_{r,d}$$

is

$$\begin{aligned}
&2uv^4 + (4u^3+25u^2+29u)v^3 \\
&+ (16u^4+102u^3+206u^2+140u+14)v^2 \\
&+ (18u^5+137u^4+385u^3+484u^2+258u+44)v \\
&+ 4u^6+38u^5+140u^4+252u^3+230u^2+104u+24.
\end{aligned}$$

Every coefficient is nonnegative. This proves (C.38). Summation in d, with $t_{r,1} = (r-1)/r$, gives

$$J_r \le \frac{2r^3}{r-1}\frac{t_{r,1}}{2} = r^2. \tag{C.39}$$

For the remaining sum, direct calculation gives

$$(I_2, I_3, I_4) = \left(6, 4, \frac{25}{4}\right), \qquad (J_2, J_3, J_4) = \left(0, 6, \frac{43}{4}\right).$$

Thus (14.27) holds for $r \le 4$. Also, $I_5 = 24/5 \le 5$ and $I_6 = 329/60 \le 6$. For $r \ge 7$, $\binom{r}{p} \ge \binom{r}{2}$ when $2 \le p \le r/2$, so

$$I_r \le \frac{3(r+1)}{r} + \frac{3(r+1)(r-2)}{r(r-1)} \le r.$$

Combine this with (C.39) to finish the proof. □

Appendix D. Weight Realizability for Orthogonal Quartics

The necessity in Proposition 8.12 follows from the quartic isoparametric classification and these compatibility lemmas.

Lemma D.1 (A block of weight one)**.** *Let E be a nonzero homogeneous quartic on $\mathbb{R}^N$, with $R = |x|^2$, satisfying $|\nabla E|^2 = 16RE$ and $\Delta E = 8R$. Then $E = (x^T P x)^2$ for a symmetric matrix P with $P^2 = \mathrm{Id}$ and $\operatorname{tr} P = 0$. In particular, its quadratic square root is nondegenerate.*

Proof. The norm identity gives $E \geq 0$. Set $q = \sqrt{E}$ on a connected component of $\{E > 0\}$. The equations give

$$|\nabla q|^2 = 4R, \qquad \Delta q = 0, \qquad |\operatorname{Hess} q|^2 = \tfrac{1}{2}\Delta|\nabla q|^2 = 4N.$$

Since all second derivatives of q are harmonic, applying the Laplacian once more yields $0 = \Delta|\operatorname{Hess} q|^2 = 2|\nabla^3 q|^2$. Thus q is locally a polynomial of degree at most two. Homogeneity removes its affine terms, and polynomial continuation gives globally $E = (x^T P x)^2$, where $P = P^T$, $P^2 = \mathrm{Id}$, and $\operatorname{tr} P = 0$. In particular, this quadratic form is nondegenerate. □

Lemma D.2 (Joining compatible Clifford blocks)**.** *Let (P_α) and (Q_β) be two nonempty families of symmetric Clifford matrices on the same Euclidean space; a one-matrix family means a symmetric involution. If*

$$A = \sum_\alpha (x^T P_\alpha x)^2, \qquad B = \sum_\beta (x^T Q_\beta x)^2, \qquad A + B \leq R^2,$$

then their union is a Clifford system.

Proof. Fix $P = P_\alpha$ and $Q = Q_\beta$. On either eigenspace of P, $(x^T P x)^2 = R^2$, so $x^T Q x = 0$ there. Polarization shows that the two diagonal blocks of Q relative to these eigenspaces vanish. Hence $PQ + QP = 0$ for every cross pair. For the eigenspace description of Clifford systems, see [13, Section 3.9, pp. 164–165]. □

Classification input. For a triple satisfying (8.20), put $\ell = r_0 + r_1 + r_2 - 1$, so the ambient dimension is 2ℓ. For $r_i \geq 2$, Remark 8.10 gives

$$\Phi_i = R^2 - 2E_i, \qquad |\nabla \Phi_i|^2 = 16R^3, \qquad \Delta \Phi_i = 8(\ell + 1 - 2r_i)R.$$

The nonzero blocks exclude $\Phi_i = \pm R^2$. By the eikonal classification [85, Theorem 1 and the discussion of $m = 4$], any other nongenuine possibility has, up to sign and orthogonal coordinates, the form $R^2 - 8uv$, where $u = |x'|^2$, $v = |x''|^2$, and $u + v = R$. Its Laplacian is radial only when the two block dimensions both equal ℓ. The corresponding E_i is then $(u - v)^2$ or $4uv$, of weight 1 or ℓ, respectively. Both are excluded by $2 \leq r_i \leq \ell - 1$. The ordered multiplicities of Φ_i are consequently $(r_i - 1, \ell - r_i)$. The completed classification says that, up to sign and orthogonal coordinates, every such polynomial is OT–FKM, apart from the homogeneous unordered multiplicities $(2, 2)$ and $(4, 5)$; see [15, 16] and the summary in [36, Section 1].

Lemma D.3 (Exclusion of the exceptional triple)**.** *There is no triple satisfying* (8.20) *with weights* $(1, 5, 5)$.

Proof. Suppose such a triple exists. Then $\ell = 10$. By the classification input above, a weight-five block has multiplicities $(4, 5)$ and is the exceptional homogeneous example [15]. On $W = \Lambda^2 \mathbb{C}^5$, let $P_1, \dots, P_5$ be the holomorphic quadratic Plücker coordinates of $Z \wedge Z/2$. The polynomial in [36, Section 3, equation (3.2)] is

$$\Psi = R^2 - 8 \sum_{j=1}^{5} |P_j|^2, \qquad (m_+, m_-) = (5, 4).$$

Thus our block is $E_5 = (R^2 + \Psi)/2$. By Lemma D.1, the weight-one block is q^2 for a nondegenerate real quadratic form q satisfying

$$q^2 \leq R^2 - E_5 = 4 \sum_{j=1}^{5} |P_j|^2.$$

Consequently q vanishes on the cone $\mathcal{D} = \{u \wedge v : u, v \in \mathbb{C}^5\} = \{P_1 = \cdots = P_5 = 0\}$. The equality follows from the skew normal form of a complex two-vector.

We verify directly that every real quadratic form vanishing on $\mathcal{D}$ is degenerate. Decompose it as $q(Z) = \Re h(Z) + K(Z, Z)$, with h holomorphic quadratic and K Hermitian. Replacing

$Z \in \mathcal{D}$ by $e^{it}Z$ shows that h and $K(Z,Z)$ vanish separately on $\mathcal{D}$. For fixed u, Hermitian polarization on $u \wedge \mathbb{C}^5$ gives $K(u \wedge v, u \wedge w) = 0$. Replace u by $u + ta$, $t \in \mathbb{C}$; the coefficients of t and $\bar{t}$ give $K(u \wedge v, a \wedge w) = 0$. Since decomposable vectors span W, this proves $K = 0$. Let B polarize h. Then $B(u \wedge v, u \wedge w) = 0$. The tensor $T(u, v, w, t) = B(u \wedge v, w \wedge t)$ is alternating within each pair; replacing u by $u + w$ in the preceding identity also makes it alternating in its first and third entries. Hence T is a four-form, so $h(Z) = \omega(Z \wedge Z)$ for some $\omega \in \Lambda^4(\mathbb{C}^5)^*$. Every nonzero such ω is $\iota_v \mathrm{vol}$ for a nonzero vector v. In a basis with $v = e_5$, h ignores the component $e_5 \wedge \mathrm{span}_{\mathbb{C}}(e_1, \dots, e_4)$. This gives a real subspace of dimension eight in the radical of $\Re h$. The zero form is also degenerate. This contradicts the nondegeneracy of q, excluding $(1, 5, 5)$. □

Proof of necessity in Proposition 8.12. Put $\ell = r_0 + r_1 + r_2 - 1$ and reorder the positive integer weights as $a \le b \le c$. Since $0 \le E_i \le R^2$, the preceding classification applies. Suppose $\delta(a + b - 1) \nmid \ell$. The function δ is nondecreasing and takes powers of two as values. Since $\ell - r_i \ge \ell - c = a + b - 1$, it follows that

$$\delta(\ell - r_i) \nmid \ell \qquad (i = 0, 1, 2).$$

Call a block exceptional if its genuine Cartan–Münzner polynomial has unordered multiplicities $(2, 2)$ or $(4, 5)$. Every nonexceptional block must itself be a Clifford square sum with r_i terms: weight one is covered by Lemma D.1, and the classification input above leaves only two OT–FKM orientations in higher weight. The opposite orientation would give

$$E_i = R^2 - \sum_{\alpha=1}^{\ell - r_i + 1} (x^T P_\alpha x)^2,$$

requiring $\delta(\ell - r_i) \mid \ell$ by [32, Section 3.5]. If two blocks are nonexceptional, their sum is at most R^2; Lemma D.2 produces $r_i + r_j = \ell + 1 - r_k$ Clifford matrices, again forcing $\delta(\ell - r_k) \mid \ell$, a contradiction. There must therefore be at least two exceptional blocks. For multiplicities $(2, 2)$, one has $\ell = 5$, $r_i = 3$, and two blocks leave zero third weight. For $(4, 5)$, one has $\ell = 10$, $r_i \in \{5, 6\}$, leaving only $(1, 5, 5)$, excluded by Lemma D.3. The different ambient dimensions prevent mixing the two types. This proves the required divisibility. □

D.1. **The active-sum-nine check.** For Corollary 8.13, the active pairs, up to exchange, are $(2, 7)$, $(3, 6)$, and $(4, 5)$. The table records $\delta(a + b - 1)$ after ordering the three weights as $a \le b \le c$. None of its entries divides the corresponding ℓ.

TABLE 3. The excluded weights $1 \le r_0 \le 7$ when $r_1 + r_2 = 9$.

r_0	$\ell = r_0 + 8$	$(2, 7)$	$(3, 6)$	$(4, 5)$
1	9	2	4	4
2	10	4	4	8
3	11	4	8	8
4	12	8	8	8
5	13	8	8	8
6	14	8	8	8
7	15	8	8	8

APPENDIX E. SUPPLEMENTARY ALGEBRAIC AND GEOMETRIC DETAILS

E.1. **Polarization and metrized operations in arbitrary degree.** For a homogeneous polynomial F of degree $d \ge 3$, its scalar polarization is the symmetric d-linear form $A_F =$

$D^dF/d!$, characterized by $A_F(x,\dots,x) = F(x)$ [46, Section 2.6.4, equation (2.6.5)]. Here D^dF is the constant dth differential. Its evaluation is given by the polarization formula

$$D^dF[x_1,\dots,x_d] = \sum_{I\subseteq\{1,\dots,d\}} (-1)^{d-|I|} F\Big(\sum_{i\in I} x_i\Big).$$

The alternating sum cancels terms missing some vector argument; homogeneity leaves the mixed derivative of order d.

Using the Euclidean inner product, define the symmetric $(d-1)$-linear map $\mu : V^{d-1} \to V$ by

$$\langle \mu(x_1,\dots,x_{d-1}), x_d\rangle = dA_F(x_1,\dots,x_d) = \frac{1}{(d-1)!} D^dF[x_1,\dots,x_d].$$

The complete symmetry of the right side is the metric compatibility condition. Since $DF(x)[v] = dA_F(x,\dots,x,v)$, this normalization makes $\mu(x,\dots,x) = \nabla F(x)$. For $d = 3$, the product in (9.1) satisfies $\langle xy, z\rangle = 6A_F(x,y,z)$, so $xy = 2\mu(x,y)$; its multiplication operator is consequently twice the one used here. Put $P(x) = \mu(x,\dots,x)$ and $L_x v = \mu(x,\dots,x,v)$, with $d-2$ copies of x in the latter. Polarization and Euler's identity give

$$F = \frac{1}{d}\langle P, x\rangle, \qquad \nabla F = P, \qquad \operatorname{Hess} F = (d-1)L_x, \qquad \Delta F = (d-1)\operatorname{tr} L_x. \tag{E.1}$$

Every L_x is self-adjoint. With $S = |P|^2$,

$$\mathcal{L}(F) = (d-1)\big(S\operatorname{tr} L_x - \langle L_x P, P\rangle\big). \tag{E.2}$$

In the harmonic case, $\operatorname{tr} L_x = 0$ for all x, and Bochner's formula and differentiation of S give

$$\Delta S = 2(d-1)^2 \operatorname{tr}(L_x^2), \qquad \Gamma(S,S) = 4(d-1)^2 |L_x P|^2. \tag{E.3}$$

Indeed, $\nabla S = 2(d-1)L_x P$ and $\Delta|\nabla F|^2 = 2|\operatorname{Hess} F|^2$ when $\Delta F = 0$. Thus $\mathcal{L}(F) = -qF$ is equivalent to

$$d(d-1)\langle L_x P, P\rangle = q\langle P, x\rangle$$

in this harmonic normalization. The weight q has degree $2d-4$ and need not be radial.

The polarization formula is an equivalent algebraic description of the polynomial, and the contractions above express the minimal surface equation in terms of a symmetric metrized operation. In higher degree this can organize the identities needed to compute $q, S, \Delta S, \Gamma(S,S)$ in Section 4.3. It becomes useful for a sign test when additional structure, such as Clifford relations, norm composition [71, Chapter 1, Appendix, pp. 21–22], or spectral minor identities, makes these contractions tractable. Polarization alone supplies neither a classification nor the required sign inequalities. For $d > 3$ the operation is $(d-1)$-linear, so the binary-algebra methods for cubics do not transfer automatically.

E.2. **Exceptional sets in the algebraic constructions.** For each cited potential $G = aF$, the multiplier is smooth and positive off the indicated set, and G extends continuously with the signs of F. Homogeneity gives the gradient check following Corollary 2.8.

Construction	Exceptional set Z	Ambient codimension
Theorem 6.5	$\{0\}$	N
Theorem 10.8	$\{S=0\}$, $S=\lvert\nabla F\rvert^2$	at least $2k$
Theorem 10.14	$\{z^{\#}=0\}$	$2(d+2)$
Theorem 11.5, $d=4$	$\{S=0\}$	10
Theorem 10.15	$\{\operatorname{rank}[a\ b\ c]\le 1\}$	12
Example 8.25	$\{Q=0\}$	at least $2k$
Example 8.32	$P^{-1}(0)$	at least 11
Theorem 11.7	$P^{-1}(0)$	at least $m\in\{14,26\}$
Theorem 8.22	$\{\lvert x\rvert\,\lvert y\rvert=0\}$	16
Proposition 8.31	$\{W=0\}$	14
Theorem 13.3	at least two zero factors	16
Theorem 13.6	$\{\operatorname{rank}_{\mathbb{C}} Z\le 2d-4\}$	12
Theorem I, $m\ge 2$	$\{\operatorname{rank}_{\mathbb{C}} Z\le m-2\}$	8

In the minimizing ranges, every listed set satisfies (2.12). Together with the cited sign certificates, this verifies Corollary 2.8.

E.3. **Calibrated complex cones with zero stability gap.** Dimler–Lee [27, Theorem 4.2] construct calibrated complex hypersurface cones with zero Hardy gap when degree equals complex dimension. The following extends this to all larger degrees.

Proposition E.1 (Area-minimizing complex cones without strict stability)**.** *Let $k\ge 2$, and let f be a homogeneous holomorphic polynomial of degree $d\ge k$ on $\mathbb{C}^{k+1}$, with $df(z)\neq 0$ for $z\neq 0$. Then $C=f^{-1}(0)$, with its complex orientation, is area minimizing of real dimension $2k$, and its Hardy stability gap is zero:*

$$\inf_{0\neq V\in C_c^\infty(NC_{\mathrm{reg}})}\frac{Q_C^\perp(V)}{\displaystyle\int_C r^{-2}|V|^2}=0. \tag{E.4}$$

Thus C is area minimizing but not strictly stable in the Hardy sense.

Proof. The Kähler calibration $\omega_0^k/k!$, where $\omega_0=\sum_{j=1}^{k+1}dx_j\wedge dy_j$, calibrates the complex current C [40]. Thus (2.8) gives area minimization and $Q_C^\perp\ge 0$.

Choose $g=\ell^{d-k}$, with ℓ a linear form nonzero at some point of $C\setminus\{0\}$, and take $g=1$ if $d=k$. Thus g has degree $d-k$ and does not vanish identically on C. Writing $f=u+iv$, holomorphicity gives $\nabla v=J\nabla u$: the gradients are orthogonal with equal length. Hence

$$V_g=\frac{\operatorname{Re} g\,\nabla u+\operatorname{Im} g\,\nabla v}{|\nabla u|^2}\quad\text{on } C_{\mathrm{reg}} \tag{E.5}$$

is normal and satisfies $df(V_g)=g$. Near each compact subset of C_{reg}, the holomorphic equations $f-tg=0$ form a smooth family of complex, hence minimal, hypersurfaces for small real t. Differentiating the equation and minimality at $t=0$ gives normal velocity V_g and $J_C^\perp V_g=0$.

Since $\deg g=d-k$ and $\deg\nabla u=d-1$,

$$V_g(r\theta)=r^{1-k}W(\theta),\qquad \theta\in\Lambda=C\cap\mathbb{S}^{2k+1},$$

where W is smooth and not identically zero on the compact smooth link. The degree $1-k=-(\dim_{\mathbb{R}}C-2)/2$ is the critical Hardy degree. Multiplying $J_C^\perp V_g=0$ by χ^2V_g and integrating by parts in (2.3) yields

$$Q_C^\perp(\chi V_g)=\int_C|\nabla_C\chi|^2|V_g|^2 \tag{E.6}$$

for every smooth compactly supported scalar cutoff on C_{reg}. Take a nonzero $\zeta\in C_c^\infty((-2,-1))$ and, for $L>1$, set $\chi_L(r)=\zeta(\log r/L)$. Writing $b=\int_\Lambda|W|^2>0$ and using $d\mu_C=$

$r^{2k-1}\,dr\,d\mu_\Lambda$, we obtain

$$Q_C^{\perp}(\chi_L V_g) = \frac{b}{L}\int_{\mathbb{R}} |\zeta'|^2, \qquad \int_C r^{-2}|\chi_L V_g|^2 = bL\int_{\mathbb{R}} |\zeta|^2.$$

The quotient is $L^{-2}\int|\zeta'|^2/\int|\zeta|^2$, which tends to zero. Together with stability, this proves (E.4). □

Appendix F. Computational Verification

The accompanying exact-arithmetic package and its execution record are archived in the fixed release

`HongbinCui/laplacian-subcalibrations: verification-20261012-v15`.

The same verification files are supplied in the ancillary archive `verification-package.zip`. Using Theorem 3.10, the programs check identities and exact sign certificates. The index records tested orders and distinguishes checks conditional on prescribed closure tables from checks in Euclidean coordinates. The rational Bernstein certificate in Appendix C proves positivity on its entire interval. The determinant recurrence and bounds are proved in arbitrary order in Section 14; bounded-order computational checks only supplement those proofs.

The Wolfram Language suite requires Mathematica 12.0 or later and runs by evaluating the input cell in `StartHere.nb`, including closure identities, orthogonal quartic families, the complex Gram sextic, and determinant initial-form checks. Each run records its date, Mathematica version, operating system, source hashes, and individual results in a new directory. The README supplies reproduction instructions; `CertificateIndex.md` gives each unit's scope and statement/equation LaTeX labels. Its printed equation numbers and page locations refer to the bundled V7 manuscript snapshot; this article retains the mathematical labels with a different layout.

The author ran the complete suite in Wolfram Mathematica 12.0.0 (Windows 64-bit) on October 12, 2026: all 26 units and 7,134 checks passed. The accompanying results directory records the execution, source hashes, and individual results.

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AI disclosure. ChatGPT (OpenAI) was used to assist with language editing, notation and organization, literature searches, mathematical exploration, the preparation of computer code for symbolic and algebraic computations, and AI-assisted figure generation based on the author's sketches and instructions. The author ran the supplied verification code in Wolfram Mathematica 12.0.0 (Windows 64-bit) on October 12, 2026; the complete execution record accompanies the source files. The author takes full responsibility for the mathematical content and computational conclusions.

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